

Curvature, support certificates, and barrier complexity of conic lifts

Abstract

Conic lifts change factor counts, support-certificate ranks, affine barrier parameters, and computational interfaces in different ways. We establish quantitative relations among these resources for balls and related convex bodies. Mixed contact curvature gives a dimension-minus-two bound for proper definable cones and a complementary-rank bound for symmetric cones. For a finite repeatable dictionary of simple symmetric cones, put $B = \max a(r - 1) > 0$, where a is the Peirce constant and r the Jordan rank. The exact least worst-support certificate rank of an s -dimensional ball, $s \geq 2$, is $\lceil (s - 1)/B \rceil$, with minimization over each entire genuine certificate fiber and no global selection assumption. Global differentiable selections can impose additional topological costs. Boundary and recession tests yield delimited exact restricted-barrier frontiers and concrete intrinsic parameter formulas. For a fixed unit objective in a fixed weighted norm-tree metric, call a node active if its subtree carries nonzero objective support. The squared leading distance coefficient is the sum of active node weights plus half the inactive node weights; this determines the central route's asymptotic excess. For feasible PSD column packing, exact source allocations give a sharp first-power comparison between off-center and centered projected Hessian quotients. We also give resource ledgers, conditioned approximation and exact compilation results, and access-dependent Newton and query comparisons. The remaining bounded narrow-cap and nonsymmetric parameter intervals are stated explicitly. Movement bounds imply work bounds only under the specified charging and output contracts.

Contents

1	Introduction	7
1.1	Principal results and their scope	7
1.2	Relation to the literature	8
1.3	Conventions and reading guide	11
I	Local curvature and certificate fibers	12
2	Exact lifts and local conventions	12
2.1	Exact lifts and genuine certificates	12
2.2	Regularity obtained from the lift	13
3	Local curvature and cone resources	13
4	The exact support-certificate rank frontier	16
4.1	A uniform Peirce-perspective construction	17
4.2	Examples and the order of quantifiers	18
5	Products: certificates, compression, and private curvature	19
5.1	The support-join identity	19
5.2	Private directions in Hermitian factors	20
II	Global regularity and affine barriers	22
6	Global regularity changes the formulation frontier	22
6.1	Lorentz phases and pointwise no-sharing	22
6.2	A top-class cover argument	22
6.3	Why local and joint identities are weaker	24
7	Support-orbit rigidity and selected ranks	27
7.1	Saturation and support orbits	27
7.2	Eliminating the real order-three exception	29
7.3	The selected-rank frontier for a fixed dictionary	30
8	The standard barrier after affine restriction	33
8.1	The parameter and two boundary tests	33
8.2	A complete selection-free result at one channel	35
8.3	Arbitrarily many factors under a wide cap	36
8.4	What a remaining bounded lift would have to do	37
8.5	Counterexamples to stronger fiberwise conclusions	39
9	Exact parameters of concrete affine formulations	40
9.1	Norm trees: the root incidence determines the standard parameter	41
9.2	Grouped fixed-tail slices: coupling cannot improve the parameter	43
9.3	Bounded-fiber projection and one-block column packing	44

10	Topology of saturated contact factorizations	46
10.1	Normalized cone bases and the joint covering	46
11	Capacity equality for products of balls	48
12	Rigidity from entire boundary fibers	52
III	Formulation families and barrier comparisons	55
13	Whole rows, minimum dimension, and affine normalization	55
13.1	Contact faces and exact grouping	55
13.2	Rigidity at the absolute dimension minimum	57
13.3	What one normalization equation does imply	59
13.4	Recession normalization and barrier transfer	59
14	Complementary faces control sharing between source rows	61
14.1	A sharp model for shared scalar channels	63
15	Balance slices with sharp dimension and barrier bounds	65
16	Norm balls and weighted power cones	68
16.1	Exact counts and the dimensions of count-optimal lifts	68
16.2	Global selections and the Young construction	70
16.3	A curvature bound for generalized Euclidean power cones	72
17	Matrix balls and sparse completion cones	73
17.1	Mixed contact rank with split rows	74
17.2	Whole-row grouping and exact spectral parameters	75
17.3	Completion cones and separator credits	77
17.4	Exact equality of the reduced geometries	79
18	Nonsymmetric cones: barrier premiums and evaluation costs	79
18.1	An additive premium for fixed cone dictionaries	79
18.2	Grouped cones: duality and a parameter ledger	81
18.3	Necessary scales for characteristic barriers	82
18.4	The three-dimensional gap and failed explicit formulas	83
19	Exponential factors and entropy aggregation	85
19.1	A reusable recession certificate	85
19.2	Positive output maps and their boundary fibers	86
19.3	A sparse intrinsic Hessian	87
IV	Approximation and exact compilation	89
20	Approximation with controlled cone factors	89
20.1	A finite sampling theorem	89
20.2	Smooth near-complementarity	90
20.3	Why metric accuracy alone is insufficient	92

21	Exact compilers and the limits of representation-specific rank	92
21.1	Intersections of translates	93
21.2	Reusable summaries of perspective data	95
21.3	A two-moment relative-entropy example	96
21.4	Exponential features versus exponential-cone factors	97
22	Projection, sufficient statistics, and limits of local compilation	99
22.1	Polyhedral and ordered recourse	99
22.2	Additive quadratic and matrix sufficient statistics	101
22.3	A scalar normalization cannot repair every convolution barrier	103
22.4	Two obstructions to reusable exact summaries	103
22.5	The parameter cost of additive local logarithms	104
V	Movement in barrier metrics	106
23	Certificate ranks and movement in a fixed barrier metric	106
23.1	The weighted exposed-minor estimate	106
23.2	Composing the rank frontier with movement	108
23.3	Affine PSD pencils and spectral scales	109
24	Arbitrary barriers and the distinction between primal and dual movement	111
24.1	The feasible gap set in the product metric	111
24.2	A dimension envelope for arbitrary cone dictionaries	112
24.3	An explicit allocation on shared product balls	113
24.4	Accuracy and movement from the objective-weight distribution	114
25	Movement in concrete affine formulations	115
25.1	Barrier height and simultaneous slack contraction	116
25.2	Grouped balls and unrestricted PSD completion	116
25.3	Norm trees: uniform bounds and weighted exact central paths	118
25.4	A separate control on central-path parameter changes	120
25.5	The balance slice attains its exposed-rank scale	121
25.6	Shared spectral cones preserve the reduced movement	121
26	Exact distance coefficients for weighted norm trees	122
VI	Resource ledgers and computational contracts	125
27	Exact resource ledgers and their optimization	125
27.1	Real PSD order caps	125
27.2	Grouped Schur constructions and small-order exact frontiers	126
27.3	All finite column-packing tradeoffs	127
27.4	Joint nullity envelopes and a general resource inequality	128

28	When geometric movement composes with computational work	130
28.1	The same aggregate certificate in both estimates	130
28.2	A conditional total-work theorem	131
28.3	Attainment in specified output models	132
29	Newton systems: elimination, packing, and access	133
29.1	The common marginal barrier and its Newton equation	133
29.2	A sharp comparison away from the fiber center	135
29.3	What exact recentering requires	137
29.4	Sparse augmented Lorentz systems	140
29.5	Exact graph counts for the ball constructions	142
29.6	The full-output comparison and its boundary	143
30	Query access and classical output	144
30.1	Access conventions and an information bound	144
30.2	Exposed rank, movement, and sign output on one lift	145
30.3	Well-conditioned disk checkpoints with hard scalar readout	147
30.4	Sharp precision laws with public magnitude data	147
30.5	Several independent search diagnostics	149
30.6	The same conclusions with one cone factor	150
31	Acquiring an exact smaller formulation	151
31.1	Output and access conventions	151
31.2	Nested constraints and consensus	152
31.3	A two-dimensional family with incomparable active factors	154
31.4	A growing-dimensional fixed-objective decoder	155
31.5	Blockwise relative-entropy compilation	156
31.6	What the oracle advantage includes	158
32	Closing synthesis	158
	Appendices	160
A	Further topological restrictions	160
A.1	What the abstract tangent splitting alone implies	160
A.2	Restrictions on one everywhere saturated factor	161
A.3	Balanced real projector orbits	162
B	Restricted kernels: embeddings, locality, and flat residuals	163
B.1	Projective contact forces an embedding	163
B.2	What local tests and separated constructions can establish	165
B.3	Column ranks detect curvature-flat residuals	167
C	Global obstructions for conditioned approximate factors	169
C.1	A robust tangent splitting	170
C.2	An invariant of fixed block reparameterizations	171
C.3	Rank-one phase dropout	171
C.4	Higher-rank dropout from forbidden sphere submersions	173

D Contact sensitivity and Newton conditioning	174
D.1 A sparse change of cone coordinates	174
D.2 Bounded pointwise geometry and unbounded contact derivatives	175
D.3 What an additional sensitivity hypothesis would give	176
References	177

1 Introduction

An exact conic reformulation represents a convex body as the projection of an affine slice of a product of cones. It can change the number of factors, their dimensions, the rank of a supporting certificate, and the barrier on the affine domain. It can also change which data a Newton step needs and which outputs are inexpensive to produce. These resources are related, but none determines all the others. The purpose of this paper is to give precise comparisons, sharp formulas in specified models, and explicit examples that separate the models.

Euclidean balls and their products provide the main setting. Their slack has a simple differential structure, yet their lifts already exhibit certificate sharing, singular boundary choices, nontrivial projection fibers, and different barrier metrics. We also treat norm balls, matrix balls, balance slices, power and entropy cones, and exact constraint compilers. Throughout, an exact formula is accompanied by the class over which it is optimized. A lower bound for a chosen representation is not automatically a lower bound for every representation of its projection.

1.1 Principal results and their scope

The basic geometric calculation differentiates a nonnegative cone pairing at a primal–polar contact. A proper cone of real dimension m supplies at most $\max\{m - 2, 0\}$ independent mixed-curvature directions. In a simple Euclidean Jordan algebra, complementary primal and dual ranks p, q sharpen this count to apq , where a is the Peirce constant (Theorem 3.2). Definable local selections suffice; no globally smooth selection is needed. For a ball of dimension $s \geq 2$ and an integer cone-dimension cap $d \geq 3$, the exact minimum factor count and total cone dimension over exact definable lifts are consequently

$$k = \left\lceil \frac{s-1}{d-2} \right\rceil, \quad M = s - 1 + 2k.$$

Classical norm trees attain both bounds (Corollary 3.4).

Our central rank invariant minimizes over the *entire fiber of genuine certificates for a fixed support*. A genuine certificate represents the support slack on the whole affine slice, rather than only on selected boundary points. Fix a finite repeatable dictionary $\mathcal{K}$ of simple symmetric cones and put $B = \max_{K \in \mathcal{K}} a(K)(r(K) - 1) > 0$. For every $s \geq 2$, Theorem 4.1 proves

$$\inf_{\mathcal{L}} \max_{v \in \mathbb{S}^{s-1}} \min_{Y \in \mathcal{D}_{\mathcal{L}}(v)} \sum_i \text{rank}_J Y_i = \left\lceil \frac{s-1}{B} \right\rceil. \quad (1.1)$$

The lift is fixed before the support is chosen. For each fixed lift the lower bound holds for every certificate on a dense open set of supports with surface-null complement. A Peirce-perspective construction attains the bound in every simple family, including the Albert cone. It also shows why choosing the formulation after the objective changes the answer. Sequential compression and private curvature describe which rank costs add for product sources (Section 5).

Global selection is a different requirement. If primal boundary points and dual certificates must be chosen C^1 over the entire support manifold, saturation of a local rank bound forces a global map into a product of support orbits or cone-base boundaries. Covering and cohomology arguments then impose additional resource costs. We prove exact Lorentz frontiers and selected symmetric-cone rank frontiers, together with product saturation and whole-fiber rigidity theorems (Sections 6–12). The whole-fiber results derive regular structure from a stronger fiber hypothesis; they do not assume that arbitrary lifts have smooth selectors.

The barrier analysis distinguishes three optimization problems: the standard product barrier in the ambient cone, that same barrier after affine restriction, and the best barrier of any form on a fixed

affine domain. Boundary nullity and recession give lower tests for the restricted standard parameter. These tests yield exact minima under global selection, several selection-free exact cases, and intrinsic optima for grouped, packed, and balance slices. Norm trees have a separate root-incidence law. The results do not constitute a complete classification. For the remaining bounded narrow-cap cases in (8.6), the family is a fixed-field Hermitian order cap or a Lorentz dimension cap, and B is that cap's primitive capacity. With $s = qB + 1$ and $q \geq B + 2$, the proved interval is $q \leq \inf_{\mathcal{L}} \nu_{\text{std,slice}} \leq q + 1$. The nonsymmetric estimates in Section 18 likewise include parameter intervals, not general formulas for their optima.

Movement is always measured in a specified Hessian metric. Exposing certificates yield path-independent target-distance bounds, and the rank frontier makes these bounds depend on the permitted lift dictionary. Primal–dual movement for arbitrary logarithmically homogeneous barriers has a different endpoint and metric contract from primal movement in a displayed barrier. An explicit refinement determines the shortest-distance coefficient for every fixed weighted norm tree and fixed unit leaf objective c . If $\omega_v \geq 1$ is the barrier weight of internal node v , call v active when its subtree contains nonzero objective support. For the target $1 - c^T w \leq \epsilon$, Theorem 26.1 gives, from any fixed interior reference,

$$d(\epsilon) = \sqrt{K_c} \log(1/\epsilon) + O(\log \log(1/\epsilon)), \quad K_c = \sum_{v \text{ active}} \omega_v + \frac{1}{2} \sum_{v \text{ inactive}} \omega_v. \quad (1.2)$$

The corresponding central route has coefficient $\sqrt{\sum_v \omega_v}$. Thus inactive subtrees contribute to unavoidable distance even though their exposing dual blocks vanish, and they also produce a precisely quantified excess in central-path length. The tree, weights, and objective are fixed before ϵ tends to zero.

For packed PSD representations, exact auxiliary elimination identifies a common marginal Newton system at the fiber center. Away from that center, the source allocation equations improve a generic squared comparison to a sharp first-power transfer. At the same projected point, if the feasible residuals and their exact diagonal center obey $\mu D^0 \preceq D \preceq L D^0$, where $0 < \mu \leq 1 \leq L$, Theorem 29.3 proves

$$L^{-1} H_0 \preceq H_D \preceq \mu^{-1} H_0, \quad \kappa(H_0^{-1/2} H_D H_0^{-1/2}) \leq L/\mu. \quad (1.3)$$

A two-source feasible example attains the condition factor. Exact allocations are essential: approximate radial metadata needs its own allocation error factor. The matrix comparison alone neither identifies the reduced Newton right-hand side nor constructs a cheap preconditioner.

Finally, we specify when mathematical resources become computational costs. Exact resource ledgers distinguish ambient coordinates, free coordinates, and restricted parameters. Sparse elimination gives classical comparisons under matched access and full-output contracts. Query examples separate normalized projected states from readable values, full vectors, exact reusable compilers, and fresh scale maintenance. Some of these states need one query while classical output requires linearly many queries. Our rank-dependent work bound retains the same exposing certificate in the curvature and movement estimates and assumes an explicit fresh charge per round (Theorem 28.1). A one-shot query bound cannot simply be multiplied by a number of geometric checkpoints.

1.2 Relation to the literature

Conic lifts and slack factorizations are the starting framework of Gouveia, Parrilo, and Thomas [43, Theorem 2.4]; the survey Fawzi et al. [37] places their use among geometric and algebraic methods for concise convex descriptions. Fixed-size PSD factor counts [33], PSD rank and algebraic-boundary

Table 1: Main results by model. Each cited statement gives its full hypotheses. “Exact” refers only to the indicated optimization class.

Model and resource	Conclusion and scope
Exact definable lifts; factor count and cone dimension	Positive boundary curvature gives the dimension-minus-two bound. Ball norm trees attain the capped count and dimension minima (Theorem 3.2, Corollary 3.4).
Symmetric-cone lifts; minimum certificate rank	Entire-fiber minimization, then worst support, then lift minimization gives $\lceil (s-1)/B \rceil$ without global selection (Theorem 4.1).
Globally C^1 full-slack selections; rank and factor resources	Exact Lorentz and selected symmetric-cone frontiers; saturation forces support-orbit or cone-base topology. This is a stronger model than arbitrary affine lifts (Theorems 6.2, 7.7, 7.8).
Restricted standard barrier; arbitrary barrier on a fixed domain	Boundary/recession lower tests, delimited exact lift minima, and concrete intrinsic optima. The bounded interval (8.6) remains unresolved (Sections 8, 9, 15).
Fixed barrier metric; distance to an accurate target	Exposed-minor bounds and exact weighted tree coefficient $\sqrt{K_c}$. The separate all-LH result concerns a central start and feasible primal-dual gap endpoint (Theorems 23.1, 26.1, 24.2).
Feasible PSD packing; projected Hessian quotient	Exact centered reduction and sharp relative condition factor L/μ at the same projected point, with the source allocations fixed (Proposition 29.1, Theorem 29.3).
Specified access, output, and fresh-work contracts	Conditional work composition, sparse full-output comparisons, state/readout separations, and exact compiler acquisition costs (Sections 28–31). Fresh adversarial updates in Theorem 29.7 are not a fixed program’s Newton trajectory.

bounds [34, 36], and semidefinite extension degree [8] measure related but different resources. Extension degree minimizes the largest permitted PSD block while allowing finite products. Fawzi’s SOC obstruction and Saunderson’s short-face-chain theorem use neighborliness to exclude finite product lifts in their respective settings [32, 80]. Our local curvature bounds quantify resources when a smooth body is representable; our certificate invariant also minimizes within each fixed support fiber. Kummer’s ball bounds concern direct spectrahedral descriptions [58], which must be distinguished from projections of products of cones.

Positive curvature itself does not obstruct SOC representability. Scheiderer [81, Theorem 1.2] proves that compact convex semialgebraic sets with Nash-smooth boundary and everywhere strictly positive curvature have SOC lifts. That existence theorem allows an arbitrary finite number of order-two PSD blocks. It does not specify the factor-count frontier, support-fiber minimax, or global-selection frontiers studied here. This distinction is particularly relevant because both approaches apply to positively curved boundaries.

The Jordan classification, Peirce spaces, quadratic representations, and support manifolds are classical [31, 45, 73]. Symmetric-cone factorization algorithms provide additional context [82]. Recursive small-Lorentz-cone norm representations are also classical [13, Section 2]. We use standard covering, bundle, and cohomology tools [48, 83]; their application here depends on the precise global contact hypotheses. The claimed geometric contributions are the stated quantitative inequalities, fiber quantifiers, rigidity consequences, and attaining constructions, not these underlying tools.

Self-concordant barrier theory supplies the analytic framework [70]. Ambient optimality on symmetric cones is established in the work of Güler–Tunçel and Cardoso–Vieira [20, 47]; self-scaled barriers have a classified form [49]. Conic extension, affine restriction, and projective transformation of barriers belong to an existing theory [55]. Accordingly, the distinction between ambient and restricted parameters is not itself a claim of novelty. Our contribution is to compute the particular restricted or intrinsic quantities proved here and connect them to exact lift resources. For nonsymmetric cones we use established lower-bound and barrier constructions [22, 35, 53]. The universal barrier’s parameter bound [60] and practical natural-cone methods [24] also make clear why existence of a small parameter should be separated from derivative evaluation cost.

Nesterov–Todd and Nesterov–Nemirovski develop the barrier geometry and the comparison of central paths with Riemannian distances [69, 72]. The unpublished companion *The Cost of Following the Central Path* [21] already contains the weighted exposed-minor estimate, several distribution-sensitive movement bounds, and standard-barrier calculations for grouped, packed, and norm-tree formulations. We reproduce the proofs needed here, so no theorem depends on access to that companion. The formulation-specific developments here include composition with the certificate-rank frontier, the general affine-PSD exposure profile, and the exact active/inactive weighted-tree coefficient (1.2). We do not claim the general distinction between central-path length and shortest distance as a new observation.

The computational arguments likewise use established ingredients: search, oracle interrogation, approximate counting, and information bounds [14, 18, 66, 67, 84]; sparse cone augmentation [27]; and elimination from a supplied tree decomposition [39]. Our statements attach these tools to specified conic representations and outputs. Quantum interior-point methods and their end-to-end resource analyses [7, 26] are relevant algorithmic comparators. More recent iterative refinement improves the high-precision behavior of quantum interior-point methods [62]; we make no blanket claim that all such methods retain earlier inverse-accuracy dependencies. Our exact-field full-output comparisons and the feasible quotient theorem (1.3) have their own explicit access, feasibility, and output assumptions. In particular, the first-power improvement uses elementary positive trace inequalities and the allocation equalities; it is not a new general matrix-order principle.

The specific results we put forward as new are the selection-free entire-fiber minimax (1.1),

the active/inactive weighted-tree coefficient (1.2), and the sharp feasible allocation-aware quotient transfer (1.3), in the precise models stated above. These claims concern the formulas and their quantifiers, not priority for the general methods used to prove them. The comparisons above identify the scope of the checked primary results; absence from a literature search is not evidence of priority.

1.3 Conventions and reading guide

Table 2 collects the recurring notation. Symbols such as p, q, L have local uses in some construction and counting sections; those uses are defined where introduced. A theorem’s stated domain, metric, and order of optimization always take precedence over an informal resource name.

Table 2: Recurring notation and conventions.

Notation	Meaning
$\mathcal{L} = (K, A, \pi)$	Exact affine lift; $K = \prod_i K_i$. Relative Slater reduction is made before computing ranks and barriers (Lemma 2.1).
$\mathcal{D}_{\mathcal{L}}(v)$	Positive certificates representing $1 - \langle \pi(X), v \rangle$ for every $X \in A$, including infeasible slice points.
$s, s_a; m_i$	Projected ball dimensions; real cone-space dimensions. A Lorentz cone Q_m has total dimension m .
$r_i, a_i; p_i, q_i$	Jordan rank and Peirce constant of a simple factor; complementary primal and dual contact ranks. rank_J denotes Jordan rank; real PSD Jordan rank is ordinary matrix rank.
$B; a_i \lfloor r_i^2/4 \rfloor$	Maximum primitive capacity per dual rank in a dictionary; maximum mixed-curvature capacity of one factor. These capacities are different.
$\nu_{\text{std}, \text{slice}}$	Least parameter of the specified standard log-determinant barrier on its effective affine slice; see (8.1).
$\vartheta_{\text{opt}}(\Omega); \nu_{\text{LH}}(C)$	Infimum over all self-concordant barriers on a fixed affine domain; infimum restricted to logarithmically homogeneous barriers on a fixed cone, respectively.
d_F, ϵ	Distance in the Hessian metric of the displayed barrier; objective-gap or output accuracy as locally specified. Distances to targets minimize over paths in the stated metric domain, with the endpoint constraints of the relevant theorem.
PSD order; asymptotic notation	$H \preceq G$ means $G - H$ is positive semidefinite. Constants and fixed-instance versus joint dimension/accuracy limits are stated in each result.

Part I develops local curvature and certificate fibers. Part II treats global selection, restricted barriers, and saturation. Part III gives formulation families and nonsymmetric comparisons. Part IV addresses controlled approximation and exact compilation. Part V connects certificates to metric distance, culminating in the weighted tree theorem. Part VI gives resource ledgers and computational contracts, including the sharp PSD quotient comparison. The appendices retain topological refinements, restricted-kernel distinctions, robust approximation, and conditioning counterexamples. Each part supplies the proofs needed for its formulation claims; the contents and model table allow readers to follow a particular resource without treating every result as a theorem about the same class of lifts.

Part I

Local curvature and certificate fibers

2 Exact lifts and local conventions

Throughout, cone-dimension and matrix-order caps are integers.

2.1 Exact lifts and genuine certificates

A *proper cone* is a closed, convex, pointed cone with nonempty interior in its finite-dimensional real ambient space. Its dual is $K^* = \{y : \langle x, y \rangle \geq 0 \text{ for all } x \in K\}$. Let $C \subset \mathbb{R}^s$ be a compact convex body with nonempty interior and $0 \in \text{int } C$. Its polar body is $C^\circ = \{v \in \mathbb{R}^s : \langle x, v \rangle \leq 1 \text{ for every } x \in C\}$. An exact lift is a triple

$$\mathcal{L} = (K, A, \pi), \quad C = \pi(K \cap A),$$

where $K = \prod_{i=1}^k K_i$, A is an affine subspace, and $\pi : A \rightarrow \mathbb{R}^s$ is affine. No compactness of $K \cap A$ is assumed. A lift is *proper* if $A \cap \text{int } K \neq \emptyset$. When we speak of relative Slater feasibility, we use the following minimal-face reduction.

Lemma 2.1 (Reduction and certificate normalization). *Unrestricted free variables in a finite affine lift of a compact body can be eliminated without changing its cone factors. The cone can then be replaced by the smallest face containing its feasible points, in the span of that face, so that the resulting lift is proper. For a proper lift and every $v \in \partial C^\circ$, the set*

$$\mathcal{D}_{\mathcal{L}}(v) = \{Y \in K^* : \langle X, Y \rangle = 1 - \langle \pi(X), v \rangle \text{ for every } X \in A\} \quad (2.1)$$

is nonempty. Thus certificates satisfy the support-slack identity on the entire affine slice, including its infeasible points.

Proof. Write the original affine equalities in cone coordinates X and free coordinates z . A free direction that preserves these equalities while fixing X must have zero output direction: otherwise its entire line projects into the compact body. Consequently the output depends affinely only on X , on the affine subspace of X for which the equalities admit some z . Linear elimination gives the claimed representation.

Choose $X_0 \in \text{ri}(K \cap A)$, and let F be the minimal face of K containing X_0 . For every $X \in K \cap A$, the relative-interior property gives $X_0 - \epsilon(X - X_0) \in K \cap A$ for some $\epsilon > 0$. The face property therefore gives $X \in F$. Conversely $F \cap A \subset K \cap A$. The point X_0 belongs to $\text{ri } F$, so replacing K by F and A by $A \cap \text{span } F$ gives Slater feasibility in $\text{span } F$. Faces of a product are products of faces.

For the last assertion write $A = \{X : MX = b\}$ and extend the affine output as $\pi(X) = TX + t$. Maximizing $\langle v, TX + t \rangle$ over $K \cap A$ has value one. Slater conic duality gives a multiplier λ with $Y = M^* \lambda - T^* v \in K^*$ and $\langle b, \lambda \rangle = 1 - \langle t, v \rangle$. Hence for every $X \in A$, $\langle X, Y \rangle = \langle b, \lambda \rangle - \langle TX, v \rangle = 1 - \langle \pi(X), v \rangle$. This is the duality step in the proper-lift direction of [43, Theorem 2.4]. $\square$

For symmetric cones, every face is the cone of squares in a Peirce-one subalgebra. Jordan ranks in that face agree with ambient ranks of its members. Throughout, ranks, determinants, and barriers of a relative-Slater lift are evaluated *after* this reduction. This convention avoids requiring a positive ambient certificate when the original presentation fails Slater's condition. Faces need not themselves belong to the chosen dictionary: they are reductions of its factors, and their capacities never exceed those of the original factors.

For $C = \mathbb{B}_2^s = \{x : \|x\|_2 \leq 1\}$ its polar is itself, and a support is $v \in \mathbb{S}^{s-1}$. Define

$$r_{\mathcal{L}}(v) = \min_{Y \in \mathcal{D}_{\mathcal{L}}(v)} \sum_i \text{rank}_J Y_i, \quad R(\mathcal{L}) = \max_{v \in \mathbb{S}^{s-1}} r_{\mathcal{L}}(v). \quad (2.2)$$

Both extrema exist as extrema of nonempty subsets of a finite set of integers; compactness of the certificate fiber is unnecessary. The zero certificate is impossible because the support slack is not identically zero on C . This observation will also explain the separate interval case $s = 1$.

2.2 Regularity obtained from the lift

We call data *definable* if they are definable, allowing real parameters, in one fixed o-minimal expansion of the ordered real field. Only the universal-cone statement below needs this generality. All symmetric-cone data and selections used for the rank theorem are semialgebraic. Definable choice and finite differentiable stratification are standard [85]; in the semialgebraic setting see [16].

Lemma 2.2 (Generic regular selections). *For a proper definable lift, primal boundary lifts and normalized support certificates can be chosen definably. On a definable C^2 boundary patch, a dense relatively open subset makes their compositions with the definable normalized outward-normal contact map C^1 , in definable local coordinates. For a symmetric-cone lift of a Euclidean ball, the certificates can additionally be chosen to attain $r_{\mathcal{L}}(v)$. The common regular subset can then be chosen semialgebraic, with a lower-dimensional complement in $\mathbb{S}^{s-1}$.*

Proof. The primal fiber and the fiber (2.1) are nonempty closed definable sets; the identity on A amounts to finitely many affine equalities. Their minimum-Euclidean-norm points are unique, since both fibers are convex, and their graphs are definable by quantified formulas. For minimum Jordan rank, each condition $\sum_i \text{rank}_J Y_i = j$ is semialgebraic by spectral polynomial equations and inequalities. Partition the support sphere according to the least realized j and apply semialgebraic choice in each resulting nonempty fiber. This choice need not minimize norm and need not be globally continuous.

Finite C^1 cell decomposition, refined for all the maps at once, makes them C^1 on each full-dimensional cell. The union of the interiors of those cells is relatively open and dense; its complement has smaller dimension. In the semialgebraic case this complement has surface measure zero. A definable C^2 boundary patch and its normalized normal map are eligible for the same decomposition. Faces used in reduction are definable: for $X_0 \in \text{ri } F$, $F = \{X \in K : \exists \epsilon > 0, X_0 - \epsilon X \in K\}$. $\square$

3 Local curvature and cone resources

Let $x : U \rightarrow \partial C$ be a definable C^2 local parametrization, where $U \subset \mathbb{R}^{s-1}$, and let $n(u)$ be its outward unit normal. Set

$$y(u) = \frac{n(u)}{\langle x(u), n(u) \rangle}.$$

The denominator is positive since $0 \in \text{int } C$, and $y(u)$ lies on ∂C° . Select $X_i(u), Y_i(u)$ as in Lemma 2.2. The whole-slice certificate identity gives

$$1 - \langle x(u), y(v) \rangle = \sum_i \langle X_i(u), Y_i(v) \rangle. \quad (3.1)$$

Each summand is nonnegative and vanishes on the diagonal. The negative mixed derivative is the bilinear form

$$\mathbf{H} = Dx(0)^* Dy(0) = \frac{Dx(0)^* Dn(0)}{\langle x(0), n(0) \rangle}. \quad (3.2)$$

At a strictly positively curved point it has rank $s - 1$.

Lemma 3.1 (Universal local bound). *Let $K \subset E$ be a proper cone, $m = \dim E$, and let $X : U \rightarrow K$, $Y : U \rightarrow K^*$ be C^1 with $\langle X(u), Y(u) \rangle = 0$. The bilinear form $-\langle DX(0)[h], DY(0)[\ell] \rangle$ has rank at most $m - 2$ if $X(0), Y(0) \neq 0$, and rank zero if either is zero. In particular rays contribute no mixed curvature.*

Proof. Put $a = X(0)$ and $b = Y(0)$. When both are nonzero, nonnegativity of $\langle X(th), b \rangle$ for positive and negative t shows $DX(0)[h] \in b^\perp$. Similarly $DY(0)[\ell] \in a^\perp$. The pairing from $b^\perp$ to $(a^\perp)^*$ has kernel $b^\perp \cap \text{span}\{a\} = \text{span}\{a\}$, since $\langle a, b \rangle = 0$. Its rank is $(m - 1) - 1 = m - 2$. Pullback by the derivatives cannot increase rank. If $a = 0$, a two-sided derivative of X lies in $K \cap (-K) = \{0\}$; the case $b = 0$ is identical. $\square$

Theorem 3.2 (Curvature bounds). *Suppose a compact s -dimensional body has a relatively open C^2 boundary patch with strictly positive curvature at some point, and an exact lift by a finite product of definable proper cones of dimensions m_i . Then*

$$s - 1 \leq \sum_i \max\{m_i - 2, 0\}. \quad (3.3)$$

If the factors are simple symmetric cones of ranks r_i and Peirce constants a_i , one can choose a contact at which the complementary primal and dual ranks p_i, q_i satisfy

$$s - 1 \leq \sum_i a_i p_i q_i \leq \sum_i a_i \left\lfloor \frac{r_i^2}{4} \right\rfloor. \quad (3.4)$$

A ray has $r_i = 1$, $a_i = 0$ in this notation.

For completeness we establish the sharper local assertion before proving the theorem. In a Euclidean Jordan algebra V , write $x \circ y$ for its product, e for its identity, and $\langle x, y \rangle = \text{tr}(x \circ y)$ for the trace inner product. The cone K consists of squares; $x \succeq 0$ means $x \in K$. A Jordan frame $c_1, \dots, c_r$ gives the orthogonal Peirce decomposition $V = \bigoplus_{i \leq j} V_{ij}$, where $V_{ii} = \mathbb{R}c_i$ and $\dim V_{ij} = a$ for $i < j$ in a simple algebra. The support idempotent of $x \succeq 0$ is the sum of the frame idempotents with positive eigenvalues; their number is $\text{rank}_J x$.

Lemma 3.3 (Mixed Peirce channels). *Let $X, Y : U \rightarrow K$ be C^1 and complementary: $\langle X(u), Y(u) \rangle = 0$. If $X(0)$ and $Y(0)$ have ranks p and q , then $p + q \leq r$, and $-\langle DX(0)[h], DY(0)[\ell] \rangle$ is a positive semidefinite symmetric bilinear form of rank at most apq . No local constancy of ranks or strict complementarity is required.*

Proof. Positive orthogonal elements are Jordan complementary and diagonalize in a common Jordan frame [31, Chapters III–IV]. Thus

$$X(0) = \sum_{i \in I} \lambda_i c_i, \quad Y(0) = \sum_{j \in J} \mu_j c_j, \quad \lambda_i, \mu_j > 0, \quad I \cap J = \emptyset.$$

For every positive z in the Peirce subalgebra on the complementary indices I^c , $\langle X(th), z \rangle \geq 0$ vanishes at $t = 0$. Its derivative is zero. Since its positive cone spans that subalgebra, the derivative of X has zero projection onto it. Likewise the derivative of Y vanishes on the J^c subalgebra. The only possible overlap of the two derivatives is therefore $\bigoplus_{i \in I, j \in J} V_{ij}$, of dimension apq .

Differentiate $X(u) \circ Y(u) = 0$. Its V_{ij} component is $\mu_j(DX[h])_{ij} + \lambda_i(DY[h])_{ij} = 0$, because multiplication by either endpoint idempotent acts by one half on V_{ij} . Hence

$$-\langle DX[h], DY[\ell] \rangle = \sum_{i \in I, j \in J} \frac{\mu_j}{\lambda_i} \langle (DX[h])_{ij}, (DY[\ell])_{ij} \rangle. \quad (3.5)$$

This is the required Gram form. Empty I or J gives the zero form; alternatively use the pointed-vertex argument of Lemma 3.1. $\square$

Proof of Theorem 3.2. Reduce the lift by Lemma 2.1. The strictly positive curvature condition persists on an open subpatch. On that subpatch choose a contact in the common regular locus supplied by Lemma 2.2. Differentiating (3.1) once in each independent variable and using rank subadditivity with Lemma 3.1 gives (3.3). A reduced face has dimension no larger than its original factor, so the original dimensions give valid bounds. For symmetric cones apply Lemma 3.3 instead. The inequality $pq \leq \lfloor r^2/4 \rfloor$ follows from $p + q \leq r$. Faces of a simple symmetric cone have the same Peirce constant at smaller rank, except that rank-one faces are rays. For the Albert cone its rank-two faces are spin factors with Peirce constant eight. Their capacities likewise cannot exceed the original factor's capacity. $\square$

The number apq is the exact local channel dimension. Indeed the inner derivations $4[L_z, L_{c_i}]$, $z \in V_{ij}$, independently rotate the i, j support directions. Their exponentials are Jordan automorphisms. Applied simultaneously to $X(0), Y(0)$, their derivatives in V_{ij} are $\lambda_i z, -\mu_j z$. Thus all apq channels in (3.5) can be realized. This local fact asserts no global integrability of a specified collection of channels into an exact lift.

Write $Q_m = \{(t, z) \in \mathbb{R} \times \mathbb{R}^{m-1} : t \geq \|z\|_2\}$ for the Lorentz cone of total cone-space dimension m .

Corollary 3.4 (Exact dimension-capped ball resources). *Fix $s \geq 2$ and an integer $d \geq 3$. Among exact lifts of $\mathbb{B}_2^s$ by arbitrary definable proper cone factors of dimension at most d , the minimum number of factors and the minimum total cone-space dimension are*

$$k_{\min} = \left\lceil \frac{s-1}{d-2} \right\rceil, \quad M_{\min} = s-1 + 2k_{\min}.$$

Both minima are attained by the same Lorentz norm tree. Among symmetric cone lifts under this dimension cap, the least total ambient Jordan rank is $2k_{\min}$, again attained by that tree.

Proof. Let k_+ count factors of dimension at least two and k_1 count rays. Theorem 3.2 implies $s-1 \leq k_+(d-2)$ and $M \geq s-1 + 2k_+ + k_1$, proving the first two lower bounds. For symmetric factors only non-ray irreducible factors can contribute curvature, and each has rank at least two. Decomposing reducible factors cannot increase their dimensions or change total rank, so total rank is at least $2k_{\min}$.

For attainment we use a variable-arity form of the recursive norm representation in [13, Section 2, equation (5)]. Put $k = k_{\min}$ and write $s-1 = b_1 + \dots + b_k$, with $1 \leq b_i \leq d-2$. Form a rooted tree whose internal nodes are a chain. Node $i < k$ has b_i leaf children and the next internal node as one additional child; the last node has $b_k + 1$ leaf children. It has s leaves. Assign the coordinates of $x \in \mathbb{R}^s$ to those leaves and an auxiliary scalar t_i to each internal node. Impose

$$t_i \geq \|\text{child values of node } i\|_2 \quad \text{in } Q_{b_i+2}, \quad t_1 = 1.$$

Recursive elimination gives exactly $\|x\|_2 \leq 1$. Strict feasibility holds at $x = 0$ by choosing the nonroot scalars positive and successively smaller. The construction uses k rank-two factors and total dimension $\sum_i (b_i + 2) = s-1 + 2k$, as required. $\square$

The total ambient Jordan rank is the parameter of the standard product log-determinant before affine restriction and remains a valid parameter after restriction. It need not be the least parameter of the restricted barrier, and it does not give an intrinsic barrier lower bound for the projected ball. Keeping these distinctions explicit is essential when using curvature estimates to compare formulations.

4 The exact support-certificate rank frontier

Fix a finite dictionary $\mathcal{K}$ of simple symmetric cones, allowing arbitrary finite repetition of its members and optional nonnegative rays. The dictionary, rather than the list of factors in an individual lift, is held fixed. For a non-ray dictionary member let r and a be its rank and Peirce constant. Put

$$B = \max_{K \in \mathcal{K}} a(K)(r(K) - 1) > 0, \quad q_B(s) = \left\lceil \frac{s-1}{B} \right\rceil. \quad (4.1)$$

The capacity B differs from the maximum local curvature capacity $a[r^2/4]$. It bounds curvature *per unit dual rank*. The distinction is visible for real PSD matrices of order four: the two capacities are three and four, respectively.

Theorem 4.1 (Selection-free rank frontier). *Let $s \geq 2$. Over all exact finite relative-Slater affine lifts of $\mathbb{B}_2^s$ through $\mathcal{K}$,*

$$\inf_{\mathcal{L}} R(\mathcal{L}) = q_B(s). \quad (4.2)$$

For each fixed lift there is a dense open semialgebraic subset $U \subset \mathbb{S}^{s-1}$ with surface-null complement such that

$$\sum_i \text{rank}_J Y_i \geq q_B(s) \quad \text{for every } v \in U \text{ and every } Y \in \mathcal{D}_{\mathcal{L}}(v).$$

An explicit lift attains minimum certificate rank $q_B(s)$ at every support except one pole, where its minimum rank is one. Consequently

$$\inf_{\mathcal{L}} \text{ess inf}_{v \in \mathbb{S}^{s-1}} r_{\mathcal{L}}(v) = q_B(s). \quad (4.3)$$

Lower bound. Use Lemma 2.2 to choose a primal lift $X(u)$ of u and a minimum-total-rank certificate $Y(v)$ of $1 - \langle u, v \rangle$. On a common dense open regular subset of the sphere these are C^1 and satisfy

$$\sum_i \langle X_i(u), Y_i(v) \rangle = 1 - \langle u, v \rangle.$$

At a contact $u = v$, its negative mixed derivative is the sphere metric, of rank $s - 1$. If the paired ranks are p_i, q_i , then Lemma 3.3 gives

$$s - 1 \leq \sum_i a_i p_i q_i \leq \sum_i a_i (r_i - 1) q_i \leq B \sum_i q_i. \quad (4.4)$$

The middle inequality uses $p_i \leq r_i - 1$ whenever $q_i > 0$; when $q_i = 0$ both terms vanish. Rays contribute zero curvature. Reduced proper faces have no larger $a_i(r_i - 1)$ than the original dictionary factor. By minimality of the selected rank, (4.4) implies the asserted bound for *all* certificates at the selected support, and at every support in the regular set. The same bound gives both lower bounds in (4.2) and (4.3). $\square$

4.1 A uniform Peirce-perspective construction

For $x \in V$ put $L_x y = x \circ y$ and $P(x) = 2L_x^2 - L_{x^2}$, the quadratic representation. For an idempotent c , $P(c)$ is orthogonal projection onto $V(c, 1)$. We use the standard facts that $P(x)K \subset K$ and that $P(x)$ is a cone automorphism when x is invertible. These hold in every Euclidean Jordan algebra; see [31, Chapters II–IV] and the Peirce and quadratic representation discussion in [45, pp. 1554–1555]. Their useful rank consequence can be verified directly: approximate any x by invertible elements x_n . Cone automorphisms preserve extreme rays, so $P(x_n)c$ has rank one when c is primitive. The closedness of the set of nonnegative elements of rank at most one gives

$$P(x)c \succeq 0, \quad \text{rank}_J(P(x)c) \leq 1. \quad (4.5)$$

This argument is independent of associative matrix multiplication.

Lemma 4.2 (Primitive Peirce perspective). *Let V be simple of rank $r \geq 2$, let c be primitive, and set $d = e - c$. The half-Peirce space $W = V(c, 1/2)$ has dimension $a(r - 1)$; give it the norm induced by the trace inner product, $\|w\|^2 = \langle w, w \rangle$. For $w \in W$,*

$$X(\delta, w, z) = \delta c + \sqrt{2}w + zd \succeq 0 \iff \delta, z \geq 0, \quad \delta z \geq \|w\|^2, \quad (4.6)$$

$$\det X(\delta, w, z) = z^{r-2}(\delta z - \|w\|^2). \quad (4.7)$$

For $g \in W$ and $A > 0$ the element

$$Y = AP \left(c - \frac{g}{\sqrt{2}A} \right) c = Ac - \frac{g}{\sqrt{2}} + \frac{P(g)c}{2A} \quad (4.8)$$

is positive of rank one. Its component in $V(d, 1)$ has trace $\|g\|^2/(4A)$.

Proof. Peirce multiplication gives $w^2 = tc + u$, where $u \in V(d, 1)$ and

$$t = \langle w^2, c \rangle = \langle w, w \circ c \rangle = \frac{\|w\|^2}{2}.$$

The definition of P and $c \circ w = w/2$ give $P(w)c = u$. Its trace is $\text{tr } u = \text{tr } w^2 - t = t$. By (4.5), for $w \neq 0$ we have $u = tk$ for a primitive idempotent $k \in V(d, 1)$. Thus $w^2 = t(c + k)$. Spectral calculus shows that w lies in the Peirce-one subalgebra with identity $c + k$: the supports of w and w^2 coincide. Since $c \circ w = w/2$, we have $k \circ w = w/2$. The span of c, k, w is therefore a rank-two Jordan subalgebra, orthogonal to $d - k$, and in this subalgebra the two eigenvalues of $\delta c + \sqrt{2}w + zk$ have sum $\delta + z$ and product $\delta z - 2t = \delta z - \|w\|^2$. The remaining $r - 2$ eigenvalues are all z . This proves both displayed claims for $w \neq 0$. For $w = 0$ they follow from the diagonal element $\delta c + zd$.

Expanding the quadratic representation by its definition proves (4.8). The first expression and (4.5) show positivity and rank at most one; its c -coefficient $A > 0$ makes the rank exactly one. The calculation above with $w = g$ gives $\text{tr } P(g)c = \|g\|^2/2$, proving the trace formula. $\square$

Attainment in Theorem 4.1. Choose a dictionary member attaining B , and use its notation from Lemma 4.2. Put $q = q_B(s)$ and partition the first $s - 1$ real coordinates into q nonempty groups, each of size at most B . Isometrically embed each group into a subspace $W_G \subset W$. Write a ball point as $x = (w_1, \dots, w_q, b)$ and impose

$$X_G = (1 - b)c + \sqrt{2}w_G + z_G d \succeq 0, \quad \sum_{G=1}^q z_G = 1 + b. \quad (4.9)$$

The variables z_G are auxiliary and $w_G \in W_G$. If the constraints hold, Lemma 4.2 gives $\sum_G \|w_G\|^2 \leq (1-b)\sum_G z_G = 1-b^2$, hence $x \in \mathbb{B}_2^*$. Conversely, if $b < 1$ and $x \in \mathbb{B}_2^*$, choose $z_G \geq \|w_G\|^2/(1-b)$ with sum $1+b$; the available surplus is nonnegative. At $b = 1$ all $w_G = 0$ and any nonnegative z_G of sum two works. Thus the projection is exact. At $x = 0$, $z_G = 1/q$ all blocks are positive definite, proving Slater feasibility.

Let $v = (g_1, \dots, g_q, \eta) \in \mathbb{S}^{s-1}$, with $g_G \in W_G$. Every certificate admits the Peirce decomposition

$$Y_G = A_G c + u_G + H_G, \quad u_G \in W, \quad H_G \in V(d, 1),$$

and positivity implies $A_G \geq 0$ and $H_G \succeq 0$. Comparing coefficients on the whole affine slice (4.9) gives

$$\text{proj}_{W_G} u_G = -g_G/\sqrt{2}, \quad \text{tr } H_G = \gamma = \frac{1-\eta}{2}, \quad \sum_G A_G = \frac{1+\eta}{2}. \quad (4.10)$$

Indeed, the free variations of z_G with zero sum make their trace coefficients equal; the constant and b coefficients give the two remaining scalar identities. When $q = 1$ the same two coefficient comparisons give them directly. These conditions are also sufficient for the whole-slice identity.

If $\eta < 1$, define

$$A_G = \frac{\|g_G\|^2}{2(1-\eta)}.$$

For $g_G \neq 0$, take the rank-one completion in (4.8); its tail trace is $\|g_G\|^2/(4A_G) = \gamma$. For $g_G = 0$, take $Y_G = \gamma k_G$ with any primitive $k_G \in V(d, 1)$. The identity $\sum_G \|g_G\|^2 = 1-\eta^2$ gives the prescribed sum of the A_G . Thus every block has rank one and the total rank is q . Conversely, $\gamma > 0$ forces every block of every certificate to be nonzero, so the minimum total rank is exactly q . This includes the south pole $\eta = -1$.

At the north pole $\eta = 1$, positivity and $\text{tr } H_G = 0$ give $H_G = 0$. Positivity then gives $u_G = 0$: a positive element orthogonal to d belongs to the face $\mathbb{R}_+ c$. Concentrating $\sum_G A_G = 1$ in one block gives total rank one, which is minimal. If $W_G \neq W$, arbitrary certificates may have extra half-Peirce components orthogonal to W_G ; the favorable construction sets them to zero, while the positive tail trace still forces each block to be nonzero. No uniqueness of certificates has been asserted. The exceptional pole has surface measure zero for $s \geq 2$, proving (4.3) and finishing the theorem. $\square$

4.2 Examples and the order of quantifiers

The dictionary capacity is $r - 1$ for real PSD order r , $2(r - 1)$ for complex Hermitian order r , $4(r - 1)$ for quaternionic Hermitian order r , and $m - 2$ for a Lorentz cone Q_m . For the Albert cone $H_+^3(\mathbb{O})$ it is $8(3 - 1) = 16$. Consequently unrestricted finite repetition of Albert factors gives the exact value $\lceil (s - 1)/16 \rceil$. The proof used neither an associative representation of the Albert algebra nor an embedding into a real PSD cone.

The requirement that a lift be fixed before selecting the objective is substantive. In fact,

$$\inf_{\mathcal{L}} \sup_v r_{\mathcal{L}}(v) = q_B(s), \quad \sup_v \inf_{\mathcal{L}} r_{\mathcal{L}}(v) = 1. \quad (4.11)$$

For the second identity rotate the construction so that the prescribed v is its north pole; the first is Theorem 4.1. The universal lower bound one follows from the nonzero support slack. Likewise the theorem does not optimize the rank of a certificate *selected in advance* by an algorithm: it optimizes over the full fiber at each support. For $s = 1$, whenever at least one non-ray factor is available, choose orthogonal primitive idempotents c, k and the section $X = tc + (1 - t)k$, $0 \leq t \leq 1$, with output $2t - 1$. Its two support slacks have the rank-one certificates $2k$ and $2c$, giving frontier one; the expression $\lceil (s - 1)/B \rceil = 0$ is inapplicable.

5 Products: certificates, compression, and private curvature

Write $C = \prod_{a=1}^h \mathbb{B}_2^{s_a}$, with $s_a \geq 2$, and $M = \prod_a \mathbb{S}^{s_a-1}$. Its pure support rows are $1 - \langle x_a, v \rangle$, $v \in \mathbb{S}^{s_a-1}$. A pure-row certificate represents this affine function on the entire original lift slice. In particular, it has zero coefficients on the other projected coordinates. This condition is stronger than an identity on a selected boundary sheet.

5.1 The support-join identity

Idempotents are ordered by their associated faces. Their join $c \vee d$ is the support of $c + d$. These operations and Jordan rank are taken separately in each simple factor.

Lemma 5.1 (Compression measures new support). *For an idempotent c , put $f = e - c$. For $y \succeq 0$, let $t = \text{supp } P(f)y$. Then*

$$c \vee \text{supp } y = c + t, \quad \text{rank}_J P(f)y = \text{rank}_J(c \vee \text{supp } y) - \text{rank}_J c.$$

For positive weights, $\text{supp}(\sum_a \lambda_a y_a) = \bigvee_a \text{supp } y_a$.

Proof. We have $t \leq f$. Set $g = f - t$. Self-adjointness of $P(f)$ gives $\langle g, y \rangle = \langle g, P(f)y \rangle = 0$. Positive orthogonality implies $\text{supp } y \perp g$, hence $c \vee \text{supp } y \leq e - g = c + t$. Conversely let $k = e - (c \vee \text{supp } y)$. Then $k \leq f$ and $\langle k, P(f)y \rangle = \langle k, y \rangle = 0$, so $k \perp t$. Consequently both c and t belong to $V(e - k, 1)$, proving the reverse inclusion. Ranks add on orthogonal idempotents. Finally a positive element is orthogonal to a positive sum precisely when it is orthogonal to every summand; their complementary faces therefore agree. This proves the sum identity. $\square$

Theorem 5.2 (Additive existence of genuine certificates). *For every finite exact relative-Slater affine lift of C through the symmetric-cone dictionary of Section 4, there are directions $u_a \in \mathbb{S}^{s_a-1}$ and genuine pure-row certificates Y^a such that*

$$Q := \sum_i \text{rank}_J \left(\sum_a \lambda_a Y_i^a \right) \geq \sum_a q_B(s_a) \quad (\lambda_a > 0). \quad (5.1)$$

Every primal tuple over this simultaneous contact has total Jordan nullity at least Q . No regularity of primal or dual selections is assumed.

Proof. Choose the rows successively. Suppose the first $a - 1$ certificates have been chosen, and in factor i let $c_i = \bigvee_{j < a} \text{supp } Y_i^j$, $f_i = e_i - c_i$. Fix the previously chosen coordinates at their contacts. Their slacks vanish on the whole remaining cylinder. Nonnegativity of each block pairing therefore puts every feasible tuple on this cylinder in $\prod_i V_i(f_i, 1)_+$. Intersect the restricted affine slice with $\prod_i V_i(f_i, 1)$; this preserves its feasible set and its projection. Fix all later projected coordinates, say at zero. The resulting affine lift projects exactly onto the a th ball.

Compress every original pure-row certificate Y by $P(f_i)$. Its pairing with every tuple on this slice remains $1 - \langle x_a, v \rangle$. Reduce this restricted lift further to its minimal face $V_i(k_i, 1)_+$, where $k_i \leq f_i$. The further compressed tuple $P(k_i)Y_i$ is a certificate on the relative-Slater restricted slice. Face reduction cannot increase the dictionary capacity. The dense-open lower bound in Theorem 4.1 consequently supplies a support u_a at which every such further compression has rank at least $q_B(s_a)$. An original certificate exists by original-slice Slater duality. Choose one as Y^a . Compression cannot increase rank: by Lemma 5.1 its rank is the Jordan-rank increment of a support join, which is at most the rank of the added support (equivalently use the usual face-lattice rank

inequality). Alternatively apply a spectral decomposition of y and (4.5) termwise to $P(k)y$. Thus $\sum_i \text{rank}_J P(f_i)Y_i^a \geq q_B(s_a)$.

Lemma 5.1 identifies this sum with the increase of $\sum_i \text{rank}_J c_i$. Summing over the rows proves (5.1). Every final primal tuple is orthogonal to each chosen certificate, hence to their positive aggregate. Complementary ranks sum to at most the ambient rank, proving the nullity assertion. $\square$

The proof deliberately compresses certificates of the original lift. An arbitrary certificate of a restricted slice need not extend to that lift. Moreover, the conclusion is existence of a high-rank positive aggregate of pure-row certificates. It does not say that every certificate of the aggregate objective has this rank, or give a lower bound on the minimum rank in its full aggregate fiber. Separate copies of (4.9) give favorable aggregate certificates of rank at most $\sum_a q_B(s_a)$ at every support; this confirms the sharp scale of the construction without reversing those quantifiers.

5.2 Private directions in Hermitian factors

A different argument gives a pointwise result for selected factors. Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, $\delta = \dim_{\mathbb{R}} \mathbb{F}$, and use the real trace pairing. Suppose C^1 maps satisfy the full-row identities

$$1 - \langle x_a, v \rangle = \sum_i \langle X_i(x), Y_i^a(v) \rangle, \quad x \in M, \quad v \in \mathbb{S}^{s_a-1}. \quad (5.2)$$

The maps X_i are shared across all rows; Y_i^a depends only on its own row argument.

Theorem 5.3 (Private curvature). *Suppose the factors in (5.2) belong to $H_+^{r_i}(\mathbb{F})$, $r_i \leq R$, for an integer $R \geq 2$, and put $B = \delta(R - 1)$. At every simultaneous contact, write $p_i = \text{rank } X_i$, $U_{ia} = \text{Ran } Y_i^a(x_a)$, $U_{i,-a} = \sum_{b \neq a} U_{ib}$, and*

$$d_{ia} = \dim_{\mathbb{F}} U_{ia} / (U_{ia} \cap U_{i,-a}).$$

Then

$$s_a - 1 \leq \sum_i \delta p_i d_{ia}, \quad (5.3)$$

$$\sum_a d_{ia} \leq \text{rank} \left(\sum_a \lambda_a Y_i^a(x_a) \right) \leq \dim_{\mathbb{F}} \ker X_i, \quad \lambda_a > 0. \quad (5.4)$$

In particular, both total aggregate rank and total primal nullity are at least $\sum_a \lceil (s_a - 1)/B \rceil \geq h$.

Proof. Complementarity gives $X_i Y_i^a(x_a) = 0$. Its cylindrical form, with x_b varying and $a \neq b$, gives $(D_b X_i) Y_i^a = 0$. Decompose the underlying vector space as $\text{Ran } X_i \oplus \ker X_i$. Two-sided positivity forces the kernel–kernel compression of $D X_i$ to vanish, as in Lemma 3.3. The only part entering the row- a mixed pairing is its off-diagonal map from $\ker X_i$ to $\text{Ran } X_i$. This map annihilates $U_{i,-a}$. The corresponding derivative of Y_i^a has its other endpoint in U_{ia} ; the derivative pairing therefore factors through a real space of dimension $\delta p_i d_{ia}$. Differentiating the row identity gives the sphere metric, proving (5.3) by rank subadditivity.

Choose a complement E_{ia} of $U_{ia} \cap U_{i,-a}$ in U_{ia} . These complements are jointly independent: a relation $\sum_a w_a = 0$, $w_a \in E_{ia}$, puts each w_a in $U_{ia} \cap U_{i,-a}$, hence makes it zero. Their total dimension is therefore at most $\dim_{\mathbb{F}} \sum_a U_{ia}$, which is the aggregate rank by positivity. This sum lies in $\ker X_i$, proving (5.4). When $d_{ia} > 0$, $p_i \leq R - 1$. Thus (5.3) gives $\sum_i d_{ia} \geq \lceil (s_a - 1)/B \rceil$. Sum this inequality in a . $\square$

The sourcewise inequality (5.3) retains more information than a bound on total rank: it records which source consumes each available kernel direction. It requires differentiability only at the contact under consideration. Definable selectors of an affine lift supply such contacts generically, but need not supply them everywhere.

Example 5.4 (One factor can serve every source). Let $p = \max_a \lceil s_a/\delta \rceil$, and embed each $\mathbb{R}^{s_a}$ isometrically into $\mathbb{F}^p$. Write their images as columns w_a of W . The single-factor lift

$$Z = \begin{pmatrix} S & W^* \\ W & I_p \end{pmatrix} \succeq 0, \quad S \in H^h(\mathbb{F}), \quad S_{aa} = 1 \quad (5.5)$$

projects exactly onto C . Indeed its Schur complement $D = S - W^*W$ is positive, giving $\|w_a\| \leq 1$; conversely choose $S = W^*W + \text{Diag}(1 - \|w_a\|^2)$. At a product extreme, D has zero diagonal, hence is zero. The boundary sheet is polynomial and has nullity exactly h . For the embedded support $\tilde{v}$, put

$$Y^a(v) = \frac{1}{2}(e_a, -\tilde{v})(e_a, -\tilde{v})^*.$$

On the entire affine slice, its pairing with Z is $\frac{1}{2}(1 + \|\tilde{v}\|^2 - 2\text{Re}(w_a^*\tilde{v})) = 1 - \langle x_a, v \rangle$. These are genuine rank-one row certificates. Their ranges are independent because their first h coordinates are the distinct e_a . Thus total aggregate rank and nullity are exactly h , despite there being only one cone factor.

Part II

Global regularity and affine barriers

6 Global regularity changes the formulation frontier

The generic selectors used above provide differentiability on a dense open set. Requiring one C^1 selector on the entire sphere is a different condition. We first prove its exact cost for Lorentz factors and full product rows, including factors that vanish on open sets.

6.1 Lorentz phases and pointwise no-sharing

Use the Euclidean self-dual pairing on $Q_m = \{(t, z) : t \geq \|z\|\}$. A fixed rescaling converts this convention to the Jordan trace pairing without changing any rank or resource count. Suppose (5.2) has globally C^1 factors in Q_{m_i} , $m_i \geq 3$, and put $r_i = m_i - 2$. At contact define

$$H_i^a(x)(u, v) = -\langle D_a X_i(x)[u], D Y_i^a(x_a)[v] \rangle.$$

Then $\sum_i H_i^a$ is the sphere metric of rank $p_a = s_a - 1$. If either factor is zero, its derivative is zero by pointedness. If one is interior, its complementary factor is zero. Otherwise, on a neighborhood where both factors remain nonzero, complementarity gives

$$X_i = \alpha_i(1, q_i^a(x_a)), \quad Y_i^a = \beta_i^a(1, -q_i^a), \quad q_i^a \in \mathbb{S}^{r_i}, \quad \alpha_i, \beta_i^a > 0.$$

The cylinder identity makes the primal phase depend only on x_a there. Differentiation cancels every amplitude term and gives

$$H_i^a = \alpha_i \beta_i^a (D q_i^a)^* D q_i^a \succeq 0, \quad \text{rank } H_i^a \leq r_i. \quad (6.1)$$

If $H_i^a \neq 0$, the same primal phase is locally constant in every other source direction. Thus

$$H_i^a(x) \neq 0 \implies H_i^b(x) = 0 \quad (b \neq a). \quad (6.2)$$

This statement is pointwise. It does not assign one source to a label for all x .

6.2 A top-class cover argument

We record the topology needed in the proof. Cohomology in this subsection has real coefficients. If open sets U_a cover a space and $u_a|_{U_a} = 0$, the relative cup product implies $\prod_a u_a = 0$: lift each class to $H^*(M, U_a)$ and multiply to get a class in $H^*(M, \bigcup_a U_a) = 0$. The same argument applies to any subfamily that covers. The Künneth formula gives a nonzero product of any subset of the pulled-back top classes on a product of spheres. A proper open subset of a connected closed oriented p -manifold has zero ordinary cohomology in degree p when every component is noncompact; in particular this holds for every proper open subset of $\mathbb{S}^p$. These are the standard relative cup, Künneth, and noncompact-manifold duality facts; see [48, Sections 3.2–3.3].

Lemma 6.1 (Saturated labels have a proper phase domain). *Fix a source a with $p_a \geq 2$, and a compact set E on which its active labels are exactly J , with $\sum_{i \in J} r_i = p_a$ and $r_i < p_a$ for every $i \in J$. There is an open neighborhood P of $\pi_a(E)$ in $\mathbb{S}^{p_a}$ such that the dual phases give a local C^1 diffeomorphism*

$$(q_i^a)_{i \in J} : P \longrightarrow \prod_{i \in J} \mathbb{S}^{r_i}.$$

The set P is proper, so the source top class vanishes on $\pi_a^{-1}(P)$.

Proof. At a point of E , every active primal and dual factor is nonzero. Keep all other primal coordinates fixed and vary x_a nearby. Continuity keeps the primal factor nonzero and forces the dual factor to stay on the nonzero boundary. Hence its normalized phase is defined on an open neighborhood of x_a . Equality between the total dimension p_a and the capacity sum, together with (6.1), makes the joint phase derivative invertible. Invertibility is open. The union of these phase domains, restricted to their common invertibility locus, provides P .

If P were the whole sphere, compactness would make this local diffeomorphism a finite covering of the connected target. If any $r_i = 1$, the target has infinite fundamental group, whereas $\mathbb{S}^{p_a}$ is simply connected, which is incompatible with a finite covering. Otherwise the target is simply connected, so the covering is a diffeomorphism. There are at least two factors, since $r_i < p_a$, and its intermediate cohomology differs from that of a sphere. Both cases are impossible. The top-cohomology fact stated above completes the proof. $\square$

Theorem 6.2 (Exact smooth Lorentz resources). *Fix an integer $c \geq 1$ and allow Lorentz dimensions $3 \leq m_i \leq c + 2$. For full-row factorizations (5.2) with globally C^1 primal and dual maps, define*

$$h_a = \begin{cases} 1, & p_a \leq c, \\ \lceil (p_a + 1)/c \rceil, & p_a > c, \end{cases} \quad \tau_a = \begin{cases} p_a, & p_a \leq c, \\ p_a + 1, & p_a > c. \end{cases}$$

The exact simultaneous minima of the number of non-ray factors L , their total capacity $T = \sum_i (m_i - 2)$, and their total ambient dimension $D = \sum_i m_i$ are

$$L_{\min} = \sum_a h_a, \quad T_{\min} = \sum_a \tau_a, \quad D_{\min} = \sum_a (\tau_a + 2h_a). \quad (6.3)$$

The attaining factors come from actual Slater affine lifts with genuine whole-slice certificates. Optional rays do not improve these minima.

Proof. Call a label active for a if $H_i^a \neq 0$, and write $C_a = \sum_{i \text{ active for } a} r_i$, $N_a = \#\{i \text{ active for } a\}$. The metric identity and (6.2) give

$$C_a \geq p_a, \quad N_a \geq \lceil p_a/c \rceil, \quad \sum_a C_a \leq T, \quad \sum_a N_a \leq L.$$

Rays have zero mixed channel and may be omitted here.

Let $A = \{a : p_a > c\}$. If $T < \sum_a (p_a + \mathbf{1}_{a \in A})$, at every point some $a \in A$ has $C_a = p_a$. Thus the sets $E_a = \{C_a = p_a\}$, $a \in A$, cover M . They are closed: each active locus is open, and $C_a \geq p_a$. Partition E_a by its exact set of active labels. This is a finite clopen partition into compact sets $E_{a,J}$: all existing active labels persist nearby, and an additional label would increase C_a . Lemma 6.1 supplies a proper phase neighborhood $P_{a,J}$ for each compact piece. Choose pairwise disjoint open neighborhoods $W_{a,J}$ of the pieces, small enough to lie in $\pi_a^{-1}(P_{a,J})$. The source top class vanishes on $U_a = \coprod_J W_{a,J}$. These sets cover M , so their relative cup product vanishes, contradicting Künneth. Hence $T \geq \sum_a \tau_a$.

The pointwise count already gives $N_a \geq h_a$ except for $A_0 = \{a : p_a > c, c \mid p_a\}$. If $L < \sum_a h_a$, at every point some $a \in A_0$ has exactly p_a/c active labels. The corresponding closed loci again cover. On each exact-label piece every label must have capacity c , and their capacities sum to p_a . The same phase and relative cup argument gives a contradiction. Thus $L \geq \sum_a h_a$. Since $D = T + 2L$, this also proves the dimension lower bound.

For attainment, use a direct Q_{p_a+2} trace section $X_a = (1, x_a)$ if $p_a \leq c$, with certificate $(1, -v)$ in its row and zero in other rows. Otherwise partition all $s_a = p_a + 1$ coordinates into h_a nonempty

groups of size at most c , and impose

$$A_G(t_G, x_G) = \left(\frac{1+t_G}{2}, \frac{1-t_G}{2}, x_G \right) \in Q_{|G|+2}, \quad \sum_G t_G = 1. \quad (6.4)$$

This is equivalent to $t_G \geq \|x_G\|^2$, hence projects exactly onto the ball and is Slater at $x = 0$ with all $t_G > 0$. On the boundary choose $t_G = \|x_G\|^2$. The polynomial certificate

$$B_G(v) = \left(\frac{1+\|v_G\|^2}{2}, -\frac{1-\|v_G\|^2}{2}, -v_G \right)$$

has pairing $(t_G + \|v_G\|^2)/2 - \langle x_G, v_G \rangle$. Summing gives $1 - \langle x, v \rangle$ on the entire slice. Each group contributes one factor and capacity $|G|$. Separate factors for different sources attain all three claimed minima. $\square$

For comparison, among finite exact relative-Slater affine lifts without a global C^1 selection requirement, generic definable selectors and the pointwise argument give the exact minima

$$L_0 = \sum_a \lceil p_a/c \rceil, \quad T_0 = \sum_a p_a, \quad D_0 = T_0 + 2L_0. \quad (6.5)$$

Independent norm trees attain them by the construction in Section 2. Indeed, partition p_a into $\lceil p_a/c \rceil$ positive integers at most c ; a rooted tree with these internal excess arities has $1 + p_a = s_a$ leaves. At a common generic regular contact, pointwise no-sharing proves the matching lower bounds. Such a contact lies in the primal regular locus and the inverse images, under each source projection, of the dual regular loci; their finite intersection is dense open by definable selection. This comparison makes no claim about arbitrary unstructured factor maps. Global regularity therefore adds one capacity unit per source with $p_a > c$, and one factor exactly for those such sources with $c \mid p_a$. These are formulation resources, not intrinsic barrier parameters of the projected body. A product of L Lorentz factors has standard ambient parameter $2L$; the restrictions to the affine slices can have smaller parameters, as examined below.

6.3 Why local and joint identities are weaker

Proposition 6.3 (Joint weighted rows have a different sharp count). *For two balls of common dimension $s \geq 3$ and positive weights $\lambda_1 + \lambda_2 = 1$, the kernel*

$$S_\lambda(x, z) = \sum_{a=1}^2 \lambda_a (1 - \langle x_a, z_a \rangle)$$

on $(\mathbb{S}^{s-1})^2$ has exact globally C^1 factor count $2s - 1$ over Q_3 . In contrast, the full labelled rows require $2s$ factors.

Proof. The mixed contact metric has rank $n = 2(s - 1)$, so $L \geq n$. If $L = n$, every channel has rank one everywhere and both factors are nonzero. Their common phases $q_i : M \rightarrow \mathbb{S}^1$ lift to C^1 functions $\theta_i : M \rightarrow \mathbb{R}$ since M is simply connected. The identity $g_\lambda = \sum_i \alpha_i \beta_i d\theta_i^2$ makes $(\theta_1, \dots, \theta_n) : M \rightarrow \mathbb{R}^n$ a local diffeomorphism. Its image would be both nonempty open and compact, which is impossible. Thus $L \geq n + 1 = 2s - 1$.

For the upper bound put $Z(x) = (\sqrt{\lambda_1}x_1, \sqrt{\lambda_2}x_2) \in \mathbb{S}^{2s-1}$. Choose a unit vector q supported in the first coordinate block, and set

$$c(x) = 1 - \langle q, Z(x) \rangle > 0, \quad f(x) = \frac{Z(x) - \langle q, Z(x) \rangle q}{c(x)} \in q^\perp \cong \mathbb{R}^{2s-1}.$$

Direct expansion of squared norms gives the stereographic identity

$$1 - \langle Z(x), Z(z) \rangle = \frac{c(x)c(z)}{2} \|f(x) - f(z)\|^2.$$

For $U(t) = ((1+t^2)/2, (1-t^2)/2, t)$ and $V(t) = ((1+t^2)/2, -(1-t^2)/2, -t)$, both vectors are in Q_3 and $\langle U(t), V(u) \rangle = (t-u)^2/2$. Thus $c(x)U(f_j(x))$, $c(z)V(f_j(z))$, $1 \leq j \leq 2s-1$, give the required smooth factors. The full-row count follows from Theorem 6.2 with $c = 1$. $\square$

For $k \geq 2$ equal balls, the same lower proof gives $k(s-1) + 1$. Pairing the sources and using the preceding construction on each pair, with one coordinate-square construction on any remaining source, gives $ks - \lfloor k/2 \rfloor$ factors. We do not identify the exact count for $k \geq 3$. These are complete bounds on a restricted contact kernel; neither construction supplies the missing full labelled rows of an affine lift. The phase obstruction alone cannot close the remaining gap: a product of spheres of total dimension n embeds in $\mathbb{R}^{n+1}$. To check the last assertion, embed a sphere as an orientable hypersurface, then observe that the boundary of a small tubular neighborhood of $N^m \subset \mathbb{R}^{m+1} \subset \mathbb{R}^{m+p+1}$ is $N \times \mathbb{S}^p$; the added normal bundle is trivial. Induction proves the assertion.

Remark 6.4 (Switching must pass through open vertex sets). A Lorentz label productive for two sources must have a nonempty open primal vertex set. Indeed choose dual phase patches for both sources on which their derivatives are nonzero. Their images each contain more than one point, so smaller patches can be chosen with disjoint images. On the product of these patches cylindrical complementarity would force a nonzero primal phase to equal both images. Thus the primal factor is zero on that open cylinder. Moreover, the nonzero locus of each productive dual factor cannot be dense. Otherwise the identity saying that the primal derivative in every unrelated source direction is radial extends by continuity to all points; this excludes productivity for the second source. In particular, analyticity or empty-interior primal vertex sets make globally productive label sets disjoint. The proof of Theorem 6.2 does not need either extra hypothesis.

Example 6.5 (A codimension-two defect destroys the smooth premium). In a full binary norm tree for $\mathbb{B}_2^s$, let $t_v(x) = \|x_{S_v}\|$ for each internal node v , where S_v is its set of descendant leaves; leaves carry their signed coordinates and the root value is one. At each internal node with children v_0, v_1 take

$$A_v(x) = (t_v(x), t_{v_0}(x), t_{v_1}(x)), \quad B_v(z) = (t_v(z), -t_{v_0}(z), -t_{v_1}(z)).$$

Their pairings telescope to $1 - \langle x, z \rangle$. The dual factors are genuine certificates of the tree slice, since the same telescoping works when the primal internal variables are free. Boundary fibers are unique: the nonnegative squared cone slacks sum to $1 - \|x\|^2 = 0$, forcing every internal norm to its minimum. Both factor maps are globally Lipschitz and semialgebraic, and analytic away from the sets $x_{S_v} = 0$. Every nonroot internal node has at least two leaves, so these sets have codimension at least two in the boundary sphere when nonempty. Nevertheless the tree uses only $s-1$ copies of Q_3 . For $s \geq 3$, Theorem 6.2 requires s copies for globally C^1 factors. Thus unique boundary fibers, Lipschitz regularity, and dense-open analyticity do not imply the global smooth conclusion.

The tree also has a unique genuine dual certificate for every support $z \in \mathbb{S}^{s-1}$. To see this, write an arbitrary dual block at node v as $(\alpha_v, \beta_{v,0}, \beta_{v,1}) \in Q_3$. Matching coefficients on the entire affine slice forces

$$\alpha_{\text{root}} = 1, \quad \beta_{v,j} = \begin{cases} -z_\ell, & \text{if child } v_j \text{ is leaf } \ell, \\ -\alpha_{v_j}, & \text{if child } v_j \text{ is internal.} \end{cases}$$

Induction from the leaves, using $\alpha_v \geq (\beta_{v,0}^2 + \beta_{v,1}^2)^{1/2}$, gives $\alpha_v \geq \|z_{S_v}\|$. At the root this is equality because $\|z\| = 1$. Equality then propagates down the tree: each child head is nonnegative and at least its subtree norm, so equality in the sum of their squares forces equality for each child separately. This also handles a zero subtree, whose head and descendant coordinates must all vanish. Thus the unique tuple is exactly $B_v(z)$ above. Every nonzero block has Jordan rank one; a zero block has rank zero.

Corollary 6.6 (Exact worst pure-row certificate rank over real PSD2). *For $C = \prod_{a=1}^h \mathbb{B}_2^{s_a}$, $s_a \geq 2$, fix any positive weights $\lambda_1, \dots, \lambda_h$. Let $\mathcal{D}_{\mathcal{L},a}(u_a)$ denote the full fiber of genuine certificates of $1 - \langle x_a, u_a \rangle$ on the affine slice of a lift $\mathcal{L}$. Among finite exact relative-Slater lifts over real 2×2 positive semidefinite cones and optional rays,*

$$\inf_{\mathcal{L}} \sup_{u \in M} \sup_{\substack{Y^a \in \mathcal{D}_{\mathcal{L},a}(u_a) \\ 1 \leq a \leq h}} \sum_i \text{rank} \left(\sum_a \lambda_a Y_i^a \right) = \sum_a (s_a - 1). \quad (6.6)$$

Proof. The lower bound is Theorem 5.2 with $B = 1$. For the upper bound take independent binary norm trees, with $s_a - 1$ factors for source a . The linear isomorphism

$$(t, z_1, z_2) \mapsto \begin{pmatrix} t + z_1 & z_2 \\ z_2 & t - z_1 \end{pmatrix}$$

identifies Q_3 with the real PSD2 cone; multiplying the dual image by $1/2$ preserves the pairing and does not change rank.

Any genuine pure-row certificate for source a vanishes on all factors belonging to other sources. Indeed evaluate its identity at a feasible tuple with source a at its support contact and every other source at a Slater point of its tree. The slack is zero, and each nonnegative block pairing must therefore vanish. A positive dual element orthogonal to an interior primal element is zero. What remains on source a is its unique tree certificate, by the coefficient argument above. Consequently every positive pure-row aggregate has rank at most $\sum_a (s_a - 1)$. Equality holds on the dense open support set where every internal subtree norm is positive. This proves the upper bound and hence the formula. $\square$

The inner supremum in (6.6) is over genuine *pure-row* certificates. It neither minimizes over nor describes all certificates in the fiber of the aggregate objective.

For completeness, local saturation itself gives an exact linear-algebra statement. If positive semidefinite contact forms H_i sum to a positive metric H and $\sum_i \text{rank } H_i = \text{rank } H$, then $P_i = H^{-1/2} H_i H^{-1/2}$ are mutually orthogonal projections. Indeed write $P_i = Z_i Z_i^*$ with Z_i of full column rank; concatenating the Z_i gives a square matrix Z with $ZZ^* = I$, hence $Z^*Z = I$. This determines the orthogonal decomposition of the tangent metric at the contact, but says nothing about unused portions of the cone, higher derivatives, or the global affine slice. The global arguments above use compactness and the full two-point identities in addition to this local rank equality.

Proposition 6.7 (Everywhere differentiability already forbids saturation). *Assume the Lorentz full-row factors are everywhere Fréchet differentiable, without continuity of their derivatives, and $p_a \geq 2$. If $r_i \leq c < \max_a p_a$ for an integer $c \geq 1$, then*

$$T \geq \sum_a p_a + 1.$$

For one source this gives the same exact resource minima as Theorem 6.2. For several sources it gives the additional count $L \geq \sum_a \lceil p_a/c \rceil + 1$ whenever every p_a is divisible by c ; no full additive differentiability premium is asserted.

Proof. All pointwise phase and no-sharing calculations remain valid with pointwise derivatives. Suppose $T = \sum_a p_a =: n$. Sum the row slacks and aggregate their dual factors. At every contact, equality in the rank bound forces every factor to be nonzero and to use its full capacity. Its primal phase is therefore defined everywhere. Their product $q : M \rightarrow \prod_i \mathbb{S}^{r_i}$ is differentiable with invertible derivative, because its kernel is annihilated by the positive contact metric. The finite-dimensional local inversion theorem for everywhere differentiable maps with everywhere nonsingular derivative [78] makes q a local homeomorphism. Compactness makes it a finite covering. Since M is simply connected, circle factors are impossible; with all $r_i \geq 2$ the covering is a homeomorphism. Consequently $\prod_a (1 + t^{p_a}) = \prod_i (1 + t^{r_i})$. The smallest exponent and its coefficient determine its multiplicity; cancel that power of $1 + t^j$ and repeat. Thus the multisets of exponents agree, contradicting the cap. This proves the strict capacity bound. If every p_a is divisible by c , the baseline count L_0 satisfies $cL_0 = \sum_a p_a$, so $T \leq cL$ forces $L \geq L_0 + 1$. For one source, combine the capacity and count bounds with the grouped polynomial construction. If $c \nmid p$, its factor count is already $\lceil (p+1)/c \rceil$ by rounding; if $c \mid p$ use the extra factor. $\square$

A simple smooth example shows why analytic continuation cannot replace the cover argument. Put $h(t) = e^{-1/t^2}$ for $t > 0$ and $h(t) = 0$ for $t \leq 0$, and use U from Proposition 6.3. For distinct sources a, b , the factors

$$A(x) = h((x_a)_1)U((x_b)_1), \quad B^a(v) = h(-v_1)U(0)$$

are smooth and nonnegative Lorentz-valued, with $\langle A(x), B^a(x_a) \rangle = 0$ on the entire cylinder. The dual is not identically zero, yet the primal phase varies in source b wherever $(x_a)_1 > 0$. This refutes continuation of a cylindrical ray identity across arbitrary open zero sets. It is a counterexample to that proof step, not a full-row factorization with fewer than the required factors.

7 Support-orbit rigidity and selected ranks

7.1 Saturation and support orbits

For a simple Euclidean Jordan algebra V of rank r and Peirce constant a , write $\mathcal{I}_p(V)$ for its manifold of rank- p idempotents. Its dimension is $ap(r-p)$ and its tangent space at an idempotent c is $V(c, 1/2)$. These are classical facts about idempotent manifolds [31, 73]. We use them to identify the additional restriction imposed by globally smooth factor maps.

Theorem 7.1 (Global saturation). *Let $C \subset \mathbb{R}^{n+1}$ be a compact convex body, $n \geq 1$, containing the origin in its interior, and suppose $P = \partial C$ and $D = \partial C^\circ$ are strictly convex C^1 hypersurfaces. Suppose the full slack has a finite globally labelled factorization*

$$1 - \langle x, z \rangle = \sum_i \langle X_i(x), Y_i(z) \rangle, \quad (x, z) \in P \times D,$$

where $X_i : P \rightarrow K_i$ and $Y_i : D \rightarrow K_i$ are C^1 and K_i is a simple symmetric cone with rank r_i and Peirce constant a_i . Put $c_i = a_i \lfloor r_i^2/4 \rfloor$, with $c_i = 0$ for rays. If $\sum_i c_i = n$, each positive-capacity factor has constant balanced complementary contact ranks $p_i, r_i - p_i$, and the joint support map

$$P \longrightarrow \prod_{i:c_i>0} \mathcal{I}_{p_i}(V_i), \quad x \longmapsto (\text{supp } X_i(x))_i$$

is a finite covering. There is exactly one positive-capacity factor. It is the Lorentz cone Q_{n+2} , except that $n = 2$ also permits $\mathbb{S}_+^3$ at the level of the support-orbit topology.

Proof. Let z be the unique polar contact of x . The contact correspondence is a homeomorphism; no differentiability of that correspondence is needed. The tangent spaces are $T_x P = z^\perp$ and $T_z D = x^\perp$. Their Euclidean pairing is nondegenerate: its left kernel consists of vectors in $z^\perp \cap \text{span}\{x\}$, which is zero because $\langle x, z \rangle = 1$. Mixed differentiation in independent boundary charts gives

$$\langle u, v \rangle = \sum_i H_i(u, v), \quad H_i(u, v) = -\langle dX_i(x)[u], dY_i(z)[v] \rangle.$$

The cross-Peirce argument of Lemma 3.3, which for its rank bound uses only the two independent derivative images, gives

$$\text{rank } H_i \leq a_i p_i q_i \leq c_i, \quad p_i = \text{rank}_J X_i(x), \quad q_i = \text{rank}_J Y_i(z).$$

Nondegeneracy and $\sum_i c_i = n$ force equality throughout. For every positive-capacity factor, this gives $p_i + q_i = r_i$, the ranks are balanced, and $\text{rank } H_i = c_i$. Ranks are lower semicontinuous along the contact graph. Since the two ranks sum to r_i , each is locally constant, hence constant by connectedness.

In the following support-differential argument all indices satisfy $c_i > 0$; zero-capacity factors have $H_i = 0$ and play no role. Let $e_i(x) = \text{supp } X_i(x)$. This is C^1 : locally the zero spectral cluster is separated from the positive spectrum, and a smooth scalar cutoff equal to zero on the former and one on the latter gives e_i by spectral functional calculus. Differentiating $e_i \circ e_i = e_i$ and $X_i \circ e_i = X_i$ shows that

$$de_i[u] \in V_i(e_i, 1/2), \quad L_{X_i} de_i[u] = \frac{1}{2} \text{proj}_{V_i(e_i, 1/2)} dX_i[u].$$

Here L_{X_i} is invertible on the half-Peirce space: in a frame adapted to the positive support it acts on a cross channel by $\lambda_j/2 > 0$. Consequently $de_i[u] = 0$ precisely when the cross-Peirce component of $dX_i[u]$ vanishes. In particular

$$c_i = \text{rank } H_i \leq \text{rank } de_i \leq c_i.$$

If every de_i kills u , every $H_i(u, v)$ is zero, so $u = 0$. The joint support differential is therefore injective, and its domain and codomain have the same dimension n . The joint map is a local C^1 diffeomorphism. Its image is open and, by compactness, closed in the connected target. It is surjective and proper, hence a finite covering.

We give the topological classification explicitly. For $n \geq 2$, the source $P \cong S^n$ is simply connected, so this is the universal cover of the target. There can be no circle factor, since it would make the target fundamental group infinite. The remaining factors have compact simply connected universal covers. The universal cover of their product is the product of those covers. If there were at least two positive-dimensional factors, the mod-two top cohomology class of one factor, tensored with the unit classes of the others, would give nonzero cohomology in a degree strictly between zero and n . The Künneth formula contradicts the cohomology of S^n . Thus there is one factor. For $n = 1$, the same conclusion follows directly from the sum of the positive integer dimensions.

The simple-algebra classification now leaves the following cases. A spin factor has primitive-idempotent orbit S^{m-2} and therefore gives Q_{n+2} . For $H_r(\mathbb{R})$ the orbit is the real Grassmannian $\text{Gr}_p(\mathbb{R}^{p+q})$, $p + q = r$. At $r = 3$ its balanced orbit is $\mathbb{RP}^2$, covered by S^2 . At $r \geq 4$, balanced ranks satisfy $p, q \geq 2$. The oriented Grassmannian is simply connected by the exact sequence of

$$SO(p) \times SO(q) \longrightarrow SO(p+q) \longrightarrow \text{Gr}_p^+(\mathbb{R}^{p+q}) :$$

the induced map on fundamental groups onto $\pi_1(SO(p+q)) = \mathbb{Z}_2$ is surjective. Its kernel is nonzero: use the diagonal element when $p, q \geq 3$, and an even nonzero winding in an $SO(2)$ factor otherwise.

Exactness therefore gives nonzero π_2 of the oriented Grassmannian. Its dimension $pq \geq 4$ precludes a sphere.

The complex Grassmannian is simply connected by the fibration with isotropy $U(p) \times U(q)$ in $U(p+q)$. The map on π_1 is addition $\mathbb{Z}^2 \rightarrow \mathbb{Z}$, so its kernel gives nonzero π_2 of the Grassmannian. For $r \geq 3$ its dimension $2pq \geq 4$ precludes a sphere. For quaternionic Grassmannians, the groups $Sp(k)$ are simply connected and $\pi_3(Sp(k)) = \mathbb{Z}$, with standard inclusions preserving a generator. These facts follow inductively from $Sp(k-1) \rightarrow Sp(k) \rightarrow S^{4k-1}$ and $Sp(1) = S^3$. The map on π_3 of the isotropy inclusion is again addition $\mathbb{Z}^2 \rightarrow \mathbb{Z}$, giving nonzero π_4 of the Grassmannian. Its dimension $4pq \geq 8$ precludes a sphere. The classical-group and Grassmannian fibrations used here are described in [48, Examples 4.53–4.55]. Finally the balanced idempotent orbit of $H_3(\mathbb{O})$ is the Cayley plane: rank-one idempotents are its points and complementation identifies the rank-two orbit with it [9, Section 3.4]. Its cells have dimensions 0, 8, 16, so $H^8(-; \mathbb{Z}) = \mathbb{Z}$; see [48, Example 4.47]. It is not a sphere. Order-two real, complex, and quaternionic Hermitian algebras are already spin factors. This exhausts the classification. $\square$

7.2 Eliminating the real order-three exception

The apparent exceptional possibility is ruled out by the full ball slack. The following elementary observation avoids a classification of linear determinantal representations.

Example 7.2 (Local constant rank need not extend globally). The affine pencil

$$L(x) = \begin{pmatrix} 1+x_3 & x_1 & x_2 \\ x_1 & 1-x_3 & 0 \\ x_2 & 0 & 1-x_3 \end{pmatrix}, \quad \det L(x) = (1-x_3)(1-\|x\|^2),$$

is positive semidefinite of rank two on $S^2 \setminus \{(0,0,1)\}$. Indeed, its lower 2×2 block is positive definite and its Schur complement is zero there. Its kernel line is

$$\text{span} \left(1, -\frac{x_1}{1-x_3}, -\frac{x_2}{1-x_3} \right),$$

so its kernel map is a local diffeomorphism onto an affine chart of $\mathbb{RP}^2$ by stereographic coordinates. At the north pole, $L(0,0,1) = \text{diag}(2,0,0)$ has rank one. Thus affine behavior and constant rank on an open sphere patch do not justify global constant rank or a common kernel. This example only invalidates that local-to-global shortcut. Proposition 7.5 below excludes the global slack factorization using the full C^1 factor maps and finite affine gluing.

Lemma 7.3 (Injective kernels of an affine sphere pencil). *Let $s \geq 2$ and $X(x) = C_0 + \sum_{j=1}^s x_j C_j$ be a real symmetric matrix pencil, positive semidefinite of nullity one for every $x \in S^{s-1}$. If its kernels do not share a nonzero vector, its kernel-line map is injective.*

Proof. Suppose distinct x, y have the same kernel line, containing $k \neq 0$. The affine scalar function $f(v) = k^T X(v)k = a + b \cdot v$ is nonnegative on the unit sphere and vanishes at x and y . Nonnegativity says $a \geq \|b\|$. If $b \neq 0$, a zero is possible only when $a = \|b\|$ and $v = -b/\|b\|$, giving only one zero. Thus $a = b = 0$. For every v , positivity and $k^T X(v)k = 0$ imply $X(v)k = 0$, contradicting the absence of a common nonzero kernel vector. $\square$

Lemma 7.4 (Finite affine selection on a sphere). *Let $n \geq 2$ and let $F : S^n \rightarrow W$ be C^1 , with W finite dimensional. Suppose a dense open subset is the union of finitely many nonempty open sets U_i , and $F|_{U_i} = L_i|_{U_i}$ for ambient affine maps $L_i : \mathbb{R}^{n+1} \rightarrow W$. Then F is the restriction of one affine map.*

Proof. Merge sets with identical L_i and put $E_i = \overline{U_i}$. Continuity of values and tangent derivatives gives $j^1 F = j^1(L_i|_{S^n})$ on E_i . If distinct affine maps have the same tangent first jet at $z \in S^n$, their difference is $M(z \cdot x - 1)$ for some nonzero $M \in W$. Such a difference has matching first jet at no other sphere point: matching derivatives force the other normal line to be $\mathbb{R}z$, and matching values exclude $-z$. Thus each intersection $E_i \cap E_j$ lies in at most one point. Remove the resulting finite set Z . The connected space $S^n \setminus Z$ is partitioned into the disjoint relatively closed sets $E_i \setminus Z$; each is also relatively open because the partition is finite. Only one is nonempty. But the closure of a nonempty open U_i cannot be finite, so there was only one distinct affine map. Density finishes the proof. $\square$

Proposition 7.5 (No real order-three saturation for a ball). *There is no factorization*

$$1 - x \cdot y = \operatorname{tr}(X(x)Y(y)) + \sum_{j=1}^q a_j(x)b_j(y), \quad x, y \in S^2,$$

with $X, Y : S^2 \rightarrow \mathbb{S}_+^3$ of class C^1 , continuous nonnegative a_j, b_j , finite q , constant complementary ranks 2, 1, and a rank-one support map that is a local diffeomorphism onto $\mathbb{RP}^2$.

Proof. Exchange X, Y if necessary so that Y has rank one. Every diagonal summand is zero, hence $a_j(x)b_j(x) = 0$. For each of the finitely many continuous nonnegative functions $f = a_j, b_j$, the set $\{f > 0\} \cup \operatorname{int}\{f = 0\}$ is dense open. Intersect these sets and partition the result by their finitely many positivity/zero patterns. On each nonempty open pattern set U , for each j at least one of $a_j|_U, b_j|_U$ is identically zero. Therefore all ray terms vanish for every pair $(x, y) \in U \times U$.

The rank-one support map of Y sends U onto a projective open set. Consequently the matrices $Y(y)$, $y \in U$, span $\mathbb{S}^3$: an annihilating symmetric matrix would have a quadratic form zero on an open set of projective directions, and hence zero identically. Choose six $y_\ell \in U$ whose matrices are a basis. The equations

$$\operatorname{tr}(X(x)Y(y_\ell)) = 1 - x \cdot y_\ell, \quad x \in U,$$

force $X|_U$ to be an ambient affine pencil. There are only finitely many pattern sets, so Lemma 7.4 makes X one affine pencil on the whole sphere.

Complementarity identifies the kernel line of $X(x)$ with the support line of $Y(x)$. This map is a finite covering $S^2 \rightarrow \mathbb{RP}^2$, hence the universal double cover. Its image contains all lines, so its kernels have no common nonzero vector. But Lemma 7.3 makes the map injective, contradicting its two-point fibers. $\square$

Corollary 7.6 (Ball saturation classification). *A finite globally bi- C^1 symmetric-cone factorization of the full slack of $\mathbb{B}_2^s$, $s \geq 2$, satisfying $\sum_i a_i[r_i^2/4] = s - 1$ has exactly one positive-capacity factor, namely Q_{s+1} . Remaining factors, if any, are rays.*

Proof. Apply Theorem 7.1 and exclude its sole additional case by Proposition 7.5. $\square$

7.3 The selected-rank frontier for a fixed dictionary

For the dictionary of Section 4, define

$$\kappa_{\mathcal{H}}(s) = \begin{cases} 1, & s = B + 1 \text{ and the dictionary contains } Q_{B+2}, \\ \lceil s/B \rceil, & \text{otherwise.} \end{cases} \quad (7.1)$$

Here “contains” allows linear cone isomorphism. The formula also gives one when $s \leq B$. A dictionary of Hermitian matrices over a fixed field $\mathbb{F}$ with order cap R has $B = \delta(R - 1)$ and the exceptional branch occurs exactly when $R = 2$ and $s = \delta + 1$.

Theorem 7.7 (Exact selected-rank frontier). *For globally bi- C^1 full-slack factorizations of $\mathbb{B}_2^s$ through a fixed finite repeatable symmetric-cone dictionary, the least possible value of $\sup_v \sum_i \text{rank}_J Y_i(v)$ is $\kappa_{\mathcal{K}}(s)$. The upper bound is attained by a relative-Slater affine lift with globally smooth boundary selections and genuine whole-slice certificates.*

Proof. At every contact, with primal and dual ranks p_i, q_i ,

$$s - 1 \leq \sum_i a_i p_i q_i \leq \sum_i a_i (r_i - q_i) q_i \leq B \sum_i q_i.$$

Rays supply zero in the first two sums and cannot make any equality saturated. Unless $s - 1 = qB$ for an integer $q \geq 1$, the lower bound already rounds to $\lceil s/B \rceil$. In the remaining case suppose $Q(v) = \sum_i q_i(v) \leq q$ everywhere. Equality forces $Q \equiv q$; lower semicontinuity makes each q_i constant. Each nonzero dual factor has $q_i = 1$, $a_i(r_i - 1) = B$, and $p_i = r_i - 1$. Its primitive support orbit has dimension B . The support-differential argument above gives a finite covering from $\mathbb{S}^{qB}$ to the product of these q primitive-idempotent orbits.

If $q > 1$, a circle target gives infinite fundamental group. Otherwise the compact universal covers of the positive-dimensional target factors have nonzero intermediate product cohomology, ruling out a sphere source. If $q = 1$, the simple-algebra classification identifies the primitive orbit as a sphere for spin factors, real projective space for real Hermitian factors, complex or quaternionic projective space in the other classical cases, and the Cayley plane in the exceptional case. Complex and quaternionic projective spaces have intermediate cohomology unless their matrix order is two; the Cayley plane has its nonzero middle cohomology. These order-two cases are spin factors. For real order $R \geq 3$ the remaining map is the double covering $\mathbb{S}^{R-1} \rightarrow \mathbb{RP}^{R-1}$. All other dual factors vanish identically. Pairing the primal sheet with a basis chosen from the rank-one dual sheet makes it an ambient affine pencil, since rank-one matrices with directions in a projective open set span the symmetric matrices. Its kernel-line map has no common nonzero kernel and hence is injective by Lemma 7.3, contradicting the double cover. Thus the only saturated possibility is the stated matching spin factor. In every other case some rank is at least $q + 1$.

For attainment outside the exceptional branch, choose a dictionary member attaining B and partition all s coordinates into $g = \lceil s/B \rceil$ nonempty groups embedded in its primitive half-Peirce space. With $d = e - c$ as in Lemma 4.2, impose

$$X_G = t_G c + \sqrt{2} w_G + d \succeq 0, \quad \sum_G t_G = 1. \quad (7.2)$$

These constraints say $t_G \geq \|w_G\|^2$, so the projection is the ball and the slice is strictly feasible. Its polynomial boundary sheet has $t_G = \|w_G\|^2$. At support $v = (g_G)_G$ the rank-one completion

$$Y_G = \frac{1}{2} c - \frac{g_G}{\sqrt{2}} + P(g_G) c = \frac{1}{2} P(c - \sqrt{2} g_G) c$$

has pairing $t_G/2 - \langle w_G, g_G \rangle + \|g_G\|^2/2$. It is nonzero of rank one even when $g_G = 0$. Summing proves the genuine whole-slice slack identity and gives selected rank g everywhere. For the exceptional branch use the direct Lorentz trace section, whose boundary primal and dual factors have rank one. This proves attainment. $\square$

Theorem 7.8 (Sequential smooth ranks). *Suppose the full rows of $\prod_a \mathbb{B}_2^{s_a}$ have globally bi- C^1 factors through $\mathcal{K}$. There is a simultaneous contact u such that, for every choice of positive weights,*

$$\sum_i \text{rank}_J \left(\sum_a \lambda_a Y_i^a(u_a) \right) \geq \sum_a \kappa_{\mathcal{K}}(s_a). \quad (7.3)$$

The same bound holds for total primal nullity there. These bounds are sharp over globally selected factorizations, and their upper constructions are actual affine lifts with genuine certificates.

Proof. Use the successive joins c_i from Theorem 5.2. After fixing earlier primal coordinates at contact, cylindrical complementarity puts the primal factors in $V_i(e_i - c_i, 1)_+$. Fix future primal coordinates arbitrarily and apply the next one-ball row to the compressed dual $P(e_i - c_i)Y_i^a$. The compressed maps remain globally C^1 and satisfy the exact full two-point row identity. Lemma 5.1 identifies their total rank with the next support increment.

To apply Theorem 7.7, note that proper faces have strictly smaller per-rank capacity than their parent simple non-ray cone: classical faces have smaller order, spin faces are rays, and Albert proper non-ray faces have capacity eight rather than sixteen. Thus each remaining face has capacity at most B . If equality is needed in the rank chain of that theorem, the face must be an unreduced original cone attaining B . In particular, a newly created proper face cannot introduce a matching spin exception at capacity B . The same proof therefore gives at least $\kappa_{\mathcal{X}}(s_a)$ new rank at some next contact. Iteration and Lemma 5.1 prove the bound; complementarity gives nullity. Separate copies of (7.2) and the exceptional direct sections give equality at every contact. $\square$

In particular, under an integer real-dimension cap $d \geq 3$ on arbitrary symmetric factors, the exact selected aggregate-rank frontier is

$$\sum_a h_d(s_a), \quad h_d(s) = \begin{cases} 1, & d \geq s + 1, \\ \lceil s/(d-2) \rceil, & d < s + 1. \end{cases}$$

Indeed the per-rank capacity is at most $d - 2$, with equality only for a spin factor of dimension d . The equality analysis in Theorem 7.7 gives the lower bound, and separate grouped Lorentz sections attain it. This concerns the chosen global sheets. The nonsmooth norm-tree certificates in Example 6.5 show why the same smooth premium cannot be imposed on arbitrary affine-lift certificate fibers.

There is also a factor-aware refinement. At stage a of sequential compression, suppose the non-ray faces have ranks R_{ia} and Peirce constants α_{ia} . Define

$$g_a = \min \left\{ \sum_i q_i : q_i \in \{0, \dots, R_{ia}\}, \quad \sum_i \alpha_{ia} q_i (R_{ia} - q_i) \geq s_a - 1 \right\}.$$

The mixed-channel inequality gives at least g_a new non-ray rank at every next contact. Hence one can replace the summand in (7.3) by $\max\{g_a, \kappa_{\mathcal{X}}(s_a)\}$. This is an adaptive lower bound, not an asserted exact frontier. For at most L faces of a common Peirce constant α and order at most R , a fixed total dual rank $Q = uL + t$, $0 \leq t < L$, has capacity at most

$$\alpha[(L-t)u(R-u) + t(u+1)(R-u-1)].$$

To prove it, pad with zero ranks and increase every order to R . If two ranks differ by at least two, transferring one unit from the larger to the smaller increases their sum of $q(R-q)$. Iteration balances all ranks to u or $u+1$, proving the formula. Inverting this finite envelope gives a convenient explicit lower bound on g_a .

Corollary 7.9 (Dimension-capped smooth symmetric-cone resources). *Consider globally bi- C^1 full-slack factorizations of $\mathbb{B}_2^s$ over simple symmetric cones of real dimension at most an integer $d \geq 3$, with optional rays. Let L count non-ray factors, D the entire ambient dimension, and ρ the entire ambient Jordan rank. If $d < s + 1$, their simultaneous exact minima are*

$$L_{\min} = \left\lceil \frac{s}{d-2} \right\rceil, \quad D_{\min} = s + 2L_{\min}, \quad \rho_{\min} = 2L_{\min}.$$

If $d \geq s + 1$, the exact triple is $(1, s + 1, 2)$. The standard ambient product barrier has parameter ρ .

Proof. For every simple non-ray cone, $a[r^2/4] \leq m - 2$, with equality only for rank-two spin factors; substitution of $m = r + ar(r - 1)/2$ verifies strictness for $r \geq 3$. Write $T = \sum_{i \text{ non-ray}} (m_i - 2)$. Local curvature gives $T \geq s - 1$. If equality holds, all factors are spin and Corollary 7.6 forces the single factor Q_{s+1} . Thus in the strict-cap case $T \geq s$. Since $T \leq (d - 2)L$, $D \geq T + 2L$, and $\rho \geq 2L$, the displayed lower bounds follow. The grouped Lorentz construction attains them simultaneously. In the other case local curvature gives $T \geq s - 1$, $L \geq 1$, and hence $D \geq s + 1$, $\rho \geq 2$; the direct Lorentz section attains equality. $\square$

8 The standard barrier after affine restriction

The rank of a certificate measures one exposed support. A barrier must control every direction in its domain, including directions inside a projection fiber. We now make that distinction quantitative.

8.1 The parameter and two boundary tests

Write $L = X^\circ + T$ for the effective affine slice of a product of Euclidean Jordan algebras, after minimal-face reduction, and set $\Omega = L \cap \text{int } K$. The word effective means that redundant variables which do not change the cone tuple have been removed. For

$$F(X) = - \sum_i \log \det_i X_i,$$

its least gradient parameter on this slice is

$$\nu_{\text{std,slice}} = \sup_{X \in \Omega} \sup_{0 \neq H \in T} \frac{DF(X)[H]^2}{D^2F(X)[H, H]}. \quad (8.1)$$

The restricted Hessian is positive definite. The standard third-derivative self-concordance inequality survives affine restriction, so (8.1) is precisely the least self-concordant-barrier parameter of this specified function. It is not the infimum over other barriers on Ω . Ambient Jordan rank gives an upper bound but need not give equality.

Lemma 8.1 (Boundary order). *For every $\bar{X} \in L \cap K$,*

$$\nu_{\text{std,slice}} \geq \sum_i (r_i - \text{rank}_J \bar{X}_i). \quad (8.2)$$

In particular, the bound holds at every point of every projected boundary fiber, rather than just at a chosen boundary selection.

Proof. Let $X(t) = (1 - t)\bar{X} + tX^\circ$, $0 < t < 1$, and put $z_i = r_i - \text{rank}_J \bar{X}_i$. Apply the positive invertible quadratic representation $P((X_i^\circ)^{-1/2})$. It sends X_i° to the unit, preserves rank, and changes the determinant by a positive constant. Diagonalizing the transformed $\bar{X}_i$ therefore gives

$$\det_i X_i(t) = c_i t^{z_i} \prod_{j=1}^{r_i - z_i} ((1 - t)\lambda_{ij} + t), \quad c_i > 0, \quad \lambda_{ij} > 0.$$

If $z = \sum_i z_i$, the restriction $f(t) = F(X(t))$ satisfies $f' = -z/t + O(1)$ and $f'' = z/t^2 + O(1)$. For $z > 0$, the quotient $(f')^2/f''$ tends to z ; for $z = 0$ the claim is immediate. These statements use Jordan eigenvalues and apply also to the Albert algebra. $\square$

Lemma 8.2 (Recession and complementary boundary order). *Suppose $D \neq 0$ is a recession direction of $L \cap K$. In each factor let $c_i = \text{supp } D_i$, $f_i = e_i - c_i$, and let $C_i = P(f_i)\bar{X}_i$, where $\bar{X} \in L \cap K$. Define*

$$r = \sum_i \text{rank}_J D_i, \quad z = \sum_i (\text{rank } f_i - \text{rank}_{V_i(f_i, 1)} C_i).$$

Then $\nu_{\text{std}, \text{slice}} \geq r + z$.

Proof. The two-dimensional affine family $X(\sigma, t) = (1 - \sigma)\bar{X} + \sigma X^\circ + tD$ is interior for $0 < \sigma < 1$ and $t > 0$. Put $A(\sigma) = (1 - \sigma)\bar{X} + \sigma X^\circ$. The Jordan Schur determinant formula [45], applied to the Peirce decomposition for c_i , gives

$$\det_i(A_i(\sigma) + tD_i) = t^{r'_i} \det_{c_i} D_i \det_{f_i}(P(f_i)A_i(\sigma)) [1 + O_\sigma(t^{-1})], \quad r'_i = \text{rank}_J D_i. \quad (8.3)$$

For a zero D_i the formula means the unchanged determinant and $r'_i = 0$. The Schur correction is quadratic in the mixed Peirce part and linear in the inverse of the c_i part, hence is $O_\sigma(t^{-1})$. Alternatively, the determinant is a polynomial in (σ, t) whose leading coefficient is the positive coefficient displayed in (8.3). On every compact subinterval of $0 < \sigma < 1$, the expansion may consequently be differentiated twice, including after multiplication of each t derivative by t .

Write $f(\sigma, t) = F(X(\sigma, t))$ and $g(\sigma) = -\sum_i \log \det_{f_i}(P(f_i)A_i(\sigma))$, omitting rank-zero factors. As $t \rightarrow \infty$ with σ fixed,

$$tf_t \rightarrow -r, \quad t^2 f_{tt} \rightarrow r, \quad f_\sigma \rightarrow g', \quad f_{\sigma\sigma} \rightarrow g'', \quad tf_{t\sigma} \rightarrow 0.$$

The compressed Slater endpoint $P(f_i)X_i^\circ$ is positive definite in $V_i(f_i, 1)$ whenever this algebra has positive rank. The preceding lemma's polynomial computation gives $\sigma g' \rightarrow -z$ and $\sigma^2 g'' \rightarrow z$ as $\sigma \downarrow 0$. At each point test the affine direction $H = tD + \sigma(X^\circ - \bar{X})$. Its gradient tends, in the stated iterated limits, to $-(r + z)$ and its Hessian quadratic form to $r + z$. Equation (8.1) therefore gives $\nu_{\text{std}, \text{slice}} \geq r + z$. This derivative argument is necessary: a bounded remainder in the barrier value alone would not justify the conclusion. $\square$

Proposition 8.3 (General and regular frontiers). *For a finite repeatable symmetric-cone dictionary with $B > 0$, every relative-Slater affine lift of $\mathbb{B}_2^s$ satisfies*

$$\left\lceil \frac{s-1}{B} \right\rceil \leq \nu_{\text{std}, \text{slice}}. \quad (8.4)$$

There is a bounded lift with parameter $\kappa_{\mathcal{K}}(s)$ from (7.1). Over lifts admitting globally bi- C^1 genuine full-row selections, the exact parameter minimum for $\prod_a \mathbb{B}_2^{s_a}$ is $\sum_a \kappa_{\mathcal{K}}(s_a)$. For a single ball, every unbounded lift satisfies the stronger bound $\nu_{\text{std}, \text{slice}} \geq 1 + \lceil (s-1)/B \rceil$.

Proof. The generic full-certificate rank theorem and complementarity force a boundary tuple with nullity at least $\lceil (s-1)/B \rceil$; apply Lemma 8.1. For the upper bound use (7.2). Its determinant is $t_G - \|w_G\|^2$, so its barrier is

$$F = -\sum_{G=1}^g \log(t_G - \|w_G\|^2), \quad \sum_G t_G = 1. \quad (8.5)$$

Each summand has parameter one on its paraboloid domain. Indeed, if $h = t - \|w\|^2$, then $D^2(-\log h)[H, H] = (Dh[H]/h)^2 + 2\|H_w\|^2/h$. The third-derivative inequality follows by restriction of the rank-two Lorentz log determinant with one perspective coordinate fixed. Cauchy-Schwarz

gives the sum parameter at most g . At any sphere point set $t_G = \|w_G\|^2$. Each fixed-tail block has nullity one, so Lemma 8.1 gives equality g . For the direct spin section, $F = -\log(1 - \|x\|^2)$ has parameter one: its squared dual gradient norm is $2\|x\|^2/(1 + \|x\|^2)$, with supremum one. Products give the stated upper bound, and Theorem 7.8 gives the matching nullity lower bound.

For the last assertion, a nonempty closed unbounded convex slice has a nonzero recession direction $D \in K \cap T$; compactness of its projection implies $\pi D = 0$. Choose a generic support certificate Y of rank at least $q = \lceil (s-1)/B \rceil$ and any tuple $\bar{X}$ over that support. The whole-slice identity gives $\langle D, Y \rangle = 0$, so Y_i lies in the face complementary to $\text{supp } D_i$. Its complementarity with $\bar{X}_i$ implies that the compressed nullity z in Lemma 8.2 is at least q . Since $\sum_i \text{rank}_J D_i \geq 1$, that lemma proves the assertion. $\square$

Consequently the unrestricted single-ball minimum is already exact when B does not divide $s-1$: both endpoints are $\lceil s/B \rceil$. The same statement applies to Hermitian order caps with $B = a(R-1)$ and to Lorentz dimension caps with $B = d-2$. The remaining issue is bounded lifts at a divisible tangent dimension. If every symmetric cone of real dimension at most d is permitted, the maximum primitive capacity is $B = d-2$, attained by Q_d . Thus the exact smoothly selected standard-barrier minimum is $\lceil s/(d-2) \rceil$ for $3 \leq d < s+1$, and one for $d \geq s+1$. Separate source balls add these quantities. This dimension-cap statement allows all symmetric cone types; using a higher-rank type cannot improve the optimum.

8.2 A complete selection-free result at one channel

Theorem 8.4 (One-channel rigidity). *Let $s = B + 1$. The infimum of $\nu_{\text{std,slice}}$ over finite affine lifts through $\mathcal{K}$ equals one if the dictionary contains Q_{B+2} , and equals two otherwise. In the latter case the corresponding infimum for $(\mathbb{B}_2^{B+1})^b$ equals $2b$. There are simultaneous pure-row certificates whose every positive aggregate has total Jordan rank at least $2b$.*

Proof. Suppose first that one lift has $\nu_{\text{std,slice}} < 2$. The integral boundary orders in Lemma 8.1 imply that every boundary tuple has nullity at most one. Thus every member of every normalized genuine certificate fiber $\mathcal{D}(v)$ has rank at most one. Such a member is nonzero, since its slack is not identically zero. The positive midpoint of two elements on different product-cone rays has rank at least two. Convexity therefore puts $\mathcal{D}(v)$ on one ray. The whole-slice identity fixes its scalar, making $\mathcal{D}(v) = \{Y(v)\}$.

A Slater point provides the uniform normalization

$$0 < 1 - \|\pi X^\circ\| \leq \langle X^\circ, Y(v) \rangle \leq 1 + \|\pi X^\circ\|.$$

Strict positivity of X° bounds $Y(v)$ above and away from zero. The closed graph then makes Y continuous. On the connected sphere its one active factor is fixed. Projectivization gives a continuous injection $\mathbb{S}^B \rightarrow \mathcal{P}(V_j)$ into that factor's primitive-ray manifold: if the rays agree, evaluating the genuine identities at a lift of the ball's center fixes their scalar, and the identities then fix v .

The target dimension is $a_j(r_j - 1) \leq B$. Invariance of dimension excludes a smaller dimension, and invariance of domain makes the map a homeomorphism in equal dimension. The primitive-ray spaces are $\mathbb{RP}^{r-1}$, $\mathbb{CP}^{r-1}$, $\mathbb{HP}^{r-1}$, spheres for spin factors, and $\mathbb{OP}^2$ for the Albert algebra. The classification and cohomology used in Section 7 show that the only sphere cases are the spin factors, including the order-two classical coincidences. Thus a matching spin factor is necessary for a value below two. Its direct section attains one; otherwise the two-group construction (7.2) attains two.

For the product assertion apply sequential compression as in Section 5. At the next source, compress the entire original certificate fiber to the faces complementary to the previously chosen

support join. The resulting families are compact, convex, and have closed graphs: the original fibers are uniformly bounded by the same Slater normalization, and compression is linear. Their genuine row identities on the remaining contact cylinder ensure that zero is not in any compressed fiber. If every compressed certificate had total rank at most one, the preceding singleton and primitive-ray argument would again give a matching capacity- B spin factor in one of the remaining faces. A proper face of a simple factor has strictly smaller primitive capacity than that factor; hence no such face can arise when the original dictionary has no matching spin factor. Some compressed certificate must therefore have rank at least two. The support-join compression identity of Section 5 adds at least two new rank units at each source. The final join, and every positive aggregate of the chosen certificates, has rank at least $2b$. All tuples at that simultaneous contact are complementary to this join, so the boundary-order test gives $\nu_{\text{std},\text{slice}} \geq 2b$. Separate two-group constructions attain equality. $\square$

In particular, an Albert-cone dictionary has $B = 16$: its critical 17-dimensional ball has restricted standard-barrier minimum two. The theorem concerns the optimum over lifts. It does not say that every lift using a matching spin factor has parameter one.

8.3 Arbitrarily many factors under a wide cap

We give a geometric argument that closes a further part of the divisible case without any selection hypothesis or factor-count restriction.

Lemma 8.5 (A face-codimension budget). *Suppose a bounded relative-Slater affine slice $P = L \cap K$ projects onto $\mathbb{B}_2^s$. For any $X \in P$ over its boundary, let E_X be the linear span of the minimal face of K containing X . Then $\text{codim } E_X \geq s$.*

Proof. Write $L = X^\circ + T$. Since P is bounded and nonempty, $T \cap K = \{0\}$; also $0 \notin L$, since otherwise P would be a bounded cone with nonempty relative interior projecting onto a ball. On $\hat{L} = \text{span}\{X^\circ\} + T$ define height $\ell(tX^\circ + H) = t$. A cone point of zero height is zero. There are no cone points of negative height: otherwise adding a suitable positive multiple of X° gives a nonzero cone point of height zero; pointedness excludes cancellation to zero. Hence $\hat{L} \cap K$ is precisely the cone over P . The projection homogenizes to a linear surjection $\hat{\pi} : \hat{L} \rightarrow \mathbb{R}^{s+1}$ whose cone image is Q_{s+1} . If $k = \dim \ker \hat{\pi}$, then $\dim \hat{L} = s + 1 + k$. Every $H \in \hat{L} \cap E_X$ permits two-sided small feasible perturbations of X , because $X \in \text{ri}(K \cap E_X)$. Their images lie in the line spanned by the target extreme ray. Thus $\dim(\hat{L} \cap E_X) \leq k + 1$. Grassmann's dimension inequality gives $k + 1 \geq s + 1 + k - \text{codim } E_X$, as required. $\square$

Theorem 8.6 (Wide-cap rigidity). *Fix either Hermitian cones over one field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$ with order at most R , or Lorentz cones with dimension at most d . Put respectively $B = a(R - 1)$ or $B = d - 2$. Allow any finite number of factors and rays. If $q \geq 2$ and $B \geq q - 1$, the minimum of $\nu_{\text{std},\text{slice}}$ for affine lifts of $\mathbb{B}_2^{qB+1}$ is $q + 1$.*

Proof. Unbounded lifts were settled by Proposition 8.3. For a bounded lift suppose $\nu_{\text{std},\text{slice}} < q + 1$. All boundary tuples then have integer total nullity at most q . In a Hermitian block of order $r \leq R$ and nullity m , the minimal-face codimension is

$$\left\lceil r + \frac{ar(r-1)}{2} \right\rceil - \left\lceil r - m + \frac{a(r-m)(r-m-1)}{2} \right\rceil = m + \frac{am(2r-m-1)}{2} \leq (B+1)m.$$

A Lorentz block has codimension $d_i - 1 \leq B + 1$ at a nonzero boundary point, nullity one there, and codimension $d_i \leq 2(B + 1)$ at its vertex, nullity two there. Rays obey the same bound. Hence

$\text{codim } E_X \leq (B+1) \text{ nul } X$. Lemma 8.5 rules out nullity at most $q-1$, since $(q-1)(B+1) < qB+1$ under the stated hypothesis. Every tuple in every boundary fiber therefore has nullity exactly q .

Each compact boundary fiber is a singleton. Indeed, from a relative-interior point of a nontrivial fiber take a maximal nonzero affine chord. Its endpoint must leave the relative interior of the minimal cone face, and thus acquire an additional null eigenvalue. The resulting nullity exceeds q , a contradiction. Compactness makes the unique boundary lift $X(v)$ continuous.

On a dense open set the curvature chain is saturated:

$$qB \leq \sum_i a_i p_i q_i \leq \sum_i a_i (r_i - q_i) q_i \leq B \sum_i q_i \leq qB.$$

Exactly q cap-attaining blocks are consequently of corank one; all other blocks are interior. Along a limit of any one generic active pattern its zero eigenvalues persist. Constant total nullity excludes any additional zero eigenvalue or any new active block. Closures of distinct patterns are thus disjoint, and their finite union covers the connected sphere. One pattern is fixed everywhere.

The corresponding kernel lines define a continuous map from $\mathbb{S}^{qB}$ to $(\mathbb{F}P^{R-1})^q$, or to $(\mathbb{S}^B)^q$ in the Lorentz case. It is injective. Indeed, equal lines put the two tuples in the relative interior of the same product face. Any certificate exposing one projected point annihilates that face, so the genuine slack identity makes the other projected point identical. Source and target are compact connected manifolds of the same dimension, so invariance of domain would make the map a homeomorphism. The target has nonzero intermediate mod-two cohomology, whereas the sphere does not. This contradiction proves the lower bound. The grouped construction with $q+1$ groups attains it. $\square$

The proof also isolates a useful sufficient condition independent of cap width: a compact lift at $s-1 = qB$, $q \geq 2$, cannot have total boundary nullity exactly q at every point of every boundary fiber. Such constant nullity would give singleton fibers and the same impossible global projective-product embedding.

8.4 What a remaining bounded lift would have to do

The exact results above leave a sharply delimited classification problem. For a fixed Hermitian order cap or Lorentz dimension cap, the single-ball cases not resolved by the preceding results have

$$s = qB + 1, \quad q \geq B + 2, \quad q \leq \inf_{\text{affine lifts}} \nu_{\text{std,slice}} \leq q + 1. \quad (8.6)$$

Any lift improving the upper endpoint must be bounded. The following structural result develops the consequences of such a hypothetical improvement; none of our preceding theorems depends on resolving this remaining interval.

Proposition 8.7 (Certificate collapse and fixed generic labels). *Fix a field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, an order cap $R \geq 2$, and $B = a(R-1)$, where $a = \dim_{\mathbb{R}} \mathbb{F}$. Consider a bounded affine lift through finitely many Hermitian cones of orders at most R and optional rays, in the divisible case $s-1 = qB$, $q \geq 2$. Suppose $\nu_{\text{std,slice}} < q+1$. For $v \in \mathbb{S}^{qB}$ let $\mathcal{P}(v)$ and $\mathcal{D}(v)$ be its full primal and normalized certificate fibers. Define*

$$E_i(v) = \sum_{Y \in \mathcal{D}(v)} \text{Ran}_{\mathbb{F}} Y_i, \quad Z = \left\{ v : \sum_i \dim_{\mathbb{F}} E_i(v) \leq q-1 \right\},$$

counting a nonzero ray certificate as dimension one. Then Z is nonempty and semialgebraic, and $\dim Z \leq qB - B$. On a dense open semialgebraic set U , both fibers are singletons, strict

complementarity holds, and exactly q full-order blocks are of corank one. If $B \geq 2$, one and the same set of q block labels is active on all of U ; thus different generic label patterns cannot account for a putative improvement. The possible remaining obstruction is degeneration of this fixed pattern above Z .

Proof. The boundary-order bound gives nullity at most q at every primal boundary tuple, so every certificate has rank at most q . A finite positive average spans all the spaces $E_i(v)$, hence $\sum_i \dim E_i(v) \leq q$. Semialgebraicity follows by expressing the spanning condition with finitely many certificates (the ambient dimension bounds their number) and eliminating quantifiers.

Choose a dense open generic set where the full-fiber minimum-rank theorem holds. All certificates there have rank exactly q , and all primal tuples have nullity exactly q . The compact primal chord argument in Theorem 8.6 gives a singleton primal fiber. The dual fiber lies in a product face of rank at most q ; if it contained a nontrivial maximal chord, an endpoint would leave that face's relative interior and have rank below q . Thus it too is a singleton. On a further dense open smooth locus, equality in the curvature chain gives exactly q full-order corank-one blocks and strict complementarity.

On each smooth stratum of Z of dimension t , choose smooth primal and certificate selections after further stratification. Differentiating the slack on two copies of this stratum gives its positive definite tangent metric, of rank t . The certificate ranks sum to at most $q - 1$, so the mixed Peirce estimate gives $t \leq B(q - 1)$. This proves the dimension bound, including for the semialgebraic closure $\overline{Z}$.

For any support v , compactness gives a limit of generic primal tuples. Its nullity is at least q by persistence of zero eigenvalues, and at most q by the parameter hypothesis. If $v \notin Z$, an averaged certificate of rank q forces every primal in that fiber to have nullity q . The fiber is consequently a singleton. This singleton is the limit of every approaching sequence of generic tuples. Passing to a subsequence of fixed generic labels shows that it has exactly that pattern: an additional zero eigenvalue would exceed nullity q . Its inactive blocks are positive definite. Compactness then provides a neighborhood of v in which no generic tuple has a new active label; since each generic tuple has exactly q labels, its pattern is fixed.

If Z were empty, this argument would give a continuous singleton lift and one fixed pattern on the whole sphere. The kernel-line injection and projective-product contradiction in Theorem 8.6 would apply. Hence $Z \neq \emptyset$. If $B \geq 2$, the complement of the closed semialgebraic set $\overline{Z}$, of codimension at least two, is connected. For example, semialgebraic triangulation and a small general-position perturbation of a path avoid a subcomplex of codimension at least two. The local pattern just proved is constant on this complement. Every nonempty open generic pattern region meets the complement because $\overline{Z}$ has empty interior. There can therefore be only one such pattern. $\square$

For Lorentz cones of dimension at most $d \geq 3$, set $B = d - 2$ and retain the same boundedness and parameter hypotheses, with optional rays. The Lorentz version replaces E_i by the support face generated by the whole certificate fiber, and its dimension count by Jordan rank. Its rank-one orbit has dimension at most B . The same proof gives $\dim Z \leq qB - B$ and, for $B \geq 2$, the same fixed-label conclusion. When $B = 1$, the exceptional set may have codimension one and the connected-complement argument does not apply.

There is a useful precise statement about the topology of the exceptional fibers. The contact incidence space

$$\Gamma = \{(v, X, Y) : v \in \mathbb{S}^{qB}, X \in \mathcal{P}(v), Y \in \mathcal{D}(v)\}$$

is compact: primal compactness is assumed and the Slater normalization uniformly bounds all dual fibers. The fiber of $\Gamma \rightarrow \mathbb{S}^{qB}$ is exactly $\mathcal{P}(v) \times \mathcal{D}(v)$, since every primal is complementary to

every genuine certificate. This is a nonempty compact convex set. The Vietoris–Begle theorem for compact spaces with acyclic fibers [12] implies

$$\check{H}^k(\Gamma; \mathbb{Z}/2) \cong \check{H}^k(\mathbb{S}^{qB}; \mathbb{Z}/2). \quad (8.7)$$

Thus the incidence space itself supplies no additional cohomology. The problem is that the generic kernel-line map need not extend across rank-deficient fibers. Neither a global map nor a degree for it follows from (8.7). We do not assert such an extension.

8.5 Counterexamples to stronger fiberwise conclusions

Example 8.8 (Boundary selections can escape over a compact target). Put $\delta = 1 - x_1$ and impose three positive semidefinite blocks:

$$\begin{pmatrix} 1+x_1 & x_2 \\ x_2 & \delta \end{pmatrix} \succeq 0, \quad \begin{pmatrix} \delta & x_2 \\ x_2 & u \end{pmatrix} \succeq 0, \quad \begin{pmatrix} \delta & u \\ u & z-1 \end{pmatrix} \succeq 0.$$

The first block describes exactly $\mathbb{B}_2^2$. For every point of this ball with $\delta > 0$, the remaining blocks are feasible precisely when $u \geq x_2^2/\delta$ and $z \geq 1 + u^2/\delta$. At $(x_1, x_2) = (1, 0)$, they instead require $u = 0$ and $z \geq 1$. Thus the projection is the entire closed disk. The choice $x = 0, u = 1, z = 3$ makes all three blocks positive definite, so the lift has Slater feasibility. On the boundary $x_1 = 1 - \delta, x_2^2 = 2\delta - \delta^2$, every feasible completion satisfies

$$u \geq 2 - \delta, \quad z \geq 1 + \frac{(2 - \delta)^2}{\delta} \rightarrow \infty \quad (\delta \downarrow 0).$$

Consequently no boundary selection is locally bounded at $(1, 0)$. Compactness of the target and Slater feasibility do not imply bounded primal choices near contact; compactness assumptions on the lifted boundary therefore require separate justification.

Lemma 8.9 (A compact base removes one gradient unit). *Let F be a k -logarithmically homogeneous self-concordant barrier on a pointed cone, with positive definite Hessian on its span. On a nonempty base $\ell(X) = 1$, where ℓ is nonnegative on the cone, its restricted gradient parameter is at most $k - 1$.*

Proof. Write $g = \nabla F$, $H = \nabla^2 F$. Homogeneity gives $HX = -g$ and $X^T HX = k$. The squared dual norm of the gradient on $\ker \ell$ is its orthogonal projection in the inverse-Hessian norm:

$$\min_{\alpha} (g - \alpha \ell)^T H^{-1} (g - \alpha \ell) = k - \frac{1}{\ell^T H^{-1} \ell}.$$

The unit Dikin ellipsoid lies in the cone; since $\ell(X) = 1$ and ℓ is nonnegative there, $\ell^T H^{-1} \ell \leq 1$. This proves the claim. Compactness of the base is a convenient sufficient condition for a strictly positive height; the argument only needs the conditions stated above. $\square$

Example 8.10 (Small minimum nullity, larger global parameter). The Lorentz specialization of the perspective construction in Section 4 separates three quantities exactly. Partition $\mathbb{R}^{s-1}$ into q nonempty coordinate groups and write $x = (a, b)$. The lift is

$$W_G = \left(\frac{1-b+z_G}{2}, \frac{1-b-z_G}{2}, a_G \right) \in Q_{|G|+2}, \quad \sum_G z_G = 1 + b. \quad (8.8)$$

Equivalently, $1 - b \geq 0$, $z_G \geq 0$, and $\|a_G\|^2 \leq (1 - b)z_G$. Summing proves exact projection onto the ball; the reverse implication distributes the nonnegative residual $1 - b^2 - \|a\|^2$. All variables are bounded and $a = b = 0$, $z_G = 1/q$ is strictly feasible.

At every boundary point with $b < 1$, every inequality is an equality, $z_G = \|a_G\|^2/(1 - b)$, and each block is a nonzero extreme ray. At the north pole, the fiber is $z_G \geq 0$, $\sum_G z_G = 2$. Its relative interior again has one null eigenvalue per block, whereas a vertex has one such block and $q - 1$ zero blocks. Consequently

$$\min_{X \in \mathcal{P}(v)} \text{nul } X = q \quad \text{for every } v, \quad \max_{X \in \mathcal{P}(\text{north})} \text{nul } X = 2q - 1.$$

For completeness, at support $v = (c, \eta)$ the entire certificate fiber has blocks

$$V_G = (A_G + \gamma, A_G - \gamma, -c_G), \quad \gamma = (1 - \eta)/2, \quad \sum_G A_G = (1 + \eta)/2, \quad A_G \geq 0, \quad 4A_G\gamma \geq \|c_G\|^2.$$

Coefficient matching on the entire affine slice proves these conditions necessary and sufficient. For $\eta < 1$, all the inequalities are equalities and the certificate is unique with every block nonzero of rank one. At $\eta = 1$ it is the simplex $A_G \geq 0$, $\sum_G A_G = 1$. Thus the maximum certificate rank is q at every support, although its minimum at the north pole is one.

The restricted standard barrier is $F = -\sum_G \log((1 - b)z_G - \|a_G\|^2)$, with exact parameter $2q - 1$. For the upper bound homogenize 1 to τ in (8.8); the resulting cone has a compact $\tau = 1$ base and its restricted product barrier is $2q$ -homogeneous. Apply Lemma 8.9. The north-fiber vertex and Lemma 8.1 give the matching lower bound. Separate copies for heterogeneous source balls give respectively $\sum_a q_a$ minimum fiber nullity, $\sum_a q_a$ maximum aggregate certificate rank for strictly positive source weights, and $\sum_a (2q_a - 1)$ standard parameter. No constant mass can transfer between source certificates: a source affine slack $\mu_a - \lambda_a v_a^T x_a$ is nonnegative only if $\mu_a \geq \lambda_a$, and equality of the total constants forces equality sourcewise.

This example disproves an extra-unit lower bound with the quantifier “every primal completion at some support.” A single high-nullity completion is enough for the barrier lower bound, and is exactly what that stronger statement would miss.

Two simpler examples explain why convex fibers alone cannot globalize several projective channels. In a two-dimensional $\mathbb{F}$ -subspace U , the set $\{Y \succeq 0 : \text{Ran } Y \subset U, \text{tr } Y = 1\}$ contains all primitive rays in U , normalized to trace one, and their rank-two mixtures. At its scalar center there is no distinguished eigenline. Adding $q - 2$ fixed rank-one blocks stays within a total rank budget of q . For a primal example, encode the simplex $\sum_{j=0}^q p_j = 1$, $p_j \geq 0$, using

$$X_0 = \text{diag}(p_0, p_1), \quad X_j = \text{diag}(1, p_{j+1}), \quad 1 \leq j \leq q - 1,$$

and project to $\sigma = \sum_{j=2}^q p_j \in [0, 1]$. The fiber at $\sigma = 0$ has $p_0 = t$, $p_1 = 1 - t$: its first kernel changes between two lines at the endpoints and disappears in between. Its nullity is $q - 1$ inside the fiber and q at either endpoint. The standard barrier is $-\sum_{j=0}^q \log p_j$, with exact parameter q by Lemmas 8.9 and 8.1. This is an interval lift, not a ball lift in the critical dimension. It shows precisely why a proof for the unsettled range must use more than boundedness, convexity, and a nullity bound.

9 Exact parameters of concrete affine formulations

This section compares three representations of balls on their actual affine domains. The norm tree minimizes the number of small cone factors, but its restricted standard barrier is often less

economical than the grouped formulation. The grouped and column-packed slices also admit lower bounds for every self-concordant barrier on the same domain, including barriers with arbitrary mixed derivatives. For an open convex domain Ω in its affine hull, write $\vartheta_{\text{opt}}(\Omega)$ for the infimum of the parameters of all self-concordant barriers with positive definite Hessian on that affine hull. This infimum allows changes of the barrier, while retaining the formulation.

The standard-parameter calculations for norm trees, grouped slices, and column packing also appear in the unpublished companion manuscript *The Cost of Following the Central Path* [21]. We reproduce them with proofs to keep the present treatment self-contained. Here they support additional questions about arbitrary barriers on a fixed affine domain and resource frontiers across different lifts; the shared calculations and constructions are not claimed as separate new contributions.

9.1 Norm trees: the root incidence determines the standard parameter

Let T have $N \geq 2$ leaves and L internal nodes, each with $c_v \geq 2$ children. At an internal node v , let t_v be its axial variable and let y_v collect the values of its children: a leaf contributes a projected coordinate, and an internal child w contributes t_w . The linked cone has $t_v > \|y_v\|$ for every v . Its root slice, with $t_r = 1$, is denoted by Ω_T and projects onto $\text{int } \mathbb{B}_2^N$. Put

$$q_v = t_v^2 - \|y_v\|^2, \quad \Phi_T = - \sum_v \log q_v, \quad F_T = \Phi_T|_{t_r=1}.$$

The positive-sheet condition $t_v > 0$ is part of the domain.

Theorem 9.1 (Root-incidence law). *The least parameter of the displayed restricted standard barrier is*

$$\nu(F_T) = \begin{cases} 2L - 2, & \text{every root child is internal,} \\ 2L - 1, & \text{at least one root child is a leaf.} \end{cases} \quad (9.1)$$

For a forest of independent root-fixed trees, these parameters add.

Proof. Affine restriction of the standard Lorentz barriers gives standard self-concordance. The Hessian is positive definite, since every nonzero linked direction changes a cone coordinate. Before fixing $t = t_r$, logarithmic homogeneity has degree $2L$. With $H = D^2\Phi_T$ and e_t the root covector, homogeneity gives $H z = -D\Phi_T$ and $\|D\Phi_T\|_{H^{-1}}^2 = 2L$. Orthogonal projection onto the slice tangent space yields

$$\|DF_T\|_{D^2F_T, *}^2 = 2L - \frac{t^2}{e_t^T H^{-1} e_t}, \quad \nu(F_T) = 2L - \frac{1}{\sup \beta_T}, \quad \beta_T = \frac{e_t^T H^{-1} e_t}{t^2}. \quad (9.2)$$

The notation on the left means the dual norm induced by the restricted Hessian. The unit Dikin ellipsoid is contained in the linked cone, which lies in $t > 0$. Consequently $e_t^T H^{-1} e_t \leq t^2$: otherwise a vector of local norm less than one could move t to zero.

If a root slot is a leaf, set $t = 1$, that leaf to $\sqrt{1 - \epsilon}$, and the other root leaves to zero. Scale every internal child subtree from a fixed interior point by ϵ^2 . Then $q_r = \epsilon - O(\epsilon^4) > 0$. The direction with $h_t = 1$, $h_{\text{leaf}} = 1$, and all other components zero has

$$D^2\Phi_T[h, h] = \frac{4(1 - \sqrt{1 - \epsilon})^2}{q_r^2} \rightarrow 1.$$

The descendant terms vanish in this direction. Since $(e_t^T H^{-1} e_t)^{-1} = \min_{h_t=1} h^T H h$, the preceding limit and the Dikin bound prove $\sup \beta_T = 1$.

Suppose instead that every root child is internal, with axial values $a_i > 0$. Eliminate its strict descendants from the Hessian. The effective axial curvature of child i is at least a_i^{-2} by Dikin containment in that child's linked cone. Increasing these curvatures can only increase the final root Schur complement. It therefore suffices to consider

$$-\log(t^2 - \sum_i a_i^2) - \sum_i \log a_i.$$

Normalize $t = 1$, and put $r = \sum_i a_i^2$, $q = 1 - r$, and $Z = \sum_i a_i^4/(q + 2a_i^2)$. The child Hessian is the diagonal matrix with entries $2/q + a_i^{-2}$ plus $4aa^T/q^2$. Its cross row with the root is $-4a^T/q^2$. The rank-one inverse formula gives the root Schur complement

$$S = \frac{2(1+r) - 16Z/(q+4Z)}{q^2}. \quad (9.3)$$

Since $x/(q+2x)$ increases for $x \geq 0$, $Z \leq r^2/(q+2r) = r^2/(1+r)$. The inequality $S \geq 2$ is equivalent to $r(3-r) \geq 4(2-r)Z$; substituting this upper bound leaves $3r(1-r)^2/(1+r) \geq 0$. Hence $\beta_T \leq 1/2$. Conversely, scale all child subtrees by $\epsilon \downarrow 0$. The direction changing only the root has squared local norm $2(1+r)/(1-r)^2 \rightarrow 2$. The same variational formula proves $\sup \beta_T = 1/2$. Equation (9.2) now proves both cases. On a Cartesian product, squared dual gradient norms add; their suprema can be approached independently, proving the forest claim. $\square$

Corollary 9.2 (The best capped norm tree). *Fix an integer Lorentz dimension cap $d \geq 3$, and set*

$$A = d - 2, \quad E = N - 1, \quad L_0 = \lceil E/A \rceil, \quad \sigma = L_0 A - E.$$

Define $\chi = 1$ if $L_0 \geq 3$ and $\sigma \geq \max\{0, d - L_0\}$, and define $\chi = 0$ otherwise. Among all norm trees with $c_v + 1 \leq d$,

$$\min_T \nu(F_T) = 2L_0 - 1 - \chi. \quad (9.4)$$

Proof. The tree identity $\sum_v (c_v - 1) = E$, with $1 \leq c_v - 1 \leq A$, gives $L \geq L_0$. Any list of L_0 increments in this interval with sum E is realized by a chain of internal nodes, as in Corollary 3.4. For an all-internal root of arity c , one must additionally have $2 \leq c \leq \min(d-1, L_0-1)$. The remaining $L_0 - 1$ increments must sum to $E - (c-1)$. Such increments exist exactly when

$$L_0 \geq 3, \quad E \leq (L_0 - 1)A + \min(d-1, L_0-1) - 1.$$

The lower bound on their sum causes no further constraint, since $E \geq L_0$ and the root increment may be chosen no larger than $E - L_0 + 1$. Partitioning the remaining internal nodes into c nonempty chains realizes the required descendant forest. The displayed inequality is precisely the condition defining χ . An additional factor cannot improve the minimum: even its better root type gives $2(L_0 + 1) - 2 > 2L_0 - 1$. $\square$

The root-fixed domain has a barrier lower bound that does not depend on the displayed logarithms.

Proposition 9.3 (Intrinsic norm-tree bounds). *Every self-concordant barrier on Ω_T has parameter at least L . Every logarithmically homogeneous self-concordant barrier on the unfixed linked cone has degree at least $L + 1$. The corresponding root-fixed lower bound for an independent forest is the sum of its numbers of internal nodes.*

Proof. Choose positive leaf values with squared sum one and set t_v^* equal to the norm of the leaves below v . Each $z_v^* = (t_v^*, y_v^*)$ is a nonzero boundary vector. Intersect the linked affine slice with the product of two-planes

$$z_v = \alpha_v(t_v^*, y_v^*) + \beta_v(t_v^*, -y_v^*).$$

Cone membership is exactly $\alpha_v, \beta_v \geq 0$; its determinant is $4(t_v^*)^2 \alpha_v \beta_v$. Root fixing and linking become

$$\alpha_r + \beta_r = 1, \quad \alpha_w + \beta_w = \alpha_v - \beta_v \quad (w \text{ an internal child of } v). \quad (9.5)$$

The L variables β_v are affine coordinates: these equations determine every α_v recursively. Small positive values of all β_v give strictly positive α_v , so the section has dimension L and meets the interior of the linked slice. Nonnegativity and the equations bound all coordinates by one. At $\beta = 0$, all $\alpha_v = 1$, and exactly the L independent inequalities $\beta_v \geq 0$ are active. Every relative boundary point of this polytope lies on a cone boundary. Restriction of any barrier on Ω_T is therefore a barrier on this polytope. The independent-active-facet lower bound [70, Proposition 2.3.6] proves the claim. Product sections prove the forest extension.

The corresponding independent-facet principle for logarithmically homogeneous barriers on polyhedral cones, with one additional parameter unit at a nonzero boundary point, is due to Hildebrand [53, Theorem 6.1]. The tree section above gives the following direct argument for this application. For the conic claim let the logarithmic degree be θ . Lemma 8.9 gives a root-slice parameter at most $\theta - 1$. The first claim gives $\theta - 1 \geq L$. Logarithmic homogeneity is essential to this last argument. $\square$

Thus the intrinsic tree parameter lies between L and the value in (9.1). We do not identify these endpoints for every tree. In particular, the exact standard-barrier formula does not assert optimality among arbitrary barriers.

One tempting correction already fails on the two-block comb. The function

$$\tilde{F}(t, x, a, y, z) = -\log(t^2 - x^2 - a^2) - \log(a^2 - y^2 - z^2) + \log a$$

has logarithmic degree three. At the interior point $(3, 0, 1, 0, 0)$, in direction $(0, 0, 1, 1, 0)$, its derivatives are $D^2 \tilde{F} = 53/16$ and $D^3 \tilde{F} = -441/32$. Their squared self-concordance inequality fails, since $(441/32)^2 - 4(53/16)^3 = 45604/1024 > 0$. Removing one logarithmic axial term per link therefore does not produce a valid lower-parameter barrier by this formula.

9.2 Grouped fixed-tail slices: coupling cannot improve the parameter

For each ball a , partition its coordinates into nonempty groups $G \in \mathcal{G}_a$, and put $H = \sum_a |\mathcal{G}_a|$. Consider

$$\Omega_{\mathcal{G}} = \left\{ (u, w) : u_{aG} > \|w_{aG}\|^2, \quad \sum_{G \in \mathcal{G}_a} u_{aG} = 1 \text{ for every } a \right\}. \quad (9.6)$$

Its projection is the product of the open balls. The Lorentz realization uses $((1+u)/2, (1-u)/2, w) \in \text{int } Q_{|G|+2}$. The real PSD realization uses $\begin{pmatrix} u & w^T \\ w & I \end{pmatrix} \succ 0$. More generally, fixing the tail $z = 1$ in Lemma 4.2 gives the same domain in every simple Jordan family, for group size at most $a(r-1)$. In all these cases the restricted determinant is $u - \|w\|^2$.

Theorem 9.4 (Intrinsic grouped parameter). *The domain in (9.6) satisfies*

$$\vartheta_{\text{opt}}(\Omega_{\mathcal{G}}) = H. \quad (9.7)$$

The restricted standard barrier $F_{\mathcal{G}} = -\sum_{a,G} \log(u_{aG} - \|w_{aG}\|^2)$ attains this value.

Proof. For $f(u, w) = -\log q$, $q = u - \|w\|^2$, consider an affine line and write $A = q'/q$, $C = 2\|w'\|^2/q \geq 0$. Then

$$f'' = A^2 + C, \quad f''' = -2A^3 - 3AC, \quad 4(A^2 + C)^3 - (2A^3 + 3AC)^2 = 3A^2C^2 + 4C^3 \geq 0.$$

The Hessian is positive definite, and $(D^2f)^{-1}Df = (-q, 0)$ gives squared dual gradient norm one. Thus each summand is a one-self-concordant barrier; addition and affine restriction give the upper bound H .

For the lower bound choose positive σ_{aG} summing to one for each a , and a unit vector e_{aG} in each nonempty group. The affine section $u_{aG} = \sigma_{aG}$, $w_{aG} = t_{aG}e_{aG}$ is exactly the open box

$$\prod_{a,G} (-\sqrt{\sigma_{aG}}, \sqrt{\sigma_{aG}}).$$

Its boundary lies on the boundary of Ω_G . An arbitrary barrier, including one coupling all groups, restricts to a barrier on this H -dimensional box. The classical facet bound [70, Proposition 2.3.6] yields $\nu \geq H$. $\square$

For comparison, the product of k positive-dimensional balls itself has intrinsic parameter k . The upper bound follows by restricting the one-parameter paraboloid barriers to $u = 1$ and adding; the lower bound follows from a k -dimensional box section. Consequently the grouped domain has an intrinsic parameter excess $H - k$ over its projection. Under a genuine Lorentz cap $3 \leq d < N + 1$, a single ball uses $h = \lceil N/(d - 2) \rceil$ groups. In the notation of Corollary 9.2, put $\zeta = \mathbf{1}_{\{d-2|N-1\}}$, so $h = L_0 + \zeta$. The exact comparison is

$$\min_T \nu(F_T) - \nu(F_G) = L_0 - 1 - \chi - \zeta \geq 0. \quad (9.8)$$

Here $L_0 \geq 2$; when $L_0 = 2$, $\chi = 0$, and when $L_0 \geq 3$ the last inequality is immediate. The grouped lift uses at most one additional factor, while its parameter can be asymptotically half the tree's standard parameter. If $d \geq N + 1$, the direct Lorentz slice already has parameter one.

The same box section also gives the lower bound H for $u_{aG} > \|w_{aG}\|_p^p$, $1 < p < \infty$, with the same allocation equations. The matching upper bound just proved uses the quadratic case $p = 2$; it is not asserted for other exponents.

9.3 Bounded-fiber projection and one-block column packing

We give precise projection hypotheses because a lower bound on the projected body is not transferred by merely counting lift variables. The following is a form of the classical exact partial-minimization theorem [22, Theorem 5.2.1].

Lemma 9.5 (Parameter monotonicity under bounded-fiber projection). *Let $Q \subset E \times V$ be a closed convex set with nonempty interior, and suppose every nonempty fiber $Q_x = \{y : (x, y) \in Q\}$ is bounded. Suppose $C = \pi_E Q$ contains no line. If F is a self-concordant barrier on $\text{int } Q$ with positive definite Hessian and parameter ν , then*

$$\phi(x) = \min\{F(x, y) : (x, y) \in \text{int } Q\}$$

is a self-concordant barrier with positive definite Hessian and parameter at most ν on $\text{int } C$. The same statement holds in affine hulls after eliminating affine equations. In particular, $\vartheta_{\text{opt}}(\text{int } Q) \geq \vartheta_{\text{opt}}(\text{int } C)$.

Proof. First, inverse images in Q of bounded subsets of E are bounded. Otherwise choose $(x_j, y_j) \in Q$ with x_j bounded and $\|y_j\| \rightarrow \infty$. Normalize their differences from a fixed $q_0 \in Q$ and take a subsequence converging to a vertical unit vector $(0, v)$. Closedness and convexity imply $q_0 + t(0, v) \in Q$ for every $t \geq 0$, contradicting the bounded fiber through q_0 . This argument also proves that C is closed, by lifting a convergent sequence in C and extracting a subsequence. The standard relative-interior identity for linear images of convex sets gives $\pi_E(\text{int } Q) = \text{int } C$.

For each interior x , the fiber is bounded and its boundary lies on ∂Q . Barrier divergence and strict convexity therefore give a unique interior minimizer $y(x)$. The equation $F_y = 0$ and the implicit-function theorem show that $y(x)$ is C^2 . For a direction h , set $\hat{h} = (h, Dy(x)h)$. Differentiating stationarity gives Hessian orthogonality to every vertical direction, and hence

$$\begin{aligned} D\phi[h] &= DF[\hat{h}], \\ D^2\phi[h, h] &= D^2F[\hat{h}, \hat{h}] = \min_k D^2F[(h, k), (h, k)], \\ D^3\phi[h, h, h] &= D^3F[\hat{h}, \hat{h}, \hat{h}]. \end{aligned}$$

In the third identity the acceleration term is vertical and its Hessian pairing with $\hat{h}$ vanishes. The positive definite Schur complement proves nondegeneracy; the two defining derivative inequalities for F prove the same inequalities for ϕ . Finally, as x_j tends to a finite boundary point of C , the minimizers remain bounded by the first paragraph. Any cluster point lies on ∂Q , since an interior cluster point would project to an interior point of C . Barrier divergence therefore implies $\phi(x_j) \rightarrow +\infty$. This completes the proof. $\square$

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, let $p, b \geq 1$, and choose nonzero real linear subspaces $E_a \subset \mathbb{F}^p$. With $W = [w_1 \cdots w_b]$, $w_a \in E_a$, define

$$\Omega_{\text{pack}} = \left\{ (S, W) : S \in H^b(\mathbb{F}), S_{aa} = 1, \begin{pmatrix} S & W^* \\ W & I_p \end{pmatrix} \succ 0 \right\}. \quad (9.9)$$

Determinants for quaternionic Hermitian matrices mean Jordan determinants, and traces in the calculations below mean real traces.

Theorem 9.6 (Intrinsic column-packing parameter). *For the single-factor domain in (9.9),*

$$\vartheta_{\text{opt}}(\Omega_{\text{pack}}) = b. \quad (9.10)$$

*The restricted standard barrier $F_{\text{pack}} = -\log \det(S - W^*W)$ attains this value. This holds without any relation between the number of columns b and the row dimension p .*

Proof. The Schur complement $D = S - W^*W$ gives the exact projection $\prod_a \{w_a \in E_a : \|w_a\| < 1\}$. Indeed its diagonal entries are $d_a = 1 - \|w_a\|^2$; conversely, set $S = W^*W + \text{Diag}(d_a)$. The closed completion fibers satisfy $|D_{ac}|^2 \leq d_a d_c \leq 1$ and are bounded. The whole closed slice is bounded as well: $\|w_a\| \leq 1$, and $S \succeq 0$ with diagonal one implies $|S_{ac}| \leq 1$. It has relative interior, for example $S = I_b, W = 0$. Lemma 9.5 and the product-of-balls parameter established above show that every barrier has $\nu \geq b$.

For the upper bound work first on the unrestricted matrix epigraph $S - W^*W \succ 0$. It is an affine slice of $H_{++}^{p+b}(\mathbb{F})$, so its log-determinant is standard self-concordant. An affine shear at any base point sets its W coordinate to zero while preserving the Schur complement. At the resulting point $(D, 0)$,

$$DF[H, K] = -\text{tr}(D^{-1}H), \quad D^2F[(H, K), (H, K)] = \text{tr}(D^{-1}HD^{-1}H) + 2\text{tr}(D^{-1}K^*K).$$

The squared dual norm of this gradient is b , attained by $(H, K) = (-D, 0)$. Therefore this is a b -self-concordant barrier; the diagonal and column-subspace restrictions cannot increase its parameter. The lower bound proves equality. $\square$

For the standard barrier the partial minimizer is explicit. Hadamard's determinant inequality, with equality exactly at a diagonal positive matrix, gives

$$\min_{S: S_{aa}=1} F_{\text{pack}}(S, W) = - \sum_{a=1}^b \log(1 - \|w_a\|^2).$$

Thus column packing reduces the ambient parameter $p+b$ to b on the fixed slice, even when arbitrary coupled barriers are allowed. In contrast, the grouped slice may retain many more than b units. The one-factor count, the ambient rank, and the intrinsic slice parameter describe different costs.

Example 9.7 (Repeated boundary blocks do not bound arbitrary barriers). For $m \geq 2$, the diagonal affine slice

$$\{((1/m, x/m), \dots, (1/m, x/m)) : x \in \text{int } \mathbb{B}_2^s\} \subset Q_{s+1}^m$$

is affinely isomorphic to one ball. All m blocks reach nonzero boundary points simultaneously, and polynomial slack factors are $A_j(x) = (1, x)/m$, $B_j(z) = (1, -z)$. The restricted standard product barrier is $-m \log(1 - \|x\|^2)$ plus a constant, with exact parameter m ; its squared dual gradient norm approaches m as $\|x\| \rightarrow 1$. Yet the slice admits the one-parameter barrier $-\log(1 - \|x\|^2)$. The active conormals coincide after restriction. Consequently simultaneous boundary multiplicity alone is insufficient for an arbitrary-barrier lower bound; the polyhedral sections or projection argument above supply the additional geometry.

These parameter statements concern self-concordant barrier certificates on specified formulations. They do not assert that every algorithm takes a number of iterations proportional to the square root of the parameter, and they do not bound the conditioning or the arithmetic cost of a Newton solve.

10 Topology of saturated contact factorizations

The support-orbit argument has a counterpart for arbitrary proper cones. Here the relevant targets are the boundaries of compact cone bases. Saturation forces these initially nonsmooth boundaries to acquire smooth charts. This gives a joint covering theorem without a symmetry assumption on the cones.

10.1 Normalized cone bases and the joint covering

Let P, D be compact connected C^1 manifolds without boundary, both of dimension $n \geq 1$. Suppose $\zeta : P \rightarrow D$ is continuous and

$$s(x, z) = \sum_i \langle A_i(x), B_i(z) \rangle, \quad A_i : P \rightarrow K_i, \quad B_i : D \rightarrow K_i^*,$$

where the maps are C^1 , the cones are proper, and $s(x, \zeta(x)) = 0$. Assume that the contact pairing

$$H_x(v, w) = - \sum_i \langle DA_i(x)[v], DB_i(\zeta(x))[w] \rangle \quad (10.1)$$

is nondegenerate between $T_x P$ and $T_{\zeta(x)} D$. No differentiability of ζ is required. Write $c_i = \max\{\dim K_i - 2, 0\}$ and $I_+ = \{i : c_i > 0\}$.

Theorem 10.1 (Joint covering at capacity equality). *If $\sum_i c_i = n$, choose any $\ell_i \in \text{int } K_i^*$ for $i \in I_+$. Then A_i is nowhere zero, the boundary*

$$J_i = \partial\{a \in K_i : \ell_i(a) = 1\}$$

is a C^1 embedded hypersurface diffeomorphic to $\mathbb{S}^{c_i}$, and

$$p : P \longrightarrow \prod_{i \in I_+} J_i, \quad p_i(x) = \frac{A_i(x)}{\ell_i(A_i(x))},$$

is a finite C^1 covering. If P is simply connected, every $c_i \geq 2$ and the covering has one sheet. In particular, if $P \cong \mathbb{S}^n$, there is exactly one positive capacity, equal to n . If $P \cong \prod_{a=1}^b \mathbb{S}^{p_a}$ with $p_a \geq 2$, the multiset of positive capacities is exactly $\{p_1, \dots, p_b\}$.

Proof. At each contact every individual cone pairing is zero. Its fixed-variable first derivatives are therefore zero:

$$DA_i(T_x P) \subset B_i(\zeta(x))^\perp, \quad DB_i(T_{\zeta(x)} D) \subset A_i(x)^\perp. \quad (10.2)$$

Lemma 3.1 applies with the two tangent charts independent. Nondegeneracy and rank subadditivity force every mixed channel to have rank c_i . For $c_i > 0$ this excludes both vertices, because a two-sided derivative of a map into a pointed cone is zero at the vertex. Complementarity puts $p_i(x)$ in J_i .

For $a = A_i(x)$ and $b = B_i(\zeta(x))$, the channel factors through $b^\perp/\mathbb{R}a$, a space of dimension c_i . The quotient derivative has rank c_i , since the channel does. Direct differentiation shows

$$Dp_i(x)[v] = 0 \iff DA_i(x)[v] \in \mathbb{R}A_i(x).$$

Thus Dp_i has constant rank c_i . The base boundary J_i is initially only a topological sphere. The constant-rank theorem supplies a local c_i -dimensional transversal on which p_i is a regular embedding. Invariance of domain makes this embedded patch relatively open in J_i . Consequently $p_i(P)$ is open and compact in the connected J_i , and is all of J_i . The regular patches give its C^1 embedded structure. Radial projection from an interior point of the base is a C^1 diffeomorphism to $\mathbb{S}^{c_i}$: a supporting normal has strictly positive pairing with the radial direction, so that direction is never tangent.

If $v \in \bigcap_{i \in I_+} \ker Dp_i$, the preceding kernel formula and (10.2) give $H_x(v, w) = 0$ for every w ; zero-capacity channels also vanish. Hence $v = 0$. Source and target of p have equal dimensions, so p is a local diffeomorphism. Compactness and connectedness make it a surjective finite covering.

If P is simply connected, a submersion $p_i : P \rightarrow \mathbb{S}^1$ would lift to a real-valued function, whose maximum contradicts submersivity. Thus the target is simply connected and the cover has one sheet. A product of two positive-dimensional spheres has intermediate cohomology, which proves the single-sphere assertion. More generally, the graded indecomposable quotient $H^+/(H^+)^2$ of rational cohomology has one generator in the dimension of each sphere. Comparing this quotient on the source and target proves the multiset assertion. $\square$

For a convex body with C^1 primal and polar boundaries, the independent tangent spaces at a contact are $y^\perp$ and $x^\perp$, where $\langle x, y \rangle = 1$. Their Euclidean pairing is nondegenerate: its first-slot kernel would lie in $y^\perp \cap \mathbb{R}x = \{0\}$. Thus the theorem applies without differentiating the supporting-polarity map. It also applies to weighted product-ball contact kernels. The resulting covering need not split as a product of maps depending on individual source coordinates. Nor does this theorem identify an arbitrary saturated cone as Lorentz; it identifies its normalized boundary and the capacity profile.

Corollary 10.2 (Smooth ball frontier for arbitrary proper cones). *Let $N \geq 3$ and let $d \geq 3$ be an integer. Allow arbitrary proper cone factors of dimension at most d . Among exact ball lifts with global C^1 primal and genuine dual selections, the simultaneous minimum number of factors, total cone dimension, and parameter of a barrier on the full ambient cone product are*

$$\begin{cases} (h, N + 2h, 2h), & h = \lceil N/(d-2) \rceil, \quad d < N + 1, \\ (1, N + 1, 2), & d \geq N + 1. \end{cases}$$

The ambient-parameter conclusion includes coupled barriers, and hence also coupled logarithmically homogeneous barriers.

Proof. If $d < N + 1$, every positive capacity is smaller than $N - 1$. Theorem 10.1 excludes equality in the local bound, so their sum is at least N . If there are k such factors, $k \geq h$ and their dimensions sum to at least $N + 2k$. Every proper cone of dimension at least two has a two-dimensional linear section through its interior that is isomorphic to $\mathbb{R}_+^2$. Restricting any ambient barrier to the product of these sections gives an orthant barrier, whose parameter is at least $2k$ [70, Proposition 2.3.6]. Extra factors cannot reduce these bounds. For $d \geq N + 1$, the local bound already gives total dimension at least $N + 1$, and the same section argument gives parameter at least two. The grouped Lorentz lifts in Theorem 6.2, and the direct Lorentz lift in the second case, attain all bounds. $\square$

The following local form is useful for cover arguments involving changing active labels. It also explains why nonsmooth cone boundaries do not remove the topological obstruction.

Lemma 10.3 (Proper normalized-phase domains). *Let $p \geq 2$, let $0 < c_i < p$ with $\sum_i c_i = p$, and let J_i be the boundaries of compact convex bodies of dimensions $c_i + 1$. Suppose $U \subset \mathbb{S}^p$ is open and $q_i : U \rightarrow J_i$ are C^1 as ambient maps. If the combined derivative of $q = (q_i)_i$ is injective everywhere, then $U \neq \mathbb{S}^p$. In particular the top class of $\mathbb{S}^p$ restricts to zero on U . The same holds on the inverse image of U under any coordinate projection from a product of spheres.*

Proof. A map into J_i has derivative rank at most c_i : rank $c_i + 1$ would make its image contain an ambient open set. Therefore joint injectivity forces each component to have rank c_i . The transversal and invariance-of-domain argument above gives regular charts on their images. If U were the whole sphere, every image would be all of J_i and q would be a finite covering of their product. A circle factor contradicts simple connectivity and finite covering degree; without circle factors, the product is simply connected and its intermediate cohomology contradicts that of the sphere. Finally every proper open subset of a connected closed oriented manifold has zero ordinary top cohomology, by noncompact-manifold duality, component by component [48, Section 3.3]. $\square$

11 Capacity equality for products of balls

The single-ball support-orbit obstruction has a stronger product form. Equality forces one spin factor for each ball, even when scalar ray factors are present. The proof uses both the topology of the joint support map and the fact that each dual row depends on only one source coordinate.

Theorem 11.1 (Product saturation with scalar rays). *Let $M = \prod_{a=1}^h S^{p_a}$, where $p_a \geq 1$, and $n = \sum_a p_a$. Suppose the full row identities*

$$1 - x_a \cdot z = \sum_i \langle X_i(x), Y_i^a(z) \rangle + \sum_j \alpha_j(x) \beta_j^a(z), \quad x \in M, \quad z \in S^{p_a}, \quad (11.1)$$

have finitely many globally labelled C^1 factors. The maps X_i, Y_i^a take values in simple non-ray symmetric cones K_i , of Jordan rank r_i , Peirce constant a_i , and real dimension m_i ; the scalar factors are nonnegative. Put $c_i = a_i \lfloor r_i^2/4 \rfloor$. Then

$$\sum_i c_i \geq n.$$

If equality holds, there are exactly h non-ray factors, and after relabelling

$$K_a \cong Q_{p_a+2}, \quad Y_a^d \equiv 0 \quad (d \neq a).$$

Conversely, separate direct Lorentz sections attain equality. Thus the non-ray factors at equality have total dimension $n + 2h$ and total Jordan rank $2h$. If this cone profile fails, $\sum_i c_i \geq n + 1$.

We first establish the topological information needed in its proof. For a simple algebra V , write $\mathcal{I}_k(V)$ for the rank- k idempotent orbit, as in Section 7.

Lemma 11.2 (Product support covering). *Under equality in Theorem 11.1, the positive factors have constant balanced complementary ranks. Their support maps, complemented when necessary, form a finite covering*

$$\Sigma : M \longrightarrow \prod_i \mathcal{I}_{\lfloor r_i/2 \rfloor}(V_i). \quad (11.2)$$

The only possible non-spin factors allowed by this covering topology are $\mathbb{S}_+^3$ and $\mathbb{S}_+^4$. Their orbit universal covers are S^2 and $S^2 \times S^2$, respectively. Each of these target S^2 factors is assigned to a distinct literal S^2 factor of M : its restriction to that source factor lifts to a map of degree ± 1 , and its restriction to every other source S^2 has degree zero.

Proof. At contact put $Y_i(x) = \sum_a Y_i^a(x_a)$. Termwise nonnegativity gives $X_i(x) \circ Y_i(x) = 0$. A nonnegative differentiable scalar function has zero derivative at each zero. Hence each scalar product in (11.1) contributes zero to the mixed derivative at contact. Summing rows and differentiating the primal and dual variables independently gives the nondegenerate product-sphere metric. The Peirce rank bound yields

$$n \leq \sum_i a_i u_i v_i \leq \sum_i c_i, \quad u_i = \text{rank}_J X_i(x), \quad v_i = \text{rank}_J Y_i(x).$$

Equality forces $u_i + v_i = r_i$ and balanced ranks everywhere. Lower semicontinuity makes each rank constant. The support-differential calculation in Theorem 7.1 now applies without change: its joint differential is injective and has equal-dimensional domain and codomain. Compactness therefore gives the finite covering (11.2).

Lift this covering to universal covers. The lift is a diffeomorphism. The source universal cover is $\mathbb{R}^t \times \prod_{p_a \geq 2} S^{p_a}$, where t counts source circles. Each spin orbit is a sphere; each non-spin classical orbit is a balanced Grassmannian. The Albert orbit is $\mathbb{O}P^2$. We exclude the other possibilities explicitly.

Every positive-degree class in the mod-two cohomology of a product of spheres has square zero. In a complex Grassmannian of order at least three, the degree-two Schubert class instead satisfies $\sigma_1^2 = \sigma_2 + \sigma_{1,1} \neq 0$, omitting terms outside its rectangle. The same relation holds with degrees doubled for quaternionic Grassmannians; see [79, Section 2] for the Schubert ring and its quaternionic counterpart. These spaces are simply connected, so their square obstruction persists in the universal cover. For the Albert orbit, $H^*(\mathbb{O}P^2; \mathbb{F}_2) = \mathbb{F}_2[u]/(u^3)$, $|u| = 8$, gives the same obstruction [48, Example 4.47].

For a balanced real Grassmannian of order $r \geq 6$, both ranks $k, r - k$ are at least three. Its simply connected oriented cover has

$$\pi_2(\mathrm{Gr}_k^+(\mathbb{R}^r)) = \ker((\mathbb{Z}/2)^2 \xrightarrow{+} \mathbb{Z}/2) = \mathbb{Z}/2.$$

This follows from the fibration with fiber $SO(k) \times SO(r - k)$ in $SO(r)$; the same exact sequence proves simple connectivity. Hurewicz then gives $H_2 = \mathbb{Z}/2$, which remains a direct summand of H_2 of the universal-cover product. The source has torsion-free H_2 .

At real order five the oriented cover is the complex quadric $Q^3 \subset \mathbb{C}P^4$: an oriented orthonormal pair (u, v) maps to $[u + iv]$, identifying its plane with an isotropic complex line. Here $H^2(Q^3; \mathbb{Q})$ is one-dimensional. Indeed the same homogeneous fibration has fundamental-group map $\mathbb{Z} \times \mathbb{Z}/2 \rightarrow \mathbb{Z}/2$, $(b, e) \mapsto b + e$; its kernel is $\mathbb{Z}$, so Hurewicz computes $H_2 = \mathbb{Z}$. The hyperplane class h satisfies $\int_{Q^3} h^3 = 2$, the degree of the quadric, and hence $h^2 \neq 0$. Any square-zero degree-two class on a product containing this quadric restricts to zero on that factor. Consequently square-zero classes cannot span the product's degree-two rational cohomology. They do span degree two for a product of spheres, a contradiction. This excludes real order five without a torsion calculation in its integral cohomology ring.

The remaining non-spin possibilities are real orders three and four, whose oriented covers are S^2 and $\mathrm{Gr}_2^+(\mathbb{R}^4) \cong S^2 \times S^2$. These identifications, and the classical-group fibrations used above, are compatible with the Grassmannian descriptions in [48, Examples 4.53–4.55]. All remaining universal-cover factors are therefore spheres or contractible real lines. The indecomposable quotient of their positive-degree cohomology records one generator in each sphere dimension. It follows that the target sphere dimensions, with multiplicity, equal the source dimensions at least two. Circle multiplicities agree because the finite covering induces an inclusion of finite index on fundamental groups, whose free-abelian ranks count circles.

Finally, in the integral cohomology of a product of simply connected spheres, let $u_1, \dots, u_b$ be the degree-two sphere generators. Then

$$\left(\sum_j b_j u_j \right)^2 = 2 \sum_{j < k} b_j b_k u_j u_k.$$

The primitive square-zero degree-two classes are exactly the $\pm u_j$. The ring isomorphism induced by the universal-cover diffeomorphism thus permutes these classes up to sign. Restricting to a literal source S^2 gives the claimed degree assignment. Changing the other fixed coordinates gives homotopic restrictions in the universal cover, so these degrees hold on every such cylinder, up to orientation choices. $\square$

Lemma 11.3 (The two real exceptions violate the row identities). *Under the hypotheses of Lemma 11.2, neither $\mathbb{S}_+^3$ nor $\mathbb{S}_+^4$ occurs.*

Proof. Fix a row d at contact and let $W = \mathrm{Ran} Y_i^d(x_d)$ in a real matrix block. Complementarity along the entire cylinder gives

$$W \subseteq \ker X_i(x) \quad \text{whenever } x_d \text{ is fixed.} \quad (11.3)$$

Suppose first that i is an order-four block. Its primal support has rank two, and the two S^2 rulings of its oriented Grassmannian are assigned to distinct source factors a, b . If $W \neq 0$ for a row $d \neq a$, vary only x_a . If $\dim W = 2$, the primal plane is constant. If $\dim W = 1$, the plane lies in $W^\perp$, a fixed three-space. The inclusion

$$\mathrm{Gr}_2^+(W^\perp) \cong S^2 \longrightarrow \mathrm{Gr}_2^+(\mathbb{R}^4) \cong S_+^2 \times S_-^2$$

has degree ± 1 in both projections. To see this, represent an oriented plane by a unit simple bivector and take its self-dual and anti-self-dual parts. For planes in a fixed three-space, the two parts are related by an orthogonal map. Every map factoring through this inclusion therefore has its two degrees equal up to sign. This contradicts the degrees $(\pm 1, 0)$ on the assigned a -cylinder. Thus $Y_i^d \equiv 0$ for every $d \neq a$. Applying the same argument to the assigned b -cylinder makes all rows zero, contradicting the positive complementary dual rank. No order-four block remains.

For an order-three block assigned to source a , the primal rank is one or two. A nonzero W for $d \neq a$ either fixes its kernel line (primal rank two), or confines its range line to $\mathbb{RP}(W^\perp)$ of dimension at most one (primal rank one). The lifted map on the assigned S^2 then has degree zero, contradicting Lemma 11.2. Hence this block serves only row a .

We next show that no other non-ray block serves row a . Every other order-three block has a distinct assigned S^2 , so the preceding argument applies. Each remaining block is spin. Its sphere target can be assigned to a source sphere of the same dimension with nonzero restriction degree: in each degree at least two choose a nonzero term in the determinant of the invertible matrix on cohomological indecomposables of the universal covers. For circles use the invertible rational matrix induced on fundamental groups. These assignments are bijections; the degree-two assignments agree with the signed permutation already proved, so none of these blocks is assigned to a . Fixing row a and varying such an assigned source, a nonzero dual factor would force the spin primal support to be constant. Indeed, a rank-two cone has only one primitive idempotent orthogonal to any fixed nonzero complementary element. This contradicts its nonzero degree or winding. All other non-ray factors in row a vanish.

Fix all primal coordinates except x_a . Row a is now a factorization of the S^2 ball slack through one $\mathbb{S}_+^3$ block and finitely many scalar rays. At each contact the ray mixed derivatives vanish, so this block carries the full two-dimensional mixed metric. Its projective support map is a local diffeomorphism and its ranks are complementary one and two. Proposition 7.5 excludes precisely this factorization, exchanging its two arguments if necessary. Thus no order-three block remains either. $\square$

Proof of Theorem 11.1. Lemma 11.2 gives the capacity inequality and covering. Its remaining conclusions and Lemma 11.3 leave only spin factors. The universal-cover cohomology and fundamental-group counts give exactly one target sphere for each source sphere, of the same dimension. The determinant assignment in the preceding proof matches each target to one source with nonzero restriction degree or winding. No splitting of the covering as a product map is required.

Let block i be assigned to source a , and fix a different row d . If $Y_i^d(x_a) \neq 0$, complementarity makes the primitive support of $X_i(x)$ constant as x_a varies, by the uniqueness of the complementary primitive ray in a spin cone. Its assigned degree would then be zero. Hence $Y_i^d \equiv 0$ for $d \neq a$, as asserted. The direct separate Lorentz representations give the converse. All capacities are integers, so failure of the equality profile gives the stated unit gap. $\square$

When the scalar ray summands vanish identically, there is a shorter additional observation: the joint support map at equality is injective. If two points have the same primal supports, the dual rows at either point annihilate the primal factors at the other, and the full row identities force their every coordinate to agree. Scalar rays prevent this argument, which is why the proof above retains a covering until it has used the cylinder identities.

For Hermitian factors over a fixed field, the theorem permits equality only with order-two blocks and source sphere dimensions respectively 1, 2, 4 over $\mathbb{R}, \mathbb{C}, \mathbb{H}$. Over all symmetric cones, arbitrary spin sizes are allowed, but a cap $m_i \leq d$ makes equality impossible if some source ball has $p_a + 2 > d$. These are statements about globally C^1 full-row factorizations. They do not

impose the same equality classification on lifts whose selections exist only on generic patches. The Grassmannian topology and its cohomology are classical; the contribution here is their use with the full row identities to classify capacity equality and exclude cross-row sharing, including finite scalar ray terms. Aubrun, La Piana, and Müller-Hermes [6, Theorem 1] classify cones for which every positive linear self-map factors through a finite direct sum of Lorentz cones. Their universal quantifier over positive linear maps differs from the hypothesis here, which concerns one globally C^1 full-row slack factorization at equality in its mixed-curvature capacity. Our theorem does not assert their Lorentz-factorization property for general cones.

12 Rigidity from entire boundary fibers

Global regularity of a chosen contact sheet is a substantive hypothesis. There is, however, a different route to a global support map: constant nullity on every point of every boundary fiber makes that map intrinsic to the projection. This section gives the resulting product rigidity theorem. The hypothesis concerns the whole fiber, including points that no chosen regular sheet reaches.

Theorem 12.1 (Constant-nullity rigidity). *Let $C = \prod_{a=1}^h \mathbb{B}^{s_a}$, where $s_a \geq 2$, and put $M = \prod_a \mathbb{S}^{p_a}$, $p_a = s_a - 1$, and $n = \sum_a p_a$. Consider an exact semialgebraic affine lift in relative-Slater form,*

$$C = \pi(\mathcal{L} \cap K), \quad K = \prod_i K_i,$$

where the K_i are simple symmetric cones, rays are allowed, and $\dim K_i \leq d$ with $d \geq 3$. Free variables have been eliminated as in Lemma 2.1. Suppose the full preimage

$$E = \{X \in \mathcal{L} \cap K : \pi(X) \in M\}$$

is compact. If an integer $L \geq 1$ satisfies

$$n = L(d - 2), \quad Z(X) := \sum_i (r_i - \text{rank}_J X_i) = L \quad (X \in E), \quad (12.1)$$

then

$$\{p_a : 1 \leq a \leq h\} = \underbrace{\{d - 2, \dots, d - 2\}}_{L \text{ copies}} \quad (12.2)$$

as multisets. Exactly L cone factors have positive nullity on E ; each is Q_d and has nullity one. All other factors are interior on E and contribute zero to every genuine supporting row there. The profile (12.2) is attained by the separate direct Lorentz lift.

Proof. Projection $E \rightarrow M$ is a continuous surjection with convex, hence connected, fibers. It follows that E is connected. Indeed, a separation of the compact set E into two nonempty clopen subsets would give two disjoint compact images covering M : a connected fiber must lie on one side of the separation. This contradicts connectedness of M .

Each nullity $z_i(X) = r_i - \text{rank}_J X_i$ is upper semicontinuous and integer-valued. Near any $X \in E$, all z_i are at most their values at X . Their constant sum forces equality term by term. Thus each z_i is locally constant, and connectedness makes it constant on E . If X, X' belong to one fiber, their midpoint belongs to the same fiber. In a symmetric cone the support of a positive sum is the join of the supports. Equality of the three ranks therefore implies

$$\text{supp } X_i = \text{supp } X'_i \quad \text{for every } i.$$

The support map is continuous on a fixed-rank stratum. Since a continuous surjection from a compact space to a Hausdorff space is a quotient map, these support maps descend to continuous maps on M .

Next choose primal and genuine pure-row dual selections on a common dense open semialgebraic contact set; Lemma 2.2 supplies such selections with locally constant ranks. Summing the pure rows gives the joint kernel $\sum_a (1 - \langle x_a, v_a \rangle)$, whose mixed contact form is the product metric of rank n . Write u_i and v_i for the primal and aggregate dual Jordan ranks at one such contact. The Peirce bound gives

$$\begin{aligned} n &\leq \sum_{i:r_i \geq 2} a_i u_i v_i \leq \sum_{i:r_i \geq 2} a_i (r_i - 1) z_i \leq \sum_{i:r_i \geq 2} (m_i - 2) z_i \\ &\leq (d - 2) \sum_{i:r_i \geq 2} z_i \leq (d - 2) Z = L(d - 2) = n, \end{aligned} \tag{12.3}$$

where $m_i = \dim K_i$. For the second inequality, complementarity gives $v_i \leq z_i$ and a nonzero primal rank is at most $r_i - 1$ whenever $z_i > 0$; if the primal vanishes, its mixed contribution is zero. The third inequality follows from

$$m_i - 2 - a_i(r_i - 1) = (r_i - 2) \left(1 + \frac{a_i(r_i - 1)}{2} \right).$$

Equality throughout (12.3) implies that rays have zero nullity and every positive-nullity non-ray factor has rank two, dimension d , and primal and dual ranks one. Consequently $z_i = 1$ in exactly L factors. Constancy of each z_i extends this conclusion to all of E . The other factors have interior primal coordinates; complementarity forces their dual coordinate to vanish for every genuine pure supporting row at every point of M .

The descended supports in the active Lorentz factors define

$$\Sigma : M \longrightarrow (\mathbb{S}^{d-2})^L.$$

This map is injective. If $\Sigma(x) = \Sigma(x')$, take any genuine certificate for the row $1 - \langle \pi_a(X), x_a \rangle$ and evaluate it at any lift over x' . Equal active support rays make every active pairing zero, and all other dual coordinates are zero. Hence $1 - \langle x'_a, x_a \rangle = 0$ for every a , so $x' = x$. Source and target have equal dimension. Invariance of domain makes the image open, and compactness makes it closed. The target is connected; thus Σ is a homeomorphism. In the mod-two cohomology ring of a product of positive-dimensional spheres, the quotient of the positive-degree ideal by its square has one generator in each sphere dimension, with multiplicity. A graded ring isomorphism therefore gives (12.2), including any degree-one generators. Conversely, under that profile the lift $(1, x_a) \in Q_d$ has compact singleton fibers and one unit of nullity per source, as asserted. $\square$

For a single s -dimensional ball with $d < s + 1$ and $s - 1 = L(d - 2)$, Theorem 12.1 is an impossibility statement. For products it is a sharp classification: equality permits only the profile of separate direct factors. This conclusion requires neither a continuous chosen primal lift nor a continuous chosen dual certificate.

Definition 12.2 (Accessibility by regular contacts). A point $X \in E$ is contact-accessible if there are contact points $x^{(j)} \rightarrow \pi(X)$ and local C^1 primal and genuine pure-row dual contact selections near $x^{(j)}$, on constant-rank strata, whose primal values $X^{(j)}$ satisfy $X^{(j)} \rightarrow X$.

Corollary 12.3 (Accessible equality cases). *Retain the lift and compactness assumptions of Theorem 12.1 and suppose $n = L(d - 2)$. If every $X \in E$ is contact-accessible and the restricted standard barrier satisfies $\nu_{\text{std}, \text{slice}} \leq L$, then the profile (12.2) is necessary.*

Proof. Lemma 8.1 gives $Z(X) \leq \nu_{\text{std}, \text{slice}} \leq L$ at every $X \in E$. At each approximating regular contact the curvature chain (12.3), without its final equalities, gives $Z(X^{(j)}) \geq L$. Upper semicontinuity then yields $Z(X) \geq \limsup_j Z(X^{(j)}) \geq L$. Thus $Z = L$ throughout E , and Theorem 12.1 applies. $\square$

A standard-barrier bound alone supplies the upper bound on nullity, not accessibility or the reverse inequality at projection-singular fiber points. A single global section likewise controls only its own image. The distinction is essential: the unrestricted barrier brackets in Section 8 are not closed by this corollary.

Part III

Formulation families and barrier comparisons

13 Whole rows, minimum dimension, and affine normalization

The differential bounds above permit one slack row to use many factors. A different resource law holds when an entire family of support rows must be assigned to one factor. In that model, ordinary function rank and the dimension of a contact face give simultaneous sharp bounds without any regularity assumption. We first derive this law, then separate two conclusions that remain valid when rows may split: rigidity at minimum total dimension, and a dimension bound for projections of a single normalized product-cone base.

13.1 Contact faces and exact grouping

Let $C_a \subset \mathbb{R}^{n_a}$ be compact full-dimensional convex bodies with $0 \in \text{int } C_a$, for $a \in [k]$. Write

$$E_a = \text{ext } C_a, \quad P_a = \text{ext } C_a^\circ, \quad F_a(z) = \{x \in C_a : \langle x, z \rangle = 1\}.$$

Define the maximum contact-face dimension and its codimension by

$$e_a = \max_{z \in P_a} \dim F_a(z), \quad \lambda_a = n_a - e_a.$$

Here face dimension is affine dimension, so $1 \leq \lambda_a \leq n_a$. The maximum exists because the nonempty set of possible dimensions is a finite set of integers. For a proper cone K define

$$f(K) = \max_{0 \neq y \in K^*} \dim \text{span}(K \cap y^\perp).$$

Thus $f(K)$ counts linear dimensions of proper exposed cone faces; the zero face is allowed.

Theorem 13.1 (Whole-row dimension and face bounds). *Fix a nonempty group $G \subseteq [k]$. Suppose arbitrary maps*

$$A : \prod_{a \in G} E_a \longrightarrow K, \quad B^a : P_a \longrightarrow K^*$$

into a proper m -dimensional cone satisfy

$$\langle A(x), B^a(z) \rangle = 1 - \langle x_a, z \rangle \quad (a \in G, x \in \prod_{b \in G} E_b, z \in P_a). \quad (13.1)$$

Put $N_G = \sum_{a \in G} n_a$. Then

$$m \geq N_G + 1, \quad f(K) \geq N_G + 1 - \min_{a \in G} \lambda_a. \quad (13.2)$$

Both bounds are attained by the shared homogenization cone

$$H_G = \{(t, (y_a)_{a \in G}) : t \geq 0, y_a \in tC_a \text{ for all } a \in G\}. \quad (13.3)$$

No continuity or definability of the factor maps is required.

Proof. The sets E_a and P_a affinely span their respective spaces: their convex hulls are C_a and C_a° . Consequently the functions $1 - \langle x_a, z \rangle$, indexed by all a, z , span the constant and every coordinate of x . These $N_G + 1$ functions are linearly independent on $\prod_{a \in G} E_a$: fix all coordinates but one and use its affine span, successively for each block. By (13.1), their span has dimension at most m . This proves the first bound.

Fix a, z and restrict x to the contact cylinder

$$(F_a(z) \cap E_a) \times \prod_{b \in G \setminus \{a\}} E_b.$$

Its image under A lies in $K \cap B^a(z)^\perp$. The vector $B^a(z)$ is nonzero, since its slack is not identically zero. The extreme points of the exposed face $F_a(z)$ are precisely $F_a(z) \cap E_a$, and they affinely span that face. Thus the restricted affine functions have rank

$$1 + \dim F_a(z) + \sum_{b \in G \setminus \{a\}} n_b.$$

They are restrictions of linear functionals of $A(x)$, so this quantity cannot exceed the dimension of the span of the complementary cone face. Maximizing over a, z proves the second bound.

The cone H_G is proper and has dimension $N_G + 1$. The maps $A(x) = (1, x)$ and $B^a(z) = (1, 0, \dots, -z, \dots, 0)$ give the required factorization. To compute all its exposed faces, write $h_a(w) = \max_{x \in C_a} \langle x, w \rangle$. Its dual is

$$H_G^* = \{(\alpha, (w_a)_a) : \alpha \geq \sum_a h_a(-w_a)\}.$$

An interior dual point exposes only zero. At a nonzero dual boundary point let $J = \{a : w_a \neq 0\}$, which is nonempty. Nonnegativity of each support slack shows that the complementary face has dimension

$$1 + \sum_{a \notin J} n_a + \sum_{a \in J} \dim \arg \min_{x \in C_a} \langle x, w_a \rangle. \quad (13.4)$$

Every contact face for an arbitrary normalized boundary normal is the intersection of contact faces for extreme polar normals. Indeed, express that normal as a finite convex combination of extreme points of the polar; equality in its supporting inequality forces equality in each inequality having positive weight. Each supported block in (13.4) therefore loses at least λ_a dimensions. A single supported block with smallest λ_a , using a normal attaining e_a , attains the claimed maximum. $\square$

By a *whole-row partition* we mean a partition $\mathcal{G}$ of $[k]$ with one cone assigned to each group, factoring all of that group's rows as in (13.1). In particular, a row is not split between different assigned cones. Allowing a group's primal map to depend on other source coordinates does not weaken the lower bound: fix those coordinates and apply the theorem.

Corollary 13.2 (Exact grouping under two caps). *Let $D \geq 1$ and $F \geq 0$ be integers, and allow all proper cones with dimension at most D and $f(K) \leq F$. A group G is feasible in the whole-row model if and only if*

$$\sum_{a \in G} n_a \leq D - 1, \quad \sum_{a \in G} n_a - \min_{a \in G} \lambda_a \leq F - 1. \quad (13.5)$$

The minimum factor count is the minimum number of parts satisfying these two inequalities. For k identical bodies with dimension n and contact codimension λ , put

$$Q = \min \left\{ \left\lfloor \frac{D-1}{n} \right\rfloor, \left\lfloor \frac{F+\lambda-1}{n} \right\rfloor \right\}.$$

If $Q \geq 1$, the exact factor count is $\lceil k/Q \rceil$; if $Q = 0$, no whole-row partition is feasible.

Proof. Necessity is (13.2), and H_G proves sufficiency for every feasible group. The identical-body inequalities reduce to $|G| \leq Q$. $\square$

For strictly convex bodies, $\lambda_a = n_a$. For polytopes, $\lambda_a = 1$, since some extreme polar normal exposes a facet. For Euclidean balls, the theorem gives the exact heterogeneous shared-scale cone law

$$\dim H_G = 1 + \sum_{a \in G} n_a, \quad f(H_G) = 1 + \sum_{a \in G} n_a - \min_{a \in G} n_a.$$

Thus the face bound uses entire affine coordinate spaces, not just the tangent spaces detected by curvature.

Proposition 13.3 (Complexity of heterogeneous grouping). *For Euclidean balls with dimensions given as positive integers, deciding whether the whole-row grouping problem has a partition with at most a specified number of groups is strongly NP-complete. This holds already when $F = D - 1$.*

Proof. Under $F = D - 1$, the second inequality in (13.5) follows from the first because $\min_{a \in G} n_a \geq 1$. Consider a 3-PARTITION instance with integers $b_1, \dots, b_{3q}$, $B/4 < b_a < B/2$, and $\sum_a b_a = qB$ [40, Problem SP15]. Set $n_a = b_a$, $D = B + 1$, $F = B$, and ask for at most q groups. Their total dimension is qB , so every feasible group must have dimension exactly B and there must be exactly q groups. The size inequalities force exactly three items in each group. This is precisely the original 3-PARTITION question. The reduction preserves polynomial bounds on all integers; if explicit Euclidean-ball descriptions are required, their total coordinate count is polynomial on those strongly bounded instances. A partition is a polynomially checkable certificate for the two integer inequalities, which also proves membership in NP. The computational obstruction here is the classical packing problem, rather than an additional difficulty in testing a fixed group's conic feasibility. $\square$

13.2 Rigidity at the absolute dimension minimum

The following elementary equality argument applies to arbitrary compact bodies and allows every row to split. It is useful because a cone change at this minimum cannot improve any linear-isomorphism invariant.

Theorem 13.4 (Minimum-dimensional rigidity). *Let $C \subset \mathbb{R}^N$, $N \geq 1$, be a compact full-dimensional convex body. Every exact lift over a proper cone of total dimension M , allowing affine slices, affine projections, and free variables, has $M \geq N + 1$. At equality, after the reduction of Lemma 2.1, an invertible linear map carries the cone to*

$$\text{hom}(C) = \{(t, tx) : t \geq 0, x \in C\}$$

and its feasible slice to $\{1\} \times C$. The reduced projection is an affine isomorphism on that slice.

The same cone-isomorphism conclusion holds for any factorization of the full slack operator through a proper cone of dimension $N + 1$, without regularity assumptions on its factor maps. In particular, it holds when that cone is a product and rows split across its factors.

Proof. First consider an exact lift. Eliminate free variables and reduce to the minimal face, of dimension $M' \leq M$, with a Slater affine slice A . Its affine output has full rank on the direction space of A , so $\dim A \geq N$. If $M' \leq N$, these facts force A to be the entire cone space. Its image is an affine translate of a cone, which cannot be a full-dimensional bounded body. Thus $M' \geq N + 1$.

When $M = N + 1$, no face reduction is possible, and A must be an N -dimensional affine hyperplane: the full-space case gives the same contradiction. The output map on A is an affine

isomorphism, so $S = K \cap A$ is compact. The hyperplane does not contain zero, again because otherwise its feasible set and output would be translated cones. Write $A = \{z : \ell(z) = 1\}$. Fix $w \in A \cap \text{int } K$. If nonzero $v \in K$ satisfies $\ell(v) = 0$, then $w + tv \in S$ for all $t \geq 0$, contradicting compactness. If $\ell(v) < 0$, the nonzero vector $v - \ell(v)w \in K \cap \ker \ell$ gives the same contradiction; pointedness ensures this vector is nonzero. Therefore ℓ is strictly positive on $K \setminus \{0\}$, and S is a compact base. The affine output on A extends to a linear map $\tilde{\pi}$ on the cone space because A is a hyperplane not containing zero. The map $Tz = (\ell(z), \tilde{\pi}(z))$ is invertible: its kernel lies in the direction space of A , where the output is injective. Writing each nonzero $z \in K$ uniquely as $z = \ell(z)s$, $s \in S$, proves both claims.

For the factorization assertion, translate C so $0 \in \text{int } C$. The canonical vectors $U(x) = (1, x)$ for $x \in \text{ext } C$ and $V(z) = (1, -z)$ for $z \in \text{ext } C^\circ$ each span $\mathbb{R}^{N+1}$. The slack has rank $N + 1$, by the affine-function argument in Theorem 13.1. If $A(x), B(z)$ give another factorization in that dimension, their images span as well. A linear relation among the $U(x)$ is equivalent, by pairing with the spanning dual family, to the same relation among the $A(x)$. Hence an invertible T satisfies $A = TU$ and $B = T^{-\top}V$. Taking conic hulls gives

$$T \text{hom}(C) \subseteq K, \quad T^{-\top} \text{hom}(C)^* \subseteq K^*.$$

The extreme points of the body and its polar generate their respective compact bases, so these are the full homogenization cones. Dualizing the second inclusion reverses the first, and proves equality. $\square$

In Theorem 13.1, the labelled rows are exactly the extreme polar rows of $\prod_{a \in G} C_a$. To see this, its polar is $\{(z_a) : \sum_a h_a(z_a) \leq 1\}$. A nonzero extreme point must have one supported coordinate block, since two positive support-function contributions give a nontrivial convex decomposition into single-block points. Within that block it must be an extreme point of C_a° . Conversely each such single-block point is extreme. Thus equality $m = N_G + 1$ in that theorem forces K to be linearly isomorphic to H_G . For a fixed partition into g groups, every whole-row formulation of total dimension $\sum_a n_a + g$ therefore uses precisely those shared homogenization cones up to linear changes of coordinates. Barrier parameters are preserved by such changes. In particular, the exact shared spectral-cone values of Theorem 17.3, including its Euclidean vector-ball case, apply to every such minimum-dimensional formulation.

Corollary 13.5 (Connected extreme sets under a strict factor cap). *Suppose $\text{ext } C$ is connected. A full slack factorization through $K = \prod_{i=1}^L K_i$, with every K_i nonzero and proper, can have total dimension $N + 1$ only when $L = 1$. Consequently, if $\dim K_i \leq d < N + 1$, then*

$$M \geq N + 2, \quad L \geq \max \left\{ 2, \left\lceil \frac{N + 2}{d} \right\rceil \right\}. \quad (13.6)$$

These conclusions also hold for exact lifts, after the stated reduction convention, and permit arbitrary row splitting.

Proof. Normalize extreme rays by any strictly positive linear functional. For a product of L nonzero proper cones, the extreme rays are supported in exactly one factor; the L factor-supported pieces are nonempty and both open and closed in the normalized extreme-ray space. For $\text{hom}(C)$ that space is homeomorphic to $\text{ext } C$. Theorem 13.4 therefore excludes $L > 1$ at equality. A single factor under the strict cap cannot supply dimension $N + 1$. Integrality gives $M \geq N + 2$, and $M \leq Ld$ gives the count bound. For an unreduced exact lift with $M = N + 1$, the theorem already ensures that no reduction occurs. $\square$

This is a one-unit exclusion of the absolute minimum, not an additive charge per factor. The balance slices in Theorem 15.1 have connected extreme sets and $M = N + 2$ for arbitrarily many essential factors. They therefore preclude the stronger general claim $M \geq N + L$.

13.3 What one normalization equation does imply

Proposition 13.6 (An additive bound for one normalized base). *Let $K = \prod_{i=1}^L K_i$, where K_i is nonzero and proper of dimension m_i , and choose $\ell_i \in \text{int } K_i^*$. Let*

$$B = \{z \in K : \sum_i \ell_i(z_i) = 1\}, \quad C = \pi(B),$$

where π is affine and $\dim \text{aff } C = N$. If $\text{ext } C$ is connected, then $\sum_i m_i \geq N + L$. No irredundancy assumption is needed.

Proof. Let B_i be the compact normalized base of factor i , embedded in the product with all other coordinates zero, and put $C_i = \pi(B_i)$. Then $C = \text{conv } \bigcup_i C_i$, and $\dim \text{aff } C_i \leq m_i - 1$. Every extreme point of C belongs to some C_i , because any finite convex representation of an extreme point uses only that point. The nonempty sets $E_i = C_i \cap \text{ext } C$ form a finite relatively closed cover of the connected extreme set. Their intersection graph is connected: a graph separation would give a separation into two disjoint nonempty relatively closed finite unions. Along a spanning tree, the affine hull of each newly added E_i intersects the previous affine hull. The dimension of their combined affine hull is therefore at most the sum of their dimensions. With $I = \{i : E_i \neq \emptyset\}$ this gives

$$N = \dim \text{aff}(\text{ext } C) \leq \sum_{i \in I} \dim \text{aff } E_i \leq \sum_{i \in I} (m_i - 1) \leq \sum_{i=1}^L (m_i - 1).$$

The final inequality uses $m_i \geq 1$. Rearrangement proves the result. $\square$

An additional affine equation can create extreme points with several factor coordinates active. The factor-base cover in this proof then need not cover the extreme points of the smaller slice. This explains why the proposition and the balance-slice counterexamples are compatible.

13.4 Recession normalization and barrier transfer

The projection lemma in Section 9 requires bounded fibers. The next criterion says precisely when a single affine level can provide that hypothesis. All interiors in the statement are relative to affine hulls.

Proposition 13.7 (A criterion for barrier-preserving normalization). *Let D be a nonempty closed convex set containing no affine line, and let an affine map π satisfy $\pi(D) = C$, where C is a compact convex body. Write $R = \text{rec } D$ for its recession cone. If $R \neq \{0\}$, choose a linear functional a strictly positive on $R \setminus \{0\}$, and define*

$$m_a(x) = \min\{a(z) : z \in D, \pi(z) = x\}.$$

Every minimum is finite and attained. For any real M ,

$$\pi(D \cap \{a = M\}) = C \iff M \geq \sup_{x \in C} m_a(x). \quad (13.7)$$

Whenever this holds, the level section is compact. If the supremum is finite, every self-concordant barrier on $\text{ri } D$ induces a barrier on $\text{int } C$ with no larger parameter. If $R = \{0\}$, the same parameter conclusion holds without taking a level section.

Proof. Compactness of C implies $R \subseteq \ker \pi_0$, where π_0 is the linear part of π . The cone R is pointed because D contains no line, so a strictly positive a exists. Every nonempty sublevel $D \cap \{a \leq b\}$ is bounded. Otherwise normalize differences of an unbounded sequence from a fixed point of D and pass to a unit limit r . Closedness and convexity imply $r \in R$, while boundedness above of a on the sequence gives $a(r) \leq 0$, a contradiction. Each fiber is closed; minimizing a on a compact nonempty sublevel proves attainment and finiteness of $m_a(x)$.

Fix $0 \neq r_0 \in R$. Adding a nonnegative multiple of r_0 to a minimizer in a fiber preserves that fiber and reaches every larger a -value. This proves (13.7). The level section is closed and bounded by the same sublevel argument. If the supremum is finite, choose M greater than it and greater than $a(z^0)$ for some $z^0 \in \text{ri } D$. Adding a suitable multiple of r_0 to z^0 gives a point of the level in $\text{ri } D$. Indeed, a relative ball around z^0 remains in D after translating by a recession direction. Thus the affine restriction of the original barrier is a barrier on the relative interior of the compact level section, with the same parameter bound. Lemma 9.5 then gives a barrier on C . If $R = \{0\}$, the closed convex set D is bounded by the same normalized-sequence argument, and that lemma applies directly. $\square$

Corollary 13.8 (Arbitrary line-free lifts of a simple-vertex polytope). *Let C be an N -dimensional compact polytope with a simple vertex. Every self-concordant barrier on the relative interior of every closed line-free affine lift of C has parameter at least N , even if its fibers are unbounded. The minimum over all such lifts and barriers is N .*

Proof. Choose a lift $z_v \in D$ of each vertex v of C . For a convex combination $x = \sum_v \theta_v v$, its lifted combination belongs to D and therefore

$$m_a(x) \leq \sum_v \theta_v a(z_v) \leq \max_v a(z_v).$$

Proposition 13.7 transfers the barrier to C without increasing its parameter. At a simple vertex, the classical independent-facet lower bound gives $\nu \geq N$ [70, Proposition 2.3.6]. The universal barrier on C attains the bound N [60, Theorem 2]; it may also be used on the canonical homogenization slice, which is affinely isomorphic to C . $\square$

The simple-vertex hypothesis is essential to this argument. Choosing N independent normals at a nonsimple vertex does not reproduce its tangent cone. At e_1 of the three-dimensional cross-polytope, the four active normals are $(1, \pm 1, \pm 1)$. The three with sign pairs $(+, +), (+, -), (-, +)$ are independent, but $(-1, -1, -1)$ satisfies their three strict tangent inequalities and violates the fourth. The cited simple-vertex lower bound cannot be applied after deleting that facet.

Example 13.9 (Compact projection need not admit affine normalization). Let D be the positive-semidefinite lift of the disk in Example 8.8, with coordinates (x_1, x_2, u, z) . Its recession cone is exactly the positive z -ray. Indeed, compactness of the projected disk makes every recession direction have zero x -coordinates. The second block then requires its u -coordinate to be nonnegative, while the last homogeneous block $\begin{pmatrix} 0 & \dot{u} \\ \dot{u} & \dot{z} \end{pmatrix}$ forces $\dot{u} = 0$ and $\dot{z} \geq 0$. Conversely every such direction is a recession direction. If an affine subspace removed this ray from a nonempty section of D , that closed convex section would have zero recession cone and hence be bounded. It could not project onto the entire disk, since every completion of the boundary points approaching $(1, 0)$ in Example 8.8 has $z \rightarrow \infty$. Thus no projection-preserving affine restriction removes the recession ray, even for this semialgebraic Slater lift over three 2×2 positive-semidefinite cones. This obstructs automatic normalization; it does not disprove parameter monotonicity by some other method for general lifts with unbounded fibers.

The full-slack framework used in this section is classical [43]. The minimum-dimension statement is an elementary rigidity consequence of that framework, and the normalized-base dimension bound uses only a finite connected cover. The barrier endpoints in Corollary 13.8 are also classical; its additional content is their transfer through a given closed line-free lift, by the explicit recession-level criterion and bounded partial minimization. The whole-row grouping formulas concern a restricted formulation model and do not assert optimality over arbitrary split-row lifts.

14 Complementary faces control sharing between source rows

The whole-row results constrain an indivisible row. We now allow every row to split over several blocks, but retain globally C^1 factors of the full slack. The dimension of an exposed face then gives a different, differential limit on sharing. Its global consequence extends the Lorentz no-sharing theorem to every cone whose nonzero proper exposed faces are rays; smoothness of either cone boundary is unnecessary.

Let $M = (\mathbb{S}^p)^k$, $p = s - 1 \geq 2$, and suppose

$$1 - \langle x_a, z \rangle = \sum_{i=1}^L \langle A_i(x), B_i^a(z) \rangle, \quad x \in M, \quad z \in \mathbb{S}^p, \quad a \in [k], \quad (14.1)$$

where $A_i : M \rightarrow K_i$ and $B_i^a : \mathbb{S}^p \rightarrow K_i^*$ are C^1 and the cones are proper of dimensions $3 \leq m_i \leq d$, for an integer $d \geq 3$. Set

$$r_i = m_i - 2, \quad c = d - 2, \quad f_i = \max_{\substack{0 \neq b \in \partial K_i^* \\ K_i \cap b^\perp \neq \{0\}}} \dim(K_i \cap b^\perp), \quad \bar{f}_i = \min(f_i, k).$$

The maximum concerns dimensions, not the number of extreme rays. One may instead use any upper bound on the dimensions of the nonzero exposed faces encountered by the selected dual factors. Ray factors can be added: their mixed forms vanish and they contribute nothing to the bounds below.

Theorem 14.1 (Local face packing). *At $x \in M$ put*

$$H_i^a(u, v) = -\langle D_a A_i(x)[u], DB_i^a(x_a)[v] \rangle, \quad t_i^a = \text{rank } H_i^a, \quad I_i = \{a : t_i^a > 0\}.$$

Then $t_i^a \leq r_i$, $\sum_i t_i^a \geq p$, and

$$\sum_{a \in I_i} t_i^a \leq r_i, \quad 1 + \sum_{b \in I_i \setminus \{a\}} t_i^b \leq \dim(K_i \cap B_i^a(x_a)^\perp) \quad (a \in I_i). \quad (14.2)$$

In particular $|I_i| \leq \bar{f}_i$.

Proof. Contact and nonnegativity make every pairing $\langle A_i(x), B_i^a(x_a) \rangle$ zero. On the cylinder $x_a = u$, the function $z \mapsto \langle A_i(x), B_i^a(z) \rangle$ has a minimum at u . Consequently $\langle A_i(x), DB_i^a(u)[v] \rangle = 0$ on that entire cylinder. Differentiating in a different source direction, with the dual derivative held fixed, gives

$$\langle D_b A_i(x)[w], DB_i^a(x_a)[v] \rangle = 0 \quad (b \neq a).$$

This uses only first derivatives. The ray-quotient contact argument of Section 2 gives $t_i^a \leq m_i - 2$; at a zero factor its derivative is zero by pointedness. Mixed differentiation of the full row gives $\sum_i H_i^a = g_{\mathbb{S}^p}$, hence $\sum_i t_i^a \geq p$.

Fix one block with an active row, write $A = A_i(x) \neq 0$, and let $\Lambda_a(v)$ be the functional $w \mapsto \langle v, DB_i^a(x_a)[w] \rangle$. Choose $E_a \subset \text{im } DA_i$ of dimension t_i^a on which Λ_a is injective. The identities just proved imply $\Lambda_a A = 0$ and $\Lambda_a E_b = 0$ for $b \neq a$. Applying each Λ_a to a linear dependence proves that

$$\mathbb{R}A \oplus \bigoplus_{a \in I_i} E_a$$

is a direct sum. Fix any active row a_0 and put $b = B_i^{a_0}(x_{a_0}) \neq 0$. The nonnegative function $x' \mapsto \langle A_i(x'), b \rangle$ vanishes at x , so $\text{im } DA_i(x) \subset b^\perp$. Since $A \in b^\perp$, the entire direct sum lies in this hyperplane of dimension $m_i - 1$. Thus $1 + \sum_a t_i^a \leq m_i - 1$, proving the first inequality. When there is no active row that inequality is immediate. On the contact cylinder for row a , the primal factor lies in $F_a = K_i \cap B_i^a(x_a)^\perp$. Thus $E_b \subset \text{span } F_a$ for $b \neq a$, and $A \in F_a$. Their direct sum proves the second inequality. Since each active rank is at least one, it also gives $|I_i| \leq f_i$. $\square$

Theorem 14.2 (Global face budgets). *Define*

$$h = \begin{cases} \lceil (p+1)/c \rceil, & c < p, \\ 1, & c \geq p, \end{cases} \quad \tau = \begin{cases} p+1, & c < p, \\ p, & c \geq p, \end{cases} \quad R = \sum_i r_i, \quad K = kp + \mathbf{1}_{\{c < p\}}.$$

Every factorization (14.1) satisfies

$$\sum_i \bar{f}_i \geq kh, \quad \sum_i \bar{f}_i r_i \geq k\tau, \quad R \geq K. \quad (14.3)$$

If $f_i \leq f$ for a common $f \geq 1$, set

$$R_* = \max\{\lceil k\tau/f \rceil, K\}, \quad L_* = \max\{\lceil kh/f \rceil, \lceil R_*/c \rceil\}.$$

Then

$$L \geq L_*, \quad R \geq R_*, \quad D := \sum_i m_i \geq R_* + 2L_*. \quad (14.4)$$

Every self-concordant barrier on the ambient product has parameter at least $2L_*$. This includes coupled barriers without logarithmic homogeneity.

Proof. Write $N_a(x) = |\{i : t_i^a(x) > 0\}|$ and $C_a(x) = \sum_{i: t_i^a(x) > 0} r_i$. Pointwise, $N_a \geq \lceil p/c \rceil$ and $C_a \geq p$. Counting incidences gives

$$\sum_a N_a \leq \sum_i \bar{f}_i, \quad \sum_a C_a \leq \sum_i \bar{f}_i r_i.$$

Summing (14.2) gives $kp \leq R$. To exclude equality when $c < p$, aggregate the labelled dual maps:

$$\widehat{B}_i(z_1, \dots, z_k) = \sum_a B_i^a(z_a), \quad \sum_i \langle A_i(x), \widehat{B}_i(z) \rangle = k - \sum_a \langle x_a, z_a \rangle.$$

These are C^1 cone-valued maps on M , with contact map $z = x$ and nondegenerate mixed pairing equal to the product spherical metric. At $R = kp$, Theorem 10.1 therefore forces the positive capacities to be exactly k copies of p . This contradicts $r_i \leq c < p$. Integrality gives $R \geq kp + 1$ in this case. Zero-capacity ray factors do not affect this argument. These observations prove the last bound in (14.3), and all its bounds when $c \geq p$.

Suppose $c < p$. If $\sum_i \bar{f}_i r_i < k(p+1)$, every point belongs to some $Z_a = \{C_a = p\}$. Each Z_a is closed because activity is open and $C_a \geq p$ everywhere. Its finitely many exact-active-label pieces $E_{a,J}$ are clopen in Z_a , hence compact: existing active labels persist, whereas a new label would raise

C_a . On such a piece $\sum_{i \in J} r_i = p$. Normalize the nonzero dual factors B_i^a using an interior primal vector to obtain maps into the boundaries of compact dual-cone bases. These maps are defined near the projection of $E_{a,J}$ onto its source sphere. Indeed fix the other source coordinates at any point of the piece; nearby contact primal factors remain nonzero and force the corresponding dual factors to remain on the nonzero boundary.

The joint derivative of these normalized dual maps is injective at every projected point. If a tangent vector is killed by all normalized derivatives, each unnormalized derivative is radial and therefore annihilates every primal contact derivative. The identity $\sum_i H_i^a = g_{\mathbb{S}^p}$ then kills that tangent vector itself. Injectivity persists on an open neighborhood. Lemma 10.3 shows that this neighborhood is a proper open subset of $\mathbb{S}^p$, since every $r_i < p$ and their sum is p .

For a fixed a , choose pairwise disjoint open neighborhoods of the finitely many compact pieces $E_{a,J}$, each inside the inverse image of its proper source neighborhood. The source top class u_a vanishes on their union U_a . The U_a cover M , so the relative cup-product argument of Section 6 contradicts $u_1 \cdots u_k \neq 0$. Thus the capacity budget is at least $k(p+1)$. For the count budget, if $c \nmid p$ the pointwise count already equals h . Otherwise write $p = tc$. A putative incidence budget below $k(t+1)$ covers M by loci where some source has exactly t active labels. Every label there must have capacity c , and the same compact-piece argument applies. This proves the count bound.

Uniform face caps and $R \geq K$ imply $R \geq R_*$ and $L \geq \lceil kh/f \rceil$. Since $R \leq cL$, they also imply $L \geq \lceil R_*/c \rceil$. The dimension bound follows from $D = R + 2L$. Finally each proper cone of dimension at least two has a two-dimensional linear section through an interior point that is a pointed wedge. Its relative boundary lies in the cone boundary. The product sections form a $2L$ -orthant. Restriction of any ambient barrier and the orthant lower bound give $\nu \geq 2L \geq 2L_*$. $\square$

Corollary 14.3 (Exact ray-exposed frontier). *If every nonzero proper exposed face of each K_i is a ray, the simultaneous minima over all such dictionaries of dimension at most d are*

$$L = kh, \quad R = k\tau, \quad D = k\tau + 2kh, \quad \nu_{\text{ambient}} = 2kh.$$

They are attained by Lorentz cones and the grouped constructions of Theorem 6.2. A fixed dictionary need not contain these attaining cones. No exact interpolation for $f > 1$ is asserted.

There is a further useful tradeoff. If block i is active on $q_i \geq 2$ rows and $T_i = \sum_a t_i^a$, summing its face inequalities gives

$$(q_i - 1)T_i \leq q_i(f_i - 1), \quad T_i \leq \min\{r_i, 2(f_i - 1)\}. \quad (14.5)$$

Thus a small exposed face can further restrict a block's total mixed rank when it serves several rows; sharing never increases that rank beyond its capacity r_i . For a real PSD factor whose contact dual has nullity h_{ia} , the face inequality reads $1 + \sum_{b \neq a} t_i^b \leq h_{ia}(h_{ia} + 1)/2$; the sharper Jordan one-row capacity can be imposed simultaneously.

14.1 A sharp model for shared scalar channels

The distinction between whole rows and split rows is concrete. Define

$$P_q = \{(t, u, w) \in \mathbb{R} \times \mathbb{R}^q \times \mathbb{R}^q : t \geq 0, w_a \geq 0, tw_a \geq u_a^2\}.$$

This closed proper cone has dimension $2q + 1$. The polynomial factors

$$A(v) = (1, v, v^{\circ 2}), \quad B^a(z) = (z^2/2, -ze_a, e_a/2) \quad (14.6)$$

give $\langle A(v), B^a(z) \rangle = (v_a - z)^2/2$ on $[-1, 1]^q \times [-1, 1]$. Dual feasibility follows directly by completing a square when $t > 0$ and by continuity when $t = 0$.

Proposition 14.4 (Shared-square resources). *The cone P_q has maximum nonzero proper exposed-face dimension $2q - 1$. These dimension and face bounds are minimal for a one-block factorization of all the kernels in (14.6). Its optimal self-concordant barrier parameter is $q + 1$, attained by*

$$F_q(t, u, w) = - \sum_{a=1}^q \log(tw_a - u_a^2) + (q - 1) \log t. \quad (14.7)$$

For a partition of k source balls into g groups of sizes q_ℓ , coordinatewise use of these cones factors the full slack analytically with sg factors, scalar dimension $s(2k + g)$, maximum block dimension $2 \max q_\ell + 1$, maximum exposed-face dimension $2 \max q_\ell - 1$, and exact ambient parameter $s(k + g)$.

Proof. The dual cone consists of $\gamma \geq 0$, $\beta_a = 0$ when $\gamma_a = 0$, and $\alpha \geq \sum_{\gamma_a > 0} \beta_a^2 / (4\gamma_a)$. Let $J = \{a : \gamma_a > 0\}$. In the equality case for a nonzero dual vector, J is nonempty and the contact equations fix $u_a = -t\beta_a / (2\gamma_a)$ and $w_a = t\beta_a^2 / (4\gamma_a^2)$ for $a \in J$. The other coordinates range over a copy of $P_{q-|J|}$, so the face dimension is $1 + 2(q - |J|)$. In the strict case $t = 0$, hence $u = 0$; only w_a for $a \notin J$ remain, giving dimension $q - |J|$. These cases give the maximum $2q - 1$, including $q = 1$.

The row functions span the linearly independent family $1, v_a, v_a^2$ of size $2q + 1$, proving the dimension lower bound. On a contact cylinder $v_a = z$, the other rows linearly recover $1, v_b, v_b^2$ for $b \neq a$; a nonzero constant is also recovered from a different dual argument in row a . The contact face must therefore span at least $1 + 2(q - 1)$ dimensions.

The cone P_q is the PSD-completable partial-matrix cone on a star: the center diagonal is t , its edge entries are u_a , and the leaf diagonals are w_a . For $t > 0$ its maximum-determinant completion has determinant $t \prod_a (w_a - u_a^2/t)$. The classical completion barrier proved in Section 17 consequently is (14.7), of parameter $q + 1$. Equivalently it is the Legendre barrier of the sparse arrow-matrix log determinant. Restricting any barrier to $u = 0$ gives a $(q + 1)$ -orthant with boundary on the original boundary, so the parameter is optimal against arbitrary barriers.

For source a in a group use $v_b = x_{b,j}$ and $z = z_j$ in coordinate j . Summing gives $\frac{1}{2} \sum_j (x_{a,j} - z_j)^2 = 1 - \langle x_a, z \rangle$ on the unit spheres. Summing dimensions and parameters gives the stated ledger; the product orthant sections prove the ambient lower bound even for coupled barriers. $\square$

For comparison the rank-one real PSD factors $A(v) = (1, v)(1, v)^T$ and $B^a(z) = (e_a - ze_0)(e_a - ze_0)^T/2$ give the same scalar kernels, but expose a PSD face of dimension $q(q + 1)/2$. A whole-row version uses $v = (1, x_1, \dots, x_q)$ and sums those dual outer products over the s coordinates of a row; its dual rank is s , its nullity is $1 + (q - 1)s$, and its contact face has dimension $\binom{2+(q-1)s}{2}$. These are constructions, not optimality claims among homogeneous cones.

The homogenization of a group of q whole ball rows, by contrast, has dimension $qs + 1$ and maximum exposed-face dimension $1 + (q - 1)s$ by Theorem 13.1. Its exact ambient parameter is $q + 1$, as the vector-row specialization of Theorem 17.3. For k rows in g groups the ambient parameter is $k + g$, while fixing each scale to one gives exactly $-\sum_a \log(1 - \|x_a\|^2)$ with intrinsic parameter k , independently of grouping. Thus factor sharing, face efficiency, and the barrier on the actual slice measure different features of the formulation.

The correspondence between full slack factorizations and cone lifts is classical [43]. The resource bounds isolated here are the cross-contact direct-sum inequalities (14.2) and their conversion into the global budgets (14.3); their hypotheses explicitly include a global C^1 choice for every labelled row. The barriers used in the examples are classical spectral and PSD-completion barriers, not new barrier constructions. In particular, neither local face budgets nor a favorable ambient count alone proves an unrestricted algorithmic speedup.

15 Balance slices with sharp dimension and barrier bounds

Connected extreme points force an extra cone coordinate under a strict dimension cap, but they do not force one extra coordinate per factor. We construct slices attaining the former bound for arbitrarily many factors. Their intrinsic barrier parameters are also exact. This gives an explicit separation between dimension resources and barrier resources.

Let V_i , $1 \leq i \leq L$, be simple Euclidean Jordan algebras of rank $r_i \geq 2$, with cones of squares K_i , identities e_i , and Jordan frames $(c_{ij})_{j=1}^{r_i}$. Here $L \geq 2$. Choose a nonzero off-diagonal Peirce element $A_i \in V_i(c_{i1}, c_{i2})$, and put

$$\ell_i(X_i) = \langle A_i, X_i \rangle, \quad M = \sum_i \dim V_i, \quad \rho = \sum_i r_i.$$

Define the balance cone and its trace-one base by

$$\widehat{Z} = \left\{ X \in \prod_i K_i : \sum_i \ell_i(X_i) = 0 \right\}, \quad Z = \left\{ X \in \widehat{Z} : \sum_i \operatorname{tr} X_i = 1 \right\}. \quad (15.1)$$

All interiors and barriers below are taken in the respective affine hulls.

Theorem 15.1 (Balance slices). *The set Z is a compact convex body of dimension $M - 2$, with a path-connected extreme-point set. Each displayed factor is indecomposable, its membership constraint is essential, and it occurs in a nonconstant supporting slack. For arbitrary self-concordant barriers, including barriers coupling all blocks,*

$$\vartheta_{\text{opt}}(\operatorname{ri} \widehat{Z}) = \rho, \quad \vartheta_{\text{opt}}(\operatorname{ri} Z) = \rho - 1. \quad (15.2)$$

The product Jordan log-determinant, restricted to these domains, attains both values. Every closed convex affine lift of Z with nonempty bounded fibers has intrinsic barrier parameter at least $\rho - 1$. The conclusions include heterogeneous mixtures of classical Hermitian, spin, and Albert factors.

The elementary minimal-face criterion used below is classical in finite dimensions; see also the general injectivity formulation of Henrion, Kružík, and Weis [52, Theorem 2.6]. The ambient symmetric-cone barrier theory is standard [31, 47]. The content of the theorem is the combination of the explicit balance construction, connected extreme points, and exact parameters on its two sections, rather than a new general extremality or barrier theorem.

Proof. The point $X_i^0 = e_i/\rho$ is strictly feasible. The total trace and balance functionals are independent, so $\dim Z = M - 2$. Positivity and fixed total trace imply compactness.

We first classify and connect the extreme points. If a point X belongs to an affine section $\{X \in K : \mathcal{A}X = b\}$, it is extreme precisely when

$$\ker \mathcal{A} \cap \operatorname{span} F_K(X) = \{0\}. \quad (15.3)$$

Indeed, X is relatively interior to its minimal face, so a nonzero vector in this intersection gives a feasible segment through X . Conversely, the endpoints of any such segment lie in the minimal face. A rank- $q \geq 2$ face in a simple Jordan cone has dimension $q + a_i q(q-1)/2 \geq 3$, where a_i is the Peirce multiplicity. There are only two affine equations in (15.1). Thus each extreme point has one or two nonzero blocks, each on a primitive ray. Normalize its primitive idempotents to trace one, and write $h_i(P) = \ell_i(P)$. The constraint columns are $(1, h_i(P))$. The exact possibilities are consequently

- one block P with $h_i(P) = 0$;

- two blocks wP and $(1-w)Q$ with $h_i(P)h_j(Q) < 0$, where $w = -h_j(Q)/(h_i(P) - h_j(Q))$.

In the second case $i \neq j$. These conditions also suffice by (15.3).

We record the sign geometry of h_i . By a Jordan automorphism and positive rescaling, its defining element becomes $c_{i1} - c_{i2}$; this follows from the spectral theorem and transitivity on Jordan frames. For a classical factor $H_r(\mathbb{F})$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, primitive idempotents are $uu^*/\|u\|^2$, and their height is proportional to

$$\frac{|u_1|^2 - |u_2|^2}{\|u\|^2}.$$

The positive region, in the chart $u_1 = 1$, is $\{u_2 : |u_2| < 1\} \times \mathbb{F}^{r-2}$ and is path-connected; the negative region is obtained by interchanging coordinates. If $r \geq 3$, the zero region is path-connected: for $u_1 \neq 0$ it is $\mathbb{S}^{\dim_{\mathbb{R}} \mathbb{F}-1} \times \mathbb{F}^{r-2}$, and every such point can be joined, by letting a third coordinate tend to infinity, to a fixed ray in the subspace $u_1 = u_2 = 0$. That subspace has connected projective space (or is one point). This also joins the two chart components when $\mathbb{F} = \mathbb{R}$. For $r = 2$ the zero region is $\mathbb{S}^{\dim_{\mathbb{R}} \mathbb{F}-1}$; only the real case has two points. Every zero ray admits a path entering either strict sign region: vary $|u_2|$ in the first chart, or introduce a first or second coordinate at a ray with $u_1 = u_2 = 0$. For a spin factor, primitive idempotents form a sphere and height is a nonzero linear coordinate. Its sign regions are hemispheres, its zero region is an equator, and the same assertions hold, with a two-point equator in the three-dimensional cone.

For completeness the Albert case can be checked in coordinates, without an analogy with associative projective spaces. The chart of the octonionic projective plane with first diagonal coordinate positive is parametrized by $(x, y) \in \mathbb{O}^2$. Its primitive idempotent is

$$P(x, y) = \frac{1}{1 + |x|^2 + |y|^2} \begin{pmatrix} 1 & \bar{x} & \bar{y} \\ x & |x|^2 & x\bar{y} \\ y & y\bar{x} & |y|^2 \end{pmatrix}, \quad h(P) = \frac{1 - |x|^2}{1 + |x|^2 + |y|^2}. \quad (15.4)$$

These are the usual reduced homogeneous coordinates [51, Section 3]; the primitive-idempotent model is described in [9]. The two octonions x, y generate an associative subalgebra, which directly verifies $P^2 = P$ and $\text{tr } P = 1$. The positive region is $\mathbb{B}^8 \times \mathbb{R}^8$, and permutation of the first two frame idempotents identifies the negative region with it. The zero region in this chart is $\mathbb{S}^7 \times \mathbb{R}^8$. Outside the chart, zero height forces both first diagonal coordinates to vanish, leaving only c_{i3} : positivity makes their Peirce off-diagonal components vanish as well. A path (x, ty_0) with $|x| = |y_0| = 1$ tends to c_{i3} as $t \rightarrow \infty$. After reparametrization this is a continuous path on a closed interval, so the entire zero region is path-connected. Varying $|x|$ gives both signs near every finite zero point; keeping $|x| < 1$ or $|x| > 1$ and letting $|y| \rightarrow \infty$ gives such paths at c_{i3} .

These facts give paths of extreme points of Z as follows. From any two-block extreme point, move its positive-height ray within the positive region to a path ending at height zero, while holding the negative-height ray fixed. The displayed balance weights converge to one and zero, so this joins the point to a one-block extreme point. For prescribed zero-height rays P and Q in different blocks, choose continuous paths P_t and Q_t such that $P_0 = P$, $Q_1 = Q$, $h(P_t) > 0$ for $t > 0$, and $h(Q_t) < 0$ for $t < 1$. The positive path may be held constant after entering its sign region, and similarly for the negative path. The uniquely balanced pair extends at $t = 0, 1$ to P, Q , respectively. Finally, any two zero rays in one block can be connected through a zero ray in another block. This last step uses $L \geq 2$ and also handles each two-point zero region. Thus all extreme points lie in one path component.

We next compute barrier parameters. Let $\Phi(X) = -\sum_i \log \det X_i$. The frame-diagonal section of $\widehat{Z}$ is $\mathbb{R}_+^\rho$, because each A_i is off diagonal. It meets $\text{ri } \widehat{Z}$, and its boundary belongs to $\partial \widehat{Z}$. To see

the latter assertion, a vanished frame coordinate is detected by a nonnegative primitive functional, which is strictly positive at X^0 ; it is therefore a nonconstant supporting functional on the balance section. Its trace-one section is a simplex of dimension $\rho - 1$ with the same boundary inheritance. Restriction of any barrier to these sections preserves its parameter. The classical orthant and simplex lower bounds [70, Section 2.3.4 and Proposition 2.3.6] give respectively ρ and $\rho - 1$, even for coupled barriers.

The standard barrier gives the upper bound ρ on $\hat{Z}$. For the sharper normalized bound, let $g = \nabla\Phi$, $H = \nabla^2\Phi$, and let a be the total-trace covector. Jordan identities give

$$H^{-1}g = -X, \quad g^T H^{-1}g = \rho, \quad a^T H^{-1}a = \sum_i \text{tr}(X_i^2) \leq \left(\sum_i \text{tr} X_i \right)^2.$$

Here $H_i = P(X_i^{-1})$ and $H_i^{-1} = P(X_i)$, including in the Albert algebra. On the trace-one hyperplane the squared dual norm of the restricted gradient is exactly

$$\rho - \frac{(a^T H^{-1}g)^2}{a^T H^{-1}a} = \rho - \frac{1}{\sum_i \text{tr}(X_i^2)} \leq \rho - 1. \quad (15.5)$$

Restriction to the balance tangent can only decrease this norm. Self-concordance and boundary divergence survive affine restriction, which proves (15.2). Lemma 9.5 proves the assertion for other closed lifts with bounded fibers; no restriction on cross-block derivatives of their barriers is imposed.

Finally, simplicity makes each factor indecomposable. If membership in K_i is removed, choose $j \neq i$ and set $X_i = -Te_i/r_i$, $X_j = (T+1)e_j/r_j$, with all other blocks zero. This is an unbounded family satisfying both affine equations and all remaining cone constraints. Each factor also supports a nonconstant nonnegative slack: $X \mapsto \langle c_{i2}, X_i \rangle$ vanishes at the feasible one-block point c_{i1} and is positive at X^0 . $\square$

Corollary 15.2 (An exact strict-cap frontier). *Suppose the factors in (15.1) all have dimension d , and put $N = dL - 2$. Among full-slack factorizations of Z through products of arbitrary nonzero proper cones of dimension at most d , the minimum factor count and minimum total cone dimension are*

$$L_{\min} = L, \quad M_{\min} = N + 2 = dL.$$

These statements permit row splitting and impose no regularity on the factor maps.

Proof. For a competing product, ordinary slack rank gives $M' \geq N + 1$. Equality would identify that product with the homogenization cone of Z by Theorem 13.4. The projectivized extreme rays of a nontrivial product form a disconnected union, whereas those of the homogenization are homeomorphic to the connected set of extreme points of Z . Equality therefore requires a single factor. This violates the cap, since $d < N + 1 = dL - 1$. Consequently $M' \geq N + 2 = dL$ and $L' \geq M'/d \geq L$. The displayed relative-Slater slice gives an attaining full-slack factorization by Lemma 2.1. $\square$

Corollary 15.3 (An alternate classical moment). *For L copies of $H_r(\mathbb{F})_+$, $r \geq 2$, the preceding theorem and corollary remain valid with $A_i = \mathbf{1}\mathbf{1}^* - I_r$ instead of an off-diagonal Peirce element. Writing $\delta = \dim_{\mathbb{R}} \mathbb{F}$, the zero-height primitive rays in one block form $\mathbb{S}^{\delta(r-1)-1}$.*

Proof. This matrix has zero diagonal and signature $(1, r-1)$, with eigenvalues $r-1, -1, \dots, -1$. The Slater, barrier, and essentiality proofs therefore apply unchanged. In eigenvector coordinates (a, b) , the moment is $((r-1)|a|^2 - \|b\|^2)/(|a|^2 + \|b\|^2)$. A zero ray has the unique representative

$(1, \sqrt{r-1}u)$ with $\|u\| = 1$, proving the sphere assertion. Every strict-sign ray can be moved to a zero ray while retaining its sign until the endpoint, by scaling the two components; if one component vanishes, first introduce it. Two prescribed zero rays in distinct blocks admit an explicit extreme path. For their fixed unit vectors u , replace the representative by $(1, \lambda(c)\sqrt{r-1}u)$, where

$$\lambda(c)^2 = \frac{r-1-c}{(r-1)(1+c)}, \quad -1 < c < r-1.$$

Fix $0 < c_0 < 1$ and assign moments tc_0 and $-(1-t)c_0$. Their balance weights are $1-t$ and t . This connects the two prescribed one-block points. A second block again connects the two components when $\mathbb{F} = \mathbb{R}$, $r = 2$. Connectedness and the cap conclusion follow exactly as above. $\square$

At matched cone dimension $d = r + \delta r(r-1)/2$, the classical family and the spin-factor family built from Q_d^L both have body dimension $dL - 2$, minimum capped factor count L , and minimum total dimension dL . Their optimal direct-body barrier parameters are $rL - 1$ and $2L - 1$. These are two different bodies. The comparison proves that count and dimension do not determine intrinsic barrier cost; it does not construct a Lorentz lift of the classical body. For Albert factors the same formulas give $N = 27L - 2$, $M_{\min} = 27L$, and intrinsic parameter $3L - 1$.

The sharp family-specific combination above follows from classical minimal-face, Jordan-barrier, and simplex arguments, together with the explicit connectivity proof and minimum-dimension rigidity. No priority claim is attached to the alternate moment construction. The relation to intrinsic path length requires an additional metric calculation and is treated separately below.

Example 15.4 (A failed coupled logarithmic cancellation). In two three-dimensional Lorentz factors write $q_i = t_i^2 - x_i^2 - y_i^2$, and impose $x_1 + x_2 = 0$ and $t_1 + t_2 = 1$. The candidate $\Psi = -\log q_1 - \log q_2 + \log(q_1 + q_2)$ cancels a boundary logarithmic order, but is not self-concordant. At $(t_1, t_2, x_1, x_2, y_1, y_2) = (1/3, 2/3, 0, 0, 0, 1/3)$, the feasible direction varying only y_2 gives $D^2\Psi = 13/4$ and $D^3\Psi = 25$. Thus $(D^3\Psi)^2 > 4(D^2\Psi)^3$. For example these derivatives follow directly from $q_1 = 1/9$, $q_2 = 1/3 - 2h/3 - h^2$ along that line. This illustrates why the parameter proof above uses the restricted Hessian metric and an independent lower section, not boundary orders alone.

16 Norm balls and weighted power cones

The local bound depends on a curved boundary patch, rather than on rotational symmetry. It therefore gives the same unrestricted dimension frontier for every ℓ_p ball with $1 < p < \infty$. The global frontier also persists, but its proof uses the primal and polar boundaries as independent C^1 manifolds. This distinction matters at coordinate zeros.

16.1 Exact counts and the dimensions of count-optimal lifts

Write

$$B_p^N = \{x \in \mathbb{R}^N : \sum_j |x_j|^p \leq 1\}, \quad K_{p,m} = \{(t, u) \in \mathbb{R} \times \mathbb{R}^{m-1} : t \geq \|u\|_p\}.$$

Fix an o-minimal expansion in which the relevant fixed powers are definable; $\mathbb{R}_{\exp}$ suffices. For rational exponents one can work entirely with semialgebraic cones and lifts.

Theorem 16.1 (Dimension-capped ℓ_p balls). *Let $1 < p < \infty$, $N \geq 2$, and let $d \geq 3$ be an integer. Among exact lifts of B_p^N through arbitrary definable proper cones of dimensions at most d , with*

arbitrary affine equations, projections, and free variables, the minimum number of factors and minimum total cone dimension are simultaneously

$$k_* = \left\lceil \frac{N-1}{d-2} \right\rceil, \quad M_* = N-1 + 2k_*. \quad (16.1)$$

Every such lift has at least k_* factors of dimension at least three, even if arbitrarily many rays are allowed.

Proof. On the boundary patch where every coordinate is nonzero, $f(x) = \sum_j |x_j|^p$ is smooth and

$$D^2 f(x) = p(p-1) \operatorname{diag}(|x_j|^{p-2}) \succ 0.$$

Its restriction to the tangent space, divided by $\|Df(x)\|$, is the positive second fundamental form. After the reductions of Lemma 2.1, Theorem 3.2 gives $N-1 \leq \sum_i (s_i - 2)_+$, where $s_i \leq d$ are the reduced factor dimensions. If k_+ factors have $s_i \geq 3$, then $k_+ \geq k_*$ and their dimensions sum to at least $N-1 + 2k_+$. Original dimensions and the total factor count are no smaller.

For attainment partition $N-1 = b_1 + \dots + b_{k_*}$ with $1 \leq b_i \leq d-2$. Make a chain of internal tree nodes: node $i < k_*$ has b_i new leaf children and node $i+1$; the last node has $b_{k_*} + 1$ leaf children. Node i requires its scalar to dominate the p -norm of its children, using K_{p, b_i+2} , and the root scalar is one. There are N leaves. Induction from the leaves shows that a node scalar dominates the p -norm of its descendant leaves. Conversely, assigning exactly those subtree norms makes every internal constraint feasible whenever $\|x\|_p \leq 1$. The projection is therefore exactly B_p^N , and the total dimension is $\sum_i (b_i + 2) = N-1 + 2k_*$. At zero leaves choose strictly positive internal scalars decreasing from the root to obtain Slater feasibility.

The cones $K_{p,m}$ are closed, pointed, full-dimensional and convex. For $p = a/b$ rational, nonnegative auxiliary variables satisfying $z_j^b = |u_j|^a$, $w^b = t^a$, $t \geq 0$ and $\sum_j z_j \leq w$ give a semialgebraic description after projection. This also handles an even denominator without changing the meaning of the power. $\square$

The tree construction is established prior work: the three-dimensional version is Krokhamal and Soberanis [57, Proposition 1]; see also Blanco and Martínez-Antón [15] for representations by specified cone families. The contribution of Theorem 16.1 is its matching lower bound against arbitrary definable proper factors and affine lifts, together with the total-dimension optimum. The polyhedral endpoints $p = 1, \infty$ have no such curved patch and are excluded. Restriction to interior two-dimensional sections of the positive-capacity factors also gives the ambient barrier lower bound $2k_*$, as in Corollary 10.2. Its attainment is asserted here only for $p = 2$.

Proposition 16.2 (All capacity excesses at the minimum count). *Under the hypotheses of Theorem 16.1, set $n = N-1$, $c = d-2$, and $\sigma = k_*c - n$. Suppose an exact lift uses exactly k_* original factors, of dimensions $m_i \leq d$, and let s_i be their dimensions after minimal-face reduction. Then every $s_i \geq 3$ and*

$$\Delta = \sum_i (s_i - 2) - n \in \{0, \dots, \sigma\}, \quad \sum_i s_i = n + 2k_* + \Delta. \quad (16.2)$$

Every indicated Δ occurs in a full-face p -norm tree lift. If c divides n , every count-optimal lift has $s_i = m_i = d$ for all i .

Proof. If one reduced factor had dimension at most two, the remaining capacity would be at most $(k_* - 1)c < n$, a contradiction. The lower curvature bound and the dimension cap now give (16.2).

For any desired Δ , start with the preceding chain and choose integers $0 \leq r_i \leq c - b_i$ with $\sum_i r_i = \Delta$. Replace K_{p,b_i+2} by K_{p,b_i+2+r_i} and fix the added norm coordinates to zero. This preserves the projection and strict feasibility, so the larger displayed cones remain their full minimal faces. If $\sigma = 0$, equality in $\sum_i (s_i - 2) \leq k_* c$ forces every $s_i = m_i = d$. $\square$

At $\Delta = 0$ the local conclusion is stronger than the arithmetic. On a patch where every coordinate is nonzero the supporting-polarity map is smooth, so the generic C^2 selections can be written on a common contact chart. Put $H_i = -DA_i^* DB_i$ for those selections $A_i(u), B_i(v)$. Each kernel $\langle A_i(u), B_i(v) \rangle$ is nonnegative and zero when $u = v$. Its Hessian at a diagonal point is positive semidefinite and annihilates (h, h) ; thus H_i is symmetric positive semidefinite. The total $H = \sum_i H_i$ is positive definite. Rank equality gives $\text{rank } H_i = s_i - 2$, and the linear-algebra argument in Section 6 gives

$$P_i = H^{-1/2} H_i H^{-1/2}, \quad P_i^2 = P_i, \quad P_i P_j = 0 \ (i \neq j), \quad \sum_i P_i = I. \quad (16.3)$$

Moreover, both quotient derivatives into $B_i^\perp / \mathbb{R}A_i$ and $A_i^\perp / \mathbb{R}B_i$ are onto: otherwise the mixed rank would be smaller than $s_i - 2$. These are local statements; they do not determine the full cones.

Example 16.3 (A saturated lift with a non-strictly-convex factor). For B_p^3 , set

$$K_1 = K_{p,3}, \quad K'_2 = \{(s, t, z) \in K_{p,3} : t \geq 0\}.$$

The two constraints

$$(t, x_1, x_2) \in K_1, \quad (1, t, x_3) \in K'_2$$

project exactly onto B_p^3 : the first already implies $t \geq 0$, so the extra halfspace in the second is redundant on the affine lift. At $x = 0$ and $0 < t < 1$ both constraints are strict. Thus both full three-dimensional factors are necessary in the curvature count, and their total capacity is $2 = 3 - 1$. Nevertheless K'_2 has the two-dimensional exposed face $\{(s, 0, z) : s \geq |z|\}$, whereas every proper nonzero face of $K_{p,3}$ is a ray. The factors are not linearly isomorphic. At every contact with $(x_1, x_2) \neq 0$, the added facet is inactive and the local cone germ is unchanged. In particular, saturation alone cannot classify the global factors or justify a target-dependent barrier premium.

16.2 Global selections and the Young construction

Let $q = p/(p-1)$. The two boundaries ∂B_p^N and ∂B_q^N are C^1 and strictly convex. A global factorization in this subsection means C^1 primal selections on the first boundary and genuine C^1 normalized dual selections on the second, satisfying the full two-point identity for $1 - \langle x, y \rangle$. The two selections are differentiated in their own boundary coordinates.

Theorem 16.4 (Global C^1 frontier for ℓ_p balls). *Let $N \geq 3$ and let $d \geq 3$ be an integer. Among exact lifts with these global selections through arbitrary proper factors of dimension at most d , the simultaneous minimum capacity $S = \sum_i (\dim K_i - 2)_+$, total factor count k , and total cone dimension M are*

$$(S, k, M) = \begin{cases} (N, h, N + 2h), & h = \lceil N/(d-2) \rceil, \quad d < N + 1, \\ (N - 1, 1, N + 1), & d \geq N + 1. \end{cases} \quad (16.4)$$

The lower bound does not require definability of the cones or selections.

Proof. At a contact $\langle x, y \rangle = 1$ the tangent spaces are $y^\perp$ and $x^\perp$. Their Euclidean pairing is nondegenerate, since its first kernel is $y^\perp \cap \mathbb{R}x = \{0\}$. Consequently $S \geq N - 1$ by Lemma 3.1. If equality holds, Theorem 10.1 allows exactly one positive-capacity factor, of dimension $N + 1$: both boundaries are C^1 diffeomorphic to $\mathbb{S}^{N-1}$. A strict cap excludes this case, so integrality gives $S \geq N$. Counting positive-capacity factors then gives $k \geq h$ and $M \geq N + 2h$. For $d \geq N + 1$, the direct lift $K_{p,N+1}$ with primal and dual maps $(1, x), (1, -y)$ attains the second line.

For the first line define, for $r \geq 1$,

$$\mathcal{P}_{p,r} = \{(u, v, w) \in \mathbb{R}_+^2 \times \mathbb{R}^r : uv^{p-1} \geq \|w\|_p^p\}. \quad (16.5)$$

This is the closed perspective epigraph of $\|w\|_p^p$, hence a proper cone of dimension $r + 2$. At $v = 0$ the definition gives $w = 0$ and $u \geq 0$, exactly the closure of its part with $v > 0$. Partition the coordinates into h nonempty groups G of sizes at most $d - 2$, and impose

$$(u_G, v_G, w_G) \in \mathcal{P}_{p,|G|}, \quad v_G = 1, \quad \sum_G u_G = 1, \quad x_G = w_G. \quad (16.6)$$

The projection is B_p^N , and $x = 0, u_G = 1/h$ is a Slater point. On the boundary the primal fiber is unique, with $u_G = \|x_G\|_p^p$. Define

$$A_G(x) = (\|x_G\|_p^p, 1, x_G), \quad B_G(y) = \left(\frac{1}{p}, \frac{\|y_G\|_q^q}{q}, -y_G\right). \quad (16.7)$$

For $v > 0$, Young's inequality gives

$$\frac{u}{p} + \frac{v\|y_G\|_q^q}{q} - \langle w, y_G \rangle \geq v \left(\frac{\|w/v\|_p^p}{p} + \frac{\|y_G\|_q^q}{q} - \langle w/v, y_G \rangle \right) \geq 0.$$

Continuity includes $v = 0$, so $B_G(y)$ belongs to the dual cone. Its coefficients are the multipliers $1/p$ for $\sum_G u_G = 1$ and $\|y_G\|_q^q/q$ for $v_G = 1$, minus the objective on w_G . Their objective value is $1/p + \sum_G \|y_G\|_q^q/q = 1$. Thus these are genuine certificates on the whole affine slice. They give

$$\sum_G \langle A_G(x), B_G(y) \rangle = 1 - \langle x, y \rangle \quad (x \in \partial B_p^N, y \in \partial B_q^N).$$

Both maps are globally C^1 because $p, q > 1$. Their total capacity, count and dimension are $N, h, N + 2h$, respectively. $\square$

The scalar case $r = 1$ of (16.5) is the power cone $P_{1/p} = \{(u, v, w) : u, v \geq 0, u^{1/p}v^{1/q} \geq |w|\}$. The corresponding standard p -norm model is recorded in MOSEK ApS [65, Section 4.2.2]. The construction and Young's inequality are not novelty claims. Theorem 16.4 proves their optimal resource count under the stated global regularity condition. For $d = 3$ it gives exactly N factors, including when the dictionary is restricted to three-dimensional power cones.

The scalar selections in (16.7) are both C^2 only when $p = 2$. If $1 < p < 2$, the primal map fails to be C^2 at coordinate zeros and the dual map is C^2 ; for $p > 2$ the roles reverse. This does not affect the preceding proof, which does not differentiate the polarity map $y_j = \text{sign}(x_j)|x_j|^{p-1}$. Every ambient barrier has parameter at least $2h$ in the strict-cap case, by the product orthant-section argument of Corollary 10.2. This is attained for $p = 2$; no such equality is asserted for the nonsymmetric perspective cones. The reduced affine domain in (16.6) also has the intrinsic barrier lower bound h : fix all allocations u_G positive with sum one, set all but one coordinate per group to zero, and

vary the remaining coordinate in $(-u_G^{1/p}, u_G^{1/p})$. This gives an h -dimensional open box through the domain interior, whose boundary lies on the domain boundary. Restriction and the box lower bound prove the assertion. The Euclidean case is attained by the grouped barrier of Section 9; the construction here makes no matching intrinsic-parameter assertion for $p \neq 2$.

Proposition 16.5 (The two-dimensional power-cone exception). *For $N = 2$, one arbitrary three-dimensional proper cone suffices for B_p^2 , with global C^1 primal and dual selections. Within the dictionary of three-dimensional power cones, the minimum total factor count is one for $p = 2$ and two for $p \neq 2$, even without requiring global selections.*

Proof. The direct $K_{p,3}$ lift proves the first assertion, and the two scalar Young factors give the upper bound in the power dictionary. A one-factor power-cone lift has total dimension three; by Theorem 13.4 its cone would be linearly isomorphic to $K_{p,3}$. We show that this is impossible unless $p = 2$.

For P_α , $0 < \alpha < 1$, the only projective boundary rays with possibly non-smooth second derivatives are its two axial rays. In the chart $u = 1$ at the first, the boundary is $v = |w|^{1/(1-\alpha)}$; in the chart $v = 1$ at the second it is $u = |w|^{1/\alpha}$. Elsewhere the boundary is smooth with positive curvature. If $\alpha \neq 1/2$, exactly one of these exponents is below two, so exactly one boundary ray is not C^2 . For $K_{p,3}$ the projective section is the p -unit circle. Near $(1, 0)$ it has graph

$$x_1 = (1 - |x_2|^p)^{1/p} = 1 - \frac{|x_2|^p}{p} + o(|x_2|^p),$$

and the same holds at its other three coordinate points. It therefore has exactly four non- C^2 rays when $p < 2$, and none when $p > 2$. An invertible linear cone map induces smooth projective coordinate changes and preserves this count. This excludes $\alpha \neq 1/2$. For $\alpha = 1/2$, the section is a nonsingular conic with nonzero second fundamental form everywhere. When $p > 2$ the four coordinate points instead have zero second fundamental form; its vanishing is preserved by projective transformations, as follows by differentiating a homogeneous supporting equation along its tangent. When $p < 2$ the C^2 obstruction already suffices. At $p = 2$ both cones are Lorentz cones, which proves the remaining case. $\square$

16.3 A curvature bound for generalized Euclidean power cones

Let $\alpha_j > 0$, $\sum_{j=1}^m \alpha_j = 1$, and set

$$g(x) = \prod_{j=1}^m x_j^{\alpha_j}, \quad C_{\alpha,k} = \{(x, z) \in \mathbb{R}_+^m \times \mathbb{R}^k : \|z\|_2 \leq g(x)\}.$$

The continuous extension of g to the nonnegative orthant is understood. It is concave, so $C_{\alpha,k}$ is a proper convex cone. It is definable in $\mathbb{R}_{\text{exp}}$, and semialgebraic for rational weights.

Theorem 16.6 (Power-cone curvature capacity). *Let $m, k \geq 1$ and $m + k \geq 3$. Every exact affine lift of $C_{\alpha,k}$ by definable proper cones of dimensions d_i satisfies*

$$\sum_i (d_i - 2)_+ \geq m + k - 2. \quad (16.8)$$

In particular, a lift using three-dimensional Lorentz cones and any number of rays requires at least $m + k - 2$ Lorentz factors. The same scalar bound $m - 1$ holds for the nonnegative geometric-mean cone $\{(x, t) : x \geq 0, 0 \leq t \leq g(x)\}$ when $m \geq 2$.

Proof. Intersect the cone with $\sum_j x_j = 1$. This is a compact convex body of dimension $m + k - 1$ in its affine hull: x ranges over the simplex and $\|z\| \leq g(x)$ is uniformly bounded. Restricting an exact lift by this affine equation introduces no cone factors. On the patch $x_j > 0$, $z \neq 0$, the boundary is $\Phi(x, z) = 0$, where $\Phi(x, z) = \|z\| - g(x)$. Write $z = g(x)\omega$ with $\|\omega\| = 1$. A tangent vector (h, ξ) obeys

$$\sum_j h_j = 0, \quad \langle \omega, \xi \rangle = Dg(x)[h].$$

The Hessian on such tangent vectors is

$$D^2\Phi[(h, \xi), (h, \xi)] = -D^2g(x)[h, h] + \frac{\|\xi\|_2^2 - \langle \omega, \xi \rangle^2}{g(x)}. \quad (16.9)$$

Direct differentiation gives the weighted variance identity

$$-D^2g(x)[h, h] = g(x) \left\{ \sum_j \alpha_j \left(\frac{h_j}{x_j} \right)^2 - \left(\sum_j \alpha_j \frac{h_j}{x_j} \right)^2 \right\}.$$

It is nonnegative and vanishes exactly when h is proportional to x . On $\sum_j h_j = 0$, this forces $h = 0$. If also the second term of (16.9) vanishes, then ξ is proportional to ω ; the tangent equation with $h = 0$ forces $\xi = 0$. Thus the boundary patch has strictly positive second fundamental form of rank $m + k - 2$. Theorem 3.2, applied in the affine hull of this compact section, gives (16.8). Minimal-face reduction can only reduce the displayed dimensions. For the nonnegative scalar cone the positive upper graph $t = g(x)$ has exactly the same Hessian on the simplex tangent; its section has dimension m and curved boundary dimension $m - 1$. The case $m = 1$ in the vector statement reduces to the ordinary Euclidean ball section. $\square$

This result is a dimension bound, not an exact count for every rational weight. Wang [87, Theorem 3.6] proves the scalar bound $m - 1$ for simple mediated-set representations, together with separate denominator-dependent bounds; his Theorem 3.15 gives $\lceil m/2 \rceil$ for arbitrary three-dimensional SOC lifts. The proof above permits arbitrary affine cone-coordinate maps, auxiliary equations and free variables, and also gives the Euclidean-output term $k - 1$. Blanco and Martínez-Antón [15, Definition 1 and Theorem 14] relate extended representations and mediated graphs; their constructive maps select triples of mediated variables. We make no priority claim for the scalar inequality from this comparison alone, and do not extend denominator-sensitive bounds beyond their established representation classes. Theorem 16.6 supplies a self-contained geometric proof in the affine-lift model used throughout this paper.

17 Matrix balls and sparse completion cones

Matrix balls illustrate two different forms of sharing. A common scale reduces the number of cone coordinates and the ambient barrier parameter. A positive-semidefinite completion cone instead preserves a sparse matrix structure. After fixing the scale or the diagonal blocks, both constructions have exactly the same barrier on the product of matrix balls. The spectral barrier and the completion barrier used below are classical [3, 24]; the results here identify their full-slack maps, sharp dimension and face constraints, and their common reduced geometry.

17.1 Mixed contact rank with split rows

For $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$ let $\delta = \dim_{\mathbb{R}} \mathbb{F}$ and use the real Frobenius pairing $\langle X, Z \rangle = \operatorname{Re} \operatorname{tr}(X^* Z)$. The extreme points of the spectral-norm ball in $\mathbb{F}^{r \times c}$, $1 \leq r \leq c$, are the coisometries

$$\mathcal{E}_{r,c} = \{X : XX^* = I_r\};$$

those of its nuclear-norm polar are $\mathcal{Z}_{r,c} = \{uv^* : \|u\| = \|v\| = 1\}$. These descriptions follow from singular-value decomposition: a contraction with a singular value below one is a nontrivial average of contractions, whereas equality on every row forces any such averaging at a coisometry to be trivial. Similarly, the nuclear unit ball is the convex hull of the rank-one unit matrices, which are exposed by suitable contractions.

Suppose a product of these extreme manifolds has a globally C^1 labelled full-row factorization

$$1 - \langle X_a, Z \rangle = \sum_i \langle A_i(X_1, \dots, X_k), B_i^a(Z) \rangle, \quad A_i \in K_i, \quad B_i^a \in K_i^*. \quad (17.1)$$

Here the nonray proper cones have real dimensions $m_i \geq 3$; ray factors, if present, contribute no mixed rank. Put

$$q_i = m_i - 2, \quad f_i = \max_{0 \neq y \in K_i^*} \dim(K_i \cap y^\perp), \quad \bar{f}_i = \min(f_i, k), \quad \kappa_a = \delta_a c_a - 1.$$

The rows may use different fields. All dimensions of cones and faces remain real dimensions.

Proposition 17.1 (Spectral contact budget). *At every contact $\langle X, Z \rangle = 1$, the mixed pairing $(H, \dot{Z}) \mapsto -\langle H, \dot{Z} \rangle$ on $T_X \mathcal{E}_{r,c} \times T_Z \mathcal{Z}_{r,c}$ has rank $\delta c - 1$. At simultaneous contacts, let*

$$t_i^a = \operatorname{rank}[(H, \dot{Z}) \mapsto -\langle D_a A_i[H], DB_i^a[\dot{Z}] \rangle], \quad I_i = \{a : t_i^a > 0\}.$$

Then

$$t_i^a \leq q_i, \quad \sum_i t_i^a \geq \kappa_a, \quad |I_i| \leq \bar{f}_i, \quad (17.2)$$

$$\sum_{a \in I_i} t_i^a \leq q_i, \quad 1 + \sum_{b \in I_i \setminus \{a\}} t_i^b \leq \dim(K_i \cap B_i^a(Z_a)^\perp) \quad (a \in I_i). \quad (17.3)$$

If $m_i \leq d$ for an integer $d \geq 3$, these imply

$$\sum_i \bar{f}_i \geq \sum_a \left\lceil \frac{\kappa_a}{d-2} \right\rceil, \quad \sum_i q_i \geq \sum_a \kappa_a. \quad (17.4)$$

In particular, a common face cap $f_i \leq f$, with $f \geq 1$, gives

$$L \geq \max \left\{ \left\lceil \frac{1}{f} \sum_a \left\lceil \frac{\kappa_a}{d-2} \right\rceil \right\rceil, \left\lceil \frac{\sum_a \kappa_a}{d-2} \right\rceil \right\}.$$

Proof. Contact implies $Xv = u$ and $X^*u = v$ by equality in the contraction inequality. Unitary changes of bases therefore give $X = (I_r, 0)$ and $u = v = e_1$. Write a coisometry tangent as $H = (A, B)$ with $A^* = -A$. The redundancy $(u, v) \mapsto (uq, vq)$ for a unit scalar $q \in \mathbb{F}$ permits the tangent gauge $u^* \dot{u} = 0$. The independent channels in $\langle H, \dot{u}e_1^* + e_1 \dot{v}^* \rangle$ are the entries A_{j1} , $j > 1$, paired with the off-axis differences of $\dot{u}$ and $\dot{v}$; the first row of B , paired with the tail of $\dot{v}$; and the imaginary entry A_{11} , paired with the imaginary entry $\dot{v}_1$. They have respectively $\delta(r-1)$, $\delta(c-r)$, and $\delta-1$ real

dimensions, and the pairing on each channel is nondegenerate. Their sum is $\delta c - 1$. This argument includes real scalar rows, where the rank can be zero. In the quaternionic case all pairings are real Euclidean pairings, so cyclic rearrangements of quaternionic traces are not required.

For clarity, the face argument does not need nondegeneracy of the full spectral pairing. At contact, each summand in (17.1) vanishes. If either factor is zero, its derivative is zero because a differentiable map into a pointed cone has zero derivative at the vertex. Otherwise the tangent-ray quotient bounds its mixed rank by $m_i - 2$. Mixed differentiation and rank subadditivity give $\sum_i t_i^a \geq \kappa_a$.

Fix i and an active row a . On the unrelated-source contact cylinder, nonnegativity gives

$$\langle D_b A_i[H_b], DB_i^a[\dot{Z}] \rangle = 0 \quad (b \neq a), \quad \langle A_i, DB_i^a[\dot{Z}] \rangle = 0.$$

Indeed the summand, as a function of the polar variable, has a minimum zero at Z_a for every point of that cylinder; differentiate this zero first derivative along source b . Choose $E_a \subseteq \text{im } D_a A_i$ of dimension t_i^a on which pairing with DB_i^a is injective. These identities show that $\mathbb{R}A_i \oplus \bigoplus_{a \in I_i} E_a$ is a direct sum. For any fixed active row a_0 , the nonzero vector $b = B_i^{a_0}(Z_{a_0})$ annihilates all of $\text{im } D A_i$: the nonnegative function $X' \mapsto \langle A_i(X'), b \rangle$ has value zero at the contact X . It also annihilates A_i , so the direct sum lies in $b^\perp$ and $1 + \sum_a t_i^a \leq m_i - 1$. If there is no active row, the total rank bound is immediate. Moreover, A_i and every E_b with $b \neq a$ lie in the span of the exposed face $K_i \cap B_i^a(Z_a)^\perp$. This proves (17.3) and $|I_i| \leq f_i$. Finally, each row needs at least $\lceil \kappa_a / (d - 2) \rceil$ active blocks. Counting these incidences and summing the total mixed-rank bounds proves (17.4). Finally $\sum_i q_i \leq (d - 2)L$, which gives the displayed count bound. $\square$

These are local necessary conditions allowing a row to be split among several factors. They assert neither a global topological premium nor attainment of every budget. In particular, the rank is $\delta c - 1$, not the dimension of the polar extreme manifold: some polar directions move inside the contact set and disappear from the mixed pairing.

17.2 Whole-row grouping and exact spectral parameters

Here take $\mathbb{F}_a \in \{\mathbb{R}, \mathbb{C}\}$, with $\delta_a = \dim_{\mathbb{R}} \mathbb{F}_a$, and orient each matrix with $r_a \leq c_a$. For a nonempty group G define

$$\mathcal{S}_G = \{(t, (X_a)_{a \in G}) : \|X_a\|_{\text{op}} \leq t\}, \quad N_G = \sum_{a \in G} \delta_a r_a c_a, \quad R_G = \sum_{a \in G} r_a, \quad \lambda_a = \delta_a(r_a + c_a - 1).$$

The dual cone is $\{(\alpha, (Z_a)) : \alpha \geq \sum_a \|Z_a\|_*\}$, under the real pairing. Thus the polynomial maps

$$A_G(X) = (1, (X_a)_{a \in G}), \quad B_G^a(Z) = (1, (0, \dots, -Z, \dots, 0))$$

represent the entire row $1 - \langle X_a, Z \rangle$.

Corollary 17.2 (Sharp whole-row spectral caps). *In the row-indivisible model, a group G can be carried by one proper cone of dimension at most D and maximal proper exposed-face dimension at most F if and only if*

$$N_G \leq D - 1, \quad N_G - \min_{a \in G} \lambda_a \leq F - 1. \quad (17.5)$$

The minimum factor count is the minimum number of parts of a partition satisfying these inequalities; the cones $\mathcal{S}_G$ attain it. For k identical $r \times c$ rows over a field of real dimension δ , one group carries at most

$$Q = \min \left\{ \left\lfloor \frac{D - 1}{\delta r c} \right\rfloor, \left\lfloor \frac{F + \delta(r + c - 1) - 1}{\delta r c} \right\rfloor \right\}.$$

If $Q \geq 1$, the exact count is $\lceil k/Q \rceil$.

Proof. For a rank-one polar support, contact fixes a 1×1 corner of X_a and kills both cross rectangles; the remaining contraction has $(r_a - 1)(c_a - 1)$ entries over $\mathbb{F}_a$. Its affine face dimension is therefore $\delta_a(r_a - 1)(c_a - 1)$, with codimension λ_a . Theorem 13.1 gives necessity. For sufficiency, the dimension of $\mathcal{S}_G$ is $1 + N_G$. More generally, a dual boundary support with nonzero blocks J of ranks h_a exposes a face of dimension

$$1 + \sum_{a \notin J} \delta_a r_a c_a + \sum_{a \in J} \delta_a (r_a - h_a)(c_a - h_a).$$

To see this, equality in spectral/nuclear duality fixes the supported singular-vector corner to $-tI_{h_a}$, kills the cross rectangles, and leaves an arbitrary contraction in the complementary rectangle. The scale contributes one dimension. Each supported block loses $\delta_a h_a (r_a + c_a - h_a)$ coordinates; this is minimized at $h_a = 1$. Consequently the maximal proper exposed-face dimension is $1 + N_G - \min_a \lambda_a$, proving sufficiency and the count formula. $\square$

Computing the heterogeneous minimum partition is strongly NP-hard: real vector rows $r_a = 1$ and a nonbinding face cap $F \geq D - 1$ give ordinary bin packing with item sizes c_a and capacity $D - 1$. The strongly bounded instances of 3-PARTITION keep even the expanded matrix dimensions polynomial. This computational statement concerns whole-row grouping, not arbitrary split-row factorizations.

Theorem 17.3 (Shared spectral cone and slice). *For a partition $\mathcal{G}$ into g nonempty groups, put $R = \sum_a r_a$ and $N = \sum_a \delta_a r_a c_a$. The product of the shared cones has real dimension $N + g$ and*

$$\vartheta_{\text{opt}} \left(\text{int} \prod_{G \in \mathcal{G}} \mathcal{S}_G \right) = R + g, \quad \vartheta_{\text{opt}} \left(\prod_a \{X_a : \|X_a\|_{\text{op}} < 1\} \right) = R. \quad (17.6)$$

Both lower bounds allow arbitrary coupled self-concordant barriers. The attaining barriers are respectively the sum over groups of

$$F_G(t, X) = - \sum_{a \in G} \log \det(t^2 I_{r_a} - X_a X_a^*) + (R_G - 1) \log t \quad (17.7)$$

and its fixed-scale restriction

$$\Phi(X) = - \sum_a \log \det(I_{r_a} - X_a X_a^*). \quad (17.8)$$

Proof. The classical spectral-cone barrier is $-\log t - \log \det(tI - XX^*/t)$, with parameter one plus the number of rows [24, Section 4.2]; the same formula for complex matrices is recorded in Coey [25, Section 2.6.1]. Apply it to the block-diagonal matrix with blocks X_a ; real blocks can be embedded in a complex matrix without changing singular values. This proves self-concordance of (17.7) and its degree $R_G + 1$ of logarithmic homogeneity.

We give an explicit lower certificate to avoid requiring logarithmic homogeneity of a competing barrier. Nesterov's recession bound says that, for an interior point x , recession directions p_j , and positive numbers a_j, b_j satisfying $x - \sum_j a_j p_j \in C$ and $x - b_j p_j \notin \text{int } C$, every self-concordant barrier of C has parameter at least $\sum_j a_j / b_j$ [35, Theorem 3.9]. Restrict the matrices of one group to their real rectangular diagonals. Writing $r = R_G$, this gives $C_r = \{(t, \xi) : t \geq \|\xi\|_\infty\}$. For $0 < \epsilon < \min(1, 2/r)$ choose

$$\begin{aligned} x &= (1, (1 - \epsilon)\mathbf{1}), & p_0 &= (1, \mathbf{1}), & p_j &= (1, \mathbf{1} - 2e_j) \quad (1 \leq j \leq r), \\ a_j &= b_j = \epsilon/2 \quad (j \geq 1), & a_0 &= 1 - r\epsilon/2, & b_0 &= 1 - \epsilon/2. \end{aligned}$$

All p_j belong to C_r , $x = \sum_{j=0}^r a_j p_j$, and $x - b_j p_j$ is on the boundary for every j . The recession bound is therefore $r + (1 - r\epsilon/2)/(1 - \epsilon/2)$, which tends to $r + 1$. For several groups, place each certificate in its own direct summand and take x to be the tuple of their interior points. The resulting single certificate gives $\sum_G (R_G + 1)$ even for a coupled barrier. These diagonal sections meet the cone interiors, so restriction of any ambient barrier to them is legitimate.

For the slice, ordinary self-concordance follows by restricting the positive-definite barrier to $\begin{pmatrix} I_r & X \\ X^* & I_c \end{pmatrix}$. At a singular-value representative $X = (\text{diag}(\sigma_j) \ 0)$, the gradient is real diagonal and this subspace is invariant under the Hessian. On its j th coordinate the first two derivatives are $2\sigma_j/(1 - \sigma_j^2)$ and $2(1 + \sigma_j^2)/(1 - \sigma_j^2)^2$. Consequently

$$\|D\Phi(X)\|_{D^2\Phi(X),*}^2 = \sum_{a,j} \frac{2\sigma_j(X_a)^2}{1 + \sigma_j(X_a)^2} < R.$$

The real diagonal section is the cube $(-1, 1)^R$, whose sharp barrier lower bound is R [70, Proposition 2.3.6]. This proves the second equality, including coupled competitors. $\square$

The cap theorem by itself optimizes factor count and dimensions in a whole-row model. Equation (17.6) gives the exact barrier costs of its displayed attaining cones; it does not infer a barrier lower bound for arbitrary higher-dimensional cones carrying the same rows. For identical rows and $g = \lceil k/Q \rceil$, these ledgers are $(M, \nu_{\text{ambient}}, \nu_{\text{slice}}) = (k\delta rc + g, kr + g, kr)$.

17.3 Completion cones and separator credits

Let G be an undirected graph on n vertices, with all diagonal entries specified. Over $\mathbb{R}$ or $\mathbb{C}$, let $\mathcal{H}_G$ be the space of Hermitian matrices supported on those entries, with the real trace pairing. The sparse positive-semidefinite cone is $K_G = \{Y \in \mathcal{H}_G : Y \succeq 0\}$. Its dual C_G identifies with the partial Hermitian matrices that admit a positive-semidefinite completion. This identification has no closure gap: in a convergent sequence of partial matrices all diagonals are bounded, and every positive-semidefinite completion obeys $|Z_{ij}|^2 \leq Z_{ii}Z_{jj}$. A convergent subsequence of completions therefore completes the limit. Also, C_G is pointed because a matrix and its negative can both be completed only if all their diagonals, and hence all entries, vanish.

Proposition 17.4 (Classical completion barrier and its exact parameter). *For arbitrary graphs $G_1, \dots, G_l$ of orders $n_1, \dots, n_l$,*

$$\vartheta_{\text{opt}} \left(\text{int} \prod_j C_{G_j} \right) = \sum_j n_j.$$

For one graph an attaining barrier, up to a constant, is

$$F_G(S) = -\log \det Z(S), \tag{17.9}$$

where $Z(S)$ is the unique maximum-determinant positive-definite completion. If the graph is chordal, its maximal cliques $\mathcal{Q}$ and a clique forest's separators $\mathcal{T}$, counted with multiplicity, give

$$F_G(S) = -\sum_{Q \in \mathcal{Q}} \log \det S_Q + \sum_{T \in \mathcal{T}} \log \det S_T. \tag{17.10}$$

Empty separators contribute zero.

Proof. The restriction of $-\log \det Y$ to $\text{int } K_G$ is an n -LH self-concordant barrier. Its signed Legendre dual

$$F_G(S) = \sup_{Y \in \text{int } K_G} \{-\langle S, Y \rangle + \log \det Y\}$$

is another n -LH self-concordant barrier [3]. Stationarity says that Y^{-1} completes S . Equivalently, stationarity of the strictly concave completion problem says that $Z(S)^{-1}$ is sparse. Boundedness of the set of completions and existence of a positive-definite completion guarantee existence; strict concavity gives uniqueness. Since Y is sparse, $\langle S, Y \rangle = \text{tr}(Y^{-1}Y) = n$, giving $F_G = -n - \log \det Z(S)$. The diagonal section of C_G is precisely $\mathbb{R}_+^n$, and it meets the interior. Across a product it is $\mathbb{R}_+^{\sum n_j}$. The sharp orthant bound proves the stated lower bound for every coupled barrier.

For a chordal graph, eliminate leaf cliques successively. If a leaf clique meets the remainder in separator T , Schur complementation maximizes the determinant by making the residual cross block zero and multiplies the determinant of the remainder by $\det S_Q / \det S_T$. Iteration gives (17.10), the classical clique-separator formula [10, 46]. The running-intersection identity $\sum_Q |Q| - \sum_T |T| = n$ gives its degree of logarithmic homogeneity. Self-concordance comes from duality, not from a general permission to subtract logarithmic barriers. $\square$

Proposition 17.5 (Block-star realization). *Consider a hub clique of size $s \geq 1$, and k disjoint leaf cliques of sizes $p_a \geq 1$, each joined completely to the hub. Put $\delta = 1$ over $\mathbb{R}$ and $\delta = 2$ over $\mathbb{C}$, and $h_\delta(v) = v + \delta v(v-1)/2$. The completion cone has real dimension and exact ambient parameter*

$$D = h_\delta(s) + \sum_a (\delta s p_a + h_\delta(p_a)), \quad \nu_{\text{ambient}}^{\text{opt}} = s + \sum_a p_a. \quad (17.11)$$

Fixing the hub and leaf diagonal blocks to identities gives the product $\prod_a \{X_a \in \mathbb{F}^{s \times p_a} : \|X_a\|_{\text{op}} \leq 1\}$. Its restricted completion barrier is exactly (17.8), with optimal parameter $\sum_a \min(s, p_a)$, and it carries every full nuclear-polar slack row.

Proof. Write the specified blocks as T , U_a , W_a on the hub, cross edges, and leaves. Chordal completion is equivalent to

$$\begin{pmatrix} T & U_a \\ U_a^* & W_a \end{pmatrix} \succeq 0 \quad \text{for every } a.$$

The count of independent entries gives D , and the graph order gives the exact parameter by Proposition 17.4. All $k-1$ clique-tree separators are the hub, so the barrier is

$$-\sum_a \log \det \begin{pmatrix} T & U_a \\ U_a^* & W_a \end{pmatrix} + (k-1) \log \det T = -\log \det T - \sum_a \log \det (W_a - U_a^* T^{-1} U_a). \quad (17.12)$$

At $T = I_s$, $W_a = I_{p_a}$, $U_a = X_a$, the Schur complement gives exactly the stated product. Sylvester's determinant identity gives $\det(I - X_a^* X_a) = \det(I - X_a X_a^*)$, so its barrier is (17.8), using the smaller order if desired. For a polar extreme $Z = uv^*$, put $b = (-u, v)$ on the hub and leaf a and zero on all other leaves. The sparse positive-semidefinite matrix $bb^*/2$ is a dual factor, and

$$\frac{1}{2} b^* \begin{pmatrix} I_s & X_a \\ X_a^* & I_{p_a} \end{pmatrix} b = 1 - \text{Re}(u^* X_a v).$$

Thus the primal identity-block section and this dual factor give all labelled full rows. The rank-one map depends only on Z : simultaneous phase changes in u, v leave bb^* invariant. $\square$

For scalar leaves, $p_a = 1$, the construction has $D = h_\delta(s) + k(\delta s + 1)$ and parameter $s + k$. It removes $\delta k(k - 1)/2$ real leaf-leaf coordinates compared with a dense Hermitian matrix of order $s + k$. Partitioning k leaves among g stars gives totals $k(\delta s + 1) + gh_\delta(s)$ and $k + gs$. In the real case, factoring T once and solving against all leaf vectors evaluates (17.12) in $O(s^3 + ks^2)$ dense arithmetic and stores $O(s^2 + ks)$ scalars. These counts concern barrier evaluation; a Newton solve with additional constraints has a separate cost.

17.4 Exact equality of the reduced geometries

After fixing the shared scales or the star's diagonal blocks, all constructions have the same retained coordinates and the same function Φ in (17.8). For one oriented block, with $M = I - XX^*$, its derivatives are

$$\nabla\Phi(X) = 2M^{-1}X, \quad D^2\Phi(X)[H] = 2M^{-1}H + 2M^{-1}(HX^* + XH^*)M^{-1}X.$$

Hence their Hessian metrics and their reduced Newton systems are identical. Specifically, with the same real affine map $\mathcal{A}$, right side b , and objective c , both systems use

$$\begin{pmatrix} D^2\Phi(X) & \mathcal{A}^* \\ \mathcal{A} & 0 \end{pmatrix}, \quad \begin{pmatrix} -D\Phi(X) - \tau c - \mathcal{A}^*y \\ b - \mathcal{A}X \end{pmatrix}.$$

This also identifies their central paths and reduced condition numbers. When orientation requires transposition or adjunction, the identification is a fixed real orthogonal coordinate change (including conjugation signs in the complex case). An oracle already supplied for one of these reduced objects therefore transfers by that public coordinate change. Constructing it from different ambient data is a separate operation: the ambient scales, completion coordinates, dual variables, and KKT systems are different. Neither the ambient parameter saving nor the sparse coordinate count alone implies a running-time or quantum-query saving.

18 Nonsymmetric cones: barrier premiums and evaluation costs

Dimension and contact capacity do not determine a cone's best barrier parameter. This section makes that distinction quantitative for the norm and perspective cones of Section 16. Throughout this section $\nu_{\text{LH}}(C)$ denotes the infimum of the parameters of logarithmically homogeneous self-concordant barriers on C . A barrier on a product may couple all its factors. The logarithmic-homogeneity hypothesis is essential in the cross-ratio arguments below.

18.1 An additive premium for fixed cone dictionaries

For $1 < p < \infty$, put $c = \min\{1/p, 1 - 1/p\}$. If $p \neq 2$, let $\gamma \in (0, 1)$ solve

$$(1 - 2c)\gamma^{1-c} + (1 - c)\gamma^{1-2c} = c, \quad h(p) = 1 + \frac{1 + \gamma^c}{1 + \gamma^{1-c}}. \quad (18.1)$$

Set $h(2) = 2$. The left side of the root equation is strictly increasing from zero to $2 - 3c > c$, so the root exists and is unique. Rearranging gives

$$h(p) - 2 = \frac{1 - 2c}{1 - c} \gamma^c. \quad (18.2)$$

Consequently $2 < h(p) < 3$ for $p \neq 2$, and $h(p) = h(q)$ when $1/p + 1/q = 1$. The one-factor bound $h(p)$ for both three-dimensional power cones and norm cones is due to Hildebrand [53, Corollaries 7.1–7.2].

We first record precisely the product argument. For a proper cone C , choose a compact affine base, a boundary basis $e_1, \dots, e_m$, and an interior point x lying in the relative interior of their simplex. On the projective line through e_a, x , let y_a be the opposite boundary contact and z_a the intersection with the span of the other basis vectors. Write $q_a = (e_a, x; y_a, z_a)$, using the convention $(a, b; c, d) = (a - c)(b - d)/((a - d)(b - c))$ and continuity at infinity. Hildebrand's Theorem 5.4 states, in particular, that

$$\sum_a q_a \geq 2 \implies \nu_{\text{LH}}(C) \geq \sum_a q_a. \quad (18.3)$$

Lemma 18.1 (Product of positive-branch certificates). *Suppose each proper cone C_i has the preceding configuration with cross-ratio sum $S_i \geq 2$. Then $\nu_{\text{LH}}(\prod_i C_i) \geq \sum_i S_i$.*

Proof. Normalize each local base by $\ell_i = 1$ and the product base by $\sum_i \ell_i = 1$. Embed each e_{ia} into its own direct summand, and choose $X = (\alpha_i x_i)_i$, where $\alpha_i > 0$ and $\sum_i \alpha_i = 1$. These embedded vectors form a boundary basis and X lies in their simplex interior. Along the line from an embedded e_{ia} through X and beyond, every other block remains a positive multiple of x_j . The first opposite boundary contact is therefore determined by block i . The missing-basis hyperplane is determined by the same block. Projection onto block i induces a projective isomorphism on this line, sending the four defining points to their local counterparts. Thus its cross-ratio is q_{ia} , unchanged by α_i . Applying (18.3) to the product gives the result. $\square$

Theorem 18.2 (Coupled nonsymmetric product premiums). *For $m_i \geq 3$ and $r_i \geq 1$,*

$$\nu_{\text{LH}}\left(\prod_{i=1}^k K_{p, m_i}\right) \geq kh(p), \quad (18.4)$$

$$\nu_{\text{LH}}\left(\prod_{i=1}^k \mathcal{P}_{p, r_i}\right) \geq kh(p). \quad (18.5)$$

At $p = 2$ both optima equal $2k$.

Proof. For $1 < p < 2$, Hildebrand's norm-cone configurations have positive cross-ratio sums tending to $h(p) > 2$. Apply Lemma 18.1 and take the limit. For $p > 2$, the signed Legendre transform $F_*(s) = \sup_{x \in \text{int } C} \{-\langle s, x \rangle - F(x)\}$ preserves the LHSC parameter and has domain $\text{int } C^*$. Since $K_{p,3}^* = K_{q,3}$, the preceding case applied to the dual product proves the assertion. This dual step is necessary: the direct $p > 2$ configurations lie on the other branch of Hildebrand's bound. In a larger norm cone, set all but two norm coordinates to zero. This linear section meets the ambient interior and is $K_{p,3}$, so barrier restriction proves (18.4).

In $\mathcal{P}_{p, r_i}$ retain one output coordinate. The resulting interior-meeting section is the scalar power cone $u^{1/p}v^{1/q} \geq |w|$. Hildebrand's power-cone configurations are on the positive branch for the exponent represented by $p \geq 2$; swapping u, v handles $p < 2$. Lemma 18.1 proves (18.5). At $p = 2$ every displayed cone is a Lorentz cone, and the product orthant lower bound and standard sum barrier both give $2k$. $\square$

The preceding theorem supplies additive lower bounds on a *fixed* product, without assuming a separable barrier. Combining it with Theorem 16.1, any dimension-capped lift of B_p^N using norm-cone factors of dimensions between three and d has ambient LH parameter at least $h(p)\lceil(N-1)/(d-2)\rceil$. Theorem 13.4 gives a dictionary-independent statement at the absolute minimum dimension: any

lift with $N + 1$ cone coordinates has lifting cone isomorphic to $K_{p,N+1}$ and thus LH parameter at least $h(p)$.

This premium cannot be assigned to arbitrary factors merely because the target is a non-Euclidean ball. For rational p , scalar power inequalities have finite Lorentz lifts [87]; applying these to $|x_j|^p \leq u_j$, $\sum_j u_j = 1$, gives an exact Lorentz-product lift of B_p^N . Its individual factors have optimal parameter two. This observation does not settle the best parameter over all dimension-capped dictionaries, since such a lift can use more factors than the minimum count.

18.2 Grouped cones: duality and a parameter ledger

The perspective cones are established constructions [22, 41]. Their exact dual is

$$\mathcal{P}_{p,r}^* = \{(a, b, z) : a, b \geq 0, \quad a^{1/p} b^{1/q} \geq p^{-1/p} q^{-1/q} \|z\|_q\}. \quad (18.6)$$

Indeed, writing $w = vs$ reduces dual membership to $a\|s\|_p^p + b + \langle z, s \rangle \geq 0$ for every s ; the conjugate of $a\|s\|_p^p$ is $\|z\|_q^q/[q(ap)^{q-1}]$ for $a > 0$, with the boundary cases obtained by closure. Thus the map $(a, b, z) \mapsto (b, a, p^{-1/p} q^{-1/q} z)$ identifies the dual with $\mathcal{P}_{q,r}$. The section $u = v = t$ also gives

$$\nu_{\text{LH}}(\mathcal{P}_{p,r}) \geq \nu_{\text{LH}}(K_{p,r+1}). \quad (18.7)$$

The same comparison holds for products, by simultaneous restriction. A dimension- m proper cone admits an m -LHSC barrier, for example its canonical barrier [54]. Therefore

$$kh(p) \leq \nu_{\text{LH}}\left(\prod_i \mathcal{P}_{p,r_i}\right) \leq \sum_i (r_i + 2). \quad (18.8)$$

These are bounds, not claimed exact values away from $p = 2$.

Here is an evaluable alternative with a larger parameter. Put $\alpha = 1/p$, $\beta = 1/q$. Chares's scalar power barrier is

$$\phi_\alpha(a, v, z) = -\log(a^{2\alpha} v^{2\beta} - z^2) - \beta \log a - \alpha \log v. \quad (18.9)$$

It has parameter three [22, Theorem 3.1.1]. Since $\mathcal{P}_{p,r}$ is represented by $\sum_i a_i = u$, $(a_i, v, w_i) \in \mathcal{P}_{p,1}$, define

$$\Psi_{p,r}(u, v, w) = \min_{\sum_i a_i = u} \sum_i \phi_\alpha(a_i, v, w_i). \quad (18.10)$$

For an interior point the fiber has a nonempty interior and lies in $0 < a_i < u$. The objective is strictly convex and diverges at its boundary. The bounded-fiber partial-minimization theorem of Section 9 therefore gives a nondegenerate self-concordant barrier. Scaling the unique minimizer proves logarithmic homogeneity with parameter $3r$. This construction and partial minimization are due to Chares; the formulas below specify the cost of this particular grouping.

At the minimizer, write $\phi_\alpha(a_i, v, w_i) + \lambda = 0$ and $\sum_i a_i = u$. For a direction $(\dot{u}, \dot{v}, \dot{w})$ set

$$A_i = \phi_{aa} > 0, \quad g_i = \phi_{av}\dot{v} + \phi_{aw}\dot{w}_i, \quad S = \sum_i A_i^{-1}.$$

Differentiation gives

$$\dot{\lambda} = -\frac{\dot{u} + \sum_i g_i/A_i}{S}, \quad \dot{a}_i = -\frac{g_i + \dot{\lambda}}{A_i}, \quad \Psi_u = -\lambda, \quad \Psi_v = \sum_i \phi_v, \quad \Psi_{w_i} = \phi_w. \quad (18.11)$$

Once the scalar multiplier and its scalar inverse evaluations are known, these identities evaluate a Hessian-vector product in $O(r)$ scalar operations. Equivalently the Hessian is an arrowhead matrix

plus a rank-one correction. Root-solving precision and its cost remain separate; this is not an exact finite-arithmetic root algorithm. Sparse-plus-low-rank Hessians for related nonsymmetric cones are established prior work [23].

For an integer cap $d \geq 3$, set $k_g = \lceil N/(d-2) \rceil$ and $k_t = \lceil (N-1)/(d-2) \rceil$. The preceding results give

Representation of B_p^N	Ambient LH optimum bounds	Scalar-power construction
Coordinate power cones	$[Nh(p), 3N]$	$3N$
Grouped perspectives	$[k_g h(p), N + 2k_g]$	$3N$
Norm-cone tree	$[k_t h(p), N - 1 + 2k_t]$	$3(N - 1 + k_t)$

For the last entry restrict a grouped barrier to $u = v = t$ in every norm-cone block; the tree contains $N - 1 + k_t$ norm coordinates in total. The existential upper endpoints use dimension-parameter barriers, whereas the last column uses (18.9). Their oracle and parameter guarantees cannot be combined as if they belonged to one barrier. At $p = 2$ the exact ambient optima improve to $2N, 2k_g, 2k_t$. The generalized-power theorem of Roy and Xiao [77] concerns Euclidean vector outputs, and supplies no constant-parameter barrier for the ℓ_p vector output in (16.5).

18.3 Necessary scales for characteristic barriers

For a proper cone $C \subset \mathbb{R}^m$, define its characteristic integral $Z_C(x) = \int_{C^*} \exp(-\langle x, y \rangle) dy$. Its logarithm is m -logarithmically homogeneous. The universal-barrier result supplies parameter m without scaling [60]; smaller scalar multiples require a separate self-concordance proof.

Theorem 18.3 (Boundary asymptotics of the characteristic family). *Let $p > 2$ and $r \geq 2$. If $\kappa \log Z_C$ is self-concordant, its LH parameter satisfies*

$$C = \mathcal{P}_{p,r} : \quad \nu \geq \frac{r+2}{3/2 + (r-1)/p} > 2, \quad (18.12)$$

$$C = K_{p,r+1} : \quad \nu \geq \frac{r+1}{1 + (r-1)/p} > 2. \quad (18.13)$$

Both right sides tend to p as $r \rightarrow \infty$. For $1 < p < 2$, the same bounds with $q = p/(p-1) > 2$ apply to barriers obtained by taking the Legendre dual of a scaled characteristic barrier of the dual cone.

Proof. Write $\alpha = 1/p$, $\beta = 1/q$. Parameterize the interior of the dual in (18.6) by

$$a = \alpha s, \quad z = -sy, \quad b = s(\beta \|y\|_q^q + \rho), \quad s, \rho > 0, \quad y \in \mathbb{R}^r.$$

The absolute Jacobian is αs^{r+1} . At $x_\delta = (1 + \delta, 1, e_1)$, integration first in ρ and then in s gives

$$Z_C(x_\delta) = \alpha \Gamma(r+1) \int_{\mathbb{R}^r} [\alpha \delta + g(y)]^{-(r+1)} dy, \quad g(y) = \alpha + \beta \|y\|_q^q - y_1. \quad (18.14)$$

Young's inequality shows that g vanishes only at e_1 . Near this point, with $y_1 = 1 + \eta$,

$$g(y) = \frac{q-1}{2} \eta^2 + \beta \|y_\perp\|_q^q + O(|\eta|^3).$$

On a sufficiently small fixed neighborhood it is bounded above and below by positive multiples of $\eta^2 + \|y_\perp\|_q^q$. On compact sets outside that neighborhood it has a positive minimum, and for large y it is bounded below by a positive multiple of $\|y\|_q^q$. For derivative orders $j = 0, 1, 2, 3$, differentiation

under the integral replaces the exponent $r + 1$ by $r + 1 + j$ and multiplies by $(-\alpha)^j(r + 1)_j$. The outside integrals are bounded uniformly as $\delta \downarrow 0$, since $q(r + 1 + j) > r$.

Inside the neighborhood substitute $\eta = \delta^{1/2}t$, $y_\perp = \delta^{1/q}z$. The resulting integrands are dominated by a constant multiple of $(1 + t^2 + \|z\|_q^q)^{-(r+1+j)}$, which is integrable: the anisotropic volume exponent is $a = 1/2 + (r - 1)/q < r + 1$. Dominated convergence gives, with $B = r + 1 - a = 3/2 + (r - 1)/p$,

$$\frac{d^j}{d\delta^j} Z_C(x_\delta) \sim (-1)^j (B)_j C \delta^{-B-j}, \quad 0 \leq j \leq 3, \quad (18.15)$$

for a constant $C > 0$. The coefficient ratios follow either by the beta integral in anisotropic radial coordinates or by differentiating that homogeneous model integral. Thus $f = \kappa \log Z_C(x_\delta)$ obeys $f'' \sim \kappa B \delta^{-2}$ and $f''' \sim -2\kappa B \delta^{-3}$. Univariate self-concordance forces $\kappa B \geq 1$. Since $\nu = (r + 2)\kappa$, this proves the first bound.

For $K_{p,r+1}$ the dual parameterization (s, sy) , $y \in B_q^r$, has Jacobian s^r , so at $\hat{x}_\delta = (1 + \delta, e_1)$,

$$Z_C(\hat{x}_\delta) = \Gamma(r + 1) \int_{B_q^r} (1 + \delta + y_1)^{-(r+1)} dy.$$

Writing $a = 1 + y_1$, the transverse section volume is $\text{vol}(B_q^{r-1})[1 - |1 - a|^q]^{(r-1)/q}$, which equals a positive constant times $a^{(r-1)/q}(1 + O(a))$ near zero. The substitution $a = \delta t$, with the same differentiated dominated-convergence argument, gives (18.15) with $B = r - (r - 1)/q = 1 + (r - 1)/p$. This proves the second bound. The strict inequalities are equivalent to $(r - 1)(1 - 2/p) > 0$. The final statement follows from (18.6), $K_{p,r+1}^* = K_{q,r+1}$, and preservation of parameter by Legendre duality. Characteristic integrals themselves need not be preserved by that duality, which is why the family is named explicitly. $\square$

The theorem constrains one barrier family, not all LH barriers. In particular its bounds do not replace the lower endpoints in the table. For $p > 2$, the perspective bound is strictly smaller than the norm bound at the same r , since their cross-multiplied difference has the sign of $(r - 1)(1/p - 1/2)$. This compares necessary scales, not actual optima.

18.4 The three-dimensional gap and failed explicit formulas

For $K_{p,3}$, the general bounds are

$$h(p) \leq \nu_{\text{LH}}(K_{p,3}) \leq 3. \quad (18.16)$$

The value is two at $p = 2$ and three at the polyhedral endpoints $p = 1$ and $p = \infty$ [53]. The optimal value in the other cases is not established here. Recent barrier-design work still treats these optimal parameters as unknown [75]. For $p > 2$, Theorem 18.3 gives the stronger family-specific necessity $\nu \geq 3p/(p + 1)$. This is the scale conjectured sufficient in Chares's numerical experiments, not a proof of sufficiency.

Proposition 18.4 (Strict separation from the cross-ratio bound). *For $p > 2$, $3p/(p + 1) > h(p)$. Consequently no scaled characteristic barrier on $K_{p,3}$ attains Hildebrand's bound. For $p < 2$ the analogous statement concerns the dual-characteristic family and the bound $3p/(2p - 1)$.*

Proof. Set $s = p - 2 > 0$ and $A = \gamma^{1/p}$. Equation (18.1) becomes $A^s(sA + s + 1) = 1$, whereas $h(p) = 2 + sA/(s + 1)$ and $3p/(p + 1) = 2 + s/(s + 3)$. The left side is increasing in A . At $A_0 = (s + 1)/(s + 3)$ its logarithm is $g(s) = \log(2s + 3) + (s + 1) \log((s + 1)/(s + 3))$. Here $g(0) = 0$ and, putting $z = 2/(s + 3)$,

$$g'(s) = \log(1 - z) + z + \frac{2z}{4 - 3z} =: H(z), \quad H'(z) = \frac{8 - 24z + 24z^2 - 9z^3}{(1 - z)(4 - 3z)^2} > 0 \quad (0 < z < 2/3).$$

The numerator decreases to zero at $2/3$: its derivative is $-3(8 - 16z + 9z^2) < 0$. Since $H(0) = 0$, $g(s) > 0$ and $A < A_0$. The desired inequality follows. Duality gives the last assertion. $\square$

Several explicit ansatzes cannot close this gap. First, a hypothetical 2-LHSC barrier F on a proper cone must be a Lorentz barrier up to a linear map and an additive constant. To see this, take h tangent to a level set of F at x , put $a = F''[h, h]$, $b = F'''[h, h, h]$, and use homogeneity to obtain

$$F''[h + tx, h + tx] = a + 2t^2, \quad F'''[h + tx, h + tx, h + tx] = b - 6at - 4t^3.$$

Self-concordance at $t = \pm\sqrt{a/2}$ forces $b = 0$. For $Q = e^{-F}$ this implies $D^3Q = 0$ on level-tangent triples; two-homogeneity gives $D^3Q[x, \cdot, \cdot] = 0$. Hence Q is quadratic. Its Hessian is positive on x and negative definite on the level-tangent space, so its signature is $(1, m-1)$. The barrier boundary condition identifies the cone with a component of $Q \geq 0$. This argument proves nonattainment at two for non-Lorentz cones; it alone would not prove a strict gap for the infimum.

For a more concrete obstruction, consider, on $K_{p,3}$,

$$F = -\alpha_0 \log A - \beta_0 \log(A^{2/p} - y^2), \quad A = t^p - |x|^p.$$

If this is an LHSC barrier, an affine line reaching a generic boundary point where $A > 0$ forces $\beta_0 \geq 1$ by univariate self-concordance. Along $(t, x, y) = (1, 1 - \delta, 0)$ the same argument forces $\alpha_0 + 2\beta_0/p \geq 1$. Its LH degree therefore satisfies $\nu = p\alpha_0 + 2\beta_0 \geq p > h(p)$ for $p > 2$. For $2 < p \leq 3$ the nonzero coefficient of $|x|^p$ at $y = 0$ also precludes C^3 regularity on the interior axis. The still simpler formula

$$-a \log(t^p - |x|^p - |y|^p) - b \log t, \quad a > 0,$$

is not C^2 on that axis for $1 < p < 2$ and has zero transverse Hessian there for $p > 2$.

Even a regular bounded correction cannot repair the latter formula. On $t = 1$, let $L(x, y) = -a \log(1 - |x|^p - |y|^p)$. For $p \leq 3$, $p \neq 2$, adding a C^3 function cannot cancel its interior-axis loss of C^2 or C^3 regularity. For $p > 3$, put $x_\delta = (1 - \delta)^{1/p}$ and $y_\delta = \delta^{1/(p-2)}$. These points are interior for small δ , and direct differentiation gives

$$L_{yy}(x_\delta, y_\delta) = ap(p-1) + o(1), \quad L_{yyy}(x_\delta, y_\delta) = ap(p-1)(p-2)\delta^{-1/(p-2)}(1 + o(1)).$$

If a correction has bounded yy and yyy derivatives along this sequence, the corrected second derivative stays bounded while the third diverges. Either convexity fails or self-concordance does. An admissible correction must therefore have singular transverse behavior near the flat boundary; no conclusion is drawn about unrestricted corrections.

For completeness the premium has explicit endpoint expansions. Write $d_0 = |1 - 2/p|$ and $w = W(e^{-1})$, where W is Lambert's function. Then

$$h(p) = 2 + 2wd_0 + 2w^2d_0^2 + O(d_0^3), \quad p \rightarrow 2, \quad (18.17)$$

$$h(p) = 3 - \frac{\log(2p) + 1}{p} + \frac{\log^2(2p) - \log(2p) + 1}{2p^2} + O\left(\frac{\log^3 p}{p^3}\right), \quad p \rightarrow \infty. \quad (18.18)$$

In particular $h(p) = 2 + w|p - 2| + O((p - 2)^2)$. To verify the expansions set $a = \gamma^c$ and $s = 1/c - 2$. The root equation is $a^s(1 + s(a + 1)) = 1$. After division of its logarithm by s , the analytic implicit-function theorem at $s = 0$ gives $a = w + \frac{1}{2}w(1 + w)s + O(s^2)$, proving (18.17). At $c \downarrow 0$, let $\ell = \log(c/2)$. Substitution in the root equation gives

$$\gamma = \frac{c}{2} \left[1 + \frac{3}{2}c(\ell + 1) + O(c^2\ell^2) \right], \quad h(p) = 3 + c(\ell - 1) + \frac{1}{2}c^2(\ell^2 + \ell + 1) + O(c^3|\ell|^3),$$

which proves (18.18). For $p = 1 + \varepsilon$, Hölder duality yields, with $L = \log(2/\varepsilon)$, $h(p) = 3 - \varepsilon(L + 1) + \frac{1}{2}\varepsilon^2(L^2 + L + 1) + O(\varepsilon^3 L^3)$.

The literature supplies the scalar cross-ratio bound, the perspective cones, the scalar power barrier, and the partial-minimization machinery. The contributions developed here are the product certificate with its fixed-dictionary consequences, and the necessary characteristic-family scales and explicit failure mechanisms. To the best of our knowledge, the dimension-dependent bounds (18.12)–(18.13) have not been stated for these families. Neither these bounds nor the parameter ledger solves the general optimal-barrier problem.

19 Exponential factors and entropy aggregation

A change in the ambient cone can improve a barrier parameter even when coupling a barrier on a fixed product cannot. The following results make this distinction exact for exponential and relative-entropy cones. Their barrier construction and lower-bound method are specializations of Fawzi and Saunderson [35]; we give the certificates and closure details needed for formulation comparisons. In particular, we do not claim a new general optimal-barrier theorem.

19.1 A reusable recession certificate

For a closed full-dimensional convex set C , let $\mathbf{n}(C)$ be the supremum of $\sum_j a_j/b_j$ over interior points x , recession directions p_j , and positive numbers a_j, b_j satisfying

$$x - \sum_j a_j p_j \in C, \quad x - b_j p_j \notin \text{int } C. \quad (19.1)$$

Nesterov’s recession theorem, in the form [35, Theorem 3.9], gives $\nu \geq \mathbf{n}(C)$ for every self-concordant barrier on C , including nonhomogeneous barriers. Certificates in separate factors combine by inserting zeros in the other coordinates. Both conditions in (19.1) survive, and consequently

$$\mathbf{n}(C_1 \times \cdots \times C_k) \geq \sum_j \mathbf{n}(C_j). \quad (19.2)$$

This is a lower-bound assertion about arbitrary coupled barriers; the usual sum rule alone supplies only an upper bound.

Lemma 19.1 (Orthant-input hypographs). *Let $n, B \geq 1$ be integers and $h : \mathbb{R}_{++}^n \rightarrow \mathbb{R}^B$ be componentwise concave and homogeneous of degree one. For*

$$H_h = \overline{\{(x, z) : x > 0, z \leq h(x)\}},$$

every self-concordant barrier has parameter at least $n + B$.

Proof. This is the orthant-output specialization of Fawzi and Saunderson [35, Theorem 3.10]. Here is its certificate. For $0 < \epsilon < 1$, set $v_i = \epsilon \mathbf{1} + (1 - \epsilon)e_i$ and $x_0 = \sum_i v_i = b \mathbf{1}$, where $b = 1 + (n - 1)\epsilon$. Concavity and homogeneity imply $d = h(x_0) - \sum_i h(v_i) \geq 0$ componentwise. Choose $T' > \max_j d_j$ and $T > T'$. At $y_0 = (x_0, h(x_0) - T \mathbf{1})$ use recession directions $(v_i, h(v_i))$ with coefficients 1 and thresholds b , and $(0, -Te_j)$ with coefficients $1 - T'/T$ and thresholds 1. Their joint subtraction leaves $(0, d - T' \mathbf{1}) \in H_h$. Each input threshold makes an input coordinate zero; each output threshold makes an output inequality tight. Hence each threshold point is noninterior. The certificate has value $n/b + B(1 - T'/T)$. First let $T \rightarrow \infty$, then $\epsilon \downarrow 0$. $\square$

Theorem 19.2 (Exact exponential-product parameter). *Let*

$$K_{\text{exp}} = \overline{\{(u, v, w) : v > 0, w \geq ve^{u/v}\}}.$$

The optimal parameter of an arbitrary self-concordant barrier on a product of R primal or dual exponential cones and P rays is $3R + P$.

Proof. Write the primal cone as the hypograph of $h(v, w) = v \log(w/v)$. Lemma 19.1 supplies certificates tending to three. The compatible barrier

$$-\log(v \log(w/v) - u) - \log v - \log w$$

has parameter three [35]. A ray has a certificate of value one and barrier $-\log t$. Equation (19.2) and the sum of these barriers give the product assertion. The map $(a, b, c) \mapsto (a - b, -a, c)$ is a linear isomorphism from K_{exp}^* onto K_{exp} : its interior formula is $a < 0, c > -a \exp(b/a - 1)$, and its closure face $a = 0, b, c \geq 0$ maps to the primal face $v = 0, u \leq 0, w \geq 0$. $\square$

The same argument gives $\sum_j (n_j + 1)$ for a product of scalar hypographs of homogeneous concave $h_j : \mathbb{R}_{++}^{n_j} \rightarrow \mathbb{R}$ whenever the compatible barriers $-\log(h_j(x) - z) - \sum_i \log x_i$ exist. This includes the usual one-sided geometric-mean hypographs. To check this example directly, let $g(x) = \prod_i x_i^{\alpha_i}$, $\alpha_i > 0, \sum_i \alpha_i = 1$. For a direction put $u_i = h_i/x_i$ and $\bar{u} = \sum_i \alpha_i u_i$. Then $-D^2 g = g \sum_i \alpha_i (u_i - \bar{u})^2$ and $D^3 g = g \sum_i \alpha_i (\bar{u} + 2u_i)(u_i - \bar{u})^2$. Hence $|D^3 g| \leq 3\|u\|_2(-D^2 g)$, the compatibility inequality with the orthant barrier. The output recession direction is essential: this reasoning gives no exact parameter for an absolute power cone. For matrix relative entropy, the published optimal parameter $2n + 1$ likewise tensorizes; restricting the inputs to diagonal matrices preserves the recession certificates and yields $\sum_j (2n_j + 1)$ for a product of these matrix epigraphs [35].

19.2 Positive output maps and their boundary fibers

Put $\phi_i(x_i, y_i) = x_i \log(x_i/y_i)$ for positive inputs. Let N, B be positive integers and $A \in \mathbb{R}_+^{B \times N}$ and retain all N input pairs and all B independent output coordinates in

$$C_A = \overline{\{(x, y, z) : x, y > 0, z \geq A\phi(x, y)\}}. \quad (19.3)$$

Zero or repeated rows and zero columns are permitted.

Theorem 19.3 (Exact intrinsic aggregation parameter). *The optimal barrier parameter of C_A is $2N + B$, attained by*

$$F_A = -\sum_{b=1}^B \log s_b - \sum_{i=1}^N (\log x_i + \log y_i), \quad s = z - A\phi(x, y) > 0. \quad (19.4)$$

If $J = \{i : A_{:,i} \neq 0\}$ and $n_+ = |J|$, an exact closed affine lift uses n_+ exponential cones, $2(N - n_+)$ input rays, and B output rays. The optimal parameter of this ambient product is $2N + n_+ + B$.

Proof. We verify the compatibility underlying the classical barrier construction. For one pair and a direction (a, b) write $u = a/x, v = b/y$. Direct differentiation gives

$$D^2 \phi = x(u - v)^2, \quad D^3 \phi = -x(u - v)^2(u + 2v).$$

Therefore $|D^3 \phi| \leq 3\sqrt{u^2 + v^2} D^2 \phi$. The common orthant barrier has second derivative $\sum_i (u_i^2 + v_i^2)$, so multiplication by a nonnegative row of A preserves the compatibility inequality. Nesterov's

compatible hypograph theorem [35, Theorem 3.3], applied to the diagonal output matrix, proves self-concordance of (19.4). Its homogeneity degree is $2N + B$; its positive-definite Hessian is also explicit below. Lemma 19.1 with $h(x, y) = -A\phi(x, y)$ proves the matching lower bound.

For the exact lift use $q_i \geq \phi_i(x_i, y_i)$ for $i \in J$, $(x_i, y_i) \geq 0$ for $i \notin J$, and

$$z = A_{\cdot J}q + r, \quad r \geq 0. \quad (19.5)$$

At the boundary, extend each active scalar entropy by $\bar{\phi}(0, y) = 0$ for $y \geq 0$, and $\bar{\phi}(x, 0) = +\infty$ for $x > 0$. The elementary bound $\phi(x, y) \geq -y/e$ shows that these terms are uniformly bounded below along convergent input sequences. If a point belongs to (19.3), choose an interior approximating sequence. Every active i has some $A_{bi} > 0$; boundedness of z_b and the lower bounds on all other terms give an upper bound on ϕ_i . Thus $x_i > 0, y_i = 0$ is impossible for active i . Lower semicontinuity then gives $z \geq A_{\cdot J}\bar{\phi}_J$. Conversely, every such inequality is approximated by positive inputs: replace each zero coordinate by ϵ ; the active entropies converge to their finite extended values, and adjust the output by their vanishing change. Choosing $q = \bar{\phi}_J$ and $r = z - A_{\cdot J}q$ proves exact finite fiber existence in (19.5). The final parameter follows from Theorem 19.2. $\square$

The zero-column correction matters for exact sets. With $N = B = 1$ and $A = 0$, $C_A = \mathbb{R}_+^3$ includes $(x, y, z) = (1, 0, 0)$; a finite exponential constraint $q \geq \phi(1, 0)$ cannot represent this point. Keeping an unused exponential cone gives only a nonclosed image with the right closure. Replacing it by two rays restores exactness.

For a partition of N terms into B nonempty blocks, one may instead sum the local epigraph coordinates within each block, without separate output rays. The exact ambient parameter is then $3N$, while the intrinsic aggregate optimum is $2N + B$. Each nonnegative block output residual can be absorbed into one local epigraph coordinate, proving surjectivity even on the boundary. The parameter difference is $N - B$. For a general positive map the lift (19.5) saves n_+ on passage to the intrinsic cone, and the ratio of the usual parameter-dependent short-step factors is

$$\sqrt{\frac{2N + n_+ + B}{2N + B}} < \sqrt{3/2}.$$

This compares parameter-based upper bounds, not actual iteration counts.

19.3 A sparse intrinsic Hessian

Let $p = (x_1, y_1, \dots, x_N, y_N)$, let J_A denote the Jacobian of $A\phi$, and put $R = \text{Diag}(s_b^{-1})$. Differentiation of (19.4) gives

$$\nabla^2 F_A = \begin{pmatrix} I & -J_A^T \\ 0 & I \end{pmatrix} \begin{pmatrix} G & 0 \\ 0 & R^2 \end{pmatrix} \begin{pmatrix} I & 0 \\ -J_A & I \end{pmatrix}, \quad (19.6)$$

where G has independent positive-definite 2×2 blocks

$$G_i = \text{Diag}(x_i^{-2}, y_i^{-2}) + \left(\sum_b A_{bi}/s_b \right) \begin{pmatrix} x_i^{-1} & -y_i^{-1} \\ -y_i^{-1} & x_i/y_i^2 \end{pmatrix}.$$

The second summand is positive semidefinite. Assembling G requires the product $A^T(s_b^{-1})_{b=1}^B$. Once G is assembled, each solve with (19.6) uses triangular substitution, two products by A or A^T , diagonal scaling, and N two-dimensional solves, for $O(N + B + \text{nnz } A)$ arithmetic operations including assembly once the point and slacks are available. This is a cone-Hessian solve; additional affine constraints still require solving their Schur system. Sparse and low-rank Hessian formulas

for ordinary relative entropy are established by Chen and Goulart [23]. Tailoring entropy cones to positive input maps, including singular images, and removing redundant determinant terms are established by He et al. [50]. Here the explicit output map records which coordinates remain independent, and the exact closure analysis identifies when the elementary lift is valid.

Part IV

Approximation and exact compilation

20 Approximation with controlled cone factors

The loss of two curvature directions per proper cone is stable under approximation only with additional control of the factors. We give two versions: a finite sampling theorem requiring no derivatives, and a local smooth theorem in which the mixed derivative of the slack remains exact. The hypotheses concern selected factors in specified Euclidean coordinates. They are stronger than approximation of the projected convex body.

20.1 A finite sampling theorem

Write $r = N - 1 \geq 1$, fix $0 < h < 1$, and put

$$v_0 = e_N, \quad v_j = he_j + \sqrt{1 - h^2}e_N \quad (1 \leq j \leq r), \quad \tau = 1 - \sqrt{1 - h^2}.$$

Suppose proper cones $K_i \subset E_i$, of dimensions m_i , admit factors $A_i(v_j) \in K_i$, $B_i(v_j) \in K_i^*$ such that

$$\left| \sum_i \langle A_i(v_j), B_i(v_\ell) \rangle - (1 - \langle v_j, v_\ell \rangle) \right| \leq \epsilon \quad (0 \leq j, \ell \leq r). \quad (20.1)$$

Let $a_i = A_i(v_0)$, $b_i = B_i(v_0)$, and let $X_i, Y_i : \mathbb{R}^r \rightarrow E_i$ have columns $A_i(v_j) - a_i$, $B_i(v_j) - b_i$, respectively. All matrix norms below are operator norms. Define

$$E_i = \frac{\sqrt{r}(\tau + \epsilon)}{\|b_i\|} \|Y_i\| + \frac{\sqrt{r}(\tau + \epsilon)}{\|a_i\|} \|X_i\| + \frac{\epsilon \|X_i\| \|Y_i\|}{\|a_i\| \|b_i\|} \quad (20.2)$$

when $a_i, b_i \neq 0$; otherwise define $E_i = \|X_i\| \|Y_i\|$.

Theorem 20.1 (Conditioned finite samples). *If $4r\epsilon + \sum_i E_i < h^2$, then*

$$N - 1 \leq \sum_i (m_i - 2)_+.$$

In particular, if all factors of dimension at least three have dimension at most an integer $d \geq 3$, at least $\lceil (N - 1)/(d - 2) \rceil$ such factors are required.

Proof. Fix a block with $a, b \neq 0$. Nonnegativity of every summand gives $0 \leq \langle a, b \rangle \leq \epsilon$ and

$$\|X^*b\|, \|Y^*a\| \leq \sqrt{r}(\tau + \epsilon).$$

Indeed, each coordinate is the difference of two numbers in $[0, \tau + \epsilon]$. Set $X_0 = P_{b^\perp}X$ and $Y_0 = P_{a^\perp}Y$. Then

$$\|X^*Y - X_0^*Y_0\| \leq \frac{\|X^*b\|}{\|b\|} \|Y\| + \|X\| \frac{\|Y^*a\|}{\|a\|}. \quad (20.3)$$

For $m \geq 2$, the product $P_{b^\perp}P_{a^\perp}$ has singular values 1 with multiplicity $m - 2$, together with $\gamma = \langle a, b \rangle / (\|a\| \|b\|)$ and 0. This follows by splitting off $(\text{span}\{a, b\})^\perp$ and computing the remaining plane; the parallel case gives the same list with $\gamma = 1$. Deleting the singular direction of size γ therefore approximates $X_0^*Y_0$ by a rank at most $m - 2$ matrix with error at most $\gamma \|X\| \|Y\|$.

Equation (20.3) proves the bound E_i . For $m = 1$, $X_0 = Y_0 = 0$, so its first two terms suffice. A zero base factor uses the zero approximant and the specified trivial error.

The mixed difference matrix of the factored kernel is $\sum_i X_i^* Y_i$. For the exact sphere slack it is $D_* = -h^2 I_r - \tau^2 \mathbf{1}\mathbf{1}^*$, whose smallest singular value is at least h^2 . Four errors per entry in (20.1) give operator error at most $4r\epsilon$. The sum of the block approximants has rank at most $\sum_i (m_i - 2)_+$ and lies within $4r\epsilon + \sum_i E_i$ of D_* . A singular matrix cannot lie at operator distance less than h^2 from D_* : apply it to a unit vector in its kernel. This proves the assertion. $\square$

A transparent sufficient condition is available when there are k blocks, $\|a_i\|, \|b_i\| \geq \rho > 0$, and $\|X_i\|, \|Y_i\| \leq Lh$. If $\epsilon \leq h^2$, then $\tau \leq h^2$, and it suffices that

$$\frac{4r\epsilon}{h^2} + \frac{4k\sqrt{r}Lh}{\rho} + \frac{k\epsilon L^2}{\rho^2} < 1. \quad (20.4)$$

For fixed r, k, L, ρ , the choice h of order $\epsilon^{1/3}$ makes the left side tend to zero, provided the increment bounds hold on these shrinking samples.

If $(1 - \eta)\mathbb{B}_2^N \subset C \subset (1 + \eta)\mathbb{B}_2^N$, $0 < \eta < 1$, choose radial boundary points $x(v) = \rho_C(v)v$ and $y(w) = \rho_{C^\circ}(w)w$. Their radial lengths lie in $[1 - \eta, 1 + \eta]$ and $[(1 + \eta)^{-1}, (1 - \eta)^{-1}]$. Consequently

$$|1 - \langle x(v), y(w) \rangle - (1 - \langle v, w \rangle)| \leq \frac{2\eta}{1 - \eta}.$$

The reduction and certificate argument of Lemma 2.1 provides a factorization of this slack from any exact affine lift of C , including at radial points that are not extreme. Thus metric accuracy supplies (20.1); it does not supply (20.2) or (20.4).

20.2 Smooth near-complementarity

For $r_0 > 0$, $\delta, H \geq 0$, define

$$\psi_{r_0}(\delta, H) = \inf_{0 < s \leq r_0} \left(\frac{\delta}{s} + \frac{Hs}{2} \right). \quad (20.5)$$

This finite-radius expression includes $H = 0$, when its value is δ/r_0 . If $H > 0$ and $\sqrt{2\delta/H} \leq r_0$, it equals $\sqrt{2H\delta}$; for $\delta = 0$ the value is zero by a limit.

Lemma 20.2 (Quantitative near-complementarity). *Let $A : U \rightarrow K$, $B : U \rightarrow K^*$ be C^2 , where $K \subset E$ is a proper cone of dimension m and U contains the closed Euclidean ball of radius r_0 about zero. Put $a = A(0)$, $b = B(0)$, $X = DA(0)$, $Y = DB(0)$, and $\delta = \langle a, b \rangle$. Suppose $a, b \neq 0$ and let*

$$H_A = \sup_{\|u\|=1, |t| \leq r_0} |\langle D^2 A(tu)[u, u], b \rangle|, \quad H_B = \sup_{\|u\|=1, |t| \leq r_0} |\langle a, D^2 B(tu)[u, u] \rangle|.$$

Write $U_1 = \|X\|$, $V_1 = \|Y\|$ and

$$\alpha = \frac{\psi_{r_0}(\delta, H_A)}{\|b\|}, \quad \beta = \frac{\psi_{r_0}(\delta, H_B)}{\|a\|}, \quad \gamma = \frac{\delta}{\|a\|\|b\|}.$$

For $m \geq 2$, X^*Y is within

$$e = U_1\beta + V_1\alpha + U_1V_1\gamma \quad (20.6)$$

of a matrix of rank at most $m - 2$. For a ray, the zero approximant has error at most $e = \alpha\beta$. If either base factor is zero, $X^*Y = 0$.

Proof. For a unit u , the function $f(t) = \langle A(tu), b \rangle$ is nonnegative, $f(0) = \delta$, and $|f''| \leq H_A$. Taylor's inequality at $-s$ or s , with sign opposite to $f'(0)$, gives $|f'(0)| \leq \delta/s + H_A s/2$. Hence $\|X^*b\| \leq \psi_{r_0}(\delta, H_A)$, and similarly for Y^*a . Apply (20.3) and its singular-value truncation to obtain (20.6). For $m = 1$ the normal components are the entire derivatives, giving $\alpha\beta$. If $a = 0$, every two-sided derivative of A lies in $K \cap (-K) = \{0\}$, so $X = 0$; the dual case is identical. $\square$

The bound (20.6) also implies the more conservative bound $U_1\beta + V_1\alpha + \alpha\beta\gamma + U_1V_1\gamma$ obtained by expanding both normal components separately. Keeping one projected factor in (20.3) removes that extra term. Both bounds are invariant under constant reciprocal scalar rescaling of A, B .

Theorem 20.3 (Conditioned smooth approximation). *Let $\mathbb{B}_2^N \subset C \subset (1 + \epsilon)\mathbb{B}_2^N$ and suppose a local C^2 factorization on a sphere chart $\sigma : U \rightarrow \mathbb{S}^{N-1}$ satisfies*

$$\sum_i \langle A_i(u), B_i(v) \rangle = h_C(\sigma(v)) - \langle \sigma(u), \sigma(v) \rangle, \quad D\sigma(0)^* D\sigma(0) = I_{N-1}. \quad (20.7)$$

Use Lemma 20.2 in each factor span and denote its dimension by s_i and its error by e_i . If $\sum_i e_i < 1$, then

$$N - 1 \leq \sum_i (s_i - 2)_+.$$

For a definable exact lift of C , such local C^2 selections exist on a nonempty open sphere patch after minimal-face reduction. The numerical error condition is an additional hypothesis.

Proof. At the chart center, nonnegativity gives $\delta_i \geq 0$ and $\sum_i \delta_i = h_C(\sigma(0)) - 1 \leq \epsilon$. Mixed differentiation of (20.7) gives the exact identity $-\sum_i DA_i(0)^* DB_i(0) = I_{N-1}$, because the support term depends only on v . Lemma 20.2 approximates this identity by a matrix of rank at most $\sum_i (s_i - 2)_+$ with error at most $\sum_i e_i$. Error less than one makes that matrix invertible.

For the existence statement, write the reduced lift as $C = \pi\{z \in K : Mz = b_0\}$ with relative Slater feasibility, absorbing an affine output offset into the dual objective if necessary. Choose the minimum-norm feasible lift of $\sigma(u)$ and the minimum-norm optimal dual multiplier $\lambda(v)$ for maximizing $\langle \sigma(v), \pi z \rangle$. Both choices exist in closed nonempty fibers; Slater dual attainment supplies the second fiber. With $B(v) = M^* \lambda(v) - \pi^* \sigma(v)$, weak duality and attainment give (20.7). Definable choice by these unique minimizers and C^2 cell decomposition give a common open patch. There is no assertion of uniform patch size or derivative bounds across families of lifts. $\square$

For example, suppose there are at most k nonzero selected blocks, $\|a_i\|, \|b_i\| \geq \mu > 0$, $\|DA_i\|, \|DB_i\| \leq L$, and $H_{A,i}, H_{B,i} \leq H$, where $H > 0$ and $\sqrt{2\epsilon/H} \leq r_0$. The sum of the errors for blocks of dimension at least two is at most

$$\mathcal{E}_k(\epsilon) = \frac{2L\sqrt{2Hk\epsilon}}{\mu} + \frac{L^2\epsilon}{\mu^2}. \quad (20.8)$$

This follows from $\sum_i \sqrt{\delta_i} \leq \sqrt{k\epsilon}$. Rays add at most $2H\epsilon/\mu^2$. Zero selected factors add zero. Thus a uniformly bounded number of blocks, positive factor norm bound, bounded derivatives, and a fixed chart radius give error $O(\sqrt{\epsilon})$. Conversely, an under-capacity family must have $\mathcal{E}_k(\epsilon) + 2H\epsilon/\mu^2 \geq 1$ (omit the last term when no rays occur). If the terms other than the square-root term sum to at most $1/2$, then

$$L\sqrt{H} \geq \frac{\mu}{4\sqrt{2k\epsilon}}.$$

This is a statement about selected factors, not an intrinsic condition number of a Newton system. Global consequences under stronger selection hypotheses appear in Appendix C.

20.3 Why metric accuracy alone is insufficient

For every $\epsilon > 0$, choose a finite δ -net $\mathcal{U} \subset \mathbb{S}^{N-1}$ with $(1 - \delta^2/2)^{-1} \leq 1 + \epsilon$. The polytope

$$P = \{x : \langle u, x \rangle \leq 1 \text{ for all } u \in \mathcal{U}\}$$

contains $\mathbb{B}_2^N$. For $x \neq 0$ choose u within δ of $x/\|x\|$. Then $\langle u, x/\|x\| \rangle \geq 1 - \delta^2/2$, so $\|x\| \leq 1 + \epsilon$ for $x \in P$. One nonnegative slack per inequality gives an exact ray-only lift of P . Therefore no positive Hausdorff or multiplicative error threshold alone preserves the non-ray capacity sum. The number of scalar inequalities is a separate resource. In the lifted polyhedral Lorentz-cone approximation model, Ben-Tal and Nemirovski prove matching bounds of order $N \log(1/\epsilon)$ on that resource for small ϵ [13, Theorem 1.1 and Proposition 3.1]. For a realization as an orthant of that dimension, its optimal ambient barrier parameter equals the number of slacks. This is a property of that realization, not of an arbitrary custom cone for its projection.

A different obstruction occurs at fixed error even when every block is counted. For $d \geq 3$,

$$\mathbb{B}_2^{2(d-1)} \subset \mathbb{B}_2^{d-1} \times \mathbb{B}_2^{d-1} \subset \sqrt{2}\mathbb{B}_2^{2(d-1)}.$$

The outer ratio and Hausdorff distance $\sqrt{2} - 1$ are attained by pairs of unit vectors. Two Q_d factors represent the product, whereas the exact ball requires at least $\lceil (2d - 3)/(d - 2) \rceil = 3$ factors of dimension at most d .

Even bounded factors and arbitrarily small sampled slack error do not replace relative conditioning in Theorem 20.1. For its two-point sample ($r = 1$), fix $\alpha > 0$ and put $\beta = \tau/\alpha$. In $\mathbb{R}_+^2$ take

$$A(v_0) = B(v_0) = (\alpha, \alpha), \quad A(v_1) = (0, \alpha + \beta), \quad B(v_1) = (\alpha + \beta, 0).$$

The kernel values are $2\alpha^2, \tau + \alpha^2, \tau + \alpha^2, 0$, so the error tends to zero. Its mixed difference is nevertheless $-2\tau = -h^2 - \tau^2$, exactly the sphere value, although $(m - 2)_+ = 0$. Apply the diagonal automorphism $D = \text{diag}(\sqrt{\beta/\alpha}, \sqrt{\alpha/\beta})$ to A and D^{-1} to B . All factors become bounded as $\alpha \downarrow 0$ and the pairings stay unchanged. The relative increment hypothesis still fails. This finite example does not by itself construct a global fixed-size approximation of a ball.

Approximate slack factorizations and their geometric approximation properties are classical. Gouveia, Parrilo, and Thomas bound inner and outer polytope approximations using the maximum column and row Euclidean norms of a finite slack-matrix error; they also obtain approximate factorizations from lifted approximations of nested polyhedra [44, Sections 2.7 and 4.2]. The results here add explicit local conditioning hypotheses at curved contacts that preserve the $\dim K_i - 2$ rank bound. They do not assert that a metric approximation admits those selections or that failure of the hypotheses imposes an algorithmic cost.

21 Exact compilers and the limits of representation-specific rank

A formulation can become smaller because several input constraints have one exact replacement, because a source-dependent function has a finite summary, or because auxiliary variables compose simpler functions. These are different operations. We give precise criteria for the first two and show why exponential-feature rank does not obstruct the third. Throughout this section, a *compiler* means a map from the specified input data to an exact representation with the specified output format. A restriction on that format is part of the theorem.

21.1 Intersections of translates

For a proper cone $K \subset \mathbb{R}^d$, write $x \leq_K y$ if $y - x \in K$. The following is the finite-dimensional vector-lattice characterization of simplicial cones, expressed as an exact compilation criterion. The order-theoretic result is classical; see, for example, Németh [68, Introduction] and the references there.

Proposition 21.1 (One-translate compilation). *Let $d \geq 1$. The identity*

$$(a - K) \cap (b - K) = c - K$$

has a solution c for every pair $a, b \in \mathbb{R}^d$ if and only if $K = T\mathbb{R}_+^d$ for an invertible linear map T . In that case, for any finite family,

$$\bigcap_{i=1}^N (a_i - K) = Tm - K, \quad m_j = \min_i (T^{-1}a_i)_j. \quad (21.1)$$

If the output apex must be one of the two input apices, the universal pairwise property holds if and only if $d = 1$.

Proof. The intersection identity says precisely that c is the greatest lower bound $a \wedge b$. Negation supplies least upper bounds, so the induced order is a vector lattice. Here is a finite-dimensional proof of the resulting simpliciality. Distinct extreme rays of K are disjoint in the lattice sense: a positive element below representatives of both rays must lie on both and therefore vanish. The lattice decomposition identity

$$0 \leq z \leq x + y \implies z = (z \wedge x) + (z - x)^+, \quad 0 \leq z \wedge x \leq x, \quad 0 \leq (z - x)^+ \leq y$$

shows by induction that finite sums of disjoint groups of extreme rays are disjoint. Any linear dependence among distinct extreme-ray representatives would equate two nonzero positive sums over disjoint groups, which is impossible. Thus there are at most d extreme rays. A compact base of a closed pointed cone is the convex hull of its extreme points; since K spans $\mathbb{R}^d$, there are exactly d such rays and they form a basis. This gives $K = T\mathbb{R}_+^d$. Coordinatewise minima prove the converse and (21.1).

An input apex can always be retained exactly when every pair is comparable. This is equivalent to $K \cup (-K) = \mathbb{R}^d$. For $d \geq 2$, the unit sphere is connected, whereas its intersections with K and $-K$ would be disjoint nonempty closed sets covering it. Pointedness gives the contradiction. A one-dimensional proper cone has a total order. $\square$

The number of incomparable input apices need not control the size of the output: even an antichain of N apices in a simplicial cone is summarized by d coordinate minima. For explicit classical output and coherent access to an individual transformed entry $(T^{-1}a_i)_j$, quantum minimum finding [28] gives

$$Q = O(d\sqrt{N} \log(d+1)), \quad Q = \Omega(d\sqrt{N}), \quad R = \Theta(dN). \quad (21.2)$$

Here Q and R denote bounded-error quantum and randomized entry-query complexities, respectively, and success means that all d output coordinates are correct. Exact comparison and retrieval of an encoded entry are included in the oracle contract. For the upper bound, run each minimum search with failure probability at most $1/(3d)$. For the quantum lower bound, restrict each column to all ones or to a single hidden zero. The output supplies d independent search-decision answers, to

which the adversary direct-sum theorem applies [2, Theorem 4]. For the randomized lower bound, couple an input with independent equiprobable all-one/single-zero columns to the input with any one column changed. If an algorithm succeeds on the full answer vector with probability at least $2/3$, each column must be distinguished with constant advantage; a uniform hidden zero then forces $\Omega(N)$ expected probes in each column. Summing over columns gives $\Omega(dN)$. The logarithm in the quantum upper bound accounts for simultaneous success; no exact $\Theta(d\sqrt{N})$ upper bound is asserted here. If the supplied oracle instead returns a physical coordinate of a_i and T^{-1} is dense, forming one transformed entry can require d source queries.

For the canonical logarithmic barriers, the scenario product has parameter Nd and the compiled translate has the optimal parameter d . On identical-apex instances, a radial curve whose common transformed slack is $\alpha\mathbf{1}$ has lengths

$$\sqrt{Nd}|\log(\alpha_1/\alpha_0)| \quad \text{and} \quad \sqrt{d}|\log(\alpha_1/\alpha_0)|,$$

respectively. This follows directly from the Hessians of $-N\sum_j \log s_j$ and $-\sum_j \log s_j$. It compares these barriers, not all algorithms for the underlying optimization problem.

A factorwise radial restriction permits the same operation for arbitrary proper cones. If $K = \prod_{j=1}^B K_j$, $e_j \in \text{int } K_j$, and $a_i = (r_{i1}e_1, \dots, r_{iB}e_B)$, then

$$\bigcap_i (a_i - K) = \prod_{j=1}^B (r_{*j}e_j - K_j), \quad r_{*j} = \min_i r_{ij}. \quad (21.3)$$

Indeed, the translates in each factor are nested in their scalar radius. The corresponding query bounds are (21.2) with $d = B$. Given barriers of parameters ν_j on the factors, the compiled product has a barrier of parameter $\sum_j \nu_j$, whereas the repeated scenario barrier has parameter $N\sum_j \nu_j$. No symmetry of K_j is needed.

Proposition 21.2 (Lorentz-translate obstruction). *For every N , there are $a_1, \dots, a_N \in \mathbb{R}^3$ such that every finite identity*

$$\bigcap_{i=1}^N (a_i - Q_3) = \bigcap_{\ell=1}^m (c_\ell - Q_3)$$

satisfies $m \geq N$.

Proof. Choose distinct $v_i \in \mathbb{S}^1$ and $0 < \rho < R$, and set $a_i = (0, \rho v_i)$. On the plane $t = -R$, the intersection is $\bigcap_i B(\rho v_i, R)$. The point $z_i = -(R - \rho)v_i$ satisfies

$$\|z_i - \rho v_i\| = R, \quad \|z_i - \rho v_j\|^2 = (R - \rho)^2 + \rho^2 + 2\rho(R - \rho)\langle v_i, v_j \rangle < R^2 \quad (j \neq i).$$

Hence each original circle contributes an open boundary arc. An alternative translate intersects this plane in a disk. Its radius must be positive because the common intersection contains an open disk. Every point of an original open arc lies on the boundary of at least one alternative disk. Two distinct circles intersect in at most two points, so finitely many alternative circles can cover that arc only if one coincides with the original circle. Distinct original circles therefore require distinct alternative disks. $\square$

This obstruction already occurs in a cone of Jordan rank two. Copying the three-dimensional variable at each constraint and imposing consensus along a path gives a formulation whose graph of variable blocks is a path. Thus a bound on cone rank or on this block graph does not bound the surviving number of translates. The proposition concerns same-space, same-oriented translates only. It gives no lower bound for lifts with auxiliary variables, different cone maps, or custom barriers.

21.2 Reusable summaries of perspective data

Let $h : (0, \infty) \rightarrow \mathbb{R}$ be continuous. Its *dilation rank* is

$$R_h = \dim \text{span}\{r \mapsto h(\lambda r) : \lambda > 0\},$$

possibly infinite. Consider the fixed-source perspective sum

$$\Phi(U, V) = \sum_{i=1}^N w_i c_i U h\left(\frac{d_i}{c_i} \frac{V}{U}\right), \quad U, V, w_i, c_i, d_i > 0. \quad (21.4)$$

A reusable summary must determine this function for every $U, V > 0$; it is not merely an answer at one query point.

Theorem 21.3 (Continuous summary dimension). *If $R_h = R < \infty$, the source dependence in (21.4) is summarized by R scalar linear moments. Conversely, choose any q linearly independent dilates $h(\lambda_1 r), \dots, h(\lambda_q r)$. If a continuous map $E : \mathbb{R}_{++}^q \rightarrow \mathbb{R}^m$ admits a deterministic decoder recovering $\sum_i a_i h(\lambda_i r)$ for every $r > 0$ and every $a \in \mathbb{R}_{++}^q$, then $m \geq q$. Consequently R_h is the exact minimum summary dimension for these unrestricted positive-coefficient families; if $R_h = \infty$, no fixed finite dimension suffices.*

Proof. Choose a basis $h_1, \dots, h_R$ of the dilation space and write $h(\lambda r) = \sum_{\ell} m_{\ell}(\lambda) h_{\ell}(r)$. Then

$$M_{\ell} = \sum_i w_i c_i m_{\ell}(d_i/c_i), \quad \Phi(U, V) = U \sum_{\ell=1}^R M_{\ell} h_{\ell}(V/U).$$

The coefficient functions m_{ℓ} are continuous: choose R evaluation points giving an invertible evaluation matrix and solve for the coefficients using the continuous values $h(\lambda r_j)$.

For the lower bound, equality $E(a) = E(b)$ would give equality of the reconstructed functions, hence $a = b$ by linear independence. Thus E is injective. If $m < q$, append $q - m$ zeros to obtain a continuous injection from the open set $\mathbb{R}_{++}^q$ into $\mathbb{R}^q$. Invariance of domain [48, Theorem 2B.3] says its image is open, whereas the image lies in a proper coordinate subspace. This is impossible. $\square$

Decoder continuity is unnecessary. Encoder continuity and the finite-dimensional real output are essential hypotheses; arbitrary set-theoretic encodings into one infinite-precision real are excluded. For a fixed restricted family of coefficients, the lower bound need not equal R_h .

Proposition 21.4 (Classical dilation classification). *The rank R_h is finite if and only if*

$$h(r) = \sum_{j=1}^J r^{\rho_j} P_j(\log r), \quad (21.5)$$

where the ρ_j are distinct complex numbers and P_j are nonzero polynomials, with conjugate terms paired when h is real-valued. The empty sum represents $h = 0$, with $R_h = 0$. Otherwise $R_h = \sum_j (\deg P_j + 1)$.

Proof. Put $f(y) = h(e^y)$. Its translation span W is finite dimensional exactly when the dilation span is. Translation acts on W by a continuous group $T(t)$ of invertible matrices: continuity follows by choosing an invertible finite evaluation matrix as above. For completeness, choose small $\delta > 0$ so that $M = \int_0^{\delta} T(t) dt$ is invertible. The identity $T(s)M = \int_s^{s+\delta} T(t) dt$ proves differentiability of T . The group law then gives $T'(s) = T(s)A$ with $A = T'(0)$, so $T(s) = e^{sA}$. Evaluation at zero shows

that f is an exponential polynomial; the Jordan form of A gives (21.5). Conversely, translates of each $e^{\rho_j y} P_j(y)$ lie in the span of $y^\ell e^{\rho_j y}$ for $0 \leq \ell \leq \deg P_j$. These functions are linearly independent. Each occurs in the cyclic translation span: on this finite-dimensional span the differential operator acts with one Jordan chain of length $\deg P_j + 1$ for each distinct ρ_j , and a polynomial in that operator isolates each chain. This also gives the stated rank. $\square$

The translation-space classification is due to Anselone–Korevaar [4]; see also Engert’s extension and historical account [30]. We included the elementary continuous one-parameter proof to specify all assumptions. For example, $-\log r$ has rank two, r^α has rank one, and e^r has infinite rank. The latter follows from independence of distinct exponentials $e^{\lambda r}$ on any open interval. The contribution of Theorem 21.3 is the exact continuous-summary interpretation, rather than a new classification of translation spaces. It concerns source evaluation, not arbitrary conic lift size or barrier parameter.

21.3 A two-moment relative-entropy example

For positive definite Hermitian matrices use the homogeneous, uncorrected Umegaki convention $D(X\|Y) = \text{tr } X(\log X - \log Y)$. Let the weights and positive scalar scales be fixed, and define

$$A = \sum_i w_i c_i, \quad B = \sum_i w_i c_i \log(c_i/d_i), \quad D_{\text{eff}} = A e^{-B/A}. \quad (21.6)$$

Then

$$\sum_i w_i D(c_i X \| d_i Y) = A D(X\|Y) + B \text{tr } X = D(AX \| D_{\text{eff}} Y). \quad (21.7)$$

Indeed, scalar multiplication adds $(\log c)I$ to a matrix logarithm; no commutativity of X and Y is required. The identity extends to the positive-semidefinite closure with the usual condition $\text{supp } X \subseteq \text{supp } Y$ for finite relative entropy. If that condition fails, both sides are $+\infty$. If $X = 0$, both sides vanish.

After acquisition of A, B , all value and derivative evaluations use one relative-entropy term and no further access to the scenarios. The scalar restriction, in which $-\log r$ has dilation rank two, shows that two continuous real statistics are necessary for general positive coefficient/scale families. If A is fixed in advance, only B needs to be stored. For $c_i = 1$ and $\sum_i w_i = 1$, $D_{\text{eff}} = \prod_i d_i^{w_i}$.

The convention matters. For $D_+(X\|Y) = D(X\|Y) + \text{tr } Y - \text{tr } X$ and $C = \sum_i w_i d_i$, the exact identity instead is

$$\sum_i w_i D_+(c_i X \| d_i Y) = D_+(AX \| D_{\text{eff}} Y) + (C - D_{\text{eff}}) \text{tr } Y.$$

The compiler preserves a weighted sum in one objective or epigraph; it does not preserve separate scenario inequalities or separate outputs. Nor does its source-free online evaluation apply when the scales change with the optimization variables.

The relative-entropy epigraph barrier of parameter $2n+1$ for $n \times n$ matrices is established by Fawzi and Saunderson [35]. Positive-map composition and removal of redundant input log determinants are treated more generally by He et al. [50]. Equation (21.7) is an elementary scaling identity; its role here is to make an exact source summary explicit. It is not a new barrier construction.

Acquiring the moments is a separate task. For a uniform average of bounded coherently accessible entries, quantum amplitude estimation uses $O(\min\{N, \eta^{-1}\})$ queries for constant-success additive error η , compared with the randomized upper bound $O(\min\{N, \eta^{-2}\})$ [18]. This statement presumes that preparing the uniform index state and computing an entry are included in the access model; arbitrary implicit weights are not free. On Boolean entries, exact acquisition, or error strictly below

$1/(2N)$ in the mean, recovers the Hamming weight and requires $\Omega(N)$ quantum queries [67]. If A is known exactly, an estimate $\hat{B}$ changes the epigraph boundary by precisely $(\hat{B} - B) \text{tr } X$. Since $X \succeq 0$, one-sided bounds on B therefore give globally valid inner and outer epigraphs.

A small barrier parameter alone does not supply source-free evaluation. For $z \in \{0, 1\}^N$, let

$$\phi_z(x_1, x_2) = \sum_{i=1}^N D(x_1 \| e^{-z_i} x_2) = Nx_1 \log(x_1/x_2) + |z|x_1.$$

The epigraph on $x_1, x_2 > 0$ is a linear image of a single scalar relative-entropy cone: use $D(Nx_1 \| Nx_2) = Nx_1 \log(x_1/x_2)$ and shift the epigraph coordinate by $|z|x_1$. Thus

$$F_z(t, x_1, x_2) = -\log(t - \phi_z(x_1, x_2)) - \log x_1 - \log x_2$$

is a three-parameter logarithmically homogeneous self-concordant barrier, up to an additive constant [35]. Under the promise $|z| \in \{0, 1\}$, its value at the common interior point $(2, 1, 1)$ differs by exactly $\log 2$. An oracle returning that barrier value with error less than $\log 2/3$ solves unique-search decision and requires $\Omega(\sqrt{N})$ quantum or $\Omega(N)$ randomized source queries. The lower bound is on initial source access: it does not apply after $|z|$ has been learned and stored.

21.4 Exponential features versus exponential-cone factors

Use the coordinate convention

$$K_{\text{exp}} = \text{cl}\{(a, b, c) : b > 0, c \geq be^{a/b}\}.$$

In particular, $(a, 1, c) \in K_{\text{exp}}$ means $c \geq e^a$. An exact *signed exponential representation* of a function on an open set is an identity $f(z) = \sum_{j=1}^M \beta_j e^{\langle \lambda_j, z \rangle}$ with real nodes λ_j and real coefficients β_j . Its rank is the minimum number of nonzero terms, allowing arbitrary competing nodes.

Proposition 21.5 (Unbounded feature rank with three cones). *For every integer $m \geq 1$, the function*

$$F_m(z) = \left(\frac{1 + e^{z/m}}{2} \right)^m$$

has signed exponential rank $m + 1$ on every nonempty open interval, but its full epigraph has an exact lift using three exponential cones and one scalar linear inequality:

$$(z/m - s, 1, u), (-s, 1, v), (ms, 1, t) \in K_{\text{exp}}, \quad u + v \leq 2. \quad (21.8)$$

All affine coefficients have $O(\log(m + 1))$ -bit rational encodings.

Proof. The first two cones and the inequality admit u, v exactly when $s \geq \log((1 + e^{z/m})/2)$. The last cone then projects exactly to $t \geq F_m(z)$, with equality attainable. On the other hand,

$$F_m(z) = 2^{-m} \sum_{j=0}^m \binom{m}{j} e^{jz/m}.$$

Distinct real exponentials are linearly independent on an open interval: differentiating a finite zero combination at any interior point gives a Vandermonde system. Comparing any competing representation with this one therefore forces all $m + 1$ displayed nodes to remain. $\square$

Here F_m is the moment-generating function of $m^{-1} \sum_{j=1}^m B_j$ for independent Bernoulli(1/2) variables, so even restricting to probability measures supported on $[0, 1]$ does not restore a feature-rank bound in terms of the number of cones. The log-sum-exp and outer-exponential operations used in (21.8) are standard modeling constructions [64, Sections 5.2.5–5.2.6]. The separation comes from comparing this composition with the exact signed-feature output contract.

Theorem 21.6 (Bounds for exact exponential lifts of a product-cosh graph). *Let $r \geq 1$, $K > 0$, and $f_r(z) = \prod_{i=1}^r \cosh z_i$. Its signed exponential rank on every nonempty open subset of $\mathbb{R}^r$ is 2^r . Its epigraph has a lift with $2r + 1$ exponential-cone factors and r scalar inequalities. Conversely, any exact affine lift of its epigraph over $[-K, K]^r$ through $K_{\text{exp}}^R \times \mathbb{R}_+^P$, with arbitrary affine projections and free auxiliary variables, satisfies*

$$R \geq r. \quad (21.9)$$

Consequently the minimum exponential-cone factor count, with scalar inequalities uncharged, lies between r and $2r + 1$. For any such original ambient product, every self-concordant barrier has parameter at least $3R + P \geq 3r + P$, and the ambient optimum is $3R + P$.

Proof. The expansion $f_r(z) = 2^{-r} \sum_{\epsilon \in \{-1, 1\}^r} e^{\langle \epsilon, z \rangle}$ has 2^r distinct nodes. To prove independence against any competing finite node set, choose a direction not perpendicular to any pairwise node difference, and restrict the identity to a short line segment inside the given open set. The one-dimensional Vandermonde argument then applies.

For the upper bound impose, for each i ,

$$(z_i - s_i, 1, u_i), (-z_i - s_i, 1, v_i) \in K_{\text{exp}}, \quad u_i + v_i \leq 2, \quad \left(\sum_i s_i, 1, t \right) \in K_{\text{exp}}.$$

Eliminating u_i, v_i gives $s_i \geq \log \cosh z_i$, and eliminating s gives exactly $t \geq f_r(z)$. Restricting to the cube adds only scalar inequalities.

For the lower bound choose $T > 1$ and intersect the given epigraph with $t \leq T$. Its image is the compact convex body

$$C = \{(z, t) : z \in [-K, K]^r, f_r(z) \leq t \leq T\} \subset \mathbb{R}^{r+1}.$$

It has nonempty interior. In a neighborhood of $(0, 1)$ its lower boundary is the graph of f_r , and

$$\nabla^2 f_r(z) = f_r(z) \left[\text{diag}(\text{sech}^2 z_i) + (\tanh z)(\tanh z)^T \right] \succ 0. \quad (21.10)$$

Its second fundamental form is therefore positive definite in all r tangent directions. The intersection adds one nonnegative-ray factor and does not add an exponential factor. After translation to put an interior point at the origin, Lemma 2.1 eliminates free variables and passes to the minimal face. These sets are definable in the real exponential structure; the generic selection argument and Theorem 3.2 apply. Each retained face of an exponential cone has dimension at most three and thus contributes at most one unit to $\sum_i \max\{\dim K_i - 2, 0\}$; every ray contributes zero. The r -dimensional positive-curvature patch gives $r \leq R$. Finally, Theorem 19.2 gives the exact ambient parameter $3R + P$ for the *original* product. The artificial truncation ray is used only in the curvature argument and is not charged to that original product. $\square$

The same proof gives $R \geq r$ for any definable convex epigraph with an r -dimensional C^2 graph patch having positive definite Hessian, provided that a compact polyhedral truncation retains that patch. It is important to keep the epigraph coordinate in this argument: a level-set section of f_r alone has only $r - 1$ curved directions. Theorem 21.6 makes no optimality claim for the constant 2 in its constructive upper bound. Its lower bound follows from curvature, independently of exponential-feature rank.

Proposition 21.7 (Polyhedral obstruction to an approximate transfer). *Let $D \subset \mathbb{R}^r$ be a compact polytope and let $f > 0$ be convex and C^2 on a neighborhood of D . For every $0 < \varepsilon < 1$, a finite piecewise-affine convex function p satisfies*

$$(1 - \varepsilon)f \leq p \leq f \quad \text{on } D.$$

In particular,

$$\text{epi}(p/(1 - \varepsilon)) \subseteq \text{epi } f \subseteq \text{epi } p \quad \text{over } D,$$

and the two approximating epigraphs use only scalar linear inequalities.

Proof. Let $m_f = \min_D f > 0$ and $L_f = \sup_D \|\nabla^2 f\|_{\text{op}}$. If $L_f = 0$, one tangent is exact on D . Otherwise choose a finite δ -net $\mathcal{N} \subset D$, with $\delta^2 \leq 2\varepsilon m_f / L_f$, and set

$$p(z) = \max_{x \in \mathcal{N}} \{f(x) + \langle \nabla f(x), z - x \rangle\}.$$

Convexity gives $p \leq f$. Taylor's theorem on the segment between z and a net point within δ gives $f(z) - p(z) \leq L_f \delta^2 / 2 \leq \varepsilon f(z)$. The epigraph of a finite maximum of affine functions is their halfspace intersection, giving the claim. $\square$

Thus uniform relative approximation on a fixed cube can use zero exponential cones when arbitrarily many scalar inequalities are free. Charging those inequalities, or imposing the conditioned contact hypotheses of the preceding approximation results, changes the question. Neither exact signed-feature rank nor the exact curvature count alone is an approximation lower bound. Together, the three compiler models identify which representation or access restriction supplies each conclusion; none yields a lower bound on all optimization algorithms.

22 Projection, sufficient statistics, and limits of local compilation

An exact reformulation may discard the original recourse coordinates and optimize over their projection. This changes the computational interface: a small first-stage barrier need not provide all recourse variables, and a small existential barrier need not have a cheap reusable oracle. We make these distinctions explicit. Farkas elimination, quadratic epigraph barriers, and quantum spectral approximation are established tools. The results below combine them under precise access assumptions and identify limitations of two specific compilation rules. They make no claim that a small projected barrier preserves the full product-slack geometry.

22.1 Polyhedral and ordered recourse

Let $W \in \mathbb{R}^{q \times p}$ and $T \in \mathbb{R}^{q \times k}$ be public, and consider

$$x \in X_0, \quad W y_i = h_i - T x, \quad y_i \geq 0 \quad (1 \leq i \leq N). \quad (22.1)$$

Write $H = \text{range } W$ and $C = \{z \in H : W^T z \geq 0\}$. This polyhedral cone is pointed in H : if $z, -z \in C$, then $W^T z = 0$ and $z \in H$, whence $z = 0$. Let $z_1, \dots, z_L$ generate its extreme rays; $L = 0$ is allowed.

Proposition 22.1 (Shared recourse directions). *The projection of (22.1) is exactly*

$$x \in X_0, \quad h_i - T x \in H \quad (i \leq N), \quad a_\ell^T x \geq \beta_\ell \quad (\ell \leq L), \quad (22.2)$$

where $a_\ell = -T^T z_\ell$ and $\beta_\ell = \max_i(-z_\ell^T h_i)$. On the affine space defined by the displayed equalities, suppose X_0 has a ν_0 -barrier G , and remove inequalities that are identically satisfied. If the remaining inequalities have a common strict feasible point in $\text{ri } X_0$, then

$$G(x) - \sum_{\ell=1}^L \log(a_\ell^T x - \beta_\ell)$$

is a barrier of parameter at most $\nu_0 + L$ on that relative domain.

Proof. Farkas' lemma gives $Wy = h_i - Tx$, $y \geq 0$, precisely when $z^T(h_i - Tx) \geq 0$ for every z satisfying $W^T z \geq 0$. The lineality space of this dual cone is $H^\perp$; it imposes exactly $h_i - Tx \in H$. In H , the extreme rays generate C , so only their inequalities remain. Taking the maximum over i gives (22.2). The barrier statement follows by affine restriction and addition of logarithmic half-space barriers. $\square$

If W has full row rank, the membership equalities disappear. With a coherent block oracle for h_i , exact comparisons at the supplied data precision, and public rays, quantum maximum finding computes all thresholds with joint constant success probability in $\tilde{O}(L\sqrt{N})$ block queries [28]. Here and below the tilde allows amplification logarithms. Derivatives of the compiled barrier then need no scenario queries. A path-following iteration bound also requires a supplied initialization and the usual accuracy and initial-gap factors; it is not a consequence of the barrier parameter alone. Without full row rank, the compatibility equalities must be supplied or checked as well.

There is also a local reconstruction statement, with an essential strictness hypothesis. Suppose $h_i - Tx \in \text{ri}(W\mathbb{R}_+^p)$, equivalently there exists $y > 0$ with $Wy = h_i - Tx$. Then

$$y_i^c(x) = \underset{Wy=h_i-Tx, y>0}{\text{argmin}} \left\{ \frac{1}{2} \|y\|^2 - \sum_{j=1}^p \log y_j \right\} \quad (22.3)$$

exists uniquely. Indeed, the objective is coercive at infinity, diverges at the coordinate boundary, and has positive-definite Hessian. Let x vary in the affine compatibility space $\mathcal{A}_i = \{x : h_i - Tx \in \text{range } W\}$. For a tangent direction h to $\mathcal{A}_i$, its KKT derivative is, after applying the same independent-row reduction to W and T ,

$$\begin{bmatrix} I + \text{diag}(y_j^{-2}) & W^T \\ W & 0 \end{bmatrix} \begin{bmatrix} Dy[h] \\ D\lambda[h] \end{bmatrix} = \begin{bmatrix} 0 \\ -Th \end{bmatrix}.$$

The matrix is invertible by positive definiteness and full row rank. Thus this recourse rule is smooth on the relative open subset of $\mathcal{A}_i$ with strictly positive feasible recourse, and depends only on the requested block and x . Computing it requires a local convex solve, not one arithmetic operation. Mere nonnegative feasibility does not justify (22.3); at a boundary point one must instead solve a local feasible problem or first determine its minimal face.

For varying $W_i \in \mathbb{R}^{q \times p}$, suppose first that each has full row rank. Its pointed dual cone has at most $\binom{p}{q-1}$ extreme rays. An extreme ray has $q-1$ independent active inequalities: otherwise its minimal face has dimension at least two. Each such subset specifies a line, and pointedness permits at most one orientation of that line. This proves the count, including $q=1$. For rank $r_i < q$, apply the argument in $\text{range } W_i$, obtaining $\binom{p}{r_i-1}$ rays when $r_i > 0$, and retain the orthogonal compatibility equalities separately. A rank-zero block supplies only equalities. Consequently, for fixed q a local block query suffices, with polynomial local arithmetic, to enumerate the inequalities defining that block's projection.

Here is a deliberately conditional algorithmic consequence. Suppose these inequalities and any constraints for X_0 define a full-dimensional LP in k effective variables, with finite optimum; assume coherent row access, the data precision assumptions, and an initial point sufficiently close to the central path required by Apers and Gribbling [5, Theorems 1.3 and 6.1]. Their algorithm returns an explicit feasible ϵ -optimal first-stage vector. With fixed q and a polynomial number of public constraints for X_0 , simulating each row query locally gives $\tilde{O}(\sqrt{N} \text{poly}(p, k))$ block queries. This is a reduction to that algorithm; initialization, facial reduction, and full recourse output are additional tasks.

Finally, if recourse has the ordered form $h_0 + \alpha_i e - Tx \in K$, where K is a proper cone and $e \in K$, it reduces exactly to the single constraint with $\alpha_* = \min_i \alpha_i$. This follows because $(\alpha_i - \alpha_*)e \in K$, and the minimum is attained. For a symmetric cone of rank r , its standard barrier adds at most r to the first-stage parameter. The threshold again takes $\tilde{O}(\sqrt{N})$ queries. The scalar family $x \geq z_i$, $z_i \in \{0, 1\}$, shows the familiar OR lower bounds $\Omega(\sqrt{N})$ quantum and $\Omega(N)$ randomized [14] for determining the threshold to error below $1/2$. These statements concern ordered offsets; arbitrary offsets have no such single-threshold reduction.

22.2 Additive quadratic and matrix sufficient statistics

Consider the closed formulation

$$r_i = A_i x - b_i, \quad t_i \geq \|r_i\|^2, \quad \tau \geq \sum_i t_i, \quad x \in X_0. \quad (22.4)$$

It uses N rotated Lorentz factors and one ray. Its canonical ambient parameter is $2N + 1$, before adding a barrier for X_0 . Its exact projection is $\tau \geq q(x) = \sum_i \|A_i x - b_i\|^2$.

Proposition 22.2 (Quadratic compilation). *Set $C_i = [A_i, -b_i]$, $\xi = (x, 1)$, and*

$$M = \sum_i C_i^T C_i, \quad Q = \sum_i A_i^T A_i, \quad g = \sum_i A_i^T b_i.$$

Then $q(x) = \xi^T M \xi$. On the quotient by $\ker Q$, the function $f(x, \tau) = -\log(\tau - q(x))$ is a barrier of exact parameter one. On unquotiented variables, if G is a ν_0 -barrier for X_0 , then $G + f$ is a barrier of parameter at most $\nu_0 + 1$. With $s = \tau - q(x)$ and $d = 2(Qx - g)$, the derivatives of f are

$$\nabla f = (d, -1)/s, \quad \nabla^2 f = \begin{bmatrix} 2Q/s & 0 \\ 0 & 0 \end{bmatrix} + (d, -1)(d, -1)^T/s^2. \quad (22.5)$$

Proof. Expansion gives the sufficient statistic and the derivatives. Since $\ker Q = \bigcap_i \ker A_i$, q is constant along the deleted directions. For a line direction h , put $a = Ds[h]$ and $b = -D^2 s[h, h] \geq 0$. Direct differentiation gives

$$Df[h] = -a/s, \quad D^2 f[h, h] = a^2/s^2 + b/s, \quad D^3 f[h, h, h] = -2a^3/s^3 - 3ab/s^2.$$

For $u, v \geq 0$, $2u^3 + 3uv \leq 2(u^2 + v)^{3/2}$, as is seen by squaring and subtracting, leaving $3u^2 v^2 + 4v^3 \geq 0$. Apply this with $u = |a|/s$, $v = b/s$ to obtain self-concordance. Also $(Df[h])^2 \leq D^2 f[h, h]$. The Hessian is positive definite on the quotient, and the function diverges at the epigraph boundary. The direction that varies only τ attains equality in the gradient inequality, so the parameter is exactly one. Addition of G removes any remaining flat directions and gives the asserted parameter bound. $\square$

This is the familiar quadratic epigraph barrier, not a new barrier formula. Its relevance here is the exact replacement of many product factors by a single statistic, while retaining local recourse access. Querying block i recovers $r_i = A_i x - b_i$ and $t_i = \|r_i\|^2$. This supplies a feasible completion for every projected feasible (x, τ) ; strict recourse can be obtained by distributing positive coupling slack. Materializing all N recourse scalars requires $\Omega(N)$ output operations, whereas one requested block uses one block query plus local arithmetic.

The statistic may also be approximated safely. Let C be the stack of all C_i , with m rows and $k+1$ columns. The quantum spectral approximation theorem of Apers and Gribling [5, Theorem 3.1] gives, with high probability, $\widetilde{M}$ satisfying

$$(1 - \delta)\widetilde{M} \preceq M \preceq (1 + \delta)\widetilde{M}$$

in $\widetilde{O}(\sqrt{m(k+1)}/\delta)$ row queries, for $0 < \delta < 1$. Row sparsity affects arithmetic cost separately; rank deficiency is allowed. Taking $\widehat{M} = (1 + \delta)\widetilde{M}$ yields

$$M \preceq \widehat{M} \preceq \frac{1 + \delta}{1 - \delta} M. \quad (22.6)$$

Thus $\tau \geq \xi^T \widehat{M} \xi$ implies original feasibility. For pure least-squares minimization over a fixed X_0 , an exact minimizer of the inflated objective has original objective at most $(1 + \delta)/(1 - \delta)$ times the original infimum, whenever a minimizer exists; an additive solve error adds the same error to this bound. Subsequent optimization uses only the explicit statistic. When $m = Np$, this gives the source's $\widetilde{O}(\sqrt{Np(k+1)}/\delta)$ row bound; it is a block-query bound only when the specified block oracle simulates one row query at constant query cost. The case $A_i = 0$, $b_i \in \{0, 1\}$, with zero or one nonzero b_i , reduces objective approximation with error below $1/2$ to OR, and gives the corresponding quantum and randomized scenario-query lower bounds.

The matrix counterpart is equally exact. Let $R_i(x) = B_{i0} + \sum_{j=1}^k x_j B_{ij} \in \mathbb{R}^{d_i \times s}$ and let $R(x)$ stack the $R_i(x)$. Delete directions h with $DR[h] = 0$.

Proposition 22.3 (Matrix quadratic compilation). *On $Y - R(x)^T R(x) \succ 0$ in the effective variables,*

$$f(x, Y) = -\log \det(Y - R(x)^T R(x))$$

is a barrier of exact parameter s . No barrier for this domain has parameter below s . Without quotienting, addition of a ν_0 -barrier for X_0 gives a barrier of parameter at most $\nu_0 + s$. All data enter through the positive-semidefinite block Gram matrix

$$\mathcal{M}_{j\ell} = \sum_i B_{ij}^T B_{i\ell}, \quad 0 \leq j, \ell \leq k,$$

of order $(k+1)s$.

Proof. Put $S = Y - R^T R$. Along a direction, set

$$A = S^{-1/2} D S S^{-1/2}, \quad B = 2S^{-1/2} (DR)^T D R S^{-1/2} \succeq 0.$$

Differentiation gives

$$Df = -\operatorname{tr} A, \quad D^2 f = \operatorname{tr} A^2 + \operatorname{tr} B, \quad D^3 f = -2 \operatorname{tr} A^3 - 3 \operatorname{tr}(AB).$$

The bounds $|\operatorname{tr} A^3| \leq \|A\|_F^3$ and $|\operatorname{tr}(AB)| \leq \|A\|_F \operatorname{tr} B$, together with the scalar inequality in Proposition 22.2, prove self-concordance. Moreover $(\operatorname{tr} A)^2 \leq s \operatorname{tr} A^2 \leq s D^2 f$. The Hessian vanishes

only if $DR = 0$ and $DY = 0$, so is positive definite in the effective variables. The boundary divergence is immediate. Varying only Y in direction S gives decrement squared s . For the lower bound against any barrier, fix x and restrict $Y = R(x)^T R(x) + \text{diag}(u_1, \dots, u_s)$. This is the positive orthant, with its coordinate boundary in the boundary of the full domain, so its parameter lower bound is s . Finally, expanding $R^T R$ gives $\sum_{j,\ell} \xi_j \xi_\ell \mathcal{M}_{j\ell}$, with $\xi_0 = 1$. $\square$

Equivalently, this is a fixed-identity Schur-complement slice of a PSD cone; the restriction credits the fixed block. Applying spectral approximation to the stack $[B_{i0}, \dots, B_{ik}]$ and using (22.6) gives a uniform Loewner upper bound on $R(x)^T R(x)$, by congruence with $\xi \otimes I_s$. Thus the same safety argument compiles a matrix epigraph. It does not by itself give a relative objective guarantee for an arbitrary SDP objective.

22.3 A scalar normalization cannot repair every convolution barrier

The preceding reduction uses an additive quadratic statistic. A natural alternative for general blocks is to convolve block measures and take the Legendre transform of the resulting log-Laplace function. The following obstruction concerns exactly that construction and scalar normalizations that depend only on the number of blocks.

Proposition 22.4 (Convolution normalization obstruction). *Let μ be uniform probability measure on $[0, 1]$, with log-Laplace function f and conjugate $F = f^*$. For N independent variables, consider the conjugate of the log-Laplace function of their average, multiplied by a positive scalar a_N . Suppose this rule must be a standard self-concordant barrier for both of the following collections: N copies of μ , and one copy of μ with $N - 1$ point masses at zero. Then $a_N \geq 1$. Its barrier parameter on the replicated collection is therefore at least N .*

Proof. Independence gives $f_{\text{av}}(\theta) = Nf(\theta/N)$ for identical copies, hence $f_{\text{av}}^*(x) = NF(x)$. For the second collection it gives $f_{\text{av}}(\theta) = f(\theta/N)$, hence $f_{\text{av}}^*(x) = F(Nx)$. For $\theta < 0$,

$$\begin{aligned} f(\theta) &= \log(1 - e^\theta) - \log(-\theta), & f'(\theta) &= -1/\theta + O(e^\theta), \\ f''(\theta) &= \theta^{-2} + O(e^\theta), & f'''(\theta) &= -2\theta^{-3} + O(e^\theta). \end{aligned}$$

As $x = f'(\theta) \downarrow 0$, Legendre differentiation gives $F' = \theta$, $F'' = 1/f''(\theta)$ and $F''' = -f'''(\theta)/(f''(\theta))^3$. Therefore

$$\frac{(F')^2}{F''} \longrightarrow 1, \quad \frac{|F'''}{2(F'')^{3/2}} \longrightarrow 1.$$

Affine rescaling $x \mapsto Nx$ does not change either ratio. Multiplying a function by a_N divides the second ratio by $\sqrt{a_N}$, so self-concordance for the second collection forces $a_N \geq 1$. On the first collection, the first ratio tends to $a_N N$, which is a lower bound on its barrier parameter. $\square$

The degenerate blocks are part of the hypothesis. The proposition neither rules out a rule using the individual distributions nor supplies a lower bound for all barriers of the projected interval. In particular, it does not obstruct a nonfactorized entropic or universal barrier. We claim only the stated normalization impossibility, rather than novelty for the classical conjugacy identity used in its proof.

22.4 Two obstructions to reusable exact summaries

Quadratic sufficient statistics have bounded dimension when the first-stage dimension is fixed. This fails for arbitrary exponential sums even in one first-stage variable.

Proposition 22.5 (Continuous exact summaries). *Fix distinct real $a_1, \dots, a_N$ and let $q_w(x) = \sum_i w_i e^{a_i x}$, with $w \in \mathbb{R}_{++}^N$. Suppose a continuous statistic $S : \mathbb{R}_{++}^N \rightarrow \mathbb{R}^d$ determines the entire strict epigraph of q_w . Then $d \geq N$. The decoder need not be continuous.*

Proof. Equal statistics give equal epigraphs and hence equal functions. Their first N derivatives at zero give the Vandermonde system $\sum_i (w_i - v_i) a_i^j = 0$, $0 \leq j < N$, so $w = v$. Thus S is injective. If $d < N$, composing with the coordinate inclusion $\mathbb{R}^d \hookrightarrow \mathbb{R}^N$ contradicts invariance of domain, since its image lies in a subspace with empty interior. $\square$

The next statement concerns preprocessing queries rather than the dimension of a continuous statistic. An offline summary is called *reusable* if, after preprocessing, it can answer all the stated requests with joint success probability at least $2/3$, using no further input queries. It may be classical or quantum, provided reuse has this operational meaning. This excludes interpreting a single consumable quantum state as an oracle for arbitrarily many later answers.

Proposition 22.6 (Reusable boundary oracles). *Let $z \in \{0, 1\}^N$, $\gamma = 1/4$, and define the planar epigraph*

$$E_z = \{(x, t) : t \geq \varphi_z(x)\}, \quad \varphi_z(x) = x^2 + \max_{1 \leq i \leq N} (2ix - i^2 + \gamma z_i). \quad (22.7)$$

Each inequality has a constant-size rotated-SOC representation. A reusable summary giving all values $\varphi_z(i)$ to absolute error less than $\gamma/3$ requires $\Omega(N)$ quantum preprocessing queries to the bit oracle for z . The same lower bound holds for reusable membership answers at the public points $(i, 2i^2 + \gamma/2)$ with the promise of vertical margin $\gamma/2$.

Proof. At $x = i$, the i th affine term exceeds term $j \neq i$ by $(i - j)^2 + \gamma(z_i - z_j) \geq 3/4$. Thus $\varphi_z(i) = 2i^2 + \gamma z_i$. Either oracle recovers every bit. For completeness, a T -query quantum algorithm on an N -bit input has final state in the span of at most $D_T = \sum_{j=0}^T \binom{N}{j}$ fixed vectors: expand its amplitudes in the characters of the input signs, whose degree increases by at most one per query. Purification handles intermediate measurements and randomness. An ensemble of 2^N states in a space of dimension at most D_T can be identified, on a uniform input, with probability at most $D_T/2^N$, because each density matrix is bounded above by the identity on that space and the decoding POVM sums to that identity. For $T \leq N/4$, the binomial entropy bound makes this probability tend to zero. Constant successful recovery therefore requires $T = \Omega(N)$, as in oracle interrogation [84]. $\square$

Rescaling the exposed locations to $x = i/N$ and the affine terms to $2Nix - i^2$ preserves the gap. Replacing the common quadratic by x^r on $x > 0$, for fixed rational $r > 1$, or by e^x , gives the same recovery proof with constant-size power or exponential constraints. The locations then stay in a bounded interval, so the common exponential term causes no large-value precision issue. This is a restriction on reusable preprocessing. Computing a single requested boundary value can still use fresh $\tilde{O}(\sqrt{N})$ maximum-finding queries. It also does not rule out approximate summaries under a weaker norm or an input distribution that changes the recovery promise.

22.5 The parameter cost of additive local logarithms

The same exposed arcs impose a distinct barrier obstruction, even though E_z has only two coordinates.

Proposition 22.7 (Additive local barrier cost). *For the slacks $s_i(x, t) = t - x^2 - 2ix + i^2 - \gamma z_i$ in (22.7), suppose*

$$F(x, t) = - \sum_{i=1}^N w_i \log s_i(x, t), \quad w_i > 0,$$

is a standard self-concordant barrier with parameter ν on $\text{int } E_z$. Then $w_i \geq 1$ for every i , and $\nu \geq \sum_i w_i \geq N$. Nevertheless, this line-free planar domain admits a barrier of parameter two.

Proof. At $(i, \varphi_z(i) + \epsilon)$, the i th slack is ϵ and all other slacks stay at least $3/4$. In direction $h = (0, 1)$,

$$D^2 F[h, h] = w_i \epsilon^{-2} + O(1), \quad |D^3 F[h, h, h]| = 2w_i \epsilon^{-3} + O(1).$$

The self-concordance ratio tends to $1/\sqrt{w_i}$, forcing $w_i \geq 1$. For fixed x and $t \rightarrow \infty$,

$$\frac{(DF[h])^2}{D^2 F[h, h]} = \frac{(\sum_i w_i / s_i)^2}{\sum_i w_i / s_i^2} \longrightarrow \sum_i w_i.$$

The gradient inequality implies the parameter bound. The epigraph is full-dimensional and contains no affine line, because its boundary grows quadratically in either horizontal direction and is bounded below on each vertical line. The universal-barrier theorem in dimension two therefore supplies a two-parameter barrier [60]. $\square$

The upper and lower statements concern different permitted formulas. A barrier that combines scenarios through a global construction can have parameter independent of N ; a weighted sum of these local logarithms cannot. Together with the reusable-oracle lower bound, this explains why an existential projected barrier, an explicit sufficient statistic, and a cheap oracle are three separate requirements.

Part V

Movement in barrier metrics

23 Certificate ranks and movement in a fixed barrier metric

A lower bound on a barrier parameter does not by itself give a lower bound on primal movement. The missing information is which part of the cone an objective forces toward its boundary. We now connect the certificate ranks proved above to this part of the metric. The weighted exposed-minor estimate below is also established in the companion manuscript [21]; we include its proof to make the connection independent of that manuscript. The Jordan identities are classical [31], as is the conversion from bounded local steps to Riemannian distance [72]. The contribution here is their composition with the lift-dictionary and selected-certificate frontiers, with the same certificate retained in both arguments.

For a barrier F with positive-definite Hessian on its effective affine domain Ω , write

$$\|h\|_{F,x}^2 = D^2F(x)[h, h], \quad d_F(x, y) = \inf_{\gamma} \int_0^1 \|\dot{\gamma}(t)\|_{F,\gamma(t)} dt.$$

The infimum is over piecewise C^1 curves in Ω joining the two points. For a set $T \subset \Omega$, put $d_F(x, T) = \inf_{y \in T} d_F(x, y)$, with value $+\infty$ if T is empty. A free affine direction on which the slack does not change is removed before defining this metric.

23.1 The weighted exposed-minor estimate

Let $K = \prod_i K_i$ be a product of symmetric cones with their Jordan trace inner products. Let $\Omega = \text{int } K \cap \mathcal{A} \neq \emptyset$ for an affine subspace $\mathcal{A}$. Suppose an affine objective ℓ has finite supremum ℓ_* and a genuine certificate $s = (s_i) \in K$ satisfying

$$g(x) := \ell_* - \ell(x) = \sum_i \langle s_i, x_i \rangle \quad (x \in \Omega). \quad (23.1)$$

Thus this equality holds on the entire feasible affine slice. For each nonzero s_i , let $c_i = \text{supp } s_i$, $q_i = \text{rank}_J s_i$, and let $\det_{c_i}$ denote the determinant in the Peirce algebra with unit c_i . The expressions below omit factors with $q_i = 0$.

Theorem 23.1 (Weighted exposed-minor distance). *Use the restricted barrier*

$$F(x) = - \sum_i \alpha_i \log \det_i x_i, \quad \alpha_i \geq 1, \quad Q_\alpha = \sum_i \alpha_i q_i > 0.$$

For a fixed reference $x^0 \in \Omega$, define

$$\Delta_\alpha = Q_\alpha \left[\prod_{i: q_i > 0} \frac{\{\det_{c_i}(s_i) \det_{c_i}(P(c_i)x_i^0)\}^{\alpha_i}}{\alpha_i^{\alpha_i q_i}} \right]^{1/Q_\alpha}. \quad (23.2)$$

Then $\Delta_\alpha > 0$ and, for every $\varepsilon > 0$,

$$d_F(x^0, \{x : g(x) \leq \varepsilon\}) \geq \sqrt{Q_\alpha} [\log(\Delta_\alpha/\varepsilon)]_+. \quad (23.3)$$

Here $[t]_+ = \max(t, 0)$. All inactive coordinates and all feasible completion variables remain available to the path. No strict complementarity or boundedness of those variables is assumed.

Proof. For an idempotent c of rank q , put $a = P(c)x$ and $\phi_c(x) = -\log \det_c a$. The element a is interior to its Peirce cone: $x \succeq \delta e$ implies $a \succeq \delta c$. The gradient and inverse Hessian of $-\log \det x$ are $-x^{-1}$ and $P(x)$, respectively. Since $P(c)$ is self-adjoint and $a^{-1} \in V(c, 1)$,

$$D\phi_c(x)[h] = -\langle a^{-1}, h \rangle, \quad \|D\phi_c(x)\|_{-\log \det, *}^2 = \langle a^{-1}, P(x)a^{-1} \rangle = q.$$

The last equality follows from the fundamental identity

$$P(c)P(x)P(c) = P(P(c)x) = P(a), \quad P(a)a^{-1} = a, \quad \langle a^{-1}, a \rangle = q.$$

These are Jordan identities, so they include the exceptional algebra. For $\Phi(x) = \sum_{i:q_i>0} \alpha_i \phi_{c_i}(x_i)$, the full product metric consequently gives $\|D\Phi\|_{F,*}^2 = Q_\alpha$. Restricting a covector and the metric to an affine tangent subspace can only decrease its dual norm. Hence

$$|D\Phi(x)[h]| \leq \sqrt{Q_\alpha} \|h\|_{F,x} \quad (h \in T_x \Omega). \quad (23.4)$$

For a feasible endpoint x , set $g_i = \langle s_i, x_i \rangle > 0$ on the active factors. In $V(c_i, 1)$, the positive element $P(s_i^{1/2})P(c_i)x_i$ has trace g_i and determinant $\det_{c_i}(s_i) \det_{c_i}(P(c_i)x_i)$. Spectral AM–GM therefore gives

$$\det_{c_i}(P(c_i)x_i) \leq \det_{c_i}(s_i)^{-1} (g_i/q_i)^{q_i}.$$

If $\sum_i g_i \leq \varepsilon$, maximizing $\sum_i \alpha_i q_i \log(g_i/q_i)$ over these positive g_i gives $g_i = \varepsilon \alpha_i q_i / Q_\alpha$. It follows that

$$\prod_i \det_{c_i}(P(c_i)x_i)^{\alpha_i} \leq (\varepsilon/Q_\alpha)^{Q_\alpha} \prod_i \alpha_i^{\alpha_i q_i} \det_{c_i}(s_i)^{-\alpha_i}.$$

Comparison with the reference point yields $\Phi(x) - \Phi(x^0) \geq Q_\alpha \log(\Delta_\alpha/\varepsilon)$. Integrating (23.4) along any feasible curve proves the result. $\square$

Corollary 23.2 (Bounded-move consequence). *Suppose a procedure starts at x^{in} with $d_F(x^{\text{in}}, x^0) \leq D_0$ and makes T rounds, each with metric displacement at most $B_0 > 0$, before attaining gap at most ε . Then*

$$T \geq \frac{[\sqrt{Q_\alpha} \log(\Delta_\alpha/\varepsilon) - D_0]_+}{B_0}. \quad (23.5)$$

In particular, for an integer $m \geq 1$ and $0 < R < 1$, at most m feasible forward chords of starting local norm at most R per round allow $B_0 = -m \log(1 - R)$. An initial local error $\|x^{\text{in}} - x^0\|_{F,x^0} \leq \rho < 1$ allows $D_0 = -\log(1 - \rho)$. More generally, if the procedure makes M chords, where $M \geq 1$ is an integer, with respective starting norms $r_j < 1$ and $D = [\sqrt{Q_\alpha} \log(\Delta_\alpha/\varepsilon) - D_0]_+$, then

$$\sum_{j=1}^M -\log(1 - r_j) \geq D, \quad \max_j r_j \geq 1 - e^{-D/M}. \quad (23.6)$$

Proof. The triangle inequality proves (23.5). For a chord $x + th$, the standard self-concordant Hessian comparison gives $\|h\|_{F,x+th} \leq \|h\|_{F,x}/(1 - t\|h\|_{F,x})$. Its integral is at most $-\log(1 - \|h\|_{F,x})$. Summing these bounds proves the variable-chord assertion as well; see also [72, Lemma 3.2 and Corollary 3.2]. $\square$

The square-root coefficient is sharp on the full product cone. A quadratic-representation isometry sends x^0 to the unit. In a Jordan frame diagonalizing the transformed s_i , multiply exactly its

q_i positive-support coordinates by t and keep all remaining coordinates one. This curve has gap $t \sum_i \langle s_i, e_i \rangle$ and length $\sqrt{Q_\alpha} \log(1/t)$ from the unit. Thus, on the full cone,

$$\lim_{\varepsilon \downarrow 0} \frac{d_F(x^0, \{g \leq \varepsilon\})}{\log(1/\varepsilon)} = \sqrt{Q_\alpha}. \quad (23.7)$$

In this statement the certificate is transformed with the coordinates; its pairing, and hence the gap, is unchanged. An affine slice can exclude the displayed curve, so (23.7) does not assert a matching upper bound for every slice.

Hauser and Güler's classification [49, Theorem 5.5] says that every self-scaled barrier on a symmetric cone is, up to an additive constant, of the displayed form on its irreducible factors, with $\alpha_i \geq 1$. Consequently the theorem covers all self-scaled barriers on a fixed symmetric cone. Changing the scales changes both Q_α and Δ_α ; retaining the unweighted Δ after rescaling would be incorrect. The classification gives no analogous assertion for arbitrary coupled barriers on an affine slice.

23.2 Composing the rank frontier with movement

For the unit scales, put $Q = \sum_i q_i$ and write Δ for (23.2). For a fixed lift, reference, objective, and certificate, Theorem 23.1 gives

$$\liminf_{\varepsilon \downarrow 0} \frac{d_F(x^0, \{g \leq \varepsilon\})}{\log(1/\varepsilon)} \geq \sqrt{Q}. \quad (23.8)$$

Its application to earlier rank results has the following exact quantifiers.

Corollary 23.3 (Dictionary consequences). *Let the fixed symmetric-cone dictionary have capacity B from (4.1), and use its standard product barrier on the relative-Slater operational lift. For a lift of $\mathbb{B}_2^s$, on a dense open full-measure set of supports, (23.8) has right-hand side at least $\sqrt{\lceil (s-1)/B \rceil}$. For every lift of $\prod_a \mathbb{B}_2^{s_a}$, there exist directions u_a and genuine pure-row certificates Y^a such that every fixed choice of positive weights λ_a gives an objective $\ell(x) = \sum_a \lambda_a \langle u_a, \pi_a x \rangle$ for which the right-hand side is at least*

$$\sqrt{\sum_a \left\lceil \frac{s_a - 1}{B} \right\rceil}.$$

Under the globally bi- C^1 genuine-selection hypotheses of Theorem 7.7, some support of a single ball instead has coefficient at least $\sqrt{\kappa_{\mathcal{X}}(s)}$.

Proof. For one ball choose any genuine certificate at a support in Theorem 4.1's regular set. For the product take exactly the certificate $s_i = \sum_a \lambda_a Y_i^a$ supplied by Theorem 5.2. It represents the full gap $\sum_a \lambda_a (1 - \langle u_a, \pi_a x \rangle)$ and has the asserted aggregate rank. In the smooth case use the genuine selected certificate whose rank is bounded below in Theorem 7.7. Apply (23.8) in each case. $\square$

The product assertion only needs existence of one such aggregate certificate. It neither says that every aggregate certificate has this rank nor minimizes over its whole dual fiber. A high-rank certificate suffices because its determinant estimate holds for every accurate primal endpoint. For self-scaled barriers the same statements hold with the actual weighted $Q_\alpha \geq Q$. The constants Δ_α can depend on the lift, objective, reference, and certificate; the corollary is a fixed-instance asymptotic statement, not a uniform finite-accuracy estimate over all lifts. In particular one cannot replace Q in this argument by a larger ambient or restricted barrier parameter.

23.3 Affine PSD pencils and spectral scales

A complementary formulation uses a compression of an affine pencil. It is useful when an exposed minor is known directly, including in completion formulations. The convexity behind the contraction is classical matrix-fractional convexity [17, Example 3.4]. The matrix-ball specialization and its rank-profile interpretation also appear in [21].

Proposition 23.4 (Principal-minor contraction). *Let $L(x) = L_0 + Ax \succ 0$ be a real symmetric or complex Hermitian affine pencil and $F = -\log \det L$ on an effective affine domain with positive-definite Hessian. For an injective matrix R with $r \geq 1$ columns put $\psi_R(x) = -\log \det(R^*L(x)R)$. Then*

$$D^2\psi_R \preceq D^2F, \quad |D\psi_R(x)[h]| \leq \sqrt{r}\|h\|_{F,x}. \quad (23.9)$$

If the compressed function has the sharper restricted gradient bound $|D\psi_R[h]| \leq \sqrt{\vartheta} (D^2\psi_R[h, h])^{1/2}$ for $\vartheta > 0$, the second constant can be replaced by $\sqrt{\vartheta}$. Consequently

$$d_F(x^0, x) \geq \frac{1}{\sqrt{\vartheta}} \left[\log \frac{\det(R^*L(x^0)R)}{\det(R^*L(x)R)} \right]_+, \quad (23.10)$$

with $\vartheta = r$ always permitted.

Proof. First take R isometric and complete it to a unitary basis. In that basis write $L = \begin{pmatrix} A & B \\ B^* & C \end{pmatrix}$, where $A = R^*LR$. If R is square there is nothing to prove. Otherwise

$$F - \psi_R = -\log \det(C - B^*A^{-1}B).$$

The Schur complement is matrix concave because the matrix-fractional map $B^*A^{-1}B$ is matrix convex: test the scalar convex function $(Bv)^*A^{-1}(Bv)$ against each fixed vector v . The scalar matrix-fractional convexity follows directly from its Schur-complement epigraph, over either field. Composition with the convex, Loewner-decreasing function $-\log \det$ is convex. This proves the Hessian inequality. Also, with $T = A^{-1/2}(DA[h])A^{-1/2}$,

$$D\psi_R[h] = -\operatorname{tr} T, \quad D^2\psi_R[h, h] = \operatorname{tr}(T^2), \quad |\operatorname{tr} T| \leq \sqrt{r} \sqrt{\operatorname{tr}(T^2)}.$$

A thin QR factorization of an arbitrary injective R changes ψ_R only by a constant, proving the general case. Integration proves (23.10), also with any stated sharper constant. $\square$

For example, for the matrix-ball pencil $L(X) = \begin{pmatrix} I_p & X \\ X^* & I_q \end{pmatrix}$, compression to $1 \leq r \leq \min(p, q)$ chosen left coordinates and all right coordinates has determinant $\det(I_r - U^*XX^*U)$, where U is the p by r matrix of the selected orthonormal left coordinate vectors. Its restricted gradient parameter is r , although its pencil order is $r + q$. Thus one should use $\vartheta = r$, not its unreduced pencil order. A covector bound remains valid when this compressed Hessian has inactive null directions.

Theorem 23.5 (Exposed spectral profile). *In Proposition 23.4, let $W \succeq 0$ have rank $r \geq 1$, and suppose the target satisfies $\operatorname{tr}(WL(x)) \leq \varepsilon$. Write $W = QDQ^*$ with $Q^*Q = I_r$ and $D \succ 0$, and set*

$$Z_0 = D^{1/2}Q^*L(x^0)QD^{1/2}, \quad d_1 \geq \dots \geq d_r > 0$$

for its eigenvalues. Then

$$d_F(x^0, x) \geq \max_{1 \leq m \leq r} \sqrt{m} \left[\log \frac{m(d_1 \dots d_m)^{1/m}}{\varepsilon} \right]_+. \quad (23.11)$$

The same bound divided by $-\log(1 - R_0)$ is a lower bound on the number of feasible forward chords of local norm at most $0 < R_0 < 1$.

Proof. Take top- m orthonormal eigenvectors U_m of Z_0 and use the injective compression $R_m = QD^{1/2}U_m$. Its reference determinant is $\prod_{j \leq m} d_j$. At the endpoint,

$$\mathrm{tr}(R_m^* L(x) R_m) \leq \mathrm{tr}(W L(x)) \leq \varepsilon, \quad \det(R_m^* L(x) R_m) \leq (\varepsilon/m)^m.$$

Apply Proposition 23.4 with $\vartheta = m$, then maximize over m . The chord statement follows from Corollary 23.2's chord estimate. $\square$

For $m = r$ this is the unweighted exposed-minor estimate. Smaller compressions can avoid a severe loss from tiny positive eigenvalues. The quantities d_i include both the exposing matrix and the reference slack; if $L(x^0) = I$, they are precisely the positive eigenvalues of W .

Here are two explicit asymptotics of the right side of (23.11), denoted $\mathcal{E}_r(\varepsilon)$. They describe that lower envelope; they are not asserted to be exact distances for an arbitrary pencil. Put $\Lambda = \log(1/\varepsilon)$. For $d_i = e^{-a(i-1)}$, $a > 0$, and any fixed $\delta > 0$, if $r \geq (2/(3a) + \delta)\Lambda$ for all sufficiently small ε , then

$$\mathcal{E}_r(\varepsilon) = \left(\frac{2}{3} \sqrt{\frac{2}{3a}} + o(1) \right) \Lambda^{3/2}. \quad (23.12)$$

For $d_i = i^{-\beta}$, $\beta > 1$, put $C_\beta = e^{(2-\beta)/(\beta-1)}$. If $r \geq (C_\beta + \delta)\varepsilon^{-1/(\beta-1)}$, then

$$\mathcal{E}_r(\varepsilon) = \left(2(\beta-1) \sqrt{C_\beta} + o(1) \right) \varepsilon^{-1/[2(\beta-1)]}. \quad (23.13)$$

To verify (23.12), maximize $\sqrt{m}[\Lambda + \log m - a(m-1)/2]_+$ over positive real m . Its positive maximizer satisfies

$$m_* = \frac{2}{3a}(\Lambda + \log m_* + a/2 + 2) = \frac{2\Lambda}{3a} + O_a(\log \Lambda).$$

The bracket there is $am_* - 2$. Rounding m_* to an integer gives (23.12). For the polynomial case, Stirling's estimate gives

$$\log \frac{m(m!)^{-\beta/m}}{\varepsilon} = \Lambda - (\beta-1) \log m + \beta + O_\beta(\log(m+1)/m).$$

The error after multiplication by $\sqrt{m}$ is bounded uniformly in $m \geq 1$. The corresponding continuous expression is maximized at $m_* = C_\beta \varepsilon^{-1/(\beta-1)}$, with bracket $2(\beta-1)$, proving (23.13). In both cases integer rounding is a lower-order error. If the rank does not contain this scale, the truncated maximum in (23.11) is the applicable bound. In particular, for fixed finite r , its leading behavior as $\varepsilon \downarrow 0$ is $\sqrt{r} \log(1/\varepsilon)$.

The orders of these lower bounds are attained by particular diagonal pencils. For each finite list d_i , take

$$L(q) = \mathrm{Diag}(q_1, \dots, q_r, 2 - q_1, \dots, 2 - q_r), \quad q \in (0, 2)^r, \quad q^0 = \mathbf{1}, \quad W = \mathrm{Diag}(d_1, \dots, d_r, 0, \dots, 0).$$

The target is $\sum_i d_i q_i \leq \varepsilon$. On $0 < q \leq 1$ define

$$\varrho(q) = \int_q^1 \sqrt{t^{-2} + (2-t)^{-2}} dt, \quad \log(1/q) \leq \varrho(q) \leq \sqrt{2} \log(1/q).$$

The coordinates $\varrho(q_i)$ turn the product metric into Euclidean metric. Set $q_i^* = \min\{1, \varepsilon/(rd_i)\}$. This endpoint has gap at most ε , and a straight segment in the ϱ coordinates gives a feasible path of length at most

$$\sqrt{2} \left(\sum_{i=1}^r [\log(rd_i/\varepsilon)]_+^2 \right)^{1/2}. \quad (23.14)$$

Choose $r = \lceil c\Lambda \rceil$ with fixed $c > 2/(3a)$ in the geometric case. Every summand in (23.14) is $O_a(\Lambda^2)$, so the upper order is $\Lambda^{3/2}$. In the polynomial case take $r = \lceil c\varepsilon^{-1/(\beta-1)} \rceil$, $c > C_\beta$. Then $\log(rd_i/\varepsilon)$ is a bounded constant minus $\beta \log(i/r)$, whose positive-part square has a finite integral on $(0, 1]$. The sum is $O_\beta(r)$, giving the matching order $\varepsilon^{-1/[2(\beta-1)]}$. Constants here may depend on the fixed c and decay parameter. These examples concern arbitrary feasible paths in specially chosen box spectrahedra, with rank growing at the specified rate; they do not supply matching central paths or matching upper bounds for all affine pencils.

24 Arbitrary barriers and the distinction between primal and dual movement

The rank-sensitive movement bounds use a specified symmetric-cone metric. A different conclusion holds uniformly over logarithmically homogeneous barriers if movement is measured in the full primal–dual metric. We give the classical geometric argument explicitly and then combine it with the formulation bounds. The dimension envelope and the objective-distribution calculations below also appear in the companion manuscript [21, Sections 5, 9, and 10]; they are included to make the different metric and endpoint contracts precise.

24.1 The feasible gap set in the product metric

Let F be a ν -logarithmically homogeneous standard self-concordant barrier on a proper cone K . Define the negative-pairing conjugate and product barrier by

$$F_*(s) = \sup_{x \in \text{int } K} \{-\langle s, x \rangle - F(x)\}, \quad \widehat{F}(x, s) = F(x) + F_*(s).$$

Consider the strictly feasible conic pair

$$\min\{\langle c, x \rangle : Ax = b, x \in K\}, \quad \max\{\langle b, y \rangle : A^T y + s = c, s \in K^*\}.$$

Write $z(\eta) = (x(\eta), s(\eta))$ for its exact centers, with $s(\eta) = -\eta^{-1}F'(x(\eta))$ and gap ν/η . Distances for $\widehat{F}$ allow all curves in the ambient product interior; requiring intermediate affine feasibility can only increase them. A dot denotes differentiation in $\lambda = \log \eta$.

Proposition 24.1 (Classical allocation of central velocity). *Put $H = F''(x)$, $g = H^{-1/2}F'(x)$, and let Π project orthogonally onto $H^{1/2} \ker A$. Then*

$$\begin{aligned} \nu_P(\eta) &:= \|\dot{x}\|_{F,x}^2 = \|\Pi g\|^2, & \nu_D(\eta) &:= \|\dot{s}\|_{F^*,s}^2 = \|(I - \Pi)g\|^2, \\ \nu_P(\eta) + \nu_D(\eta) &= \nu. \end{aligned} \tag{24.1}$$

After removing redundant equality rows,

$$\nu_P = \eta^2 c^T M c, \quad M = H^{-1} - H^{-1} A^T (A H^{-1} A^T)^{-1} A H^{-1}.$$

For $p(\eta) = \langle c, x(\eta) \rangle$ and $d(\eta) = \langle b, y(\eta) \rangle$,

$$-\eta^2 p'(\eta) = \nu_P(\eta), \quad \eta^2 d'(\eta) = \nu_D(\eta). \tag{24.2}$$

If both objectives converge to the finite optimum p^* , then

$$p(\eta) - p^* = \int_\eta^\infty \frac{\nu_P(u)}{u^2} du, \quad p^* - d(\eta) = \int_\eta^\infty \frac{\nu_D(u)}{u^2} du.$$

Proof. Homogeneity and conjugacy give $Hx = -F'(x)$, $\|g\|^2 = \nu$, and $F''(s) = \eta^2 H^{-1}$. Differentiation gives $\eta \dot{s} = F'(x) - H\dot{x}$. Thus $v = H^{1/2}\dot{x}$ and $d_0 = \eta H^{-1/2}\dot{s}$ add to g . Primal and dual feasibility put these vectors in $H^{1/2} \ker A$ and its orthogonal complement, respectively. This proves the speed formulas. Differentiating primal stationarity gives $\dot{x} = -\eta M c$; $MHM = M$ proves the displayed Schur formula and the first identity in (24.2). Differentiate $p - d = \nu/\eta$ for the second, and integrate under the stated convergence assumption. $\square$

This is the orthogonal decomposition underlying Nesterov–Todd’s constant-speed primal–dual path [72, Theorems 4.1 and 5.2], not a new general central-path identity.

Theorem 24.2 (Classical distance to a feasible gap sublevel). *Let $z_0 = z(\eta_0)$ have gap $\Delta_0 = \nu/\eta_0$, and let $\mathcal{Z}_\epsilon$ be the strictly primal–dual feasible pairs with gap at most $0 < \epsilon < \Delta_0$. Then*

$$\sqrt{\nu/2} \log(\Delta_0/\epsilon) \leq d_{\hat{F}}(z_0, \mathcal{Z}_\epsilon) \leq \sqrt{\nu} \log(\Delta_0/\epsilon). \quad (24.3)$$

Consequently, if each counted round contains at most $m \geq 1$ chords, with m an integer, of starting product–Dikin norm at most $0 < R < 1$, a trajectory from z_0 to $\mathcal{Z}_\epsilon$ requires

$$T \geq \frac{\sqrt{\nu/2} \log(\Delta_0/\epsilon)}{m \log(1/(1-R))}. \quad (24.4)$$

Intermediate chord endpoints need not satisfy the affine equations.

Proof. At $z_c = z(\eta)$, conjugacy gives $F'_*(s_c) = -\eta x_c$ and $\hat{F}(z_c) = \nu \log \eta - \nu$. For any feasible $z = (x, s)$, $\langle x - x_c, s - s_c \rangle = 0$, since the first difference belongs to $\ker A$ and the second to $\text{range } A^T$. Hence

$$D\hat{F}(z_c)[z - z_c] = -\eta\{\langle s_c, x - x_c \rangle + \langle x_c, s - s_c \rangle\} = -\eta\{\langle x, s \rangle - \nu/\eta\}.$$

At $\eta = \nu/\epsilon$, convexity shows that z_c minimizes $\hat{F}$ on $\mathcal{Z}_\epsilon$. Every target therefore has barrier increase at least $\nu \log(\Delta_0/\epsilon)$ from z_0 . The ambient dual norm of $D\hat{F}$ is exactly $\sqrt{2\nu}$; integration along any joining curve proves the lower bound. The central route has constant speed $\sqrt{\nu}$ by Proposition 24.1, proving the upper bound. For a chord $z + th$ with $\|h\|_z = r < 1$, self-concordant Hessian comparison bounds its speed by $r/(1-tr)$; integration gives length at most $-\log(1-r)$. Sum over the counted chords. $\square$

This endpoint-set argument is Nesterov–Todd’s Theorem 5.1(c), with their chord comparison and primal–dual geometry [72, Section 3 and Theorems 5.1–5.2]. An initial metric error D_0 subtracts D_0 from the numerator’s distance bound, by the triangle inequality. The theorem says nothing about unrestricted long moves, primal-only endpoints, or query counts.

24.2 A dimension envelope for arbitrary cone dictionaries

Theorem 24.3 (Dimension-only barrier charge). *Let a compact full-dimensional body $C \subset \mathbb{R}^D$, $D \geq 1$, have an exact affine lift over a product of proper cones. After operational face reduction and deletion of zero factors, suppose all factors have dimension at most an integer $d \geq 2$. Define*

$$\Psi_d(h) = 2\lfloor h/d \rfloor + \min\{h \bmod d, 2\}.$$

Every possibly coupled logarithmically homogeneous standard self-concordant barrier on this operational product has

$$\nu \geq \Psi_d(D+1) \geq 2(D+1)/d. \quad (24.5)$$

Thus the coefficient $\sqrt{\nu/2}$ in (24.4) is at least $\sqrt{\Psi_d(D+1)/2}$.

Proof. The operational ambient dimension M is at least $D + 1$, by Theorem 13.4. This also follows from the ordinary rank $D + 1$ of the body's affine slack operator after the operational reductions of Section 3. Let r_0 factors be rays and q_0 have dimension at least two. In each nonray factor a two-dimensional linear section through an interior point is a proper planar cone, hence an orthant after a linear change of coordinates. Their product with all the rays is an interior orthant section of dimension $r_0 + 2q_0$. Restriction of the barrier preserves its homogeneity degree and self-concordance, so the sharp orthant lower bound gives $\nu \geq r_0 + 2q_0$ [53, 70]. This argument permits arbitrary coupling. The cap gives $r_0 + dq_0 \geq M \geq D + 1$. For $h = ad + b$, $0 \leq b < d$, the minimum of $r + 2q$ over nonnegative integers with $r + dq \geq h$ is $2a + \min(b, 2)$: for $q \leq a$ its minimum is $h - (d - 2)q$, while $q \geq a + 1$ costs at least $2a + 2$. Finally $\min(b, 2) \geq 2b/d$. $\square$

The envelope is an exact integer consequence of these two charges; it does not assert an attainable lift frontier for every body. No definability or contact regularity is needed. For a product of balls it uses $D = \sum_a s_a$, whereas on a positively curved D -body with definable factors, Theorem 3.2 gives the usually stronger bound

$$\nu \geq 2 \left\lceil \frac{D-1}{d-2} \right\rceil \quad (d \geq 3).$$

Under the hypotheses of Theorem 6.2, the exact smooth Lorentz factor count $\sum_a h_a$ instead gives $\nu \geq 2 \sum_a h_a$ and coefficient $\sqrt{\sum_a h_a}$. For the single shared product-ball cone below, the sharp charge is $k + 1$, and the coefficient is $\sqrt{(k + 1)/2}$. These substitutions concern the ambient cone and its conjugate, not the restricted standard barrier on a fixed-scale slice.

24.3 An explicit allocation on shared product balls

For positive dimensions s_a , consider

$$\mathcal{H} = \{(t, y_1, \dots, y_k) : t \geq \|y_a\|_2 \ (1 \leq a \leq k)\}.$$

Theorem 17.3, specialized to vector rows, gives the optimal $(k + 1)$ -logarithmically homogeneous barrier

$$F(t, y) = - \sum_a \log(t^2 - \|y_a\|^2) + (k - 1) \log t. \quad (24.6)$$

Its fixed-scale restriction is $f(y) = - \sum_a \log(1 - \|y_a\|^2)$. For unit vectors u_a , weights $w_a \geq 0$, and objective $-\sum_a w_a \langle u_a, y_a \rangle$ on $t = 1$, the centers are

$$y_a(\eta) = r(\eta w_a) u_a, \quad r(z) = \frac{z}{\sqrt{1 + z^2 + 1}}. \quad (24.7)$$

Indeed $2r/(1 - r^2) = \eta w_a$. Differentiation in $\log \eta$ gives $\dot{r} = r(1 - r^2)/(1 + r^2)$; the radial Hessian is $2(1 + r^2)/(1 - r^2)^2$. Consequently

$$\nu_P(\eta) = q_{\text{eff}}(\eta) := \sum_a \left(1 - \frac{1}{\sqrt{1 + (\eta w_a)^2}} \right), \quad \nu_D(\eta) = k + 1 - q_{\text{eff}}(\eta). \quad (24.8)$$

The quantity q_{eff} counts objective directions visible at scale $1/\eta$, with a continuous transition between zero and one per factor.

With $w_1 = 1$ and all other weights zero, any interval $0 < \eta_0 < \eta_1$ satisfies

$$L_P \leq \log(\eta_1/\eta_0), \quad L_D \geq \sqrt{k} \log(\eta_1/\eta_0), \quad d_{\widehat{F}}(z(\eta_0), z(\eta_1)) \geq \sqrt{(k + 1)/2} \log(\eta_1/\eta_0).$$

From the primal analytic center $\eta = 0$ to $\eta_1 \geq 1$, the length is at most $1/\sqrt{2} + \log \eta_1$, since $q_{\text{eff}}(\eta) \leq \eta^2/2$ near zero. There is also a unique-optimizer version. For fixed $0 < \epsilon \leq 1$, set $\nu = k + 1$, $w_1 = 1$, $w_a = \epsilon^2/\nu^2$ for $a > 1$, and $\eta_1 = \nu/\epsilon$. Then all weights are positive, the terminal full gap is ϵ , and throughout $1 \leq \eta \leq \eta_1$,

$$q_{\text{eff}}(\eta) \leq 1 + \frac{\epsilon^2}{2\nu}, \quad L_P(1, \eta_1) \leq \sqrt{1 + \frac{\epsilon^2}{2\nu}} \log(\nu/\epsilon).$$

Thus even unique exposure does not imply a parameter-only primal distance bound uniform over objectives. The weights here depend on ϵ ; for any fixed instance with all $w_a > 0$, eventually $q_{\text{eff}}(\eta) \rightarrow k$. The general distinction between primal and product geometry is classical [69, 72]; the explicit specializations above are also recorded in [21, Section 9].

24.4 Accuracy and movement from the objective-weight distribution

The preceding example has a positive counterpart. Write

$$Q_j(\eta) = \sum_a \min\{(\eta w_a)^j, 1\} \quad (j = 1, 2), \quad E(\eta) = \sum_a w_a [1 - r(\eta w_a)].$$

For $z \geq 0$, direct scalar calculation gives

$$(1 - 1/\sqrt{2}) \min(z^2, 1) \leq 1 - (1 + z^2)^{-1/2} \leq \min(z^2/2, 1), \\ (2 - \sqrt{2}) \min(z, 1) \leq z[1 - r(z)] \leq \min(z, 1).$$

For the first lower bound, use $[1 - (1 + z^2)^{-1/2}]/z^2 = [\sqrt{1 + z^2}(1 + \sqrt{1 + z^2})]^{-1}$ on $(0, 1]$, and monotonicity on $[1, \infty)$. For the second, use monotonicity of $1 - r(z)$ below one and of $1 + z - \sqrt{1 + z^2}$ above one. Both lower constants are attained at $z = 1$. Summation and integration give

$$(2 - \sqrt{2})Q_1(\eta)/\eta \leq E(\eta) \leq Q_1(\eta)/\eta, \tag{24.9}$$

$$\sqrt{1 - 1/\sqrt{2}} \mathcal{J}(\eta_0, \eta_1) \leq L_P(\eta_0, \eta_1) \leq \mathcal{J}(\eta_0, \eta_1), \quad \mathcal{J} = \int_{\eta_0}^{\eta_1} \sqrt{Q_2(\eta)} \frac{d\eta}{\eta}. \tag{24.10}$$

The integral converges at $\eta_0 = 0$. After sorting the positive weights, on an interval with j weights satisfying $\eta w_a \geq 1$, write $B_j^2 = \sum_{a>j} w_a^2$. Then $Q_2 = j + \eta^2 B_j^2$ and, when $j, B_j > 0$, an antiderivative is

$$\sqrt{j + \eta^2 B_j^2} + \sqrt{j} \log \frac{\eta B_j}{\sqrt{j} + \sqrt{j + \eta^2 B_j^2}}.$$

The cases $j = 0$ and $B_j = 0$ give ηB_0 and $\sqrt{j} \log \eta$, respectively.

Proposition 24.4 (Head and tail estimates). *Assume $1 \geq w_1 \geq \dots \geq w_k \geq 0$, $0 < \epsilon \leq 1$, and $1 \leq m \leq k$. If $\sum_{a>m} w_a \leq \epsilon/2$, the center at $\eta_f = 2m/\epsilon$ is primal ϵ -accurate and*

$$L_P(0, \eta_f) \leq \sqrt{(m + \epsilon^2/4)/2} + \sqrt{2m} \log(2m/\epsilon). \tag{24.11}$$

Alternatively, if $\nu = k + 1$ and $\sum_{a>m} w_a^2 \leq \epsilon^2 m/\nu^2$, then at $\eta_g = \nu/\epsilon$ the full central gap is ϵ and

$$L_P(0, \eta_g) \leq \sqrt{(m + m\epsilon^2/\nu^2)/2} + \sqrt{3m/2} \log(\nu/\epsilon). \tag{24.12}$$

Proof. The head error is at most m/η_f , the tail error at most its weight sum. For $\eta \leq 1$, speed is at most $\eta\|w\|_2/\sqrt{2}$, contributing $\|w\|_2/\sqrt{2}$ to length. For $1 \leq \eta \leq \eta_f$ use $q_{\text{eff}} \leq Q_1 \leq m + \eta \sum_{a>m} w_a \leq 2m$, and $\|w\|_2^2 \leq m + \epsilon^2/4$. For the second claim, throughout $0 < \eta \leq \eta_g$, $q_{\text{eff}} \leq m + (\eta^2/2) \sum_{a>m} w_a^2 \leq 3m/2$; the same early-time integration and $\|w\|_2^2 \leq m + m\epsilon^2/\nu^2$ prove the result. $\square$

The second estimate bounds only the primal part of a full-gap route. Equation (24.8) assigns the complementary energy to the dual part. All tail weights can be positive in either estimate. If at least m weights are visible on an interval, (24.10) gives the matching central-arc lower bound $\sqrt{(1 - 1/\sqrt{2})m \log(\eta_1/\eta_0)}$.

For completeness, these arc estimates give an explicit bounded-move schedule. Set $\mathcal{L}(\eta) = \int_0^\eta \sqrt{q_{\text{eff}}(u)} du/u$ and choose successive centers at increments of at most $\log(1 + R)$ in $\mathcal{L}$, where $0 < R < 1$. For a nonzero objective this scalar function is strictly increasing, so the parameters are uniquely specified. The standard self-concordant distance comparison $\log(1 + \|y - x\|_x) \leq d_f(x, y)$ implies that each endpoint chord has starting Dikin norm at most R ; its distance is at most the intervening arc length [72, Section 3]. The number of chords is at most $\lceil L_P / \log(1 + R) \rceil$. Evaluation, numerical inversion of $\mathcal{L}$, reading weights, and materializing the centers have separate costs. This is a mathematical movement schedule, not a running-time guarantee for computing exact real parameters.

As a concrete scale separation, take $w_a = 2^{-(a-1)}$ and $m = \lceil \log_2(4/\epsilon) \rceil \leq k$. The first estimate gives $L_P(0, 2m/\epsilon) = O(\log^{3/2}(1/\epsilon))$ as $\epsilon \downarrow 0$. This is also the order of this central arc: on $2^{j-1} \leq \eta \leq 2^j$, $1 \leq j \leq m$,

$$j \leq Q_2(\eta) \leq j + 4/3.$$

Each interval contributes $\Theta(\sqrt{j})$, giving $\Theta(m^{3/2})$ up to 2^m ; the remaining interval to $2m/\epsilon$ contributes $O(\sqrt{m} \log m)$ because $q_{\text{eff}} \leq 2m$ there. The conclusion concerns central arclength and allows k to grow with requested accuracy. Fixed finite k eventually has all its positive weights visible. These distribution laws and their broader spectral versions are prior companion results [21, Section 5]; their role here is to prevent an ambient parameter or factor count from being substituted for the actual primal motion of a chosen formulation.

Remark 24.5 (A pointwise obstruction to a parameter-only argument). Even Dikin containment, determinant growth, and the gradient inequality alone do not supply the missing primal coefficient. In the cube at $x_\delta = (1 - \delta)\mathbf{1}$, $0 < \delta \leq 1$, put $P = \mathbf{1}\mathbf{1}^T/k$ and

$$H_\delta = \frac{2}{k\delta^2}P + \frac{2}{\delta^2}(I - P), \quad g_\delta = \frac{\mathbf{1}}{k\delta}.$$

Then $(H_\delta^{-1})_{ii} = \delta^2(1 - 1/(2k))$, so its unit ellipsoid fits inside every cube facet, while $\|\mathbf{1}\|_{H_\delta} = \sqrt{2}/\delta$, $\det H_\delta = 2^k/(k\delta^{2k})$, and $g_\delta^T H_\delta^{-1} g_\delta = 1/2$. These pointwise data are not asserted to integrate to a global barrier. The genuine product-ball examples above already disprove a uniform parameter-only primal statement; the matrices explain why these weaker pointwise tests cannot rescue it. No assertion about all barriers for uniformly nondegenerate objectives follows either way.

25 Movement in concrete affine formulations

The ambient cone count does not determine distance in the restricted barrier metric. We make this distinction explicit for grouped balls, norm trees, balance slices, and shared spectral cones. The barrier-height argument and the grouped, packed, and unweighted tree central paths below are also proved in the companion manuscript [21]. We include proofs to make the comparisons self-contained.

The new all-objective distance refinement for trees is given in the next section. Throughout, distance means the infimum of lengths of piecewise C^1 curves in the indicated open affine slice, with its displayed Hessian metric; the distance to a target set is the infimum over its points. We write $[u]_+ = \max\{u, 0\}$.

25.1 Barrier height and simultaneous slack contraction

Lemma 25.1 (Two endpoint estimates). *Let F be a barrier whose restricted gradient satisfies $\|DF\|_{F,*} \leq \sqrt{\nu}$. For every reference x^0 and target set E ,*

$$d_F(x^0, E) \geq \frac{[\inf_{y \in E} F(y) - F(x^0)]_+}{\sqrt{\nu}}. \quad (25.1)$$

If, in addition, $F = -\sum_{i=1}^H \log q_i$ for positive C^2 concave functions on a convex domain and the Hessian is positive definite, then

$$d_F(x, y) \geq \left(\sum_i \left| \log \frac{q_i(y)}{q_i(x)} \right|^2 \right)^{1/2}. \quad (25.2)$$

In particular, if $\prod_i q_i(y) \leq (A\epsilon/H)^H$ throughout E_ϵ and $\bar{q}_0 = (\prod_i q_i(x^0))^{1/H}$, then

$$d_F(x^0, E_\epsilon) \geq \sqrt{H} \left[\log \frac{H\bar{q}_0}{A\epsilon} \right]_+.$$

Proof. Integrate $|DF[\dot{x}]| \leq \sqrt{\nu} \|\dot{x}\|_F$ along a curve for the first assertion. Concavity gives

$$D^2(-\log q_i)[u, u] = \frac{Dq_i[u]^2}{q_i^2} - \frac{D^2 q_i[u, u]}{q_i} \geq (D \log q_i[u])^2.$$

Thus the map $x \mapsto (\log q_i(x))_i$ decreases lengths. Integration and Cauchy–Schwarz prove the remaining assertions. $\square$

Both estimates concern the whole accurate set, rather than the length of a selected central path. For later use we record their precise bounded-move consequence. A feasible chord of starting local norm $r < 1$ has length at most $-\log(1 - r)$, by integrating the self-concordant Hessian comparison. Hence if $\|x^0 - \bar{x}\|_{\bar{x}} \leq \rho < 1$, a distance lower bound D_ϵ from $\bar{x}$ implies

$$\sum_j -\log(1 - r_j) \geq [D_\epsilon + \log(1 - \rho)]_+. \quad (25.3)$$

In particular, for an integer $m \geq 1$ and $0 < R < 1$, at most m chords of local norm at most R per round require at least $[D_\epsilon + \log(1 - \rho)]_+ / [-m \log(1 - R)]$ rounds. For M chords, where $M \geq 1$ is an integer, some r_j is at least $1 - \exp(-[D_\epsilon + \log(1 - \rho)]_+ / M)$. These statements allow arbitrary computation between the counted moves; they impose a geometric restriction on the primal trajectory.

25.2 Grouped balls and unrestricted PSD completion

For each of k balls partition the coordinates into h nonempty groups, and use the grouped domain from (9.6):

$$q_{aG} = s_{aG} - \|w_{aG}\|^2 > 0, \quad \sum_G s_{aG} = 1, \quad F = -\sum_{a,G} \log q_{aG}.$$

Choose unit directions c_a , maximize $\ell = \sum_a \langle c_a, w_a \rangle$, and put $H = kh$. The optimum is k . A PSD packing puts these H group columns into arbitrary real matrix blocks W_j , pads shorter columns by zeros, and replaces the scalar residuals by

$$D_j = S_j - W_j^T W_j \succ 0, \quad F = - \sum_j \log \det D_j, \quad \sum_{G \text{ of } a} (S_{j(a,G)})_{GG} = 1.$$

All off-diagonal entries of S_j are free. Each column occurs once and contains precisely its assigned group coordinates. This includes the one-factor packing of Section 9.

Proposition 25.2 (Completion does not avoid simultaneous contraction). *For both formulations the analytic center has $w = 0$ and all residual eigenvalues equal to $1/h$. The central path minimizing $F - \tau\ell$ is given by*

$$D(\tau) = \sqrt{h^2 + \tau^2}, \quad q(\tau) = \frac{2}{D(\tau) + h}, \quad r(\tau) = \frac{\tau}{D(\tau) + h}, \quad (25.4)$$

with $w_a = r(\tau)c_a$ and $q_{aG} = q(\tau)$ or $D_j = q(\tau)I$. From every central reference $z(\tau_0)$,

$$d_F(z(\tau_0), \{k - \ell \leq \epsilon\}) \geq \sqrt{H} \left[\log \frac{Hq(\tau_0)}{2\epsilon} \right]_+. \quad (25.5)$$

This bound permits arbitrary motion in the completion variables.

Proof. For the grouped barrier, allocation stationarity makes all slacks of one ball equal, and leaf stationarity gives $2w_{aG}/q = \tau c_{aG}$. The root allocation identity is $hq + \|w_a\|^2 = 1$. Solving it gives (25.4). For packing, stationarity in off-diagonal entries of S_j makes D_j^{-1} diagonal; diagonal allocation stationarity equates its entries within each ball. The remaining equations are the same. Strict convexity on the stated affine coordinates gives uniqueness.

Let $\delta_a = 1 - \langle c_a, w_a \rangle > 0$. In either representation the sum of residual diagonal entries belonging to ball a is $1 - \|w_a\|^2 \leq 2\delta_a$. Hadamard's determinant inequality for packing, followed by AM–GM within and across balls, gives

$$\prod_{a,G} q_{aG} \leq (2\epsilon/H)^H, \quad \text{or} \quad \prod_j \det D_j \leq (2\epsilon/H)^H \quad \text{if} \quad \sum_a \delta_a \leq \epsilon.$$

The grouped barrier has parameter H . So does the unrestricted matrix epigraph barrier before allocation restriction: the shear calculation in the proof of Theorem 9.6 gives squared gradient norm equal to the number of residual columns, and restriction can only decrease it. At $z(\tau_0)$ the residual determinant product is $q(\tau_0)^H$. Lemma 25.1 proves the result, without restricting completion fibers along the path. $\square$

The blockwise version of this estimate can also retain more information than total barrier height. If c_j is the number of columns in block j , differentiation gives

$$D^2 F_j[(\dot{S}_j, \dot{W}_j), (\dot{S}_j, \dot{W}_j)] = \text{tr}((D_j^{-1/2} \dot{D}_j D_j^{-1/2})^2) + 2 \text{tr}(D_j^{-1} \dot{W}_j^T \dot{W}_j), \quad \dot{D}_j = \dot{S}_j - \dot{W}_j^T W_j - W_j^T \dot{W}_j.$$

Consequently every path from x to y has length at least

$$\left(\sum_j \frac{1}{c_j} \left| \log \frac{\det D_j(y)}{\det D_j(x)} \right|^2 \right)^{1/2}. \quad (25.6)$$

Indeed, trace Cauchy–Schwarz bounds each squared logarithmic derivative by c_j times its block metric, and integration gives the assertion. No off-diagonal entry is fixed in this argument.

For different group counts $h_a \geq 1$ and positive objective weights λ_a , set $H = \sum_a h_a$, $p_a = h_a/H$, and $\Delta_{\text{eff}} = \prod_a (\lambda_a/p_a)^{p_a}$. In both grouped and packed formulations, the analytic-center residual diagonal for source a is $1/h_a$, with off-diagonal residuals zero. The heterogeneous endpoint bound is

$$d_F \left(\bar{z}, \left\{ \sum_a \lambda_a (1 - \langle c_a, w_a \rangle) \leq \epsilon \right\} \right) \geq \sqrt{H} \left[\log \frac{\Delta_{\text{eff}}}{2\epsilon} \right]_+. \quad (25.7)$$

To verify it, put $e_a = \lambda_a (1 - \langle c_a, w_a \rangle)$. Hadamard and sourcewise AM–GM bound the residual determinant product by $\prod_a (2e_a/(\lambda_a h_a))^{h_a}$; its maximum for $\sum_a e_a \leq \epsilon$ occurs at $e_a = \epsilon p_a$. Compare with the center product $\prod_a h_a^{-h_a}$ and use Lemma 25.1. In particular, the one-block lift of order $s + k$ with one length- s column per ball has $H = k$, despite having unrestricted completion variables in its single PSD factor. For the one-column-per-ball Hermitian packing of (9.9), these endpoint and central-path statements also hold over $\mathbb{C}$ and $\mathbb{H}$, with arbitrary nonzero real column subspaces and their unit support directions. Use the real trace and, over $\mathbb{H}$, the Jordan determinant. Hermitian Hadamard gives the same diagonal product bound; Theorem 9.6 gives parameter k ; and stationarity in every real component of the free off-diagonal entries forces D^{-1} to be diagonal. Column stationarity in its real subspace then gives $w_a = r c_a$, as above. Thus the arc and tube formulas below apply to this Hermitian specialization with $h = 1$ as well.

25.3 Norm trees: uniform bounds and weighted exact central paths

Let T be a finite rooted tree whose internal nodes have at least two children. Its leaves index a Euclidean unit ball. Write $I(v)$ and $J(v)$ for the internal and leaf children of v , respectively, and set

$$q_v = t_v^2 - \sum_{u \in I(v)} t_u^2 - \sum_{j \in J(v)} w_j^2 > 0, \quad t_v > 0, \quad t_{\text{root}} = 1. \quad (25.8)$$

All blocks must be on the positive Lorentz sheet. Assign fixed weights $\omega_v \geq 1$ and use $F_T = -\sum_v \omega_v \log q_v$. Put $W = \sum_v \omega_v$, and let Ω_v be the sum of the weights of internal nodes in the subtree rooted at v , including v . For a unit leaf direction c , put $\alpha_v = \|c|_{\text{leaves below } v}\|$, so $\alpha_{\text{root}} = 1$.

Proposition 25.3 (Weighted central profile). *The unique minimizer of $F_T - \tau \langle c, w \rangle$ has*

$$\begin{aligned} q_v &= \omega_v q, & q &= \frac{2}{\sqrt{W^2 + \tau^2} + W}, & r &= \frac{\tau}{\sqrt{W^2 + \tau^2} + W}, \\ w &= r c, & t_v^2 &= \Omega_v q + r^2 \alpha_v^2. \end{aligned} \quad (25.9)$$

For $\tau > 0$ its squared speed per unit $\log \tau$ is

$$\left\| \frac{dz}{d \log \tau} \right\|_{F_T}^2 = W \left(1 - \frac{W}{\sqrt{W^2 + \tau^2}} \right). \quad (25.10)$$

The formulas allow zero leaf coefficients and zero subtree norms.

Proof. Internal-coordinate stationarity is $-2\omega_v t_v/q_v + 2\omega_{p(v)} t_v/q_{p(v)} = 0$. Since $t_v > 0$, all q_v/ω_v are equal. Leaf stationarity is $2w_j/q = \tau c_j$. The determinant identities telescope to $\sum_v q_v = 1 - \|w\|^2$, giving $Wq + r^2 = 1$ and the stated solution. The subtree identities determine the positive t_v uniquely. The weighted barrier is strictly convex on the linked root slice, so the stationary point is the unique minimizer. Differentiating restricted central stationarity gives $H_\tau z'(\tau) = \nabla \ell$; therefore squared logarithmic speed is $\tau^2 r'(\tau)$, which is (25.10). $\square$

For k independent trees of the same total weight W , unit objective weights, and $L_\omega = kW$, all profiles agree and their speeds add. The same expression holds for grouped and packed balls with $W = h$ and $L_\omega = H$. Since restricted central stationarity gives $\nabla F = \tau \nabla \ell$, their squared restricted gradient norm along the central path tends to H . Thus the standard grouped and packed barriers have exact parameter H , not just the upper bound used in Proposition 25.2. Define

$$\mathcal{A}(r) = \sqrt{2} \left[\sqrt{2} \operatorname{artanh} \frac{\sqrt{2}r}{\sqrt{1+r^2}} - \operatorname{artanh} \frac{r}{\sqrt{1+r^2}} \right], \quad 0 \leq r < 1. \quad (25.11)$$

The exact central length from the analytic center to profile r is

$$\sqrt{L_\omega} \mathcal{A}(r), \quad \mathcal{A}(r) = \log \frac{1}{1-r} + O(1) \quad (r \uparrow 1). \quad (25.12)$$

Indeed, $\tau = 2Wr/(1-r^2)$ transforms the length element into $\sqrt{2L_\omega} \sqrt{1+r^2} dr/(1-r^2)$; the substitution $y = r/\sqrt{1+r^2}$ gives (25.11). For $0 < \epsilon \leq k$, the central route to gap ϵ uses $r = 1 - \epsilon/k$. At $\epsilon = k$ its length is zero; larger tolerances are already met at the analytic center. An arc of length a has endpoint displacement at most $e^a - 1$ in its starting local norm: along unit metric arclength, self-concordance bounds norm transport by e^s , and integration gives this estimate. Partitioning the route into arcs of length $\log(1+R)$, where $0 < R < 1$, therefore gives at most

$$\left\lceil \frac{\sqrt{L_\omega} \mathcal{A}(1 - \epsilon/k)}{\log(1+R)} \right\rceil \quad (25.13)$$

feasible chords of starting local norm at most R .

There is also a uniform finite-accuracy estimate that retains heterogeneous block counts and objective weights.

Proposition 25.4 (Entropy scale for different trees). *For source a , let $W_a = \sum_v \omega_{av}$ and let $\lambda_a > 0$ multiply its unit support objective. Put*

$$L_\omega = \sum_a W_a, \quad p_a = W_a/L_\omega, \quad \Delta_{\text{eff}} = \prod_a (\lambda_a/p_a)^{p_a}.$$

If ν is any valid gradient parameter for the displayed restricted barrier $F = -\sum_{a,v} \omega_{av} \log q_{av}$, then from its analytic center $\bar{z}$,

$$d_F \left(\bar{z}, \left\{ \sum_a \lambda_a (1 - \langle c_a, w_a \rangle) \leq \epsilon \right\} \right) \geq \frac{L_\omega}{\sqrt{\nu}} \left[\log \frac{\Delta_{\text{eff}}}{2\epsilon} \right]_+. \quad (25.14)$$

The product-cone value $\nu = 2L_\omega$ is always valid. For unit barrier weights, write b_a for the number of internal nodes. One may use the exact parameter $\nu = \sum_a \nu_{T_a}$ from Theorem 9.1, where $\nu_{T_a} = 2b_a - 2$ if the root has only internal children and $\nu_{T_a} = 2b_a - 1$ otherwise.

Proof. At the center $q_{av}^0 = \omega_{av}/W_a$. Write $e_a = \lambda_a(1 - \langle c_a, w_a \rangle)$. Telescoping gives $\sum_v q_{av} = 1 - \|w_a\|^2 \leq 2e_a/\lambda_a$. Weighted AM–GM within a tree gives

$$\prod_v \left(\frac{q_{av}}{q_{av}^0} \right)^{\omega_{av}} \leq (2e_a/\lambda_a)^{W_a}.$$

Under $\sum_a e_a \leq \epsilon$, the product on the right is maximized at $e_a = \epsilon p_a$. Thus $F(y) - F(\bar{z}) \geq L_\omega \log(\Delta_{\text{eff}}/(2\epsilon))$. Apply Lemma 25.1. $\square$

For unit objective weights, $\Delta_{\text{eff}} = \exp(H(p))$, where $H(p) = -\sum_a p_a \log p_a$ is Shannon entropy in natural units. In general $\Delta_{\text{eff}} \leq \sum_a \lambda_a$, with equality exactly when the objective weights are proportional to W_a . For k unweighted identical trees with b internal nodes, $L = kb$, the bound is $(L/\sqrt{\nu})[\log(k/(2\epsilon))]_+$. Since $L \leq \nu < 2L$, it and (25.13) give matching $\Theta_R(\sqrt{L} \log(k/\epsilon))$ chord counts for $0 < \epsilon \leq k/4$, with constants independent of the tree. The lower estimate applies to arbitrary feasible paths. The exact leading coefficient for a fixed objective, which can be smaller than the central-path coefficient, requires the next section's analysis.

25.4 A separate control on central-path parameter changes

The endpoint bounds do not require accepted iterates to remain near a central path. When they do, one also obtains a bound on each individual parameter increment. This retains a distinct refinement of the grouped short-step analysis, and its proof applies to the tree profile as well.

Proposition 25.5 (Tube parameter increments). *Use one of the identical-source profiles above, with scalar parameter $W \geq 1$, $L_\omega = kW$, and objective optimum k . Fix $a > 0$, $0 < R < 1$, an integer $m \geq 1$, and*

$$\gamma_a = 1 + a - \sqrt{1 + a^2}, \quad b_a = \sqrt{1 - (1 + a^2)^{-1/2}}, \quad 0 \leq \rho < \gamma_a.$$

Suppose $aW \leq \tau_0 \leq \dots \leq \tau_T$, accepted points satisfy $\|x_j - z(\tau_j)\|_{z(\tau_j)} \leq \rho$, and each transition uses at most m feasible chords of starting norm at most R . Put

$$B_\rho = \frac{\rho}{(1 - \rho)^2}, \quad A_{\rho,R,m} = (1 - R)^{-m} - 1 + B_\rho(1 + (1 - R)^{-m}).$$

Then every round satisfies

$$\log \frac{\tau_{j+1}}{\tau_j} \leq \frac{A_{\rho,R,m}}{(1 - \rho)b_a \sqrt{L_\omega}}. \quad (25.15)$$

If the final objective gap is at most ϵ , and $\Delta = L_\omega/\tau_0$, then

$$T \geq \frac{(1 - \rho)b_a \sqrt{L_\omega}}{A_{\rho,R,m}} \left[\log \frac{(\gamma_a - \rho)\Delta}{\epsilon} \right]_+. \quad (25.16)$$

Proof. All covectors are restricted to the affine hull. Hessian comparison and integration from $z(\tau_j)$ to x_j give

$$\|\nabla F(x_j) - \tau_j \nabla \ell\|_{x_j,*} \leq B_\rho.$$

For one R -chord, the gradient change in its starting dual norm is at most $R/(1 - R)$; transporting a dual norm backwards across that chord costs at most $(1 - R)^{-1}$. A geometric sum over at most m chords, including both endpoint residuals, consequently gives

$$(\tau_{j+1} - \tau_j) \|\nabla \ell\|_{x_j,*} \leq A_{\rho,R,m}.$$

The exact speed formula and the tube comparison give $\tau_j \|\nabla \ell\|_{x_j,*} \geq (1 - \rho)b_a \sqrt{L_\omega}$. Use $\log u \leq u - 1$ to prove (25.15). For $u = \tau/W$, the central gap g satisfies $\tau g/L_\omega = 1 + u - \sqrt{1 + u^2} \geq \gamma_a$. The objective change from a center to a point in its ρ -tube is at most $\rho \sqrt{L_\omega}/\tau \leq \rho L_\omega/\tau$. Thus accuracy forces $\tau_T \geq (\gamma_a - \rho)L_\omega/\epsilon$. Sum (25.15) to finish. $\square$

25.5 The balance slice attains its exposed-rank scale

Return to Z in (15.1), with total Jordan rank $\rho = \sum_i r_i$, and use the restricted standard barrier $\Phi = -\sum_i \log \det X_i$. Choose the frame primitive $c = c_{11}$ and put $S = (e_1 - c, e_2, \dots, e_L)$ and $Q = \rho - 1$. The gap $g(X) = \langle S, X \rangle = 1 - \langle c, X_1 \rangle$ has unique zero $(c, 0, \dots, 0)$: positivity and vanishing pairing force every block into the corresponding complementary face, and trace one fixes its coefficient. This point is feasible because all balance moments vanish on the chosen frame.

Proposition 25.6 (Exact leading distance on the balance slice). *Let $X_i^0 = e_i/\rho$ and $g_0 = Q/\rho$. For $0 < \epsilon < g_0$,*

$$\sqrt{Q} \log \frac{g_0}{\epsilon} \leq d_\Phi(X^0, \{g \leq \epsilon\}) \leq \sqrt{Q} \log \frac{g_0}{\epsilon} + \log(\rho(1 - \epsilon)). \quad (25.17)$$

Thus its exact fixed-instance leading coefficient is $\sqrt{\rho - 1}$.

Proof. The certificate S has exposed rank Q , determinant one on its support, and exposed determinant ρ^{-Q} at X^0 . Theorem 23.1 therefore gives scale $\Delta = Q/\rho = g_0$ and the lower bound. More explicitly, the negative logarithm of the exposed determinant has dual gradient norm at most $\sqrt{Q}$ on the slice, and spectral AM–GM bounds that determinant at gap ϵ by $(\epsilon/Q)^Q$.

The central path minimizing $\Phi + \tau g$ has

$$X_1 = ac + b(e_1 - c), \quad X_i = be_i \ (i > 1), \quad a + Qb = 1, \quad \frac{1}{b} = \tau + \frac{1}{a}.$$

These are the full-slice stationary equations: its balance moments vanish, and $\nabla \Phi + \tau S$ is the total-trace normal $-e/a$. Strict convexity gives uniqueness and also proves the center assertion at $\tau = 0$. Writing $g = Qb$ gives

$$\tau = \frac{Q - (Q + 1)g}{g(1 - g)}, \quad g(0) = g_0, \quad g(\tau) \sim Q/\tau.$$

The frame-diagonal metric has central length

$$\int_\epsilon^{g_0} \sqrt{\frac{Q}{g^2} + \frac{1}{(1 - g)^2}} dg.$$

Bound its integrand above by $\sqrt{Q}/g + 1/(1 - g)$ to obtain the upper bound in (25.17). $\square$

Theorem 15.1 independently proves that every barrier on this body has parameter at least $\rho - 1$. The metric statement above concerns the displayed standard barrier. Equality of these two numerical quantities does not prove the same distance bound for every optimal coupled barrier.

25.6 Shared spectral cones preserve the reduced movement

Use the real or complex matrix balls and orientations of Section 17, with $r_a \leq c_a$, and put $R_0 = \sum_a r_a$. Their common reduced barrier is

$$\Phi(X) = -\sum_a \log \det(I_{r_a} - X_a X_a^*).$$

Let $C_a = (I_{r_a} \ 0)$, and maximize $\ell(X) = \sum_a \operatorname{Re} \operatorname{tr}(C_a^* X_a)$, whose optimum is R_0 .

Proposition 25.7 (Grouping-independent spectral distance). *For every shared-cone grouping and its fixed-identity chordal block-star realization,*

$$d_\Phi(0, \{R_0 - \ell \leq \epsilon\}) \geq \sqrt{R_0} \left[\log \frac{R_0}{2\epsilon} \right]_+. \quad (25.18)$$

The exact central path is $X_a(\tau) = r(\tau)C_a$, where $r(\tau) = \tau/(\sqrt{1+\tau^2}+1)$. Its length to gap $0 < \epsilon < R_0$ is $\sqrt{R_0}\mathcal{A}(1 - \epsilon/R_0)$. Consequently the exact fixed-instance leading distance coefficient is $\sqrt{R_0}$.

Proof. Let $\sigma_{ai} < 1$ be the singular values of X_a . Von Neumann's trace inequality gives $\sum_{a,i} (1 - \sigma_{ai}) \leq R_0 - \ell(X)$. Since $1 + \sigma_{ai} \leq 2$, AM–GM gives

$$\prod_a \det(I - X_a X_a^*) \leq (2\epsilon/R_0)^{R_0} \quad \text{when } R_0 - \ell(X) \leq \epsilon.$$

Theorem 17.3 gives the restricted parameter R_0 , and $\Phi(0) = 0$; apply Lemma 25.1. The barrier gradient at rC_a is $2rC_a/(1 - r^2) = \tau C_a$, proving full stationary optimality. Along the path the barrier is $-R_0 \log(1 - r^2)$, and its length is exactly the scalar arc already computed. The two bounds have the same leading coefficient. Finally, Proposition 17.5 identifies the restricted barrier literally with Φ , so the metric and every displayed distance and path formula are identical in those formulations. $\square$

Vector rows recover products of Euclidean balls. Maximal sharing can therefore collapse the cone count without changing this reduced path geometry. Equations (25.3) and (25.18) give the same bounded-chord lower bound for all these descriptions. This conclusion uses their common barrier after affine elimination; it supplies neither a query lower bound nor a cost for evaluating the ambient Newton system.

26 Exact distance coefficients for weighted norm trees

The finite-accuracy estimate above is uniform over objectives. For a fixed objective, one can determine the leading distance coefficient more precisely. An internal subtree on which the objective vanishes contributes half its barrier weight to the squared coefficient. This contribution is invisible to the rank of a nonzero exposing certificate, because the corresponding dual block is zero. It nevertheless cannot be avoided: the subtree must shrink toward the vertex of its Lorentz cone.

We use the norm-tree domain Ω_T of Section 9. Write V for its internal nodes, $b = |V|$, and w for its vector of leaf coordinates. Fix weights $\omega_v \geq 1$ and the displayed barrier

$$F_\omega = - \sum_{v \in V} \omega_v \log q_v, \quad W = \sum_{v \in V} \omega_v.$$

It is a self-concordant barrier with positive definite restricted Hessian. All distances in this section refer to this Hessian, with the root fixed to one and all linking equations imposed. Fix a unit vector c in the leaf space, put $g(w) = 1 - c^T w$, and define

$$\alpha_v = \|c\|_{\text{leaves below } v}, \quad V_+ = \{v : \alpha_v > 0\}, \quad V_0 = V \setminus V_+, \quad K_c = \sum_{v \in V_+} \omega_v + \frac{1}{2} \sum_{v \in V_0} \omega_v.$$

The root belongs to V_+ ; the set V_0 is a union of complete subtrees.

Theorem 26.1 (Weighted norm-tree distance). *For every fixed $x_0 \in \Omega_T$, as $\epsilon \downarrow 0$,*

$$d_{F_\omega}(x_0, \{x \in \Omega_T : g(w) \leq \epsilon\}) = \sqrt{K_c} \log(1/\epsilon) + O(\log \log(1/\epsilon)). \quad (26.1)$$

More precisely, the lower bound has an $O(1)$ remainder; the displayed $O(\log \log(1/\epsilon))$ is needed only for our upper construction. The constants may depend on the fixed tree, weights, objective and reference.

Proof. We first construct a potential that bounds every feasible path. For the primal Lorentz block $x_v = (t_v, y_v)$, define the Euclidean dual vector

$$s_v = (\alpha_v, -(\alpha_u)_{u \in I(v)}, -(c_j)_{j \in J(v)}).$$

Here $I(v)$ and $J(v)$ are the internal-child and leaf-child sets. The identity $\alpha_v^2 = \sum_{u \in I(v)} \alpha_u^2 + \sum_{j \in J(v)} c_j^2$ says that s_v is a nonzero boundary dual vector for $v \in V_+$, and is zero otherwise. Consequently

$$\beta_v = \langle s_v, x_v \rangle > 0 \quad (v \in V_+), \quad \sum_{v \in V_+} \beta_v = 1 - c^T w = g(w).$$

The sum telescopes, including all internal links. Thus each $\beta_v \leq g$.

There is also a uniform bound $q_z \leq C_c g$ for every inactive node $z \in V_0$. To see this, first let $u \in V_0$ have an active parent v . Write $e_v = ((\alpha_u)_{u \in I(v)}, (c_j)_{j \in J(v)})/\alpha_v$, which is a unit vector. The coordinate occupied by t_u is orthogonal to e_v . Since $\|y_v\| < t_v$ and $t_v \leq 1$,

$$t_u^2 \leq \|y_v - (e_v^T y_v) e_v\|^2 \leq t_v^2 - (e_v^T y_v)^2 \leq 2t_v(t_v - e_v^T y_v) = \frac{2t_v \beta_v}{\alpha_v} \leq \frac{2g}{\alpha_v}.$$

Every descendant axis is at most t_u , so its determinant is at most t_u^2 . Taking the maximum of $2/\alpha_v$ over the finitely many active parents of inactive subtrees proves the assertion.

For clarity, the needed dual-norm identities can be checked without Jordan normalization. For $f(x) = -\log(x^T J x)$ on a Lorentz cone, $J = \text{Diag}(1, -I)$ and $q = x^T J x$, direct inversion gives

$$(D^2 f(x))^{-1} = x x^T - \frac{q}{2} J.$$

If $s \neq 0$ is a boundary dual vector, then $s^T J s = 0$ and $\|D \log(s^T x)\|_{(D^2 f)^{-1}}^2 = 1$. Logarithmic homogeneity likewise gives $\|D f\|_{(D^2 f)^{-1}}^2 = 2$. It follows that the potential

$$\Psi(x) = - \sum_{v \in V_+} \omega_v \log \beta_v - \frac{1}{2} \sum_{v \in V_0} \omega_v \log q_v$$

has squared dual gradient norm exactly K_c in the ambient product metric $D^2 F_\omega$. Restriction to the linked root slice can only decrease this norm: its value is the supremum over fewer tangent directions. At every point with $g \leq \epsilon$, the preceding determinant and pairing bounds imply

$$\Psi(x) - \Psi(x_0) \geq K_c \log(1/\epsilon) - O(1).$$

Integrating $|D\Psi[h]| \leq \sqrt{K_c} \|h\|_{F_\omega, x}$ along an arbitrary path proves the claimed lower bound.

For the upper bound put $A = |V_+|$, $Z = |V_0|$, and, for large λ , set $h = e^{-\lambda}$ and prescribe

$$q_v = \begin{cases} h, & v \in V_+, \\ h/\lambda^2, & v \in V_0, \end{cases} \quad r^2 = 1 - Ah - Zh/\lambda^2, \quad w = rc, \quad t_v^2 = r^2 \alpha_v^2 + \sum_{z \in V \text{ below } v} q_z. \quad (26.2)$$

The last sum includes v . Choose the positive square roots. The subtree identities give precisely the prescribed determinants, and the root equation gives $t_{\text{root}} = 1$. Thus this is an interior feasible curve. Its gap is $1 - r = Ah/2 + O(h/\lambda^2 + h^2)$, since $A \geq 1$.

A dot will denote differentiation with respect to λ . The Lorentz metric has the identity

$$D^2 f(x)[\dot{x}, \dot{x}] = \left(\frac{\dot{q}}{q}\right)^2 - \frac{2(\dot{t}^2 - \|\dot{y}\|^2)}{q}. \quad (26.3)$$

At an active node, t_v tends to $\alpha_v > 0$ and $\dot{t}_v = O(h)$. The same holds for active child axes and for leaf-coordinate derivatives; inactive child derivatives are $O(\sqrt{h}/\lambda)$. Since $q_v = h$, this block contributes $1 + O(h + \lambda^{-2})$ to the unweighted squared speed. At an inactive node, all its leaf coordinates vanish and all its descendants are inactive. Its whole block is a fixed interior Lorentz vector multiplied by $\sqrt{h}/\lambda$. Its squared speed is therefore exactly $2(1/2 + 1/\lambda)^2$. Adding the weighted contributions gives

$$\|\dot{x}\|_{F_{\omega},x}^2 = K_c + O(1/\lambda).$$

Starting the curve at any sufficiently large fixed λ_0 , its length to λ is $\sqrt{K_c}\lambda + O(\log \lambda)$. Connect x_0 to the starting point by an interior line segment, whose length is finite. Along the curve, $\lambda = \log(1/g) + O(1)$, which proves the upper bound. $\square$

Corollary 26.2 (Central-path excess and independent products). *For the weighted tree, the central route from its analytic center to gap ϵ has length $\sqrt{W} \log(1/\epsilon) + O(1)$. Its ratio to the shortest distance to that target tends to $\sqrt{W/K_c}$. It is asymptotically optimal in leading coefficient exactly when every internal subtree has nonzero objective support.*

For independent trees and the objective $\sum_a \lambda_a c_a^T w_a$, with fixed $\lambda_a > 0$ and unit c_a , the squared distance coefficient is $\sum_a K_{c_a}$. Independent factors with zero objective weight contribute zero.

Proof. The weighted central-path formulas of the preceding section give the first length. The ratio follows from Theorem 26.1; $K_c = W$ holds exactly when V_0 is empty. For products, sum the lower potentials. Each individual gap is at most ϵ/λ_a , which changes only the fixed remainder, and the squared dual norms add. For the upper bound synchronize the parameter λ in (26.2) across all positively weighted trees. The total objective gap is a fixed positive constant times $e^{-\lambda}(1 + o(1))$, and squared speeds add. Keep the remaining independent factors at their reference points. $\square$

For example, take a tree with two internal nodes, with the objective supported entirely on a root leaf. For unit weights, the root is active and its internal child is inactive. The shortest-distance coefficient is $\sqrt{3/2}$, whereas the central coefficient is $\sqrt{2}$. If the objective also has nonzero support in the child subtree, both coefficients are $\sqrt{2}$. Thus the coefficient need not vary continuously as a component of the objective tends to zero: the asymptotic here holds with the objective fixed before accuracy tends to zero.

The classical study of barrier metrics and central-path distances [69, 72], and the uniform norm-tree estimates in Central-path companion [21], provide the context for this calculation. The specific result here is the explicit weighted coefficient K_c , including objective-free subtrees, for this fixed family of linked Lorentz barriers. It does not identify the best barrier on the tree domain, nor assert an iteration lower bound without a restriction on the metric distance traversed per accepted step.

Part VI

Resource ledgers and computational contracts

27 Exact resource ledgers and their optimization

Cone dimension, ambient barrier parameter, restricted barrier parameter, and free affine coordinates measure different aspects of a formulation. This section records exact integer tradeoffs. Its optimization statements concern the indicated resource models; none by itself bounds running time. The lifts and their slack factorizations are those of Sections 5–9. Their interpretation uses the factorization framework of Gouveia et al. [43]; the balancing and integer optimization arguments below are elementary consequences of the curvature and nullity bounds proved here.

27.1 Real PSD order caps

Let $N \geq 4$ and $R \geq 2$ be integers, and consider exact lifts of $\mathbb{B}_2^N$ through real PSD blocks of orders $2 \leq r_i \leq R$, with globally bi- C^1 full-slack contact selections. Put

$$c_r = \lfloor r^2/4 \rfloor, \quad d_r = r(r+1)/2, \quad M = \sum_i d_{r_i}, \quad \rho = \sum_i r_i.$$

Optional ray factors add resources and no curvature, so they may be omitted from the lower ledgers. The ambient product cone has optimal barrier parameter ρ , including among coupled barriers. This does not identify the parameter after its affine equations are imposed. By Corollary 7.6,

$$\sum_i c_{r_i} \geq N. \quad (27.1)$$

Indeed the local lower bound is $N - 1$, and equality would require the single matching spin factor Q_{N+1} , unavailable among these real PSD cones when $N \geq 4$.

Proposition 27.1 (Exact integer consequences). *Define $K_R(N), V_R(N), F_R(N)$ as the minima of, respectively, $\sum_r a_r$, $\sum_r r a_r$, and $\sum_r d_r a_r$ over nonnegative integers $a_2, \dots, a_R$ satisfying $\sum_r c_r a_r \geq N$. Every lift in the stated class satisfies*

$$L \geq K_R(N), \quad \rho \geq V_R(N), \quad M \geq F_R(N).$$

Writing $N = qc_R + t$, $0 \leq t < c_R$, gives

$$K_R(N) = \lceil N/c_R \rceil, \quad (27.2)$$

$$V_R(N) = qR + \begin{cases} 0, & t = 0, \\ \lceil 2\sqrt{t} \rceil, & t > 0. \end{cases} \quad (27.3)$$

$$F_R(0) = 0, \quad F_R(n) = \min_{2 \leq r \leq R} \{d_r + F_R((n - c_r)_+)\}. \quad (27.4)$$

In particular,

$$\begin{aligned} \rho &\geq \max\{\lceil RN/c_R \rceil, \lceil 2\sqrt{N} \rceil\}, \\ M &\geq \max\{\lceil (2 + 1/\lfloor R/2 \rfloor)N \rceil, 2N + \lceil \sqrt{N} \rceil\}. \end{aligned} \quad (27.5)$$

Two explicit cases are

$$F_4(N) = 10\lfloor N/4 \rfloor + 3(N \bmod 4), \quad F_5(N) = \lceil 5N/2 \rceil \quad (N \geq 4).$$

These are exact integer ledgers. For $R \geq 4$, they are not asserted to be attainable lift frontiers.

Proof. The count formula is immediate. Extend c_r to nonnegative integer r . Its increments $c_r - c_{r-1} = \lfloor r/2 \rfloor$ are nondecreasing. Moving one unit from a smaller positive order to a larger order below R therefore increases or preserves capacity. Repeating, a list of total order $S = \ell R + j$, $0 \leq j < R$, has capacity at most $\ell c_R + c_j$. This relaxation permits zero and one as orders, which only strengthens its use as an upper bound on capacity. The smallest order with $c_r \geq t > 0$ is $\lceil 2\sqrt{t} \rceil$. Maximal blocks and this last block attain (27.3); the capacity bound just proved excludes every smaller total order. Selecting the first block of an optimal multiset proves both inequalities in (27.4).

For $r = 2j$ or $2j + 1$ one has $d_r/c_r = 2 + 1/j$, whereas c_r/r increases with r . These give the cap-dependent bounds in (27.5). Also $N \leq \sum_i r_i^2/4 \leq \rho^2/4$ and $d_r = 2c_r + \lceil r/2 \rceil$, proving the other two bounds. For $R = 4$, orders two and three cost three per unit of capacity; order four costs ten for four units. Using the maximal possible number of complete four-unit blocks and paying three per remaining unit is optimal; overshooting by another four-unit block costs ten rather than at most nine. For $R = 5$, the ratio lower bound is $5/2$. Every even integer at least four is a nonnegative combination of four and six, realized by orders four and five at that ratio. For odd $N \geq 5$, add one order-two block to a realization of $N - 1$. This attains the ceiling. $\square$

27.2 Grouped Schur constructions and small-order exact frontiers

Partition $[N]$ into nonempty groups G_j of sizes $g_j \leq R - 1$. The formulation

$$\begin{pmatrix} s_j & x_{G_j}^T \\ x_{G_j} & I_{g_j} \end{pmatrix} \succeq 0, \quad \sum_{j=1}^k s_j = 1 \quad (27.6)$$

projects exactly onto $\mathbb{B}_2^N$. At a boundary point its allocation is $s_j = \|x_{G_j}\|^2$. Polynomial contact factors are

$$A_j(x) = \begin{pmatrix} \|x_{G_j}\|^2 & x_{G_j}^T \\ x_{G_j} & I \end{pmatrix}, \quad B_j(y) = \frac{1}{2}(1, -y_{G_j})^T(1, -y_{G_j}),$$

since $\sum_j \text{tr}(A_j(x)B_j(y)) = \frac{1}{2}\|x - y\|^2 = 1 - x^T y$ on the unit sphere. Thus this construction meets the global hypotheses. For a prescribed $\lceil N/(R - 1) \rceil \leq k \leq N$, write $N = ak + t$, $0 \leq t < k$. Balancing the group sizes gives the exact minimum ambient dimension within this construction:

$$(L, \rho, M, \nu_{\text{slice}}) = (k, N + k, (k - t)d_{a+1} + td_{a+2}, k). \quad (27.7)$$

Here $\nu_{\text{slice}} = k$ is even the intrinsic parameter of this fixed affine domain, by Theorem 9.4. Strict convexity of d_{g+1} proves balancing. The minimum count is $\lceil N/(R - 1) \rceil$, whereas the minimum dimension is $3N$:

$$d_{g+1} - 3g = (g - 1)(g - 2)/2 \geq 0,$$

with equality for groups of size one or two. Public identity elimination leaves $N + k$ variables and one equality. In the unreduced cone encoding there are $1 + \sum_j d_{g_j}$ affine equations, giving the same affine dimension $N + k - 1$.

The lower ledgers and this construction prove the following simultaneous exact ambient triples in the global contact-selection class:

real PSD dictionary	$L_{\min}$	$M_{\min}$	$\rho_{\min}$
$r \leq 2$	N	$3N$	$2N$
$r \leq 3$	$\lceil N/2 \rceil$	$3N$	$\lceil 3N/2 \rceil$.

(27.8)

For the fixed complex order-two dictionary, the corresponding triple is

$$(\lceil N/2 \rceil, 4\lceil N/2 \rceil, 2\lceil N/2 \rceil) \quad (N \geq 4); \quad (27.9)$$

for the fixed quaternionic order-two dictionary it is

$$(\lceil N/4 \rceil, 6\lceil N/4 \rceil, 2\lceil N/4 \rceil) \quad (N \geq 6).$$

The direct exceptions are $(1, 4, 2)$ for complex $N = 3$ and $(1, 6, 2)$ for quaternionic $N = 5$. Indeed these cones are Q_4, Q_6 , with capacities two and four; the strict capacity increment and fixed-tail construction of Section 7 give the claims. Dimensions here count full dictionary cones even for the final partially filled coordinate group.

27.3 All finite column-packing tradeoffs

Let $C = (\mathbb{B}_2^s)^b$, where $b \geq 2$ and $s \geq 3$ are integers. Choose an integer $1 \leq p \leq s$ and split each source into $h = \lceil s/p \rceil$ nonempty groups of at most p coordinates. Pad columns by zeros and pack the $H = bh$ columns into ℓ nonempty blocks. Block j has a $p \times c_j$ matrix W_j and a free symmetric matrix S_j :

$$\begin{pmatrix} S_j & W_j^T \\ W_j & I_p \end{pmatrix} \succeq 0, \quad \sum_{\gamma \text{ belonging to } a} (S_{j(\gamma)})_{\gamma\gamma} = 1 \quad (a = 1, \dots, b). \quad (27.10)$$

Every column belongs to one source group and appears once.

Proposition 27.2 (Exact finite packing ledger). *Write $H = q\ell + t$, $0 \leq t < \ell$. For fixed p, ℓ , balanced column counts minimize maximum order, ambient cone dimension M , and free coordinate count V . The resulting ledger is*

$$\begin{aligned} L &= \ell, & r_{\max} &= p + \lceil H/\ell \rceil, & \rho &= p\ell + H, \\ M &= (\ell - t)d_{p+q} + td_{p+q+1}, & V &= bs + (\ell - t)d_q + td_{q+1}, & \nu_{\text{slice}} &= H. \end{aligned} \quad (27.11)$$

The last equality is the exact parameter of the restricted standard barrier; in fact the fixed affine domain has intrinsic parameter H . The count V precedes the b independent allocation equations. Under an order cap R , all undominated resource vectors of this fixed-width family are obtained by enumerating

$$1 \leq p \leq \min(s, R - 1), \quad \lceil H/(R - p) \rceil \leq \ell \leq H \quad (27.12)$$

and removing dominated vectors.

Proof. Schur complementation gives $D_j = S_j - W_j^T W_j \succeq 0$. The allocation identities imply each source has squared norm at most one. Conversely any source points in the balls admit diagonal residuals with the required nonnegative sums. This proves exactness. Moving a column from a block with at least two more columns to a smaller block decreases both d_{p+c} and d_c sums, and cannot increase maximum order. This proves balancing and all counts except the barrier equality.

The barrier $-\sum_j \log \det D_j$ has parameter at most $\sum_j c_j = H$ by the matrix-epigraph calculation in Theorem 9.6. Project the closed slice onto all W_j and all diagonal entries of S_j . Its image is exactly the grouped domain of Theorem 9.4, with H nonempty groups. Fibers are bounded: $|D_{uv}|^2 \leq D_{uu}D_{vv}$ and each diagonal residual is at most one. There is a strictly feasible point at $W = 0$ and positive allocation to every column. Lemma 9.5 therefore gives an intrinsic lower bound H , proving equality. Finally, balanced maximum order is at most R exactly in (27.12); widths $p > s$ only add public padding and resources. $\square$

For example the exact minimum count and ambient rank in this family are

$$\min_p \left\lceil \frac{H_p}{R-p} \right\rceil, \quad \min_p \left\{ H_p + p \left\lceil \frac{H_p}{R-p} \right\rceil \right\}, \quad H_p = b \lceil s/p \rceil.$$

Minimizing M or a proxy such as $M\sqrt{H_p}$ requires the same finite enumeration over ℓ ; substituting a balanced continuous aspect ratio would discard the final-block overshoot. A different answer results if only free affine coordinates are charged. For every $\theta \geq 0$, put $p_* = \min(s, R-1)$ and $H_* = b \lceil s/p_* \rceil$. Then

$$\min V \nu_{\text{slice}}^\theta = (bs + H_*)H_*^\theta. \quad (27.13)$$

Indeed splitting a column block of sizes $c_1 + c_2$ saves $c_1 c_2$ free coordinates, so for fixed p the unique optimal column allocation is $\ell = H_p$. The function $(bs + H)H^\theta$ increases with $H > 0$. Thus the largest feasible p is optimal; smaller widths on the same ceiling plateau tie it. These are the unshared grouped Schur blocks.

Without a cap, direct Lorentz lifting has $(L, \rho, M, \nu_{\text{slice}}) = (b, 2b, b(s+1), b)$. Every packing instead obeys

$$M \geq (2p+1)H_p \geq b(2s+1), \quad H_p \geq b, \quad (27.14)$$

since $d_{p+c} = (2p+1)c + (p-c)(p-c+1)/2$ and the last term is nonnegative for integer $p-c$. Thus the direct Lorentz formulation strictly improves cone dimension and weakly improves the restricted parameter; $p = s$ ties the latter. Nevertheless one PSD factor is possible with minimum order within this family

$$r_{\text{one}} = \min_{1 \leq p \leq s} \{p + b \lceil s/p \rceil\}, \quad r_{\text{one}} < 2b \iff b > s.$$

If $\lceil s/p \rceil \geq 2$, the order exceeds $2b$; otherwise $p = s$ and the order is $s+b$. This proves the equivalence.

Under the same raw order cap $3 \leq R < s+1$, grouped Lorentz blocks use $h_L = \lceil s/(R-2) \rceil$ groups per source and have

$$(L, \rho, M, \nu_{\text{slice}}, V) = (bh_L, 2bh_L, bs + 2bh_L, bh_L, bs + bh_L).$$

Grouped PSD Schur blocks instead use $h_S = \lceil s/(R-1) \rceil$, with $(L, \rho, \nu_{\text{slice}}, V) = (bh_S, bs + bh_S, bh_S, bs + bh_S)$ and balanced dimension from (27.7), summed over sources. They weakly improve the free-coordinate and restricted-parameter counts, although a PSD cone of order R contains d_R scalar coordinates and a Lorentz cone of dimension R only R . Under a matched scalar-coordinate cap D , PSD order is only about $\sqrt{2D}$. In the regime where source and last-block effects are negligible, grouped Lorentz has $(L, \nu_{\text{slice}}, \rho, M) \sim (bs/D, bs/D, 2bs/D, bs)$, whereas balanced PSD packing has $(2bs/D, \sqrt{2}bs/\sqrt{D}, 2\sqrt{2}bs/\sqrt{D}, 2bs)$. Cross-source sharing, rather than scalar-coordinate economy, is the packing construction's distinctive advantage in this comparison.

27.4 Joint nullity envelopes and a general resource inequality

Return to globally bi- C^1 full product-slack factorizations over L real PSD blocks of orders at most R . Write $n = b(s-1)$ and let $\nu = \nu_{\text{std}, \text{slice}}$. At a simultaneous contact let $c_i = \text{nullity } X_i$. The mixed derivative has rank n , so

$$n \leq \sum_i c_i(r_i - c_i), \quad \sum_i c_i \leq \nu. \quad (27.15)$$

The second inequality is Lemma 8.1. For integers $z \geq 0$ define the largest capacity allowed by a nullity budget as

$$\mathcal{C}_R(L, z) = \max \left\{ \sum_i c_i(R - c_i) : 0 \leq c_i \leq R, c_i \in \mathbb{Z}, \sum_i c_i \leq z \right\}.$$

Put $m = \lfloor R/2 \rfloor$, $z' = \min(z, Lm)$ and $z' = qL + t$, $0 \leq t < L$. Concavity and balancing give the exact expression

$$\mathcal{C}_R(L, z) = (L - t)q(R - q) + t(q + 1)(R - q - 1). \quad (27.16)$$

For completeness, replacing $c_i > m$ by $R - c_i$ preserves capacity and reduces its budget; below m all marginal increments are positive. Using the budget up to Lm and balancing therefore gives precisely (27.16). Together with the sourcewise selected-rank bound of Theorem 7.8, this proves the coupled necessary conditions

$$\lfloor \nu \rfloor \geq b \lceil s/(R - 1) \rceil, \quad \mathcal{C}_R(L, \lfloor \nu \rfloor) \geq n. \quad (27.17)$$

The first bound is attained by grouped Schur lifts. Neither condition asserts realizability of every integer point. For an actual nullity total $z \leq LR/2$, Cauchy–Schwarz also gives $n \leq Rz - z^2/L$. Here $n > 0$ forces $z > 0$ and $Rz - n \geq z^2/L > 0$, so $L \geq z^2/(Rz - n)$.

Theorem 27.3 (Resource product bound). *Let $J_\omega = \sum_i r_i^\omega$. Under the preceding contact and standard-barrier hypotheses, for $\theta \geq 0$ and $0 \leq \omega \leq 2 + \theta$,*

$$J_\omega \nu^\theta \geq C_\theta \frac{n^{1+\theta}}{R^{2+\theta-\omega}}, \quad C_\theta = \frac{(2 + \theta)^{2+\theta}}{(1 + \theta)^{1+\theta}}. \quad (27.18)$$

The equality aspect ratio in its continuous capacity relaxation is $p : c = (1 + \theta) : 1$, where $p = r - c$.

Proof. For $c_i > 0$, write $p_i = r_i - c_i = x_i r_i$. Maximizing $x(1 - x)^{1/(1+\theta)}$ on $[0, 1]$ gives

$$p_i c_i \leq K_\theta R^{(2+\theta-\omega)/(1+\theta)} r_i^{\omega/(1+\theta)} c_i^{\theta/(1+\theta)}, \quad K_\theta = \frac{1 + \theta}{2 + \theta} (2 + \theta)^{-1/(1+\theta)}.$$

Zero-capacity terms can be discarded. For $\theta > 0$, sum and apply Hölder with exponents $1 + \theta$ and $(1 + \theta)/\theta$. Use (27.15) and $\sum c_i \leq \nu$, then raise to power $1 + \theta$. Since $K_\theta^{-(1+\theta)} = C_\theta$, the result follows. At $\theta = 0$ it is simply the sum of $p_i c_i \leq R^{2-\omega} r_i^\omega / 4$. The shape maximum is unique at $x = (1 + \theta)/(2 + \theta)$. For $\omega < 2 + \theta$ equality also requires full order R in every contributing block; at the endpoint scale is neutral. $\square$

For $\theta = 1/2$ write $C_{1/2} = (25/6)\sqrt{5/3}$. Since $M \geq (1 + 1/R)J_2/2$, the theorem gives

$$M\sqrt{\nu} \geq \frac{25}{12} \sqrt{\frac{5}{3}} (1 + 1/R) \frac{n^{3/2}}{\sqrt{R}}. \quad (27.19)$$

Divisible packings with $p + c = R$, $p : c = (1 + \theta) : 1$ and $H = bs/p$, $L = H/c$ attain (27.18) with bs in place of $b(s - 1)$. Their ratio to the lower bound is $(s/(s - 1))^{1+\theta}$. The cone-coordinate comparison has exactly the same ratio because all blocks have full order. Thus the relaxation is asymptotically attained along such divisible families as $s \rightarrow \infty$. When $b \lceil s/(R - 1) \rceil$ constrains the optimizer, or when divisibility fails, use (27.17) and the finite enumeration instead.

The same aspect ratio follows directly by optimizing uniform full-order packing: $J_\omega H^\theta = (bs)^{1+\theta} R^\omega / (p^{1+\theta} c)$. For static resources $\theta = 0$ it is $1 : 1$; for the common proxy multiplier $\sqrt{\nu}$ it is

3 : 2. If $\omega > 2 + \theta$, larger scale increases this proxy. The upper cap need not bind, and there is no universal aspect ratio once source-width, lower-size, and integrality constraints intervene. In particular the 3 : 2 conclusion does not automatically apply to dense cubic algebra.

There is a different optimum even within unshared PSD formulations. For divisible Schur groups of common width g and $A = bs$,

$$L = \nu = A/g, \quad \rho = A(1 + 1/g), \quad M\sqrt{\nu} = A^{3/2} \frac{(g+1)(g+2)}{2g^{3/2}}.$$

The first two resources times $\sqrt{\nu}$ decrease with g , whereas the last has its unique integer minimum at $g = 4$, if available, with value $15A^{3/2}/8$. To verify this, consecutive values decrease exactly when $(g+1)^5 - g^3(g+3)^2 = -g^4 + g^3 + 10g^2 + 5g + 1 \geq 0$; this holds for $g = 1, 2, 3$ and fails thereafter. The polynomial is negative at four and has negative derivative thereafter. Heterogeneous groups cannot improve the value when $4 \mid s$: at fixed mean width u , convexity balances widths between two neighboring integers. The interpolated cost is $au + b$ with $a > 0$ and $b \leq 0$, so the derivative of $(au + b)u^{-3/2}$ changes sign, if at all, from positive to negative. There is no interior minimum. Its minimum is therefore at an integer width, where the preceding comparison applies. If four is unavailable, the largest allowed width below four is optimal in the divisible model; all remaining cases are handled by the exact integer ledger.

28 When geometric movement composes with computational work

The resource inequalities in Section 27 become work lower bounds only under an explicit cost premise. The point requiring proof is that the rank paying for curvature is also the rank paying for movement. A larger ambient or restricted barrier parameter cannot be substituted for that rank in a movement lower bound.

28.1 The same aggregate certificate in both estimates

Consider the full rows of $C = \prod_{a=1}^b \mathbb{B}_2^{s_a}$, with $s_a \geq 2$, and globally bi- C^1 selections as in (5.2). Suppose each dual row is a genuine certificate on the whole feasible affine slice. Choose positive weights λ_a and a simultaneous contact u . The objective and its certificate are

$$\ell(x) = \sum_a \lambda_a u_a^T x_a, \quad \ell_* = \sum_a \lambda_a, \quad S_i = \sum_a \lambda_a Y_i^a(u_a).$$

Write $n = \sum_a (s_a - 1)$, $p_i = \text{rank}_J X_i(u)$, $q_i = \text{rank}_J S_i$ and $Q = \sum_i q_i$. For simple factors with rank r_i and Peirce constant a_i , the mixed-channel bound gives

$$n \leq \sum_{i \text{ non-ray}} a_i p_i q_i, \quad p_i + q_i \leq r_i. \quad (28.1)$$

To see this directly, the weighted slack has mixed derivative $-\sum_a \lambda_a \langle h_a, k_a \rangle$, a nondegenerate form on an n -dimensional tangent space. At contact, positivity makes $X_i(u)$ and S_i complementary. The differentiated mixed pairing factors through the cross Peirce space between their supports, of dimension $a_i p_i q_i$. The ranks add to at most r_i by complementarity; rank subadditivity proves (28.1). In real PSD notation this is the space $\text{Hom}(\text{Ran } S_i, \text{Ran } X_i)$; its transpose is not a second independent channel. The whole-slice certificate identity allows Theorem 23.1 to use exactly this Q .

There is also an explicit factor-aware consequence for a common Jordan family with Peirce constant $a > 0$, orders at most an integer $R \geq 2$, and at most an integer $L \geq 1$ non-ray factors. With $\mathcal{C}_R$ as in (27.16), set

$$Q_{\min}(n; L, a, R) = \min\{z \in \mathbb{Z}_{\geq 0} : a\mathcal{C}_R(L, z) \geq n\}. \quad (28.2)$$

It exists only if $n \leq aL\lfloor R^2/4 \rfloor$; otherwise there is no such factorization. In the feasible case every aggregate certificate above has $Q \geq Q_{\min}$. Optional rays can contribute to Q but not to capacity, and do not affect the assertion. For the non-ray rank Q_0 , Cauchy–Schwarz gives $n \leq a(RQ_0 - Q_0^2/L)$, whence also

$$Q \geq \frac{L}{2} \left(R - \sqrt{R^2 - \frac{4n}{aL}} \right). \quad (28.3)$$

Thus at high curvature loading the movement coefficient is of order $\sqrt{LR}$, even if the number of factors is only L . For an exact fixed non-ray rank $Q_0 \leq LR$, its untruncated maximum capacity is $a[(L-t)u(R-u) + t(u+1)(R-u-1)]$ with $Q_0 = uL + t$. For a rank *budget*, the truncation in $\mathcal{C}_R$ is necessary: capacity is maximized by stopping at $L\lfloor R/2 \rfloor$. All these envelopes are integer consequences of curvature, not claims that each balanced rank vector integrates to a lift.

These bounds can also be imposed at every stage of the sequential compression in Theorem 7.8. For a dictionary $\mathcal{K}$ whose non-ray faces retain the common Peirce constant a and order bound R , the same proof selects a simultaneous contact with

$$Q \geq \sum_{a'=1}^b \max\{\kappa_{\mathcal{K}}(s_{a'}), Q_{\min}(s_{a'} - 1; L, a, R)\}. \quad (28.4)$$

There are at most L non-ray faces at every stage, and each compressed full row has mixed rank $s_{a'} - 1$; newly exposed ranks add by Lemma 5.1. These facts prove the displayed bound. It applies to each fixed classical Hermitian family and to a fixed spin family; Albert faces also retain Peirce constant eight when non-ray. If these factors also have real dimension at most d , the proved bound $h_d(s_{a'})$ of Section 7 is an alternative one-row bound and can be combined with $Q_{\min}$ in the same way. For mixed families use the adaptive face ranks and constants in that section, instead of assigning them an artificial common constant.

28.2 A conditional total-work theorem

Specialize to real PSD blocks $2 \leq r_i \leq R$ and the standard restricted barrier $F = -\sum_i \log \det X_i$. A method starts at a fixed interior reference x^0 . Every counted round moves intrinsic distance at most $B > 0$, and every round is charged at least $a_0 J_\omega$ work, where $a_0 > 0$ and $J_\omega = \sum_i r_i^\omega$. The charge is an assumption about what the method must process or output in each round. It is not implied by access to sparse entries or quantum oracles.

Theorem 28.1 (Work under a fresh per-round charge). *For the support objective and certificate above, let $\Delta > 0$ be (23.2) with all weights $\alpha_i = 1$. If $0 < \varepsilon < \Delta$ and $0 \leq \omega \leq 5/2$, every such method reaching objective gap at most ε has total charged work*

$$\mathcal{W} \geq \frac{a_0}{B} \frac{25}{6} \sqrt{\frac{5}{3}} \frac{n^{3/2}}{R^{(5-2\omega)/2}} \log \frac{\Delta}{\varepsilon}. \quad (28.5)$$

If each round instead costs at least $a_0 M$, where $M = \sum_i d_{r_i}$, then

$$\mathcal{W} \geq \frac{a_0}{B} \frac{25}{12} \sqrt{\frac{5}{3}} (1 + 1/R) \frac{n^{3/2}}{\sqrt{R}} \log \frac{\Delta}{\varepsilon}. \quad (28.6)$$

A round of at most an integer $m \geq 1$ feasible chords of starting local norm at most $0 < \eta < 1$ satisfies the distance premise with $B = m \log(1/(1 - \eta))$.

Proof. Apply the algebraic proof of Theorem 27.3 to $p_i \leq r_i - q_i$ and (28.1), taking $\theta = 1/2$. It gives

$$J_\omega \sqrt{Q} \geq \frac{25}{6} \sqrt{\frac{5}{3}} \frac{n^{3/2}}{R^{(5-2\omega)/2}}.$$

Theorem 23.1 bounds the endpoint distance below by $\sqrt{Q} \log(\Delta/\varepsilon)$. The triangle inequality therefore requires at least this distance divided by B rounds. Multiplication by the stipulated charge proves (28.5). The inequality $M \geq (1 + 1/R)J_2/2$ proves the coordinate version. The chord estimate is the classical self-concordant Hessian comparison already used in Corollary 23.2. $\square$

28.3 Attainment in specified output models

For homogeneous source dimensions s suppose the integers allow $p = 3R/5$, $c = 2R/5$, $h = s/p$ and $L = bh/c$. The packed lift has $H = bh$ columns and full order- R blocks. For unit source directions v_a , maximize $\sum_a v_a^T x_a$. The full fiber-centered path of Proposition 25.2 has

$$x_a = r(\tau)v_a, \quad r(\tau) = \frac{\tau}{\sqrt{h^2 + \tau^2} + h}, \quad D_j = q(\tau)I_c, \quad q(\tau) = \frac{2}{\sqrt{h^2 + \tau^2} + h}.$$

At its support contact, the c dual column vectors in each block are linearly independent because their allocation coordinates are different basis vectors. Hence $q_i = c$ and $Q = H$. Its resource product is exactly

$$J_\omega \sqrt{H} = \frac{25}{6} \sqrt{\frac{5}{3}} \frac{(bs)^{3/2}}{R^{(5-2\omega)/2}}. \quad (28.7)$$

Its distance from the analytic center to accuracy ε is at least $\sqrt{H}[\log(b/(2\varepsilon))]_+$, and its central arc has length $O(\sqrt{H} \log(b/\varepsilon))$ for $0 < \varepsilon \leq b/4$. Thus bounded-distance checkpoints attain the movement order with the resource ratio $(s/(s-1))^{3/2}$ to (28.5). The instance-dependent reference scale in that theorem is retained; no uniform scale over all lifts is asserted.

For a concrete dense-output upper cost, preprocess the fixed objective columns V_j and their Gram matrices $G_j = V_j^T V_j$ once. At each checkpoint the entire lifted point is

$$W_j = rV_j, \quad S_j = r^2 G_j + qI_c.$$

Writing all ambient cone entries therefore takes $O(M)$ exact real arithmetic operations and writes per checkpoint, including the fixed identity blocks if the output contract requires them. This proves a matching *postprocessing and serialization* bound for that path, with preprocessing and scalar evaluation charged separately. It does not require a fresh Gram computation at every round. An explicit bounded-distance schedule also avoids a hidden optimization subproblem: from $\tau = 0$ to h , use increments $O(h/\sqrt{H})$; subsequently use constant increments $O(1/\sqrt{H})$ in $\log \tau$ until $\tau = H/\varepsilon$. The squared log-parameter speed is $H(1 - h/\sqrt{h^2 + \tau^2}) \leq H$, and before h the ordinary speed is at most $\sqrt{H}/2/h$. Also $b(1 - r(\tau)) \leq H/\tau$. This verifies both its step budget and its endpoint accuracy. Square-root and exponential evaluations, or their chosen precision implementations, are separate scalar costs; there is no hidden dense linear solve here.

A charge of one record per factor (J_0), or r_i records per factor (J_1), is a different output model. If those records consist of the public scalars and references to preprocessed fixed directions and Gram matrices, they can be refreshed in $O(J_0)$ or $O(J_1)$ word operations. Equation (28.7) then describes matching charged record work for this representation. It does not claim that all bs projected

coordinates, or all dense factors, can be produced in that cost. Conversely, the full-coordinate model really does require fresh writes of every output entry in every counted round. Counting only free coordinates instead leads to (27.13), whose optimizer is grouped Schur rather than 3 : 2 packing.

Finally, none of these fresh charges follows from one-shot query complexity. Fixed input data can be preprocessed and reused, and public constant cone blocks can be added without changing the problem. For example, let $v_{aj} = \sigma_{aj}/\sqrt{s}$ with hidden signs $\sigma_{aj} \in \{-1, 1\}$. One phase-oracle query prepares

$$\frac{1}{\sqrt{bs}} \sum_{a,j} \sigma_{aj} |a, j\rangle.$$

Every nonzero projected center and projected predictor on the displayed path is a public scalar multiple of this same vector. Fresh preparation costs one hidden query per checkpoint, independently of the number of PSD factors; it does not cost $\Omega(L)$ queries. This says nothing about auxiliary Gram-state preparation, final classical readout, or other solver subroutines. It demonstrates why a one-shot lower bound cannot be multiplied by a movement count without a further temporal or output premise.

29 Newton systems: elimination, packing, and access

Cone resources do not determine the cost of a Newton equation. The projected variables, the auxiliary iterate, the chosen elimination, and the requested output all matter. We establish this point for the grouped and column-packed ball formulations. In particular, feasible auxiliary allocations improve the off-center comparison from a quadratic to a sharp linear dependence on residual eccentricity.

29.1 The common marginal barrier and its Newton equation

Let $C = \prod_{a=1}^b B_2^{s_a}$, let $n = \sum_a s_a$, and partition each source vector x_a into $h_a \geq 1$ nonempty coordinate groups. Pad a group by public zero coordinates when necessary, and arrange the resulting columns in real matrices $W_\ell \in \mathbb{R}^{p_\ell \times c_\ell}$. Every projected coordinate occurs once. Let E_a^ℓ be the diagonal projector onto the columns of block ℓ assigned to source a . These projectors are orthogonal and sum to the identity in each block; source a has h_a columns in total. The packed formulation is

$$Z_\ell = \begin{pmatrix} S_\ell & W_\ell^T \\ W_\ell & I_{p_\ell} \end{pmatrix} \succ 0, \quad \sum_\ell \text{tr}(E_a^\ell S_\ell) = 1, \quad F = - \sum_\ell \log \det Z_\ell. \quad (29.1)$$

All additional constraints and the objective in this subsection depend only on x . Write $D_\ell = S_\ell - W_\ell^T W_\ell$ and $\rho_a = 1 - \|x_a\|^2 > 0$.

Proposition 29.1 (Packing-independent marginal and centered system). *For fixed $x \in \text{int } C$, the unique minimum of F on its auxiliary fiber has diagonal residuals*

$$(D_\ell^0)_{\gamma\gamma} = d_a := \rho_a/h_a \quad (\gamma \text{ assigned to } a). \quad (29.2)$$

The marginal barrier and its Hessian are

$$\Phi(x) = - \sum_a h_a \log \rho_a + \sum_a h_a \log h_a, \quad (29.3)$$

$$H_0(x) = \bigoplus_a \left(\frac{2h_a}{\rho_a} I_{s_a} + \frac{4h_a}{\rho_a^2} x_a x_a^T \right). \quad (29.4)$$

Its self-concordant barrier parameter is exactly $\sum_a h_a$. The grouped paraboloid formulation with the same groups has the same marginal. At their fiber centers, both formulations give the same projected Newton matrix and right-hand side for every common affine constraint $Ax = d$, linear objective, multiplier, and projected iterate.

Proof. The Schur determinant identity gives $\det Z_\ell = \det D_\ell$. For fixed diagonal, Hadamard's inequality uniquely maximizes the determinant at a diagonal positive matrix. The allocation equations become $\sum_{\gamma \in \Gamma_a} d_\gamma = \rho_a$; AM–GM uniquely maximizes their product at $d_\gamma = \rho_a/h_a$. The same AM–GM calculation applies to the grouped slacks $t_\gamma - \|x_\gamma\|^2$. Differentiation gives (29.4). Each summand is h_a times the standard unit-ball barrier, and its squared dual gradient norm is $2h_a\|x_a\|^2/(1 + \|x_a\|^2)$. This tends to h_a at the boundary. Additivity and simultaneous boundary approach prove exactness of the parameter.

For completeness, the equality extends to the uncondensed residual equations. For a direction $(A_\ell, U_\ell) = (\delta S_\ell, \delta W_\ell)$ put $B_\ell = A_\ell - W_\ell^T U_\ell - U_\ell^T W_\ell$. Direct differentiation along this affine direction gives

$$D^2 F[(A, U)]^2 = \sum_\ell \left[\text{tr}(D_\ell^{-1} B_\ell D_\ell^{-1} B_\ell) + 2 \text{tr}(D_\ell^{-1} U_\ell^T U_\ell) \right]. \quad (29.5)$$

At diagonal D^0 , off-diagonal entries of B are independent positive quadratic modes. They have zero gradient, objective, and constraint coefficients. The diagonal modes $r_\gamma = B_{\gamma\gamma}$ obey

$$\sum_\gamma \frac{r_\gamma^2}{d_\gamma^2} + 2 \sum_\gamma \frac{\|u_\gamma\|^2}{d_\gamma}, \quad \sum_{\gamma \in \Gamma_a} (r_\gamma + 2x_\gamma^T u_\gamma) = 0. \quad (29.6)$$

These are precisely the grouped residual equations, including their linear terms $-\sum_\gamma r_\gamma/d_\gamma$ and the common objective. Additional rows $Au = 0$ coincide. Eliminating r and the allocation multipliers therefore gives the same matrix and right-hand side. $\square$

With no additional affine coupling, the graph of (29.6), with one allocation multiplier per source, is a disjoint union of stars. Solving these stars, or applying the diagonal-plus-rank-one inverse in (29.4), costs $O(n + \sum_a h_a) = O(n)$ arithmetic. The graph of a fully materialized generic Hessian is instead a union of s_a -cliques. Both descriptions are correct; they measure different representations of the same operator. Adding constraint hubs for A gives a graph G on projected variables, source hubs, and constraint hubs. Attaching the r_γ leaves gives width $\max\{1, \text{tw}(G)\}$; this graph, rather than the PSD packing layout, controls the corresponding sparse solve.

Corollary 29.2 (Matched projected-oracle equivalence). *Supply both formulations with the same complete canonical oracles for their common projected data and reduced Newton equation, at the same precision. Require the same projected direction, or its normalized quantum state when the direction is nonzero. Then their minimum classical and quantum query complexities agree exactly. This equivalence also holds along any adaptive procedure that uses only these projected records and recenters the auxiliary fibers before each Newton call.*

Proof. Proposition 29.1 identifies both the operator and the right-hand side, in the same coordinates. Forward each query to the identical oracle; retain all other operations, measurements, and stopping decisions. The simulation works in both directions. Induction on calls gives the adaptive statement. A common charged compiler from common raw data likewise gives identical preparation work. $\square$

Here “same oracle” includes its action on unused input and ancilla subspaces. Equality of an encoded matrix block alone would permit a unitary completion to reveal extra information. Thus

the corollary is an exact interface statement, not an oracle construction at no cost. It preserves normalization, spectrum, right-hand-side norm, solution overlap, and output accuracy. In residual coordinates the centered PSD KKT system is the grouped system direct-summed with a positive diagonal off-diagonal block and a zero right-hand side; public zero padding also identifies those solution states. The original δS output is different: reconstructing it requires $W^T U + U^T W$. Its cost is not covered by the projected equivalence. If all $h_a = h$, changing h multiplies the unconstrained marginal Hessian by h . It changes its metric scale but not its condition number or normalized inverse state for a fixed right-hand side. This illustrates why fewer cone factors alone do not establish a state-generation advantage.

29.2 A sharp comparison away from the fiber center

Theorem 29.3 (Exact quotient and allocation-aware transfer). *At a strictly feasible point of (29.1), define*

$$z_a(U) = \sum_{\ell} \text{tr}(E_a^{\ell} W_{\ell}^T U_{\ell}), \quad M_{ab}(D) = \sum_{\ell} \text{tr}(E_a^{\ell} D_{\ell} E_b^{\ell} D_{\ell}). \quad (29.7)$$

The quadratic form obtained by eliminating all auxiliary Hessian directions is

$$Q_D(U) = 2 \sum_{\ell} \text{tr}(D_{\ell}^{-1} U_{\ell}^T U_{\ell}) + 4z(U)^T M(D)^{-1} z(U). \quad (29.8)$$

Let H_D denote this form on the projected coordinates. If $\mu D_{\ell}^0 \preceq D_{\ell} \preceq L D_{\ell}^0$ for all ℓ , where $0 < \mu \leq 1 \leq L$, then

$$L^{-1} H_0 \preceq H_D \preceq \mu^{-1} H_0, \quad \kappa(H_0^{-1/2} H_D H_0^{-1/2}) \leq L/\mu. \quad (29.9)$$

The dependence and the condition factor are sharp. The comparison survives restriction to any common projected tangent space. For any full-row-rank matrix A , the dual normal matrices satisfy

$$\mu A H_0^{-1} A^T \preceq A H_D^{-1} A^T \preceq L A H_0^{-1} A^T. \quad (29.10)$$

Proof. Equip symmetric residual directions with $\langle B, C \rangle_D = \sum_{\ell} \text{tr}(D_{\ell}^{-1} B_{\ell} D_{\ell}^{-1} C_{\ell})$. The allocation functional $B \mapsto \sum_{\ell} \text{tr}(E_a^{\ell} B_{\ell})$ has Riesz representative $(D_{\ell} E_a^{\ell} D_{\ell})_{\ell}$. The Gram matrix of these representatives is $M(D)$. It is positive definite: for $E_{\ell}(\theta) = \sum_a \theta_a E_a^{\ell}$,

$$\theta^T M(D) \theta = \sum_{\ell} \|D_{\ell}^{1/2} E_{\ell}(\theta) D_{\ell}^{1/2}\|_F^2,$$

and its vanishing forces every $\theta_a = 0$, since each source has a column. The tangent allocation constraint is $\sum_{\ell} \text{tr}(E_a^{\ell} B_{\ell}) = -2z_a(U)$. Its minimum squared D -norm is $4z^T M^{-1} z$, proving (29.8) from (29.5).

The sharper comparison uses feasibility, not just tensor-product order. Although $E_{\ell}(\theta)$ may be indefinite, $E_{\ell}(\theta) D_{\ell} E_{\ell}(\theta) \succeq 0$. Positive trace pairing therefore gives

$$\mu \sum_{\ell} \text{tr}(D_{\ell}^0 E_{\ell} D_{\ell} E_{\ell}) \leq \theta^T M(D) \theta \leq L \sum_{\ell} \text{tr}(D_{\ell}^0 E_{\ell} D_{\ell} E_{\ell}).$$

The diagonal matrices D_{ℓ}^0 and E_{ℓ} commute. Exact source allocations, $\sum_{\ell} \text{tr}(E_a^{\ell} D_{\ell}) = \rho_a$, yield

$$\sum_{\ell} \text{tr}(D_{\ell}^0 E_{\ell} D_{\ell} E_{\ell}) = \sum_a \theta_a^2 d_a \rho_a = \theta^T M_0 \theta, \quad M_0 = \text{diag}(\rho_a^2 / h_a). \quad (29.11)$$

Hence $\mu M_0 \preceq M(D) \preceq L M_0$. Both terms in (29.8) now obey the first-power bounds L^{-1} and μ^{-1} . At the same W , their vector $z(U)$ is unchanged, so these inequalities prove (29.9). Restriction preserves order; inversion followed by congruence proves (29.10). Sharpness is established in Example 29.4 below. $\square$

It is essential to distinguish this Hessian elimination from first minimizing the barrier value over a fiber. The latter always replaces D by D^0 . It is also essential to retain the eliminated gradient. For minimizing $F + \eta\langle c, x \rangle$ at a feasible packed point, the reduced Newton gradient is the linear functional

$$g_D[U] = 2\rho^T M(D)^{-1} z(U) + \eta\langle c, U \rangle. \quad (29.12)$$

Indeed the Newton quadratic before auxiliary elimination is

$$\frac{1}{2} \|B\|_D^2 - \sum_{\ell} \text{tr}(D_{\ell}^{-1} B_{\ell}) + \sum_{\ell} \text{tr}(D_{\ell}^{-1} U_{\ell}^T U_{\ell}) + \eta\langle c, U \rangle.$$

Completing the square replaces B by $B - D$ and its allocation by $-2z(U) - \rho$. Its minimum is $\frac{1}{2}(2z + \rho)^T M^{-1}(2z + \rho)$, plus the displayed U terms and a constant. This proves (29.12). At D^0 it is $\sum_a 2h_a \langle x_a, U_a \rangle / \rho_a + \eta\langle c, U \rangle$, the marginal gradient. Away from the center, a spectral comparison of matrices alone does not identify Newton directions or their normalized states. Homogeneous additional tangent constraints are common to both systems; nonzero affine infeasibility has additional right-hand-side terms.

Example 29.4 (Sharpness and fixed-projection obstruction). Pack two sources as two columns of one block, take $W = 0$, and put

$$D = \begin{pmatrix} 1 & \xi \\ \xi & 1 \end{pmatrix}, \quad D^0 = I_2, \quad |\xi| < 1.$$

The source equations hold exactly. Formula (29.8) reduces to

$$Q_D(U) = \frac{2}{1 - \xi^2} (\|u_1\|^2 + \|u_2\|^2 - 2\xi u_1^T u_2).$$

The relative eigenvalues are $1/(1 + \xi)$ and $1/(1 - \xi)$. Thus $\mu = 1 - |\xi|$, $L = 1 + |\xi|$, and the relative condition is exactly L/μ . For objective columns $(v, 0)$, $\|v\| = 1$, the projected Newton directions are $-\eta(v, \xi v)/2$ and $-\eta(v, 0)/2$ at D and D^0 . Their normalized states have trace distance $|\xi|/\sqrt{1 + \xi^2}$. Replacing ξ by $-\xi$ leaves all projected input records unchanged but makes the two states have trace distance $2|\xi|/(1 + \xi^2) \rightarrow 1$. Consequently projected data alone cannot uniformly recover unrestricted off-center Newton states. This is a one-iterate obstruction; it asserts no lower bound on the number of outer iterations.

Corollary 29.5 (Approximate allocations). *Keep W and its exact center D^0 fixed, but allow positive residuals $\hat{D}$ with allocations*

$$(1 - \zeta)\rho_a \leq \sum_{\ell} \text{tr}(E_a^{\ell} \hat{D}_{\ell}) \leq (1 + \zeta)\rho_a, \quad 0 \leq \zeta < 1.$$

If $\mu D^0 \preceq \hat{D} \preceq L D^0$, with $0 < \mu \leq 1 \leq L$, their homogeneous-tangent Hessian quotient obeys

$$\frac{H_0}{L(1 + \zeta)} \preceq H_{\hat{D}} \preceq \frac{H_0}{\mu(1 - \zeta)}. \quad (29.13)$$

In particular relative residual error $\|(D_{\ell}^0)^{-1/2}(\hat{D}_{\ell} - D_{\ell}^0)(D_{\ell}^0)^{-1/2}\|_2 \leq \delta < 1$ gives condition at most $((1 + \delta)/(1 - \delta))^2$ without exact allocations, and at most $(1 + \delta)/(1 - \delta)$ when the allocations are exact.

Proof. The middle trace in (29.11) lies between $(1 - \zeta)\theta^T M_0 \theta$ and $(1 + \zeta)\theta^T M_0 \theta$. Repeat the proof and use the weaker of the resulting bounds for the two nonnegative terms of Q . Spectral error δ implies allocation error at most δ by positive trace pairing with each projector. Set $\mu = 1 - \delta$, $L = 1 + \delta$, and $\zeta = \delta$ or 0 , respectively. $\square$

The linear improvement in Theorem 29.3 is specific to the feasible allocation structure. Its proof uses elementary positive trace inequalities; it is not a new general matrix-order principle. Neither version supplies free access to D^{-1} , $M(D)^{-1}$, or $H_0^{-1/2}$. Those constructions and any encoding normalization remain separate costs.

29.3 What exact recentering requires

The center has an inexpensive implicit representation. Store (W, d) with the semantics $S_\ell = W_\ell^T W_\ell + \text{Diag}(d_\ell)$. Computing all squared source norms and all labels $d_\gamma = \rho_a/h_a$ uses $O(n)$ arithmetic and storage, independently of packing, because $\sum_a h_a \leq n$. The source equations hold exactly by $\|x_a\|^2 + h_a d_a = 1$. For B -bit common-denominator dyadic coordinates the squared norms and labels are exact rationals of $O(B + \log n)$ bits. Fast integer arithmetic gives the conservative bound $\tilde{O}(nM(B + \log n))$, where $M(t)$ is the cost of multiplying t -bit integers. This represents the exact center of the stored interior rational iterate. It does not require subtracting rounded Gram matrices. Unrelated rational denominators, inverse evaluation, and requested numerical accuracy have their own bit costs.

A reversible value oracle for ρ_a similarly supplies d_γ with one scale query and reversible division by the public h_a . Together with blockwise direction-state access, it permits the common rank-one compiler for (29.4); radii and certified normalization bounds must be supplied or computed. In particular normalized state access to x_a alone does not provide this scale. The same complete state-preparation unitary can prepare the direction of $r_0 v$ and $r_1 v$, while their $1 - r_j^2$ differ by arbitrarily large factors. No algorithm with only that unitary can return both scales to a fixed relative accuracy.

Proposition 29.6 (Cost of missing radial metadata). *For coordinate-bit access to a source of dimension $s \geq 2$, producing a classical center scale to relative error $\delta < 1/3$ can require $\Omega(\sqrt{s})$ quantum or $\Omega(s)$ randomized queries. For b independent sources, returning all scales with joint success at least $2/3$ requires $\Omega(b\sqrt{s})$ and $\Omega(bs)$, respectively. These orders are attained on the promise used in the proof. For $k \mid s$, $1 \leq k \leq s/2$, the one-source promise can instead have smaller slack $\rho = k/s$, with tight costs $\Theta(\rho^{-1/2})$ quantum and $\Theta(\rho^{-1})$ randomized.*

Proof. Take $x_i = z_i/\sqrt{s}$ with $z_1 = 0$ public and all other bits one, except for at most one additional hidden zero. The slack is $1/s$ or $2/s$. Their relative- δ intervals are disjoint precisely when $\delta < 1/3$, so recovering the scale solves promised unstructured search. The standard search bounds apply [14, 18].

For clarity, the quantum direct sum has a short adversary proof. Index one source's promise by the no-mark input and its $s - 1$ single-mark inputs. The star adjacency matrix Γ has norm $\sqrt{s - 1}$, and filtering by any queried bit leaves norm one. For b sources use $\sum_{a=1}^b I^{\otimes(a-1)} \otimes \Gamma \otimes I^{\otimes(b-a)}$. Its norm is $b\sqrt{s - 1}$; filtering a query in one source retains only that summand. Its entries vanish between inputs having the same output vector. The adversary lower bound for function evaluation [59] gives the claim. For randomized algorithms, give each source an independent fair no-mark/single-uniform-mark promise. Joint success implies constant average success for each output, and the usual search indistinguishability argument requires a constant fraction of the $s - 1$ expected queries per such source; summing proves $\Omega(bs)$. On this promise, exact amplitude amplification for the known

single-mark case followed by verification distinguishes zero from one mark with $O(\sqrt{s})$ queries and no error. Applied sourcewise it attains joint success without an amplification logarithm. Scanning attains the randomized order.

Finally partition coordinates into blocks of size k . One public block is zero; all candidate blocks are one except possibly one hidden zero block. The two slacks are k/s and $2k/s$. A representative coordinate is an oracle for its block mark, reducing to search on $s/k - 1$ labels. This number is within a constant factor of s/k in the stated range. $\square$

The joint-output bound also applies to a standalone compiled value interface whose output can be read on all labels with the same joint correctness guarantee and with no further coordinate-input queries. It does not describe one online coherent evaluation: the promised search can be controlled by the source label at $O(\sqrt{s})$ input queries per evaluation. Those queries remain charged on every invocation. The following fresh update family supplies a temporal bound under an explicit output contract.

A *compile-and-commit maintainer* returns, before the next epoch, either a classical record of all source scales or a complete reusable unitary for their encoded numerical values. The latter must allow every basis-label value to be copied out, using the unitary and its inverse, with no further raw-update queries and with the entire backing workspace restored exactly, including correlations with other retained registers. More precisely, each committed object has fixed value strings v_a ; compute-copy-uncompute on label a acts as

$$|a, z\rangle|\psi\rangle \mapsto |a, z \oplus v_a\rangle|\psi\rangle$$

on its backing state and any retained reference. The strings encode numbers $\hat{d}_a$. Correctness means that, with joint probability at least $2/3$, all committed values at all epochs obey $|\hat{d}_a - d_a| \leq \delta d_a$. A state consumed or disturbed by reading its values does not meet this contract. Rational encodings allow exact values even for $\delta = 0$; fixed-point encodings require enough bits for the requested positive error.

Theorem 29.7 (Fresh-update scale maintenance). *Let $b, T \geq 1$, $s \geq 2$, and $0 \leq \delta < 1/3$. For the packing (29.1) with b sources of dimension s and public column counts h_a , there is a family of T fresh coordinate-bit update batches, each zero or one-sparse per source, on which compile-and-commit maintenance has tight worst-case raw-query costs*

$$Q = \Theta(Tb\sqrt{s}), \quad R = \Theta(Tbs) \quad (29.14)$$

for quantum and randomized algorithms, respectively. Each epoch starts from the same public interior base and is reset after commitment. For $k \mid s$, $1 \leq k \leq s/2$, a replicated-block version has k -sparse updates and smaller slack $\rho_ = k/s$, with costs*

$$Q = \Theta(Tb\rho_*^{-1/2}), \quad R = \Theta(Tb\rho_*^{-1}). \quad (29.15)$$

The lower bounds allow all epoch oracles from the start, superposed queries over epoch, source, and coordinate, and arbitrary persistent workspace. The upper bounds have zero error on these promises.

Proof. Use $\bar{x} = (0, 1, \dots, 1)/\sqrt{s}$. Independently for each pair (t, a) , the update is either $u_{t,a} = 0$ or $u_{t,a} = -e_j/\sqrt{s}$ at one hidden $j \in \{2, \dots, s\}$. The raw oracle returns the corresponding mark bit at a queried coordinate. After the update,

$$\rho_{t,a} = 1 - \|\bar{x} + u_{t,a}\|^2 \in \{1/s, 2/s\}, \quad d_{t,a} = \rho_{t,a}/h_a.$$

Both cases are strictly interior, and Proposition 29.1 gives these exact PSD fiber-center labels. Their relative-error intervals are disjoint because $1 + \delta < 2(1 - \delta)$. Undo each batch after commitment; inverse-update access uses the same bits and reveals no new information. A public reset only strengthens the model.

Read every committed label before continuing the maintainer, restoring its workspace in the quantum case. Thresholding recovers $m = Tb$ independent presence bits with joint success $2/3$ and no extra raw queries. Apply the star adversary from Proposition 29.6, with tensor factors now indexed by (t, a) . Its Kronecker sum has norm $m\sqrt{s-1}$ and every coordinate filter has norm one. It is entrywise nonnegative and vanishes between inputs with the same output vector, so the finite-output spectral adversary bound applies [2, Theorems 3–4]. The proof already uses simultaneous oracle access and allows cross-epoch query scheduling.

For the randomized bound, independently choose no mark with probability $1/2$ and a uniform single mark otherwise. Fix the randomness and all other pairs, and let L count distinct candidates queried when this pair has no mark. The no-mark and random-mark transcripts coincide unless the mark lies in these L locations. Averaging over the fixed data, the total variation distance is at most $\mathbb{E}L/(s-1)$. Marginal success at least $2/3$ forces $\mathbb{E}L \geq (s-1)/3$. Since the no-mark case has probability $1/2$, this pair contributes at least $(s-1)/6$ expected queries under the product distribution. Summing over the m pairs proves the worst-case randomized bound.

Exact amplitude amplification designed for one marked coordinate, followed by verification, decides zero versus one mark in $O(\sqrt{s})$ queries [18, Theorems 4 and 16]. Verification always rejects in the zero-mark case. Search each pair and store its exact rational scale; no joint-success amplification logarithm is needed. Scanning gives the classical upper bound.

For the margin version, partition coordinates into s/k blocks of size k , with one public zero block in the base. An update zeros either no candidate block or one whole candidate block. The slacks are k/s and $2k/s$. A coordinate query reveals only its block's mark bit, and one representative coordinate per block suffices. Replace $s-1$ throughout by $s/k-1 = \Theta(s/k)$, which is at least one in the stated range. $\square$

Corollary 29.8 (Universal invocation versus compilation tradeoff). *On the same promises, suppose a service must support any adaptive client making at most r_t value invocations at epoch t , including coherent label inputs in the quantum case. Invocations implement the reusable value interface above, but may query the raw input; all preprocessing and invocation queries are charged. Require joint correctness with probability at least $2/3$ over the entire interaction. For prescribed nonnegative integer budgets r_t , the tight worst-case costs are*

$$Q = \Theta\left(\sqrt{s} \sum_{t=1}^T \min\{r_t, b\}\right), \quad R = \Theta\left(s \sum_{t=1}^T \min\{r_t, b\}\right). \quad (29.16)$$

In the margin version replace $\sqrt{s}, s$ by $\rho_^{-1/2}, \rho_*^{-1}$, respectively.*

Proof. Choose the permitted client that reads $\min\{r_t, b\}$ distinct basis labels at each epoch. Fix all other update pairs to zero. Its transcript recovers $\sum_t \min\{r_t, b\}$ independent search bits, so the same adversary and product-distribution proofs give the lower bounds. For the upper bound, if $r_t < b$, run the exact promised search controlled by the source label on each invocation. Coherently verify, copy the deterministic presence bit, and uncompute the search workspace; this is a clean value unitary at $O(\sqrt{s})$ queries per call. If $r_t \geq b$, search all sources once and keep a classical table for query-free reversible lookup. Classical scans give the analogous strategies. Apply the same choice to candidate-block representatives for the margin version. $\square$

This universal-service contract is stronger than a particular downstream algorithm's structured calls. One online coherent evaluation has no b factor; a service supporting at least b arbitrary calls reaches the compilation cost. Neither result concerns arbitrary one-shot state output or finite-precision gate complexity.

For an explicit sparse list, the exact identity

$$\rho_a(x + \alpha u) = \rho_a(x) - 2\alpha x_a^T u_a - \alpha^2 \|u_a\|^2 \quad (29.17)$$

instead maintains scales in $O(\sum_a |\text{supp } u_a|)$ arithmetic, given access to the current coordinates and a list of affected sources. In particular one-sparse lists cost $O(b)$ per epoch. The lower bound charges discovery of hidden support, not sparsity itself. Nor does a snapshot lower bound imply a factor T for one fixed oracle: if every round requests the same promised OR bit, compute and cache it once in $\Theta(\sqrt{s})$ quantum or $\Theta(s)$ randomized queries. All T subsequent requests then use no raw queries.

Proposition 29.9 (Fixed central-path scale caching). *For the uncoupled product of balls, minimize $\Phi(x) + \eta \sum_a c_a^T x_a$ with fixed c_a and $\eta \geq 0$. Put $q_a = \|c_a\|$ and $\tau_a = \eta q_a / h_a$. The unique center is*

$$\rho_a(\eta) = \frac{2}{1 + \sqrt{1 + \tau_a^2}}, \quad x_a(\eta) = -\frac{\eta \rho_a(\eta)}{2h_a} c_a, \quad d_a(\eta) = \frac{\rho_a(\eta)}{h_a}. \quad (29.18)$$

For $s_a \geq 2$, its source Hessian has radial-to-tangential eigenvalue ratio $2/\rho_a(\eta) - 1 = \sqrt{1 + \tau_a^2}$.

Proof. Stationarity is $2h_a x_a / \rho_a + \eta c_a = 0$. Substitution in $\rho_a = 1 - \|x_a\|^2$ gives $\rho_a + \tau_a^2 \rho_a^2 / 4 = 1$; its positive solution is (29.18), also when $q_a = 0$ or $\eta = 0$. It is interior, and strict convexity of Φ gives uniqueness. Equation (29.4) gives eigenvalues $2h_a / \rho_a$ and $2h_a / \rho_a + 4h_a(1 - \rho_a) / \rho_a^2$, proving the ratio (equal to one at $x_a = 0$). $\square$

Thus supplied reversible access to the fixed norms q_a evaluates a requested scale with $O(1)$ norm queries and charged scalar arithmetic; uncomputing norm scratch may require an inverse query. Explicit objective data permits all norms to be acquired once in $O(n)$ real arithmetic, including square roots, and reused for every η . Requested precision and reversible arithmetic still have their own costs. Additional affine constraints or inexact neighborhood iterates need not satisfy this stationarity equation. The fresh batches in Theorem 29.7 have not been embedded in the Newton trajectory of one fixed conic program: such an embedding and a contract recovering its independent outputs would be needed to multiply a query lower bound by an iteration count. The closed-form uncoupled path witnesses reuse of fixed data, and supplies no such embedding.

Explicit dense auxiliaries have a different output cost: $\Omega(\sum_\ell c_\ell^2)$ scalar writes and ordinary Gram formation cost $O(\sum_\ell p_\ell c_\ell^2)$. An implicit ambient encoding can avoid these writes: put $K_\ell = (W_\ell \ I_{p_\ell})$ and write $Z_\ell = K_\ell^T K_\ell + \text{Diag}(d_\ell, 0)$. Products of a supplied α_K -normalized encoding of K_ℓ and its adjoint, followed by a positive linear combination with an α_d -normalized diagonal encoding, give normalization $\alpha_K^2 + \alpha_d$. This costs a constant number of supplied encoding calls. Constructing these encodings, their norms, and an inverse remains charged; no statement about a flattened Gram-matrix output state follows.

29.4 Sparse augmented Lorentz systems

Low-rank augmentation of cone Hessians is classical, including in ECOS and nonsymmetric-cone implementations [23, 27]. We use it to make the classical comparator explicit for the formulations above. A graph below means the fixed structural graph of the augmented scalar matrix; special numerical zeros may delete edges. Its treewidth is the minimum, over tree decompositions, of the largest bag size minus one.

For a Lorentz block $x = (t, z)$, let $q = t^2 - \|z\|^2 > 0$ and $J = \text{diag}(1, -I)$. Its standard Hessian is

$$H_x = -\frac{2J}{q} + \frac{4(Jx)(Jx)^T}{q^2} = D_x + w_x w_x^T. \quad (29.19)$$

Thus the equality-form Newton matrix and its one-hub augmentation are

$$K_0 = \begin{pmatrix} H & A^T \\ A & 0 \end{pmatrix}, \quad \widetilde{K}_0 = \begin{pmatrix} D & W & A^T \\ W^T & -I & 0 \\ A & 0 & 0 \end{pmatrix}. \quad (29.20)$$

Here W has one embedded w_x column per cone. Eliminating the negative identity gives K_0 , and the augmented right-hand side is $(g_x, 0, g_y)$. There is no assumption of a positive diagonal in D .

Proposition 29.10 (Exact field-operation comparison). *Suppose $A \in \mathbb{R}^{p \times M}$ has full row rank, there are k Lorentz blocks, and a compact width- τ tree decomposition of the structural graph of $\widetilde{K}_0$ is supplied. Then a Newton equation can be solved in $O((M + p + k)(\tau + 1)^2)$ exact field operations, after $O(M + \text{mz } A)$ coefficient formation. Compactness here means $O(M + p + k)$ bags; processing a larger supplied decomposition is an additional input cost.*

Proof. Positive definiteness of H and full row rank of A imply nonsingularity of both matrices in (29.20). Replace each vertex of each bag by its row and column copies. The resulting decomposition of the matrix's row-column bipartite graph has width at most $2\tau + 1$: every off-diagonal entry has both endpoints in some original bag, each diagonal entry is covered, and connectedness of each copy's bags is preserved. Apply the exact low-treewidth linear-system algorithm of Fürer et al. [39, Corollary 3]. Its bound is $O((m + n)(w + 1)^2)$ for a supplied compact width- w bipartite decomposition. It allows nonzero pivots chosen during elimination and algebraic cancellations, so the zero or negative diagonal entries here cause no difficulty. Formula (29.19) uses only field operations and gives the formation cost. $\square$

A positive-diagonal version is useful when a reusable symmetric factorization is wanted. If $r = \|z\| > 0$ and $e = z/r$, put

$$D = \frac{2}{q}I, \quad a = \frac{\sqrt{2r(t+r)}}{q} \begin{pmatrix} 1 \\ -e \end{pmatrix}, \quad v = \frac{\sqrt{2r(t-r)}}{q} \begin{pmatrix} 1 \\ e \end{pmatrix}. \quad H_x = D + aa^T - vv^T. \quad (29.21)$$

At $r = 0$ take $a = v = 0$. The identity follows by restricting $H_x - D$ to the span of $(1, 0)$ and $(0, e)$, where its matrix is $4q^{-2} \begin{pmatrix} r^2 & -tr \\ -tr & r^2 \end{pmatrix}$. Moreover $v^T D^{-1} v = 2r/(t + r) < 1$, so $D - vv^T \succ 0$. With embedded update and downdate columns denoted by U, V , respectively, the regularized augmentation is

$$\begin{pmatrix} D & V & U & A^T \\ V^T & I & 0 & 0 \\ U^T & 0 & -I & 0 \\ A & 0 & 0 & -\varpi I \end{pmatrix}, \quad \varpi > 0. \quad (29.22)$$

Grouping (x, V) as positive variables and (U, y) as negative variables shows symmetric quasidefiniteness: the first diagonal block has Schur complement $D - VV^T \succ 0$, and the second is negative definite. Every symmetric permutation therefore admits an exact pivot-free LDL^T factorization [86]. A supplied width- τ elimination order gives $O((M + p + k)(\tau + 1)^2)$ real arithmetic, $O((M + p + k)(\tau + 1))$ storage, and width-linear subsequent solves. Square roots in (29.21) distinguish this branch from the field-operation result.

This is not an unconditional numerical stability statement. Writing $K_\varpi = K_0 - \varpi \text{diag}(0, I)$, the resolvent identity gives, for $d_0 = K_0^{-1}g \neq 0$ and $\varpi\|K_0^{-1}\|_2 < 1$,

$$\frac{\|K_\varpi^{-1}g - d_0\|}{\|d_0\|} \leq \frac{\varpi\|K_0^{-1}\|_2}{1 - \varpi\|K_0^{-1}\|_2}. \quad (29.23)$$

Working precision, small pivots, growth, and residual certification still need bounds. Quasidefiniteness rules out exact zero-pivot breakdown, not arbitrarily small pivots. Also, two hubs cannot generically be reduced to one while retaining a positive diagonal base: when $m \geq 3$ and every coordinate of Jx is nonzero, off-diagonal consistency in $H = D_+ + \epsilon aa^T$ forces $\epsilon = 1$ and $a_i^2 = 4(Jx)_i^2/q^2$. The axial entry of D_+ is then $-2/q$. This only proves minimality within that diagonal-plus-one-signed-rank-one class.

The two-hub graph can be bounded from coarse incidence. Make one vertex for each cone and each equality row, joining them when the row uses a coordinate of that cone. If this graph has width w and each scalar column of A has at most one nonzero, the expanded graph has width at most $\max\{2w + 1, 3\}$. Replace every cone vertex in a coarse bag by its two hubs. For each used coordinate attach a bag consisting of that coordinate, its two hubs, and its single row, to a bag covering the corresponding coarse edge. For unused coordinates omit the row. These bags cover all edges and preserve the connected-bag condition. Thus a coarse forest gives width at most three regardless of cone dimension.

29.5 Exact graph counts for the ball constructions

For N projected coordinates grouped into k nonempty groups, set $q_j = s_j - \|w_j\|^2$ and impose $\sum_j s_j = 1$. The reduced barrier has Hessian blocks

$$\text{diag}(0, 2I/q_j) + q_j^{-2}(1, -2w_j)(1, -2w_j)^T. \quad (29.24)$$

Their one-hub graph has the paths $\lambda - s_j - h_j$ and leaves $h_j - w_{ji}$. It is a tree with $N + 2k + 1$ vertices and $N + 2k$ edges. The zero s_j diagonal prevents an unqualified scalar leaf-pivot claim; Proposition 29.10 instead supplies the exact solve. Restricting (29.21) by $(s, w) \mapsto ((1 + s)/2, (1 - s)/2, w)$ gives positive base $\text{diag}(1/q, 2I/q)$ and a positive downdated base by congruence. The two-hub reduced graph has $N + 3k + 1$ vertices, $2N + 3k$ edges, and width two. Bags containing the two hubs and one coordinate, together with bags $\{\lambda, s_j\}$, prove the upper bound; each nonempty group supplies a cycle for the lower bound.

For comparison, retain all raw Lorentz coordinates (u_j, v_j, w_j) and equations $u_j + v_j = 1$, $\sum_j (u_j - v_j) = 1$. The exact structural counts are

representation	vertices	edges	treewidth
raw grouped, one hub	$N + 4k + 1$	$N + 6k$	2
raw grouped, two hubs	$N + 5k + 1$	$2N + 8k$	3
reduced grouped, one hub	$N + 2k + 1$	$N + 2k$	1
reduced grouped, two hubs	$N + 3k + 1$	$2N + 3k$	2

To verify the raw widths, the one-hub graph has $K_{2,3}$ on axial coordinates u_j, v_j and the hub, local row, and global row. Bags $\{u_j, v_j, a\}$ for these three vertices a , plus hub-coordinate leaf bags, give width two. The two-hub graph has a subdivision of K_4 : its branch vertices are the two hubs and u_j, v_j ; a row connects the two axial vertices and one w coordinate connects the two hubs. A width-three decomposition uses the four branch vertices as a core, the two axial vertices with the

local and global rows as another bag, and bags of the two hubs with each w coordinate. Different groups join through the global row. Counting incidences gives the edge totals.

The direct ball formulation $(t, x) \in Q_{N+1}$, $t = 1$, has a one-hub tree with $N + 3$ vertices; eliminating t gives a star with $N + 1$ vertices. A binary norm tree with $L = N - 1$ internal nodes has $3L$ raw cone coordinates, L hubs, $L - 1$ link rows, and one root row. It is connected with $5L$ vertices and $3L + 2(L - 1) + 1 = 5L - 1$ edges, hence a tree. Its rows are independent because each link or root equality uses its own cone-coordinate slots. These counts also explain why the smooth coordinatewise construction ($k = N$) has raw width two but reduced width one. Smooth contact selections do not impose an intrinsic width-two cost.

All these isolated-ball Newton equations have linear exact arithmetic cost. The ambient and restricted barrier parameters and the movement bounds proved earlier remain different. Thus replacing a large cone by many small cones can increase the standard short-step certificate and the bounded-Dikin movement cost without improving this linear solve. For products of cone epigraphs, fixing their axial coordinates to one reduces to products of balls; the earlier additive curvature bound forces at least $k(s - 1)$ copies of Q_3 to represent Q_{s+1}^k , even under joint reformulation. Their ambient product barrier has parameter at least $2k(s - 1)$ by its orthant section. Separate binary trees attain the count. This compares the generic ambient-barrier certificate with the $2k$ certificate for the direct product, not all algorithms' iteration counts.

29.6 The full-output comparison and its boundary

The preceding exact solves give a precise replacement principle. Suppose an outer algorithm requests a classical direction or updated iterate in every round, its theorem accepts any direction satisfying a stated Newton residual contract, and the current coefficients and right-hand side are classically evaluable at the cost charged to its quantum access model. If fixed compact decompositions of widths τ_t are supplied for its expanded systems, replacing the quantum Newton module by Proposition 29.10 costs

$$\sum_t O((M_t + p_t + k_t)(\tau_t + 1)^2) \quad (29.25)$$

exact field operations, plus coefficient formation and any decomposition input cost. The outer guarantee is preserved because the exact solution meets its residual contract. This need not reproduce the quantum algorithm's random trajectory; the comparison requires the outer guarantee and the structural width bound for every admitted trajectory. Dense materialization already requires $\Omega(M_t + p_t)$ scalar writes per round. When $p_t + k_t = O(M_t)$ and $\tau_t = M_t^{o(1)}$, the Newton module therefore gives no polynomial dimensional saving under this matched full-output, exact-arithmetic contract. The centered packed family and the grouped or norm-tree ball family have $O(n)$ or $O(N)$ replacements. Dense external affine coupling can invalidate their width bounds. A finite-precision replacement additionally needs a stable implementation and certification at the prescribed tolerance.

There is a complementary classical route for sparse well-conditioned systems. For a consistent full-column-rank system $Kd = r$ and zero initial guess, conjugate gradients on its normal equations, implemented by matrix-vector products as CGLS or LSQR [74], has the exact-arithmetic residual estimate

$$\|Kd_j - r\|_2 \leq 2 \left(\frac{\kappa_2(K) - 1}{\kappa_2(K) + 1} \right)^j \|r\|_2.$$

This follows from the usual CG energy-norm bound, because that norm is $\|K(d_j - d_*)\|_2$ and $\sqrt{\kappa_2(K^T K)} = \kappa_2(K)$. If one pair of products by K, K^T costs $\mathcal{M}$, including current entry evaluation, a relative residual $0 < \epsilon_{\text{lin}} < 1$ costs $O(\mathcal{M}\kappa_2(K) \log(2/\epsilon_{\text{lin}}))$. The case $\kappa_2(K) = 1$ terminates

after one step. No normal matrix need be materialized. Sparse bounded-size conic blocks fit this comparison only when their actual augmented entries are cheaply available. Quantum-only encodings and tall implicit Hessian formation are different access models; in particular this argument does not replace the row-sampling work in Apers and Gribbling [5].

The classical algorithms, signed-rank augmentations, and search bounds used here are established tools. The formulation-specific conclusions are the packing-independent centered equation, its sharp feasible off-center transfer, and the exact access and output conditions under which the same classical Newton work applies. Published quantum SDP and portfolio IPM analyses explicitly account for conditioning, data access, and classical readout [7, 26]; the present comparison is confined to the displayed structured families. In a block-diagonal SDP embedding the matrix order is the sum of block orders, not their maximum. Substituting a block cap for that sum in a generic solver theorem is therefore unjustified. State, sample, scalar, or coherent-iterate output can have different costs, as the following query results demonstrate.

30 Query access and classical output

A formulation can have few cone factors, a simple Newton system, and an inexpensive normalized quantum state, while its requested classical output still requires reading most of the input. We make this distinction on the same instances used for the geometric results. The query ingredients are classical: the polynomial method, search, approximate counting, and quantum information bounds [11, 14, 18, 66, 67]. The contribution here is their combination with specified conic lifts, barriers, and output contracts. We do not claim new Boolean query bounds.

30.1 Access conventions and an information bound

The input is a string $b \in \{0, 1\}^n$, with $\sigma_j = (-1)^{b_j}$, accessed through the clean indexed oracle

$$O_b|j, z\rangle = |j, z \oplus b_j\rangle.$$

Quantum and randomized query complexities below have worst-case success probability at least $2/3$. Public arithmetic, magnitude preparation, index routing, numerical precision, and writing the output are separate costs. An explicit classical vector means a list of real coordinates at the accuracy specified, rather than a circuit, a sampling data structure, or an amplitude state. For a nonzero real vector v , its amplitude state is $|v\rangle = \|v\|^{-1} \sum_j v_j |j\rangle$ in an ordinary complex Hilbert space.

Whenever sample-and-query access is mentioned, it supplies coordinate queries, squared-coordinate samples, and the specified global norm. Coherent preparation access has a fixed completion $U_b = O_\sigma U_{\text{pub}}$, with O_σ the diagonal sign action on specified slots. Both U_b and its inverse then have constant raw-query cost. The action on all other slots is public. A reference slot on which the oracle acts trivially may be added, so controlled phase queries and clean bit queries are equivalent up to a constant. No free norms of hidden-dependent projections, free postselection probabilities, or arbitrary hidden-dependent unitary completions are included.

Lemma 30.1 (Explicit approximate sign recovery). *Suppose a T -query algorithm, on each $\sigma \in \{-1, 1\}^n$, returns a classical string with at most αn errors with probability at least $2/3$, where $0 \leq \alpha < 1/4$ is fixed. Then $T = \Omega_\alpha(n)$.*

Proof. Purify the algorithm and defer its measurements. Induction on its clean bit queries gives

$$|\psi_\sigma\rangle = \sum_{S \subseteq [n], |S| \leq T} \left(\prod_{j \in S} \sigma_j \right) |a_S\rangle.$$

Indeed a bit query, in its target's $|+\rangle, |-\rangle$ basis, either preserves a coefficient or multiplies it by one σ_j ; input-independent unitaries preserve degree. Thus the ensemble of final states is supported on a subspace of dimension $D_T \leq \sum_{j=0}^T \binom{n}{j}$. For uniform σ , the expected fraction of erroneous decoded bits is at most $d = (1 + 2\alpha)/3 < 1/2$. If $\hat{\sigma}$ is the decoded string, elementary conditional entropy and concavity of binary entropy H_2 give

$$H(\sigma \mid \hat{\sigma}) \leq \sum_j H(\sigma_j \mid \hat{\sigma}_j) \leq nH_2(d).$$

Holevo's information bound therefore implies

$$n[1 - H_2(d)] \leq I(\sigma; \hat{\sigma}) \leq \log_2 D_T.$$

For $T \leq n/2$, the latter is at most $nH_2(T/n)$, proving a positive linear lower bound; $T > n/2$ already suffices. These are standard entropy arguments underlying quantum random access bounds [66]; the Fourier span also appears in oracle interrogation [84]. $\square$

We will also use the standard bounded-error Boolean-mean laws

$$Q(n, \delta) = \Theta(\min\{n, \delta^{-1}\}), \quad R(n, \delta) = \Theta(\min\{n, \delta^{-2}\}) \quad (0 < \delta \leq c_0), \quad (30.1)$$

for a sufficiently small universal $c_0 > 0$. Quantum upper bounds follow from amplitude estimation, capped by reading all bits; Nayak–Wu's separated central-weight bound gives the matching lower bound [18, 67]. For the randomized lower bound, use Bernoulli inputs of means $1/2 \pm C\delta$ when $\delta \geq C'/\sqrt{n}$. Concentration makes an accurate empirical-mean estimate distinguish the two distributions, while the divergence of a T -query transcript is $O(T\delta^2)$. This gives $\Omega(\delta^{-2})$. Applying this bound at $\delta = C'/\sqrt{n}$ and monotonicity gives $\Omega(n)$ at finer accuracy. Constants handle the finite small- n cases. At precision strictly below half an adjacent integer-weight gap, parity gives a direct linear lower bound [11]; under the promise of zero or one mark, search instead gives $\Theta(\sqrt{n})$ quantum and $\Theta(n)$ randomized queries [14, 18].

30.2 Exposed rank, movement, and sign output on one lift

Fix a simple EJA of rank $r \geq 2$ attaining dictionary capacity $B = a(r-1) > 0$. Let $n = qB$, $s = n+1$, $q \geq 2$, and use full half-Peirce groups in the lift (4.9). Put

$$g_\sigma = n^{-1/2}(\sigma_1, \dots, \sigma_n), \quad v_\sigma = (g_\sigma, 0), \quad X_G^0 = c + q^{-1}d.$$

The reference is Slater feasible; it need not be the analytic center.

Theorem 30.2 (A common instance with different output costs). *For every hidden string, the minimum total Jordan rank of a genuine certificate for v_σ on this lift is q . In the restricted standard Jordan log-determinant metric,*

$$d_F(X^0, \{X : 1 - \langle v_\sigma, \pi X \rangle \leq \epsilon\}) \geq \sqrt{q} \log(1/\epsilon), \quad 0 < \epsilon < 1. \quad (30.2)$$

The projected optimizer amplitude state is prepared by one sign query. Real-coordinate states of its structured lifted optimizer and of every positive-multiplier standard-barrier central point also use one sign query, conditional on preparation of their public magnitude states. For every fixed $0 < \epsilon < 1/8$, explicit feasible projected or lifted ϵ -optimal output requires $\Theta_\epsilon(n)$ queries in either the quantum or randomized model.

Proof. Every group has $\|g_G\|^2 = 1/q$. The unique projected optimizer has $w_G = g_G$, $b = 0$, and feasibility forces $z_G = 1/q$. Set

$$A = \frac{1}{2q}, \quad Y_G = Ac - \frac{g_G}{\sqrt{2}} + \frac{P(g_G)c}{2A}.$$

Lemma 4.2 gives a positive rank-one element with tail trace $1/2$. Direct pairing gives $\sum_G \langle Y_G, X_G \rangle = 1 - g_\sigma^T w$ on the whole affine slice. Every certificate has nonzero half-Peirce coefficient in each full group, so every block is nonzero, proving the exact minimum rank q . The positive eigenvalue of Y_G is $(q+1)/(2q)$. Its primitive support idempotent u_G has c -mass $1/(q+1)$ and d -mass $q/(q+1)$, hence

$$\det_{u_G} Y_G = \frac{q+1}{2q}, \quad \det_{u_G} (P(u_G)X_G^0) = \frac{2}{q+1}.$$

Thus the scale in Theorem 23.1 is $\Delta = q[(1/q)^q]^{1/q} = 1$, proving (30.2).

The optimizer state is $n^{-1/2} \sum_j \sigma_j |j\rangle$. For central points, the determinant formula (4.7) and stationarity in w_G give $w_G = t_G g_G$ with $t_G > 0$. The objective and barrier depend on group orientations only through these inner products and norms. Equal group norms and strict convexity imply common public values $t_G = t$ and $z_G = z$, and a public b . All remaining coordinates are public. Phase kickback therefore supplies the hidden signs in one query after public magnitude preparation. For the Albert algebra, this statement encodes its 27 real coordinates; it assumes neither octonionic quantum amplitudes nor an efficient Jordan-product circuit.

For any feasible projected point x of support gap at most ϵ ,

$$\|x - v_\sigma\|^2 \leq 2(1 - v_\sigma^T x) \leq 2\epsilon.$$

Each incorrectly decoded sign costs at least $1/n$ in this squared error, so at most $2\epsilon n$ signs are wrong. Lemma 30.1 applies with $\alpha = 2\epsilon < 1/4$. Reading all signs constructs the optimizer and its structured lift, proving the upper bound. Projection transfers the lower bound to explicit lifted output. $\square$

The movement theorem implies at least $\sqrt{q} \log(1/\epsilon)/[-\log(1-R)]$ feasible chords of starting local norm at most $0 < R < 1$. This bound and the query lower bound are simultaneous requirements; they do not multiply. The query statement permits reuse of all information learned at earlier points.

The quantifier reversal (4.11) has an access interpretation. Let $u = (n^{-1/2}\mathbf{1}, 0)$, choose a public orthogonal U_0 sending u to the pole, and set

$$U_\sigma = U_0(D_\sigma \oplus 1).$$

Then $U_\sigma v_\sigma$ is the rank-one pole support. One coherent use of this objective-dependent rotation costs one sign query plus public gates. An explicit classical matrix approximation with operator-norm error strictly below $1/2$ reveals every sign from the diagonal of $U_0^T \tilde{U}_\sigma$ and costs $\Theta(n)$ queries. The cost of using a reformulation therefore depends on whether it is implemented coherently or written explicitly; the geometric quantifier reversal alone cannot prescribe that cost.

30.3 Well-conditioned disk checkpoints with hard scalar readout

For $N \geq 2$, minimize $\sum_i c_i^T z_i$ over $(\mathbb{B}_2^2)^N$, where $c_i = e_1$ if $b_i = 0$ and $c_i = -e_2$ if $b_i = 1$. For

$$F(z) = -\sum_i \log(1 - \|z_i\|^2), \quad r(a) = \frac{a}{\sqrt{1+a^2}+1},$$

the central point is $z_i(\eta) = -r(\eta)c_i$. The Hessian of one block is

$$H(z) = \frac{2}{1 - \|z\|^2} I + \frac{4}{(1 - \|z\|^2)^2} z z^T. \quad (30.3)$$

Its radial-to-transverse eigenvalue ratio is $\sqrt{1+\eta^2}$, so $\kappa(F''(z(2))) = \sqrt{5}$, independently of N . At this checkpoint the scalar Hessian is diagonal; its structural graph has only disjoint constant-size components. If $r_0 = r(2)$ and $w = |b|$, the fixed affine observable

$$Nr_0 + \sum_i (z_i)_1 = r_0 w \quad (30.4)$$

requires $\Theta(N)$ quantum and randomized queries at any fixed error strictly below $r_0/2$. Under $w \in \{0, 1\}$ the complexities are instead $\Theta(\sqrt{N})$ and $\Theta(N)$. These follow directly from parity and search, and reading all bits gives the first upper bound.

The objective optimum itself is the public value $-N$; the observable (30.4) is a solution readout. Explicit feasible output of total objective gap less than $1/8$ is also query-linear, since

$$\|z_i + c_i\|^2 \leq 2(1 + c_i^T z_i)$$

places each block within distance $1/2$ of its optimizer, and the two optimizer prototypes $-e_1, e_2$ are separated by $\sqrt{2}$. In contrast, $|z(\eta)\rangle$ is obtained for any $\eta > 0$ by querying a uniform index into an axis register and applying the corresponding public sign, using one clean bit query. The norm $\sqrt{N}r(\eta)$ is public. Ordinary raw objective sample-and-query access adds no information: blocks are uniformly sampled and their occupied axis is learned with one query. Access to a norm of an arbitrary coordinate subvector would be stronger, for example revealing $\sqrt{N-w}$ for the first-coordinate objective part.

The affine functional in (30.4) has Euclidean norm $\sqrt{N}$. Normalizing it to norm one changes its required error to $O(N^{-1/2})$. Likewise a constant local-neighborhood guarantee at the checkpoint does not ensure the fixed scalar accuracy: its dual local norm is $\Theta(\sqrt{N})$, so a generic local-norm error guarantee must be $O(N^{-1/2})$. The lower bound concerns the specified scalar output.

30.4 Sharp precision laws with public magnitude data

Impose $y_i = x_i$ in each disk and use raw objective pairs

$$\tilde{c}_i = \frac{\Gamma}{\sqrt{10N}}(1, 3\sigma_i), \quad \Gamma > 0.$$

All raw magnitudes and the raw global norm Γ are public. In the orthonormal coordinates $(x_i, y_i) = t_i(1, 1)/\sqrt{2}$, the domain is $[-1, 1]^N$ and the reduced coefficient is

$$\beta_i = \frac{\Gamma}{\sqrt{5N}} \begin{cases} 2, & b_i = 0, \\ -1, & b_i = 1. \end{cases}$$

Hence

$$\text{OPT}(b) = -\frac{2\Gamma\sqrt{N}}{\sqrt{5}} + \frac{\Gamma}{\sqrt{5N}}w. \quad (30.5)$$

Theorem 30.3 (Product-domain precision law). *For $0 < \epsilon \leq c_0 \Gamma \sqrt{N}$, with $c_0 > 0$ a sufficiently small universal constant, estimating (30.5) to additive error ϵ has worst-case complexities*

$$Q = \Theta\left(\min\left\{N, \frac{\Gamma\sqrt{N}}{\epsilon}\right\}\right), \quad R = \Theta\left(\min\left\{N, \frac{\Gamma^2 N}{\epsilon^2}\right\}\right). \quad (30.6)$$

The same laws hold for the ordinary objective evaluated at the exact central multiplier $\eta_\Gamma = 2\sqrt{5N}/\Gamma$. The reduced Hessian has condition at most four along the entire central path. Its normalized optimizer state is one-query preparable. Normalized projected checkpoint and central-predictor states are heralded exact in $O(1)$ expected queries, or have constant error in $O(1)$ worst-case queries. Explicit feasible output of objective gap below $\Gamma/\sqrt{5N}$ requires $\Theta(N)$ queries.

Proof. Equation (30.5) is an affine rescaling of w/N with scale $\Gamma\sqrt{N/5}$. Equation (30.1) proves (30.6), with constants absorbed into c_0 . The reduced barrier is $-\sum_i \log(1 - t_i^2)$ and its central point is $t_i = -\text{sign}(\beta_i)r(\eta|\beta_i|)$. Its scalar Hessian at force a is

$$h(a) = \sqrt{1 + a^2}(\sqrt{1 + a^2} + 1).$$

For $a > 0$, $h(a)/a^2 = s/(s - 1)$, $s = \sqrt{1 + a^2}$, is decreasing; thus $h(2a)/h(a) \leq 4$, and continuity handles $a = 0$. At η_Γ , the forces are four and two. Subtracting the public all-zero baseline from the central objective gives

$$L_b(\eta_\Gamma) + \frac{2\Gamma\sqrt{N}}{\sqrt{5}}r(4) = \frac{\Gamma}{\sqrt{N}}\Delta_{\text{cp}}w, \quad \Delta_{\text{cp}} = \frac{\sqrt{17} - \sqrt{5}}{2\sqrt{5}} > 0. \quad (30.7)$$

This is another affine rescaling of w/N by $\Theta(\Gamma\sqrt{N})$.

The optimizer coordinates are $-\sigma_i$, so phase kickback gives their normalized state. Checkpoint magnitudes are the two public constants $r(4), r(2)$. Query a uniform index, rotate a flag with amplitude proportional to the desired signed magnitude, uncompute the input bit, and postselect the flag. The success probability is uniformly bounded below by $(r(2)/r(4))^2 > 0$. Repetition gives heralded exact preparation with constant expected queries; a fixed number of repetitions gives constant error. The central predictor is $t_i = -\beta_i/h(\eta|\beta_i|)$, so the same argument at the specified checkpoint applies to its two nonzero public magnitudes. It is a statement about this right-hand side, not arbitrary Newton systems. Finally, any wrong coordinate sign incurs gap at least $\min_i |\beta_i| = \Gamma/\sqrt{5N}$. Below that threshold every sign is correct and all bits are recovered; parity supplies the lower bound. $\square$

Raw squared-coordinate sampling uses uniform i , then probabilities $1/10, 9/10$ for the two coordinates, and a returned sign costs one query. The canonical coherent completion phases only the second coordinate of a public magnitude preparation. These interfaces preserve the theorem. Granting $\|\beta\|^2 = \Gamma^2(4N - 3w)/(5N)$ for free would reveal the answer and does not preserve it.

Moving the signs into the constraints. Equivalently, use the public objective $\Gamma(1, 3)/\sqrt{10N}$ in every disk and impose the hidden equalities $y_i = \sigma_i x_i$. Their matrix has disjoint rows $(-\sigma_i, 1)$, hence $EE^T = 2I$. Row norms, singular values, column degrees, and squared sampling laws are all public. The hidden orthonormal coordinate $(x_i, y_i) = t_i(1, \sigma_i)/\sqrt{2}$ gives exactly the preceding coefficients and query laws. The canonical coherent matrix completion phases the first-coordinate slots of a public preparation; its calls have constant raw-query cost. A coherent use of the hidden basis costs $O(1)$ queries; its explicit classical materialization costs $\Theta(N)$. The ambient optimizer pairs

are $(-\sigma_i, -1)/\sqrt{2}$, and thus admit one-query amplitude preparation after a public coordinate split. At the checkpoint they are $(-r(4), -r(4))/\sqrt{2}$ or $(r(2), -r(2))/\sqrt{2}$, with the heralded preparation just proved. These assertions concern projected variables; they give neither the normalization of a hidden vector for free nor a full lifted-state oracle.

Comparison with a single direct ball. Keep the same raw objective and public equalities $y_i = x_i$, but replace the product of disks by the one ball $\mathbb{B}_2^{2N}$. The reduced domain is $\mathbb{B}_2^N$, its standard restricted barrier has parameter one, and

$$\text{OPT}_{\text{ball}}(b) = -\Gamma\phi(w/N), \quad \phi(u) = \sqrt{(4 - 3u)/5}.$$

Since $3\sqrt{5}/20 \leq |\phi'(u)| \leq 3\sqrt{5}/10$, the value laws become

$$Q = \Theta(\min\{N, \Gamma/\epsilon\}), \quad R = \Theta(\min\{N, \Gamma^2/\epsilon^2\}) \quad (0 < \epsilon \leq c_0\Gamma). \quad (30.8)$$

At multiplier $1/\Gamma$, the central value is $-\Gamma\phi(u)r(\phi(u)) = -\Gamma(\sqrt{1 + \phi(u)^2} - 1)$. Its derivative in u is $3\Gamma/[10\sqrt{1 + \phi(u)^2}]$, bounded above and below by positive multiples of Γ , so the same laws hold. The reduced Hessian condition is at most $3/\sqrt{5}$ by (30.3); it is diagonal plus rank one. Its normalized optimizer and central states are $|\beta\rangle$, prepared by projecting the raw objective state onto the public equality subspace. The probability is $\|\beta\|^2/\Gamma^2 \in [1/5, 4/5]$, giving heralded exact constant expected queries. Explicit feasible gap below $\Gamma/(8\sqrt{5}N)$ decodes all signs: its squared distance to $-\beta/\|\beta\|$ is at most $2\epsilon/\|\beta\|$, whereas every optimizer coordinate has magnitude at least $1/(2\sqrt{N})$. The equivalent public-objective hidden-equality encoding also applies.

For this direct ball, the fixed metric distance from zero to the ϵ -accurate set is at least $\lceil \log(\Gamma/(2\sqrt{5}\epsilon)) \rceil_+$: if $d = -\beta/\|\beta\|$ and gap $g = \|\beta\|(1 - d^T t)$, then $1 - \|t\|^2 \leq 2g/\|\beta\|$, and the barrier gradient has dual norm at most one. Thus logarithmic movement and the value-query law hold on the same direct-ball instances. The half-power difference between (30.6) and (30.8) concerns two different feasible bodies. It is not a reformulation separation for one optimization problem. Moreover, the sliced product domain is a box, so full-product disk factor-minimality results do not apply to that slice.

30.5 Several independent search diagnostics

There is also a useful intermediate output contract between one scalar and a dense vector. Let k source balls have even dimension $s \geq 4$, and let each have one hidden marked index $j_a \in [s]$. Define $\sigma_{ai} = 1 - 2\mathbf{1}_{i=j_a}$ and minimize $\sum_{a,i} \sigma_{ai} w_{ai}$. Use the globally smooth cap-optimal grouped Lorentz formulation (6.4): for an integer cap $3 \leq d < s + 1$, put $\ell = \lceil s/(d - 2) \rceil$, partition each ball into balanced nonempty groups, and use

$$q_{aG} = t_{aG} - \|w_{aG}\|^2 > 0, \quad \sum_G t_{aG} = 1, \quad F = -\sum_{a,G} \log q_{aG}.$$

For $d \geq s + 1$, take $\ell = 1$ and the direct ball barrier. Stationarity at $\eta_0 = 4\ell/(3\sqrt{s})$ gives exactly

$$q_{aG} = \frac{3}{4\ell}, \quad w_{ai} = -\frac{\sigma_{ai}}{2\sqrt{s}}, \quad t_{aG} = \frac{3}{4\ell} + \frac{|G|}{4s}. \quad (30.9)$$

Indeed $w = -\eta_0 q \sigma / 2$ and $\ell q + s(\eta_0 q / 2)^2 = 1$. For the public first half $H \subset [s]$, the affine diagnostic $p_a = \sqrt{s} \sum_{i \in H} w_{ai} + |H|/2$ equals $\mathbf{1}_{j_a \in H}$.

Returning all k diagnostics with error less than $1/3$ has query laws $\Theta(k\sqrt{s})$ quantum and $\Theta(ks)$ randomized, whereas their central projected amplitude state is prepared by one phase query. To verify the quantum lower bound, on one block use the adversary matrix joining all indices in H to all in its complement. Its norm is $s/2$; masking by a single query gives norm $\sqrt{s/2}$. For k blocks, sum its tensor-factor copies. They share a top eigenvector, giving numerator $ks/2$; a query masks all terms except its own block, giving the same denominator. The spectral adversary bound [56] proves the claim. The matching upper bound finds all k marks among ks positions: knowing the number remaining, exact amplitude amplification uses $O(\sqrt{ks/j})$ queries at the j -remaining stage, whose sum is $O(k\sqrt{s})$ [18]. This avoids a joint success penalty. For randomized algorithms under independent uniform marks, a randomly chosen block, with all other blocks simulated for free, has expected query cost T/k . Its half-location bit requires $\Omega(s)$ expected queries, hence $T = \Omega(ks)$. Scanning gives the upper bound.

The checkpoint is also well conditioned after an explicit public scaling. Write $z = \sqrt{\ell} \delta t$, $y = \delta w$, and $A = 2\sqrt{\ell} w_G$. Dividing the Hessian quadratic form by ℓ gives

$$\frac{16}{9}(z - A^T y)^2 + \frac{8}{3}\|y\|^2, \quad \|A\|^2 = \ell|G|/s < 2.$$

The coupled two-dimensional matrix, multiplied by nine, has determinant 384 and trace below 72, so all blocks have common eigenvalue bounds $384/(72 \cdot 9)$ and $72/9$. The condition is at most $27/2$, also after restriction to $\sum_G z_{aG} = 0$. The direct-ball case has condition $5/3$. This is a scaled restricted-Hessian statement, not a condition bound for an arbitrary augmented KKT matrix. A dense classical vector still needs $\Omega(ks)$ writes; a list of the k exceptional indices preserves the search advantage. These different output contracts explain why the linear arithmetic Newton solves do not settle the query comparison.

30.6 The same conclusions with one cone factor

The full disk family has the one-factor PSD formulation

$$\begin{pmatrix} S & W^T \\ W & I_2 \end{pmatrix} \succeq 0, \quad S_{ii} = 1, \quad D = S - W^T W, \quad F = -\log \det D, \quad (30.10)$$

where the columns of W are the disks. The cone order is $N + 2$. For fixed W , the fiber center is $D = \text{diag}(1 - \|w_i\|^2)$ by Hadamard's inequality, so the marginal barrier is exactly the disk barrier, with slice parameter N . Consequently all its projected central states and readouts above are unchanged. Dense auxiliary Gram coordinates $S = W^T W + D$ are not covered by the easy projected-state statement.

Starting at $W = 0, D = I$, any feasible endpoint of gap at most ϵ in the full disk problem satisfies

$$\det D \leq \prod_i (1 - \|w_i\|^2) \leq (2\epsilon/N)^N.$$

The height estimate of Lemma 25.1, with parameter N , therefore gives

$$d_F \geq \sqrt{N}[\log(N/(2\epsilon))]_+.$$

For the public-objective hidden-equality version, let $a_i = |\beta_i|$ and $g_i = a_i(1 - t_i t_i^*)$. Since $a_i \geq \Gamma/\sqrt{5N}$ and $1 - t_i^2 \leq 2g_i/a_i$, the corresponding bound is

$$d_F \geq \sqrt{N} \left[\log \frac{\Gamma\sqrt{N}}{2\sqrt{5}\epsilon} \right]_+. \quad (30.11)$$

Both bounds allow arbitrary feasible off-diagonal completions along the path, not just fiber centers. If the initial point is within center local norm $0 \leq \rho < 1$, subtract $-\log(1 - \rho)$ from the lower distance, then take the positive part. Dividing by $m[-\log(1 - R)]$ gives a round lower bound when each round has at most $m \geq 1$ integer-counted feasible chords of starting local norm at most $0 < R < 1$.

Alternatively the same disks are the fixed-scale slice of the single cone

$$\mathcal{H}_{N,2} = \{(t, z) : \|z_i\| \leq t \ (1 \leq i \leq N)\}.$$

Theorem 17.3 gives its exact optimal ambient LH barrier parameter $N + 1$, attained by

$$F_{\mathcal{H}}(t, z) = - \sum_i \log(t^2 - \|z_i\|^2) + (N - 1) \log t.$$

At $t = 1$ the restricted barrier is again the same one. The public-objective hidden-equality problem has strict primal feasibility at $t = 1, z = 0$ and strict dual feasibility: its dual cone consists of (M, v) with $M \geq \sum_i \|v_i\|$, so choose a sufficiently large positive M and the public objective blocks for v_i . For *any* ambient LH barrier, Theorem 24.2 therefore yields, from an exact primal–dual central point of gap Δ_0 to a strictly feasible point of gap $0 < \epsilon < \Delta_0$,

$$T \geq \frac{\sqrt{(N + 1)/2} \log(\Delta_0/\epsilon)}{m[-\log(1 - R)]}. \quad (30.12)$$

Here chords use the primal–dual product metric; intermediate affine feasibility is not required. If the start is at the public multiplier $\eta_0 = 2\sqrt{5}$ and $\epsilon < (N + 1)/\eta_0$, the numerator can be replaced by $\sqrt{(N + 1)/2} \log((N + 1)/(\eta_0\epsilon))$. Indeed $\Delta_0 = \nu/\eta_0$, $\nu \geq N + 1$, and $\sqrt{\nu} \log(\nu/(\eta_0\epsilon))$ is increasing on this range. The easy-state and diagonal-Hessian assertions hold for the displayed barrier, whereas (30.12) holds for every ambient LH barrier with its own metric and central start.

At $\Gamma = \sqrt{N}$ and fixed sufficiently small error, these explicit formulations thus combine one cone factor, $\Theta(N)$ scalar or explicit output queries, constant-query projected states, and $\Omega(\sqrt{N} \log N)$ bounded-movement rounds under the stated metric contracts. A model charging both resources obtains their maximum as a lower bound. Multiplying them would require a separate theorem forcing fresh query work at every round; these fixed-input families provide no such theorem. This is consistent with quantum optimization analyses that separately charge data access, linear solving, and classical output [5].

31 Acquiring an exact smaller formulation

The preceding formulation bounds concern a representation once it is known. Removing redundant constraints can require information about the input. We give families in which this acquisition cost, the resulting barrier, and its metric can all be calculated. The search bounds are classical quantum-query results; the point is their precise connection with an exact conic reformulation. In particular, an existential small barrier need not provide a cheap way to evaluate that barrier from the supplied data.

31.1 Output and access conventions

A source query returns one indexed input symbol, either classically or coherently in the usual reversible oracle. Public arithmetic and the preparation of index registers do not count as source queries. Their gate and bit costs are separate. All bounded-error statements require success

probability at least $2/3$ on the complete output, on every promised input. An *exact reusable compiler* produces a finite classical description of the exact projected body, usable without further source queries. It can, for example, return its retained inequalities and a barrier for their intersection. A wrapper that continues calling the source oracle does not meet this output convention. Lower bounds also cover a classical description with exact membership or boundary evaluation, provided that this evaluation requires no further source queries. Computation using the description may be arbitrarily expensive in a query lower bound.

Lemma 31.1 (Acquisition of independent marked positions). *Let $b \geq 1$ and $m \geq 2$. In each of b public blocks of m bits, exactly one bit is one. Returning all marked positions has quantum query complexity $\Theta(b\sqrt{m})$ and randomized query complexity $\Theta(bm)$. If exactly k of N bits are marked, with k known, all marks can be found with certainty in $O(\sqrt{Nk})$ quantum queries.*

Proof. Exact amplitude amplification with a known number of marked positions finds a mark in $O(\sqrt{n/t})$ queries when $t > 0$ of n positions are marked [18, Theorem 4]. Apply this separately to the blocks. For known total cardinality, suppress previously found indices and sum $O(\sqrt{N/t})$ for $t = k, k-1, \dots, 1$. There is no accumulation of failure probabilities in these exact-query upper bounds.

For completeness, index block inputs by $r \in [m]^b$ and use the nonnegative adversary matrix

$$\Gamma = \sum_{j=1}^b I^{\otimes(j-1)} \otimes (J_m - I_m) \otimes I^{\otimes(b-j)}.$$

Its norm is $b(m-1)$. A query to position ℓ of block j retains only the j th summand, replacing $J_m - I_m$ by the adjacency matrix of the star joining ℓ to the other positions. Its norm is $\sqrt{m-1}$. The spectral adversary bound therefore gives $\Omega(b\sqrt{m-1}) = \Omega(b\sqrt{m})$; this is the standard direct-sum argument [2, Theorems 3–4]. Classically, under independent uniform marked positions, success probability $2/3$ for all outputs implies that probability for each individual position. Revealing the other blocks in advance cannot make a uniform marked position in an unread block less uncertain. A successful search in that block requires $\Omega(m)$ expected queries, even allowing a final unqueried guess. Summing over blocks gives $\Omega(bm)$; scanning gives the matching upper bound. $\square$

The exact quantum upper bounds use the standard query model, which permits the public rotations needed for exact amplitude amplification. An approximate gate implementation must separately account for its precision.

31.2 Nested constraints and consensus

Let $d \geq 1$ be fixed and let $r_i = 2 - b_i$, where $b \in \{0, 1\}^N$ has weight zero or one. Consider

$$\max e_1^T z_1, \quad (r_i, z_i) \in Q_{d+1}, \quad z_i = z_{i+1} \quad (1 \leq i < N). \quad (31.1)$$

Its projection is exactly $\{z : \|z\| \leq r_*\}$, where $r_* = \min_i r_i$, and its optimum is r_* . The origin is strictly feasible. The public path equalities have constant incidence; only the right-hand-side radii are hidden.

Proposition 31.2 (Exact radius compilation). *The exact projected ball and an optimizer of (31.1) can be acquired with $\Theta(\sqrt{N})$ quantum queries or $\Theta(N)$ randomized queries. The same lower bounds hold for an optimum-value estimate with additive error below $1/3$. After acquisition,*

$$f_{r_*}(z) = -\log(r_*^2 - \|z\|^2)$$

is a barrier with exact parameter one. On the all-radius-two input, the natural restricted product barrier is Nf_2 and has exact parameter N . Its central path from $z = e_1$ to $z = (2 - \epsilon)e_1$, $0 < \epsilon \leq 1/2$, has length $\Theta(\sqrt{N} \log(1/\epsilon))$, compared with $\Theta(\log(1/\epsilon))$ for f_2 .

Proof. The acquisition problem is the zero-versus-one search promise. Exact amplitude amplification, followed by verification of the returned index, decides it in $O(\sqrt{N})$ queries [18, Theorem 16]. The search lower bound is [14]; the uniform single-mark distribution gives the classical lower bound. The optimum values are separated by one. The fixed ball barrier has parameter one by the ball calculation in Section 9. Alternatively, direct inversion of its Hessian gives gradient squared dual norm $2\|z\|^2/(r_*^2 + \|z\|^2) < 1$, with boundary limit one; self-concordance follows from its Lorentz restriction. Multiplication by the integer N preserves self-concordance and multiplies this exact parameter by N . On the axis,

$$f_2''(a) = \frac{2(4 + a^2)}{(4 - a^2)^2},$$

whose square root is comparable to $(2 - a)^{-1}$ on $[1, 2)$. The length follows by integration. Reflection symmetry and strict convexity put every center for objective e_1 on this axis. $\square$

Every move of local norm at most $R \in (0, 1)$ has metric length at most $-\log(1 - R)$. Hence the displayed natural-barrier central schedules require $\Omega_R(\sqrt{N} \log(1/\epsilon))$ moves. This is a statement about that barrier and those local schedules. One cannot obtain a uniform replacement simply by dividing the original barrier by N . If exactly one radius is one and $N > 1$, its normal restriction near the tighter boundary is $-N^{-1} \log(1 - a) + O(1)$ with smooth bounded corrections. For $-c \log t$, the self-concordance ratio $|f'''|/[2(f'')^{3/2}]$ equals $c^{-1/2}$. The divided barrier therefore violates self-concordance near that boundary.

There is a separate conditioning issue in the uneliminated consensus system. At its analytic center and in one coordinate, put $D = \text{Diag}(2/r_i^2)$ and let A be the path incidence matrix. Then

$$K = \begin{pmatrix} D & A^T \\ A & 0 \end{pmatrix}, \quad \kappa(AD^{-1}A^T) = \Theta(N^2), \quad \kappa(K) = \Theta(N^2). \quad (31.2)$$

Here condition numbers use singular values and $N \geq 2$. Indeed, $I/2 \preceq D \preceq 2I$, whereas the extreme eigenvalues of AA^T have orders N^{-2} and one. For a negative eigenvalue $-t$ of K , eliminating its primal component gives $A(D + tI)^{-1}A^T y = ty$. The smallest positive solution has order N^{-2} by these same comparisons and the min-max principle; all positive eigenvalues are bounded below by $1/2$, and all eigenvalues have bounded magnitude. Thus K has the stated condition number despite its forest support. Public consensus elimination removes this factor. Equation (31.2) is consequently not a lower bound on the cost of a linear-system algorithm allowed that elimination.

More generally, suppose N constraints are partitioned into k supplied chains under inclusion, of lengths $n_1, \dots, n_k$. An ordered queried key identifies each chain's smallest member. Once that member is known, suppose it has a specified barrier of parameter ν_j . Minimum finding in each chain [28], amplified to joint success, gives

$$O\left(\log(2k) \sum_{j=1}^k \sqrt{n_j}\right) \leq O\left(\sqrt{Nk} \log(2k)\right) \quad (31.3)$$

queries and a compiled barrier with parameter $\nu_0 + \sum_j \nu_j$ when an additional public barrier of parameter ν_0 is present. Common strict feasibility and positive definiteness on the effective affine space are understood. This bound covers homothetic copies $r_i C$ of a public compact convex body containing zero: nestedness leaves only the minimum radius.

The dependence $\sqrt{Nk}$ is necessary in general. For $N = km$, take k independent interval chains $|z_j| \leq 2 - b_{j\ell}$, with each chain containing zero or one mark. Returning all optimum coordinates of $\max \sum_j z_j$ within $1/3$ returns all k OR values. The direct sum of the k star adversaries has norm $k\sqrt{m}$, while a query filters it to norm one. The lower bound is $\Omega(k\sqrt{m})$ and the classical bound is $\Omega(km)$. For this promised family the upper bound is also $O(k\sqrt{m})$, by the exact promised search in the preceding proof; no logarithmic amplification is needed. On the all-zero input the natural and compiled barriers are respectively $m \sum_j f_2(z_j)$ and $\sum_j f_2(z_j)$. Their common diagonal path from zero to total objective gap ϵ , for $0 < \epsilon \leq k/2$, has lengths

$$\Theta\left(\sqrt{N} \log(k/\epsilon)\right), \quad \Theta\left(\sqrt{k} \log(k/\epsilon)\right).$$

Additional public coupling may make the post-compilation optimization nonseparable. Its solution cost is additional to (31.3); the acquisition bound alone does not estimate it.

31.3 A two-dimensional family with incomparable active factors

For $v > 0$ write

$$q(v) = (1 + v, 3 - v^2), \quad E(q) = \{z \in \mathbb{R}^2 : q_1 z_1^2 + q_2 z_2^2 \leq 1\}, \quad u_i = \frac{i}{2(N+1)}.$$

The parameters used below lie in $(0, 1]$, so both coefficients are positive. A public sentinel is $E(q(1)) = E(2, 2)$. An input trit at position i selects $q(1)$, $q(u_i)$, or $q_i^- = (1 + u_i, 1)$, denoted respectively by $0, +, -$. Let S be the plus positions. Since the sentinel dominates all minus factors, the projected body is

$$D_S = E(q(1)) \cap \bigcap_{i \in S} E(q(u_i)). \quad (31.4)$$

Each factor is a transformed three-dimensional Lorentz slice. A path of copies of the two coordinates gives constant incidence and a scalar KKT graph of bounded treewidth: a bag can contain the two coordinates of consecutive copies, their equality multipliers, and one constant-size local cone block. Consecutive bags share only a copy. The origin is strictly feasible.

Proposition 31.3 (Pareto compilation). *If $1 \leq k \leq N/2$ and $|S| = k$ is promised, the $k + 1$ factors of (31.4) are essential in that intersection description. Exact reusable compilation has quantum query complexity $\Theta(\sqrt{Nk})$ and randomized query complexity $\Theta(N)$. These lower bounds concern an exact body description, not solution of one fixed objective or the minimum number of factors in every possible lift.*

Proof. For each retained v , put $s_v = (2v, 1)$. The identity

$$\langle q(v) - q(w), s_v \rangle = (v - w)^2$$

shows that squared coordinates $s_v / \langle q(v), s_v \rangle$ define a point on factor v and strictly inside all the others. Thus removal of v enlarges the body: a small outward perturbation still satisfies the other inequalities. It also follows that different sets of retained parameters give different bodies. An exact source-free membership description can distinguish them: evaluate it at the finitely many public witnesses obtained by these perturbations for the universe of all parameters. The perturbations may be chosen so that each witness violates just its own factor in that universe.

The upper bound is the known-cardinality search of Lemma 31.1. For the lower bound restrict to one plus position in each of k public blocks of size $m = \lfloor N/k \rfloor \geq 2$, with all other positions zero. The reusable description determines every marked position, so the lemma applies. Since $km \geq N/2$, its bounds have the claimed orders. $\square$

The retained constraints can be simultaneously active. If $v < w$ are consecutive retained parameters and $s = (v + w, 1)$, then

$$\langle q(r) - q(v), s \rangle = (r - v)(w - r).$$

Thus squared coordinates $s/\langle q(v), s \rangle$ give exactly the v and w factors tight. Their outward normals are linearly independent; a positive combination exposes this point. Its open normal cone therefore supplies objectives using both factors. The projected problem is a coupled planar intersection, not independent scalar minimum problems.

Here the cost of a barrier can change much more than the dimension. Set $F_q(z) = -\log(1 - z^T \text{Diag}(q)z)$. On the hard branch with only zeros and pluses the natural and compiled barriers are

$$F_o = MF_{q(1)} + \sum_{i \in S} F_{q(u_i)}, \quad F_c = F_{q(1)} + \sum_{i \in S} F_{q(u_i)}, \quad M = N - k + 1. \quad (31.5)$$

Their exact parameters satisfy $M \leq \nu(F_o) \leq N + 1$ and $\nu(F_c) \leq k + 1$. For the lower bound approach the sentinel boundary on the positive first axis. The other factors and all their derivatives stay bounded, and the squared gradient dual norm tends to M . The upper bounds follow by adding the parameter-one ellipsoid barriers. The planar body itself admits a barrier of parameter two [60]; the M lower bound is specific to F_o .

For objective $\max z_1$, both central paths have $z = (a, 0)$, $0 \leq a < 2^{-1/2}$. If $f_A(a) = -\log(1 - Aa^2)$, then

$$f_A''(a) = \frac{2A(1 + Aa^2)}{(1 - Aa^2)^2}.$$

For marked factors, $1 < A < 3/2$, so f_A'' is bounded above and below by positive universal constants on this whole interval. The sentinel square-root Hessian is comparable to $(2^{-1/2} - a)^{-1}$ near its endpoint. Integrating the square root of the Hessians in (31.5), and using $\max(\sqrt{x}, \sqrt{y}) \leq \sqrt{x + y} \leq \sqrt{x} + \sqrt{y}$, gives, uniformly for $0 < \epsilon \leq 1/4$,

$$L_o(\epsilon) = \Theta(\sqrt{M} \log(1/\epsilon)), \quad L_c(\epsilon) = \Theta(\sqrt{k} + \log(1/\epsilon)). \quad (31.6)$$

Lengths run from the origin to $a = 2^{-1/2} - \epsilon$. The assumption $k \leq N/2$ ensures $M \geq k$. The endpoint estimate of Lemma 25.1 also gives an $\Omega(\sqrt{M} \log(1/\epsilon))$ distance lower bound for F_o at sufficiently small universal ϵ : its barrier increase is at least $M \log(c/\epsilon)$, while $\nu(F_o) \leq M + k \leq 2M$. Thus bounded local steps incur the corresponding lower bound even if the path leaves the central curve.

31.4 A growing-dimensional fixed-objective decoder

The distinction between body compilation and a fixed objective can be removed by allowing the projected dimension to grow. Let $N = km$, $m \geq 2$, and $u_\ell = \ell/[2(m + 1)]$. In block j exactly one position ℓ_j is marked. In coordinates (y, p_j, q_j) retain the public sentinel and the marked inequality

$$4(y^2 + p_j^2 + q_j^2) \leq 1, \quad y^2 + (1 + u_{\ell_j})^2 p_j^2 + (3 - u_{\ell_j})^2 q_j^2 \leq 1. \quad (31.7)$$

Before compilation there are m hidden factors in each block, each either the sentinel or the marked factor, and one additional public sentinel. The coefficient matrices are rational with $O(\log m)$ bits, and all factors are direct Q_4 slices. The common y is the sole coupling between blocks.

Proposition 31.4 (Fixed-objective acquisition and metric separation). *The body (31.7) has dimension $2k + 1$ and $2k$ essential retained inequalities. It has a lift with $\Theta(N)$ constant-size cone factors, bounded row and column incidence, and bounded scalar KKT treewidth, all with bounds independent of k, m . Both exact reusable compilation and a coordinate-readable solution of $\max \sum_j q_j$ with coordinate error below $1/[40(m + 1)]$ require and suffice with $\Theta(\sqrt{Nk})$ quantum queries or $\Theta(N)$ randomized queries. The same lower bounds hold for an exactly feasible coordinate-readable solution with total objective gap below $1/[40(m + 1)]$.*

For the separate objective $\max y$, the natural and compiled explicit barriers have parameters of orders N and k and central lengths, from their common center to $y = 1/2 - \epsilon$,

$$\Theta\left(\sqrt{N} \log(1/\epsilon)\right), \quad \Theta\left(\sqrt{k} \log(1/\epsilon)\right) \quad (0 < \epsilon \leq 1/4).$$

Proof. The origin is strict. To make the marked factor of block j alone tight, set $y = p_j = 0$, $q_j = (3 - u_{\ell_j})^{-1}$, and all other coordinates zero. Its sentinel is strict since $3 - u_{\ell_j} > 5/2$. To make its sentinel alone tight, take $y = q_j = 0$, $p_j = 1/2$; the marked factor is strict since $1 + u_{\ell_j} < 3/2$. These points also prove essentiality after small outward perturbations. For bounded incidence, give every slot a triple copy and join triples along paths within blocks. Connect representative y copies between blocks along a path. A constant-size bag for consecutive copies, their multipliers, and each local cone block gives a tree decomposition; at block junctions only a constant number of representative coordinates need be retained. This also proves the assertion for the scalar KKT graph, including dense local cone Hessians.

For the decoding objective, every feasible point has $q_j \leq (3 - u_{\ell_j})^{-1}$. All these bounds are attained simultaneously only at

$$y = 0, \quad p_j = 0, \quad q_j^* = (3 - u_{\ell_j})^{-1}.$$

The optimum is unique. Neighboring candidate q_j^* values differ by more than $1/[18(m + 1)]$. A coordinate error below $1/[40(m + 1)]$ is less than half that spacing, so rounding recovers every mark. For a feasible point the deficits $q_j^* - q_j$ are nonnegative; a total objective gap below the same threshold bounds each deficit. Lemma 31.1 proves the lower bounds and the acquisition upper bounds. Once the marks are known, the exact optimizer takes $O(k)$ arithmetic operations. The conclusion is for readable coordinates, not the objective value alone or a normalized quantum state.

For $\max y$, reflection symmetry gives $p_j = q_j = 0$ on both central paths. Write $f_A(y) = -\log(1 - Ay^2)$. Their restrictions are exactly

$$F_o(y) = Nf_4(y) + kf_1(y), \quad F_c(y) = kf_4(y) + kf_1(y).$$

The term f_1'' is uniformly bounded on $[0, 1/2)$, whereas $\sqrt{f_4''}$ has order $(1/2 - y)^{-1}$ near the endpoint. Integration proves the length assertions. The squared gradient dual norm on this axis tends to N or k , respectively, while additivity gives upper parameters $N + k$ and $2k$. The diagonal symmetry removes mixed Hessian entries on the axis, so these are also lower bounds for the full barriers. $\square$

The post-compilation problem is only coupled through one scalar. Its explicit decoding objective is easy once the hidden constraints are known. This is an acquisition advantage; it does not require a quantum linear-system solver.

31.5 Blockwise relative-entropy compilation

The same acquisition phenomenon occurs for nonsymmetric cones. Use $D(U\|V) = U \log(U/V)$, with its closed perspective extension. Let $N = Bm$, $B \geq 1$, $m \geq 2$, and choose public values

$\gamma_1 < \dots < \gamma_m$ in $[1, 2]$ with minimum spacing $h \asymp m^{-1}$. Each block has exactly one marked position r_j . Set $d_{ji} = 1$ except $d_{jr_j} = e^{-\gamma_{r_j}}$. An oracle query returns the mark bit; the coefficient at a marked position is a public function of that position. Coefficients may be retained symbolically, so no exact transcendental-number encoding is hidden in this query convention. All source terms in block j share U_j, V_j .

The elementary identity (21.7) becomes

$$\sum_{i=1}^m D(U_j \| d_{ji} V_j) = m U_j \log(U_j / V_j) + \gamma_{r_j} U_j = D(m U_j \| m e^{-\gamma_{r_j}/m} V_j). \quad (31.8)$$

One can equivalently retain the rational/public coefficient γ_{r_j} in the middle expression. Projection of m separate epigraphs, under $t_j = \sum_i t_{ji}$, is exactly the single epigraph of (31.8). One implication follows by summing inequalities. Conversely, distribute the nonnegative aggregate slack among the m epigraph coordinates. This argument includes the closed perspective boundary whenever the aggregate value is finite; if it is infinite, neither epigraph has a point over those U_j, V_j .

Proposition 31.5 (Entropy acquisition and matched-gap lengths). *An exact reusable compiler for these B aggregate epigraphs has query complexities*

$$Q = \Theta(B\sqrt{m}) = \Theta(\sqrt{NB}), \quad R = \Theta(Bm) = \Theta(N).$$

The same lower bounds apply to a coordinate-readable, exactly feasible solution of the slice $U_j = V_j = 1$ minimizing $\sum_j t_j$, with objective excess below $h/3$. On that slice, at matched positive total gap Δ , the natural and compiled projected central points coincide. Between gaps $\Delta_0 > \epsilon > 0$ their exact barrier-metric lengths are

$$\sqrt{N} \log(\Delta_0/\epsilon), \quad \sqrt{B} \log(\Delta_0/\epsilon), \quad (31.9)$$

for their respective displayed product barriers.

Proof. The projected epigraph in block j , evaluated at $U_j = V_j = 1$, has lower boundary $t_j = \gamma_{r_j}$. Hence an exact reusable description determines all r_j ; exact membership at the public midpoints between candidate boundary values suffices. Lemma 31.1 gives both bounds and the upper algorithm. For the solution task, feasibility gives $e_j = t_j - \gamma_{r_j} \geq 0$. Total excess below $h/3$ implies $e_j < h/3$ for every j , and nearest-value rounding recovers all marks.

With $U_j = V_j = 1$, the usual relative-entropy cone barrier restricts to $-\log(t_{ji} - D(1 \| d_{ji}))$, up to a constant. At multiplier $\eta > 0$, all N source slacks equal $1/\eta$, so their total gap is N/η and each projected block slack is m/η . The compiled barrier has B slacks, each $1/\hat{\eta}$, total gap $B/\hat{\eta}$. At $\eta = N/\Delta$ and $\hat{\eta} = B/\Delta$, both project to $t_j = \gamma_{r_j} + \Delta/B$. A positive orthant logarithmic barrier has line element $\sum_i (ds_i/s_i)^2$. Along these curves it is respectively $N(d\Delta/\Delta)^2$ and $B(d\Delta/\Delta)^2$, proving (31.9). $\square$

For the natural formulation the first length is measured in its full source-slack coordinates. It is also the length of the common projected path for the marginal barrier obtained by minimizing over the source slack allocations: that marginal is $-m \sum_j \log(t_j - \gamma_{r_j})$, up to a constant. Equal multipliers do not give equal projected points; matched gaps are essential. The statement does not extend here to moving U_j, V_j . Allowing feasibility violations also changes the decoder: negative deficits must be charged in addition to the objective excess.

The two full ambient exponential-cone products have optimal barrier parameters $3N$ and $3B$, even among coupled barriers, by Theorem 19.2. On the fixed slice in Proposition 31.5, the displayed parameters are N and B . Neither ambient number is an intrinsic lower bound for every barrier on

the projected epigraph. The small relative-entropy barriers of Fawzi and Saunderson [35] and the positive-map constructions of He et al. [50] are the relevant prior barrier results. Our identity is their elementary scalar setting, not a new relative-entropy barrier theorem. The conclusions here concern acquiring the effective coefficients, decoding readable outputs, and comparing the specified metrics at equal gap.

For scalar variables, bounded incidence and bounded KKT treewidth can be retained before compilation. Arrange each block as a binary tree; carry copies of U_j, V_j down its edges and sum the epigraph variables up the same tree. Each node has only a constant number of variables and equations, and each parent–child separator has constant size. These separators give an explicit bounded-width tree decomposition. For matrix entropy variables the separator size grows with the matrix dimension; no dimension-free scalarized treewidth claim follows.

31.6 What the oracle advantage includes

Consensus reconstruction is public: a projected vector supplies every copied coordinate. For a nonzero amplitude-encoded vector, equal copying uses the public map $|z\rangle \mapsto |u_N\rangle|z\rangle$. An explicit array of all Nd copied entries still costs $\Omega(Nd)$ output operations. In the entropy example an implicit online compiler could instead search one requested block in $O(\sqrt{m})$ queries and retain access to the input; that output does not incur all-block acquisition until all blocks are requested.

There is also no free explicit-input loading theorem. With bounded fan-in $g > 1$, a readable OR output sensitive to m source bits has a backward dependency cone containing those bits, and therefore requires $\Omega(m)$ input incidences and depth $\Omega(\log_g m)$. Applying this to the independent promised OR chains gives total $\Omega(N)$ incidences and depth $\Omega(\log_g(N/k))$. Parallel classical reductions match these orders. The query improvements above consequently assume a supplied coherent oracle or an already constructed access structure.

Redundancy removal before an interior-point solve is classical presolve [42]; quantum search on partially ordered sets is also established [63]. Our chain model gives unordered access to inclusion keys, whereas the poset-search models in the latter reference have their own comparison or ordering promises. Minimum finding, amplitude amplification, and adversary direct sums are established methods. The explicit families above combine them with essential-constraint witnesses, sparse conic lifts, readable-output reductions, and exact metric comparisons. The results distinguish that conjunction from a general quantum optimization speedup, an all-barriers lower bound, or a universal claim that fewer retained inequalities always give the best lift.

32 Closing synthesis

Conic reformulation has several distinct costs. Local curvature constrains cone resources; the full certificate fiber determines the least exposing rank; global regularity can impose additional topological costs. Affine restriction can lower a standard barrier parameter, while changing the barrier on the same domain is another optimization problem. The exact formulas proved here identify these costs in their stated models. The bounded interval (8.6) and the nonsymmetric parameter intervals delimit their reach. The general integer ledgers give necessary resource bounds; their integer minima need not be attainable lift optima.

The weighted tree coefficient makes the metric distinction concrete. Active support ranks explain part of the distance, but objective-free linked subtrees contribute half their barrier weight to its squared leading coefficient. A central route pays their full weight. The PSD quotient theorem gives a different concrete lesson: retaining exact source allocations improves a spectral transfer from

squared to first powers, while changing the auxiliary point can still change the Newton right-hand side and the required data.

These geometric conclusions become computational conclusions only through a specified interface. A short description, an inexpensive normalized state, a readable solution, and a reusable exact compiler are different outputs. Fresh scale updates have a temporal query cost under the stated commitment contract; fixed data can instead be cached. Thus the paper's work bounds use explicit access and charging assumptions, and its metric bounds retain their objective, endpoint, and barrier. Keeping these conditions visible is what makes the formulation comparisons reusable.

Appendices

A Further topological restrictions

These refinements retain information about abstract tangent splittings and individual saturated channels. Joint saturation is governed by the stronger covering theorem in Section 10.

A.1 What the abstract tangent splitting alone implies

For C^2 factors with $P = D = M$ and ζ the identity, suppose the full mixed form H is positive definite. Each individual form $H_i = -DA_i^*DB_i$ is positive semidefinite: the Hessian of the nonnegative function $\langle A_i(x), B_i(z) \rangle$ at (x, x) is positive semidefinite and annihilates the tangent diagonal. Its diagonal blocks are therefore H_i , and its off-diagonal blocks are $-H_i$. If $\sum_i \text{rank } H_i = n$ everywhere, lower semicontinuity of each rank and constancy of their sum make the individual ranks locally constant. Whitening by H gives positive semidefinite matrices of total rank n summing to the identity. Factoring them as $Z_i Z_i^*$ and concatenating the n columns shows that each whitened matrix is an orthogonal projection. Thus

$$TM = \bigoplus_i E_i, \quad E_i = \text{im}(H^{-1}H_i). \quad (\text{A.1})$$

For exact globally smooth cone factors at capacity equality, Theorem 10.1 is stronger than the following necessary conditions. They remain useful when only the bundle splitting is retained.

Proposition A.1 (Ranks in a sphere tangent splitting). *Suppose $T\mathbb{S}^n = \bigoplus_i E_i$, with positive ranks r_i , and put $s = \rho(n+1) - 1$, where $\rho((2a+1)2^{4b+c}) = 8b + 2^c$ for $0 \leq c \leq 3$. Every subset sum $r_I = \sum_{i \in I} r_i$ satisfies*

$$0 < r_I \leq n/2 \implies r_I \leq s.$$

If $s < n/3$, there is a unique rank greater than $n/2$, and it is at least $n - s$. If all ranks are at most $c \leq n/2$, the splitting is impossible when $s = 0$; otherwise it needs at least $\lceil n/\min\{c, s\} \rceil$ summands. For $n \geq 3$, if a summands have rank one and b have rank two, then $a + 2b \leq s$.

Proof. For completeness, the plane-field implication can be seen from clutching. At the endpoint $2r = n$, a splitting would make $e(T\mathbb{S}^n) = e(E)e(F) = 0$, contradicting its Euler number two. For $0 < 2r < n$, both summands in a decomposition $T\mathbb{S}^n = E^r \oplus F^{n-r}$ are orientable. Their clutching maps take values in $SO(r)$ and $SO(n-r)$. If $r \leq n-r$, the two inclusions in $SO(n)$ can be conjugated into the same copy of $SO(n-r)$; the conjugations are homotopic to the identity in $SO(n)$. The sum of the resulting homotopy classes lies in the image of $\pi_{n-1}SO(n-r)$. Hence $T\mathbb{S}^n$ admits a reduction to $SO(n-r)$, which supplies r independent tangent vector fields. This is the classical plane-field argument of Steenrod [83]. Adams's theorem bounds their number by s [1]. Apply it to $\bigoplus_{i \in I} E_i$. For $n = 1$ the subset assertion is empty.

If every rank were at most $n/2$, take the first partial sum exceeding $n/2$. The preceding sum and the last added rank are each at most s . The complementary sum, when nonzero, is also at most s . This gives $n \leq 3s$ (and $n \leq 2s$ for empty complement), a contradiction. The complement of the unique large rank then has rank at most s . The capped-count assertion follows by applying the subset bound to each summand. For $n \geq 3$, rank-one bundles are trivial and oriented rank-two bundles are classified by their Euler class in $H^2(\mathbb{S}^n; \mathbb{Z}) = 0$. These summands jointly supply $a + 2b$ fields. $\square$

The largest-rank bound is sharp for abstract splittings: a maximal s -frame and its orthogonal complement have ranks s and $n-s$. The inequality $s < n/3$ holds for $n+1 \geq 3$ except $n+1 = 4, 8, 16$, as follows directly from the displayed formula for ρ . For even n , a proper splitting is impossible already from $e(E \oplus F) = e(E)e(F)$: both intermediate-degree Euler classes vanish, whereas $\langle e(T\mathbb{S}^n), [\mathbb{S}^n] \rangle = 2$. For higher ranks the precise classical test is the unstable clutching condition

$$\text{blockdiag}_*(\alpha_1, \dots, \alpha_k) = \tau_n \quad \text{in } \pi_{n-1}SO(n), \quad \alpha_i \in \pi_{n-1}SO(r_i),$$

where τ_n classifies $T\mathbb{S}^n$. Stable triviality of the tangent bundle alone does not imply this condition.

A.2 Restrictions on one everywhere saturated factor

The normalization argument in the proof of Theorem 10.1 needs only that the chosen channel have positive capacity c_i and rank c_i everywhere. It then gives an individual submersion $P \rightarrow \mathbb{S}^{c_i}$, even when the total capacity is larger than $\dim P$. The next classical sphere consequence and its product extension concern this weaker hypothesis.

Proposition A.2 (Sphere and equal-sphere-product submersions). *A C^1 submersion $\mathbb{S}^n \rightarrow \mathbb{S}^r$ with $0 < r < n$ is possible only for $(n, r) = (3, 2), (7, 4), (15, 8)$, and each pair occurs. More generally, for $q \geq 2$ and $1 \leq r < kq$, a C^1 submersion $(\mathbb{S}^q)^k \rightarrow \mathbb{S}^r$ requires*

$$r = q \quad \text{or} \quad q = 2r - 1 \quad \text{with } r \text{ even.} \quad (\text{A.2})$$

Projection realizes the first possibility when $k \geq 2$. Projection followed by a Hopf map realizes the three corresponding possibilities in the second case. No sufficiency assertion is made for the other dimension pairs in (A.2).

Proof. A C^1 submersion from a compact source has a uniformly positive surjectivity modulus. Smooth approximation preserves submersivity, so Ehresmann's theorem gives a smooth fiber bundle for purposes of an existence obstruction [29]. A target circle is excluded by lifting to $\mathbb{R}$ and taking a maximum. For a sphere total space and $r \geq 2$, the fiber is connected. Browder's theorem gives its homotopy type as $\mathbb{S}^j$, $j \in \{1, 3, 7\}$ [19, Theorem 1]. The two-row mod-two Serre spectral sequence forces $r = j + 1$: its only possible nonzero differential killing the fiber class has length $j + 1$, and must land in the sole positive base degree r . The top surviving class gives $n = 2j + 1$. The Hopf fibrations attain these pairs.

Now write $E = (\mathbb{S}^q)^k$. If $r \geq 3$, the closed fiber F is simply connected by the homotopy exact sequence. It is orientable because the base and total space are orientable. Both its rational homotopy and its cohomology are finite-dimensional, so it is elliptic. The classical formal-dimension identity [38, Theorem 32.6] is

$$\dim F = \sum_{j \geq 2} w_j d_j(F), \quad w_j = \begin{cases} j, & j \text{ odd,} \\ -(j-1), & j \text{ even,} \end{cases} \quad d_j(X) = \dim_{\mathbb{Q}} \pi_j(X) \otimes \mathbb{Q}.$$

Let a_j denote the rational rank of $\pi_j(E) \rightarrow \pi_j(\mathbb{S}^r)$. Exactness gives $d_j(F) = d_j(E) + d_{j+1}(\mathbb{S}^r) - a_j - a_{j+1}$. Since $w_j + w_{j-1}$ is two for odd j and zero for even j , substitution gives

$$\dim F = \begin{cases} kq - (r-2) - 2a_r, & r \text{ odd,} \\ kq + 2 - r - 2a_{2r-1}, & r \text{ even.} \end{cases}$$

Comparison with $\dim F = kq - r$ forces the displayed rank to equal one. An odd sphere has rational homotopy only in its own degree; an even q -sphere has it in degrees q and $2q - 1$. Thus odd r forces

$r = q$ if q is odd. The other candidate, $r = 2q - 1$ for even q , contradicts $2^k = \chi(E) = \chi(\mathbb{S}^r)\chi(F) = 0$. Even r gives $r = q$ for even q , and $q = 2r - 1$ for odd q .

It remains to treat $r = 2$, where the fiber need not be simply connected. If $q \geq 4$, a minimal Sullivan model of E has no elements in degrees two or three. Hence a model of f sends both generators of $(\Lambda(x_2, y_3), dy_3 = x_2^2)$ to zero. The homotopy-fiber model is then

$$\mathcal{M}_E \otimes \Lambda(u_1) \otimes \mathbb{Q}[v_2], \quad Du_1 = Dv_2 = 0,$$

by the standard fiber-model construction [38, Section 15]. One can obtain it explicitly by adjoining u_1, v_2 to the sphere model with $Du_1 = x_2$ and $Dv_2 = y_3 - u_1x_2$, then pulling back along the zero model map. Its cohomology is infinite-dimensional, contradicting the compact fiber. Consequently $q = 2$ or 3 , exactly the alternatives in (A.2) for $r = 2$. $\square$

In particular, one cannot apply the sphere-total-space theorem to a product source to discard, for example, $(q, r) = (11, 6)$. The rational restriction above does not decide that case. At joint capacity equality, however, Theorem 10.1 already rules out every smaller target, including genuine Hopf targets.

A.3 Balanced real projector orbits

The preceding sphere-fiber argument also restricts individual support maps into real Grassmannians. It is useful when a single symmetric-cone block is saturated but the whole collection is not.

Proposition A.3 (No sphere submersion onto a large balanced real orbit). *Let $B_r = \text{Gr}_{\lfloor r/2 \rfloor}(\mathbb{R}^r)$. For $r \geq 4$ no sphere admits a C^1 submersion onto B_r or any of its connected finite covers. For $r = 3$, the possible source dimensions are exactly two and three. For $r = 2$, they are exactly one; the maps are circle coverings. The $r = 1$ target is a point.*

Proof. For $r \geq 3$, a submersion from $\mathbb{S}^n$ must have $n \geq 2$ and lifts to the simply connected oriented Grassmannian

$$\tilde{B}_r = SO(r)/(SO(p) \times SO(t)), \quad p = \lfloor r/2 \rfloor, \quad t = \lceil r/2 \rceil.$$

The lift is onto because its image is open and compact. Smoothing and Ehresmann give a bundle with connected fiber. If the fiber has positive dimension, Browder's theorem and the two-row Serre sequence give

$$H^*(\tilde{B}_r; \mathbb{F}_2) = \mathbb{F}_2[e]/(e^{a+1}), \quad |e| = h \in \{2, 4, 8\}, \quad pt = ah, \quad \chi(\tilde{B}_r) = pt/h + 1. \quad (\text{A.3})$$

Indeed multiplication by the transgression e is an isomorphism in all base degrees below the total-space dimension, and the fundamental class fixes the last power.

For clarity, the required Euler characteristic can be computed directly. The real Grassmannian has its Schubert cells indexed by p -subsets of $\{1, \dots, r\}$; the parity generating sum of their dimensions is

$$\chi(B_r) = \begin{cases} 0, & p, t \text{ both odd,} \\ \binom{\lfloor r/2 \rfloor}{\lfloor p/2 \rfloor}, & \text{otherwise.} \end{cases}$$

The cell decomposition is classical [61, Section 6]. The formula follows by separating the subsets according to whether they contain r , giving the signed Pascal recurrence. The oriented double cover has twice this Euler characteristic. For $r = 2m$, odd m gives zero; even $m \geq 2$ gives $2\binom{m}{m/2} > m^2/2 + 1$, except that $m = 2$ is checked directly as $4 > 3$. For $r = 2m + 1$, $m \geq 4$ gives $2\binom{m}{\lfloor m/2 \rfloor} > m(m + 1)/2 + 1$. The last inequality starts at $m = 4, 5$ and propagates by the two-step

binomial recurrence; the even case starts at $m = 4$ and uses the same recurrence. At $m = 3$ the Euler characteristic is six, whereas the admissible values in (A.3) are seven and four. Thus the only numerical candidates are $r = 3$ and $r = 5$, both with $h = 2$.

For $r = 5$, the map $(u, v) \mapsto [u + iv]$ identifies $\tilde{B}_5$ with the smooth three-dimensional complex quadric. In coordinates $z_0 z_4 + z_1 z_3 + z_2^2 = 0$, the loci $z_0 \neq 0$ and $z_0 = 0, z_1 \neq 0$ are respectively $\mathbb{C}^3$ and $\mathbb{C}^2$; the remainder is $\mathbb{CP}^1 = \mathbb{C} \sqcup \{\text{pt}\}$. The resulting closed filtration and compactly supported cohomology sequence give one integral generator in each of degrees 0, 2, 4, 6. Its hyperplane class x is primitive: if $x = kz$ with $|k| \geq 2$, then k^3 would divide its cube integral, which equals two. Poincaré duality therefore supplies a degree-four generator y with $\int xy = 1$. The degree computation gives $x^2 = 2y$. Thus the unique nonzero mod-two degree-two class has square zero, contradicting (A.3) with $a = 3$.

If the fiber is zero-dimensional, the lifted map would identify a sphere with $\tilde{B}_r$. This is impossible for $r = 4$, where the target is $\mathbb{S}^2 \times \mathbb{S}^2$, and for $r = 5$ by the preceding cohomology. For $r \geq 6$, the homogeneous-space homotopy sequence gives

$$\pi_2(\tilde{B}_r) = \ker[\mathbb{Z}/2 \oplus \mathbb{Z}/2 \longrightarrow \mathbb{Z}/2] = \mathbb{Z}/2,$$

where the map is addition. This differs from $\pi_2(\mathbb{S}^{pt}) = 0$. Finally $\tilde{B}_3 = \mathbb{S}^2$ gives the two stated dimensions by the sphere-submersion result and the identity. Composing with its double cover realizes the ordinary projective-plane target. The circle case follows from the lifting-to- $\mathbb{R}$ obstruction and circle covering theory. $\square$

The topology used here—plane fields, sphere fiberings, rational formal dimension, and Grassmannian cohomology—is classical. The contribution of this section is the passage from saturated cone contact channels to normalized-base submersions and their joint covering, together with the resulting formulation consequences. These statements concern one specified globally C^1 slack factorization. They have different quantifiers from the factorization of all positive linear maps through sums of Lorentz cones studied by Aubrun, La Piana, and Müller-Hermes [6].

B Restricted kernels: embeddings, locality, and flat residuals

The full labelled rows in an affine lift carry more information than a single weighted contact kernel. This section quantifies that distinction. It also gives a nullity obstruction that remains useful when the mixed contact curvature is zero. All factorization classes used below are stated explicitly; a factorization of a restricted kernel need not extend to the full slack operator of a convex body.

B.1 Projective contact forces an embedding

Write $P_x = xx^T$ for $x \in \mathbb{S}^{r-1}$ and let $\mathbb{RP}^{r-1}$ denote the space of such rank-one projectors. Its strict-diagonal contact kernel is

$$\kappa([x], [y]) = 1 - \text{tr}(P_x P_y) = 1 - (x^T y)^2.$$

A *nowhere-zero factorization* means that every selected primal and dual factor is nonzero at every point; zero factors cannot be discarded only on an open dense subset for this definition.

Theorem B.1 (Projective phase embedding). *Let $r \geq 3$, $k \geq 1$, and $\lambda_a > 0$. Suppose that on $M = (\mathbb{RP}^{r-1})^k$ the kernel*

$$\kappa_k(u, v) = \sum_{a=1}^k \lambda_a (1 - (x_a^T y_a)^2) = \sum_{i=1}^L \langle A_i(u), B_i(v) \rangle$$

has a globally C^1 , nowhere-zero factorization with $A_i, B_i \in Q_3$. Then M has a C^1 embedding in $\mathbb{R}^L$. In particular, with $m = \lceil \log_2 r \rceil$,

$$L \geq \max\{k(r-1) + 1, k(2^m - 1)\}. \quad (\text{B.1})$$

For one projective plane, $r = 3$ and $k = 1$, the exact count in this class is four. For all r, k there is a smooth nowhere-zero factorization with

$$L = k \left(\frac{r(r+1)}{2} - 2 \right). \quad (\text{B.2})$$

Proof. Nonnegative summands and diagonal zero force complementarity in every factor. Since both factors are nonzero, their normalizations have the form

$$A_i(u) = a_i(u)(1, p_i(u)), \quad B_i(u) = b_i(u)(1, -p_i(u)), \quad a_i, b_i > 0, \quad p_i(u) \in \mathbb{S}^1.$$

The fundamental group of M is finite, so every homomorphism from it to $\pi_1(\mathbb{S}^1) = \mathbb{Z}$ vanishes. The covering lifting criterion therefore gives a global C^1 angle $\theta_i : M \rightarrow \mathbb{R}$ with $p_i = (\cos \theta_i, \sin \theta_i)$. Equivalently, $H^1(M; \mathbb{Z}) = 0$; mod-two first cohomology would not suffice. Taking the negative mixed derivative on the diagonal gives

$$2 \sum_a \lambda_a g_a = \sum_i a_i b_i d\theta_i \otimes d\theta_i,$$

where g_a is the quotient round metric on the a th projective space. Only one derivative of each factor is used. Thus $\Theta = (\theta_1, \dots, \theta_L)$ is an immersion. If $\Theta(u) = \Theta(v)$, every phase agrees and all pairings $\langle A_i(u), B_i(v) \rangle$ vanish. The strict-diagonal property then gives $u = v$. Compactness makes this injective immersion an embedding.

For clarity, the characteristic-class calculation behind the numerical bound is included. Put $n = r - 1$ and let a generate $H^1(\mathbb{RP}^n; \mathbb{F}_2)$. The tangent bundle is $\text{Hom}(\gamma, \gamma^\perp)$, where γ is the canonical line bundle, so $T\mathbb{RP}^n \oplus \varepsilon^1 \cong (n+1)\gamma$. The Whitney product formula [61] gives, for the normal bundle η of an embedding,

$$w(\eta) = (1+a)^{-r} = \sum_{j=0}^n \binom{n+j}{j} a^j.$$

The coefficient is odd exactly when the binary expansions of n and j have no common nonzero bit. This also follows directly by expanding $(1+a)^{-r}$ over $\mathbb{F}_2$. The largest such $j \leq n$ is $j_* = 2^m - r$, the complement of the first m bits of n . For the product, $w(\eta) = \prod_a (1+a_a)^{-r}$ has the nonzero monomial $\prod_a a_a^{j_*}$. Hence the normal rank is at least kj_* and $L \geq k(2^m - 1)$. Independently, a nonempty compact manifold of dimension kn cannot embed in $\mathbb{R}^{kn}$, by invariance of domain. This proves (B.1). The projective plane cannot embed in $\mathbb{R}^3$: a compact embedded hypersurface of Euclidean space is orientable, whereas $\mathbb{RP}^2$ is not. Thus its lower bound is four.

For the construction put $D = r(r+1)/2 - 1$ and use the normalized Veronese map into the unit sphere of traceless symmetric matrices,

$$X([x]) = \sqrt{\frac{r}{r-1}} \left(xx^T - \frac{I}{r} \right) \in \mathbb{S}^{D-1}, \quad \kappa(u, v) = \frac{r-1}{r} (1 - \langle X(u), X(v) \rangle).$$

Its image is a proper compact subset of that sphere. Choose a pole p outside it, set $c(u) = 1 - \langle p, X(u) \rangle > 0$, and let $F(u) = (X(u) - \langle p, X(u) \rangle p) / c(u) \in \mathbb{R}^{D-1}$. The scalar null-paraboloid maps

$$U(t) = \left(\frac{1+t^2}{2}, \frac{1-t^2}{2}, t \right), \quad V(t) = \left(\frac{1+t^2}{2}, -\frac{1-t^2}{2}, -t \right) \quad (\text{B.3})$$

satisfy $\langle U(t), V(w) \rangle = (t - w)^2/2$. The stereographic identity is

$$1 - \langle X(u), X(v) \rangle = \frac{c(u)c(v)}{2} \|F(u) - F(v)\|^2.$$

Thus $\sqrt{(r-1)/r} c(u)U(F_j(u))$ and the corresponding V factors give $D - 1$ smooth nowhere-zero channels. Concatenation with scales $\sqrt{\lambda_a}$ proves (B.2), and gives four when $r = 3$. $\square$

The bundle obstruction itself is classical. The factorization-specific step is the passage from positive complementary Lorentz phases to an injective Euclidean phase map. This statement concerns only the displayed rank-one kernel and its nowhere-zero selections. In particular, its finite upper bound is consistent with Fawzi's theorem that the full cone $\mathbb{S}_+^3$ has no finite second-order-cone lift [32].

B.2 What local tests and separated constructions can establish

Fix $q \geq 2$, $k \geq 2$, positive weights with $\sum_a \lambda_a = 1$, and

$$M = (\mathbb{S}^q)^k, \quad K(x, z) = \sum_{a=1}^k \lambda_a (1 - x_a^T z_a), \quad n = kq, \quad D = k(q + 1). \quad (\text{B.4})$$

Proposition 6.3 and its following construction give $n + 1 \leq L \leq n + \lceil k/2 \rceil$ for arbitrary globally C^1 Q_3 factorizations of this joint kernel. The following results explain both the strength and the limits of several natural attempts to close this interval.

Proposition B.2 (Exact local saturation). *Every finite collection of primal and dual points of M lies in a product of punctured-sphere charts on which K has a smooth nowhere-zero factorization with exactly n channels. The identity holds on the whole product chart, not merely on the finite samples.*

Proof. Choose poles $p_a \in \mathbb{S}^q$ outside the finite projections of the sample sets. On the punctured sphere set

$$c_a(x_a) = 1 - p_a^T x_a, \quad F_a(x_a) = \frac{x_a - (p_a^T x_a)p_a}{c_a(x_a)} \in \mathbb{R}^q.$$

For each of the q coordinates of F_a , use the two factors

$$\sqrt{\lambda_a} c_a(x_a) U((F_a)_j(x_a)), \quad \sqrt{\lambda_a} c_a(z_a) V((F_a)_j(z_a))$$

from (B.3). The stereographic chord identity gives exactly the a th summand of (B.4). There are kq channels in total. $\square$

Consequently no finite submatrix, nor a finite collection of jets of the kernel at sample pairs, distinguishes this local count from the larger global smooth count. This does not say that finitely many samples of *prescribed factor maps* determine their possible global extension.

Proposition B.3 (Partition separation). *Suppose the source coordinates are partitioned into nonempty groups $G_1, \dots, G_m$, and the channel labels into groups $I_1, \dots, I_m$, such that both factors of every channel in I_g depend only on the coordinates in G_g . Every global C^1 factorization of K of this form satisfies $L \geq n + m$. Equality is attainable for every partition into singletons and pairs.*

Proof. All diagonal pairings vanish separately. Freeze every group other than G_g at equal primal and dual arguments. The remaining channels then factor exactly $\sum_{a \in G_g} \lambda_a (1 - x_a^T z_a)$. The proof of Proposition 6.3 applies to any product of simply connected spheres: if its $q|G_g|$ -dimensional contact metric were factored with that many Q_3 channels, each channel would have rank one everywhere; the real phase map would be a local diffeomorphism from a compact manifold to $\mathbb{R}^{q|G_g|}$. Therefore $|I_g| \geq q|G_g| + 1$. Summing proves the bound. A singleton uses the $q + 1$ scalar chord maps applied to its ordinary coordinates. A pair uses the $2q + 1$ stereographic channels in Proposition 6.3. Rescale by the sum of its weights when that sum is not one. $\square$

A different restriction is a representation by finitely many *common-scale squared-distance kernels*:

$$K(x, z) = \sum_{g=1}^m \frac{r_g(x)r_g(z)}{2} \|F_g(x) - F_g(z)\|^2, \quad r_g > 0, \quad F_g : M \rightarrow \mathbb{R}^{L_g}, \quad L = \sum_g L_g, \quad (\text{B.5})$$

with globally C^1 maps. Each scalar coordinate produces one Q_3 channel by (B.3). Different summands may have the same scale function.

Proposition B.4 (Functional inertia). *Let e be the number of summands in (B.5) whose feature image is not contained in a Euclidean sphere. Then*

$$L \geq D - e \geq D - m. \quad (\text{B.6})$$

In particular an $L = n + 1$ representation in this class requires $e \geq k - 1$. Within the narrower class in which each summand is itself a positive block-additive kernel

$$\frac{r_g(x)r_g(z)}{2} \|F_g(x) - F_g(z)\|^2 = \sum_{a \in G_g} \mu_{ga} (1 - x_a^T z_a), \quad \mu_{ga} > 0, \quad \sum_{g: a \in G_g} \mu_{ga} = \lambda_a, \quad (\text{B.7})$$

the exact minimum count is $n + \lceil k/2 \rceil$, even when the groups overlap.

Proof. Set $\alpha_g = r_g \|F_g\|^2/2$, $\beta_g = r_g$, and $h_{gj} = r_g (F_g)_j$. The corresponding kernel is

$$\alpha_g(x)\beta_g(z) + \beta_g(x)\alpha_g(z) - \sum_j h_{gj}(x)h_{gj}(z).$$

Its coefficient form has at most $L_g + 1$ negative directions. If the image lies on a sphere, translate the center to zero without changing squared distances. Then α_g is a nonnegative constant multiple of β_g and the bound improves to L_g negative directions. Any pullback to the actual span of functions has no larger negative index. Conversely, the constant function and the D coordinate functions of $X(x) = (\sqrt{\lambda_a} x_a)_a$ are linearly independent, and $K(x, z) = 1 - X(x)^T X(z)$. Choose finitely many evaluation points whose feature evaluation matrix has full column rank. Their kernel matrix has exactly D negative eigenvalues. Applying the preceding upper bounds to this same finite matrix proves (B.6).

For (B.7), a nonsingleton group contributes at least $|G_g|(q + 1) - 1$ channels by the same inertia argument. A singleton contributes at least $q + 1$ by the compact phase obstruction. If $I = \sum_g |G_g|$ and t groups are nonsingletons, then $I \geq k$, $t \leq \lfloor I/2 \rfloor$, and

$$L \geq (q + 1)I - t \geq (q + 1)I - \lfloor I/2 \rfloor \geq (q + 1)k - \lfloor k/2 \rfloor.$$

Disjoint pairs and, if needed, one singleton attain this value. $\square$

In the paired construction each stereographic pair image is noncospherical. Indeed, the inverse image of a Euclidean sphere under stereographic projection is contained in a hyperplane section of the ambient unit sphere, whereas a weighted product of two full spheres affinely spans its entire ambient space. The singleton image is cospherical. Thus this construction also attains (B.6), with $e = \lfloor k/2 \rfloor$. The general class (B.5) allows m and e to be larger; the paired optimality assertion requires (B.7), not merely (B.5).

Example B.5 (A channel can depend on every source). Choose a pole $p = (p_a)_a \in \mathbb{S}^{D-1}$ with every $p_a \neq 0$ and $(\|p_a\|)_a \neq (\sqrt{\lambda_a})_a$. Then

$$\max_{x \in M} p^T X(x) = \sum_a \sqrt{\lambda_a} \|p_a\| < 1.$$

Global stereographic projection of $X(M)$ from p therefore gives $D - 1$ smooth nowhere-zero channels. To verify dependence on all sources, choose a unit $v \in p^\perp$ such that each $v_a \neq 0$ and $v_a \perp p_a$. This is possible because each block has dimension at least three. One coordinate is $f = v^T X/c$, where $c = 1 - p^T X$. The time component of its primal factor is $T = c(1 + f^2)/2$. Fix a source a . On an open subset of its sphere there are tangent variations with $d(v^T X) = 0$ and $dc \neq 0$, because v_a and p_a are independent. At points with $f \neq \pm 1$ these give $dT = (1 - f^2)dc/2 \neq 0$. Such points exist: the equation $v^T X = \pm c$ cannot hold on an open subset of the product, since its two sides would give a nontrivial affine relation among the constant and coordinate functions. This argument applies to every a . Thus one channel can depend nontrivially on all k sources. Positivity does not justify assuming source locality or pair locality in an unrestricted proof.

B.3 Column ranks detect curvature-flat residuals

The next statement uses finite-dimensional linear algebra rather than contact differentiation. For a kernel $F_a : X_a \times X_a \rightarrow \mathbb{R}$, its *column rank* is the dimension of the span of functions $F_a(\cdot, v)$ on X_a . No topology is needed in the first theorem.

Theorem B.6 (A column-rank nullity criterion). *Let $X_1, \dots, X_b$ be nonempty sets and $M = \prod_a X_a$. Suppose nonnegative kernels F_a vanish on the diagonal, have finite nonzero column ranks ρ_a , and admit the labelled real-PSD factorization*

$$F_a(x_a, v) = \sum_{i=1}^L \text{tr}(A_i(x) B_i^a(v)), \quad A_i(x), B_i^a(v) \in \mathbb{S}_+^{r_i}, \quad r_i \leq R. \quad (\text{B.8})$$

If, for every nonempty $J \subseteq \{1, \dots, b\}$,

$$1 + \sum_{a \in J} (\rho_a - 1) > \frac{R+1}{2} (|J| - 1), \quad (\text{B.9})$$

then some $x \in M$ satisfies $\sum_i \text{nullity } A_i(x) \geq b$. A sufficient uniform condition, when $\rho_a \geq \rho > 0$, is

$$(R+3)(b-1) < 2b\rho. \quad (\text{B.10})$$

For $b = 1$ there is no order-cap restriction.

Proof. Assume all total nullities are at most $b - 1$. In $\mathcal{H} = \bigoplus_i \mathbb{R}^{r_i}$ write $A = \bigoplus_i A_i$ and $B^a = \bigoplus_i B_i^a$. Diagonal contact implies $\text{Ran } B^a(x_a) \subseteq \ker A(x)$. For each a let S_a be the union of the nonzero vectors in the ranges of all $B^a(v)$. It is nonempty because $\rho_a > 0$. An independent choice $w_a \in S_a$

would, by choosing the corresponding v_a and setting $x = (v_a)_a$, put b independent vectors in $\ker A(x)$.

The independent-transversal criterion of Rado [76] therefore gives a nonempty J , with $j = |J|$, such that

$$\dim \operatorname{span} \bigcup_{a \in J} S_a \leq j - 1.$$

Here is a direct justification that also covers infinite S_a . For finite-dimensional subspaces $V_a = \operatorname{span} S_a$, independent vectors $v_a \in V_a$ exist if $\dim \sum_{a \in J} V_a \geq |J|$ for every J . Prove this by induction. If a nonempty proper subset is tight, first choose its independent vectors and then apply induction to the remaining subspaces in the quotient by its span. Otherwise every nonempty proper subset has dimension at least its cardinality plus one; select a nonzero vector in the last subspace and quotient by its span, preserving the inductive inequalities for the others. Finally, multilinearity of the exterior product expands independent choices from the V_a into at least one independent choice from the generating sets S_a . This proves the claimed criterion without a finiteness assumption on those sets.

Put $V_i = \operatorname{span}_{a \in J, v} \operatorname{Ran} B_i^a(v)$ and $m_i = \dim V_i$. The range of a block-diagonal matrix contains each block-supported range vector. Consequently the deficient span is exactly $\bigoplus_i V_i$, not merely a subspace of it, and $\sum_i m_i \leq j - 1$. All the dual matrices for $a \in J$ lie in $\bigoplus_i \operatorname{Sym}(V_i)$, whose dimension is at most

$$\sum_i \frac{m_i(m_i + 1)}{2} \leq \frac{R + 1}{2} \sum_i m_i \leq \frac{R + 1}{2}(j - 1).$$

Pairing these matrices with $A(x)$ bounds by the same number the span of all cylindrical column functions $F_a(x_a, v)$, $a \in J$. On the other hand, a relation $\sum_{a \in J} f_a(x_a) = 0$ makes each f_a constant, by varying one coordinate at a time. Its relation space has dimension at most $j - 1$, so the cylindrical span has dimension at least $\sum_{a \in J} \rho_a - (j - 1)$. This contradicts (B.9). For the uniform assertion the difference of twice the two sides is $R + 3 + j(2\rho - R - 3)$, an affine function of $1 \leq j \leq b$. It is positive at $j = 1$, and at $j = b$ positivity is exactly (B.10). For $b = 1$ this condition is automatic. $\square$

Corollary B.7 (Antipodal separation and squared slack). *Let $s \geq 3$, $X_a = \mathbb{S}^{s-1}$, and assume the kernels in (B.8) are continuous. If each $F_a(u, u) = 0$ and $F_a(u, \tau_a(u)) > 0$, where τ_a is topologically conjugate to the antipodal map, then the nullity conclusion holds whenever*

$$(R + 3)(b - 1) < 2bs. \quad (\text{B.11})$$

For the squared kernels $F_a(u, v) = (1 - u^T v)^2/2$, their exact column rank is $\rho = s(s + 3)/2$. Thus the stronger sufficient condition is

$$(R + 3)(b - 1) < bs(s + 3). \quad (\text{B.12})$$

Irrespective of either cap condition, every factorization of these squared kernels satisfies

$$\sum_i \frac{r_i(r_i + 1)}{2} \geq 1 + b \left(\frac{s(s + 3)}{2} - 1 \right). \quad (\text{B.13})$$

Proof. Choose a basis $f_1, \dots, f_{\rho_a}$ of the column space and write $F_a(u, v) = \sum_{\ell} f_{\ell}(u) c_{\ell}(v)$. The coefficient map $c : \mathbb{S}^{s-1} \rightarrow \mathbb{R}^{\rho_a}$ is continuous: select ρ_a evaluation points for which the basis evaluation matrix is invertible and recover c from those evaluations. If $c(v) = c(\tau_a(v))$, evaluation at $u = v$ contradicts the assumed separation. After conjugating τ_a to the antipodal map, the Borsuk–Ulam theorem [48, Section 2.2] implies $\rho_a \geq s$. Theorem B.6 proves (B.11).

For the squared kernel the odd parts of the columns span all s linear coordinate functions. The even parts span all quadratic forms restricted to the sphere: their coefficient matrices are proportional to $I + vv^T$. A matrix C orthogonal to every $I + vv^T$ obeys $\text{tr } C + v^T C v = 0$ for every unit v , which first makes C scalar and then zero. The restriction of homogeneous quadratic forms to the sphere is injective, and the odd and even spaces are independent. Thus $\rho = s + s(s+1)/2$. This gives (B.12). The constant function belongs to the column span, for example by averaging $F_a(\cdot, v)$ over the sphere. Coordinate column spaces are independent modulo this common one-dimensional space, so the labelled cylindrical column rank is exactly $1 + b(\rho - 1)$. Pairing with the full matrix space $\bigoplus_i \mathbb{S}^{r_i}$ proves (B.13). $\square$

These squared kernels have zero mixed contact curvature, so the preceding lower bound cannot be recovered by applying the contact Hessian estimate to them. If the selected primal tuples also belong to the closure of a relative-Slater affine slice, Lemma 8.1 converts the nullity conclusion into $\nu_{\text{std}, \text{slice}} \geq b$ for its standard barrier. This extra affine-slice hypothesis is needed for the barrier conclusion.

For example, the identity

$$1 - u^T v = \frac{1 - (u^T v)^2}{2} + \frac{(1 - u^T v)^2}{2}$$

splits ball slack into a projective term and a flat residual. The first term has the order- s factorization $\text{tr}((I - uu^T)vv^T/2)$. A construction using one such separate block for each source spends b primal nullities at every contact; if the residual uses order at most s , Corollary B.7 forces at least another b at some simultaneous contact. Moreover, with $d_s = s(s+1)/2$, (B.13) gives

$$L_{\text{residual}} \geq b + \left\lceil \frac{b(s-1) + 1}{d_s} \right\rceil > b.$$

Thus this prescribed positive splitting needs at least $2b + 1$ factors, whereas the grouped Schur construction for the unsplit full slacks uses $2b$. The unrestricted full-slack rank theorem does not assume such a splitting; the example explains why a curvature-saturating projective summand need not be an economical component of a complete formulation.

Ordinary-rank lower bounds for PSD factorizations are standard [36]. The additional step in Theorem B.6 is to combine the independent-transversal criterion with simultaneous contact and the blockwise matrix-space bound. Stating the criterion in terms of the actual column ranks preserves information that the weaker antipodal estimate $\rho_a \geq s$ discards. The conclusions concern a simultaneous primal nullity and a prescribed labelled kernel; they do not assert a lower bound on every contact or on an arbitrary barrier without further hypotheses.

C Global obstructions for conditioned approximate factors

The local rank estimates of Section 20 have global consequences when the factors have fixed labels and vary smoothly over the entire contact sphere. These hypotheses are not supplied by an arbitrary approximate lift. We first retain tangent-bundle information without requiring the selected factors to lie on cone boundaries. We then obtain stronger channelwise conclusions for nonzero boundary selections.

C.1 A robust tangent splitting

Let $n \geq 1$ and suppose proper cones K_i of dimensions $m_i \geq 2$ admit globally C^2 maps $A_i : \mathbb{S}^n \rightarrow K_i$, $B_i : \mathbb{S}^n \rightarrow K_i^*$ satisfying

$$\sum_i \langle A_i(x), B_i(y) \rangle = h_C(y) - \langle x, y \rangle, \quad \mathbb{B}_2^{n+1} \subset C \subset (1 + \epsilon) \mathbb{B}_2^{n+1}. \quad (\text{C.1})$$

Put $r_i = m_i - 2$ and assume $\sum_i r_i = n$. At x write $a_i = A_i(x)$, $b_i = B_i(x)$, $X_i = dA_i(x)$, $Y_i = dB_i(x)$. Assume uniformly that $\|a_i\|, \|b_i\| \geq \mu > 0$ and $\|X_i\|, \|Y_i\| \leq L$. Fix $0 < r_0 < \pi$ and $H > 0$ such that, for every unit $u \in T_x \mathbb{S}^n$ and $|t| \leq r_0$,

$$|\langle (A_i \circ \exp_x(tu))'', b_i(x) \rangle| \leq H, \quad |\langle a_i(x), (B_i \circ \exp_x(tu))'' \rangle| \leq H.$$

The opposite factor is held fixed at the center in these inequalities.

Theorem C.1 (Approximate saturation gives a bundle splitting). *If $\epsilon < \mu^2$, $\sqrt{2\epsilon/H} \leq r_0$, and $\mathcal{E}_k(\epsilon) < 1$, where k is the number of blocks and $\mathcal{E}_k$ is defined in (20.8), then*

$$T^* \mathbb{S}^n = \bigoplus_{i:r_i > 0} E_i, \quad \text{rank } E_i = r_i.$$

Consequently, the number of $m_i = 3$ blocks is at most $\rho(n+1) - 1$. If all blocks have dimension three, $n \in \{1, 3, 7\}$. If $n \geq 2$ is even, exactly one block has positive r_i , and that block has dimension $n + 2$.

Proof. Let $\delta_i = \langle a_i, b_i \rangle$. Equation (C.1) gives $\delta_i \geq 0$, $\sum_i \delta_i \leq \epsilon$, and $-\sum_i X_i^* Y_i = I$. Since $\gamma_i = \delta_i / (\|a_i\| \|b_i\|) < 1$, the vectors a_i, b_i are linearly independent. The spaces $W_i(x) = a_i(x)^\perp \cap b_i(x)^\perp$ therefore form continuous bundles of rank r_i . Define the canonical bilinear forms

$$R_i(u, v) = -\langle P_{W_i} X_i u, P_{W_i} Y_i v \rangle.$$

They depend continuously on x and have rank at most r_i . On W_i , both $P_{b_i^\perp} P_{a_i^\perp}$ and P_{W_i} are the identity. On the remaining plane their difference has norm γ_i . Thus the proof of Lemma 20.2 gives $\| -X_i^* Y_i - R_i \| \leq e_i$, with the sharpened error (20.6). It follows that $\|I - \sum_i R_i\| \leq \mathcal{E}_k(\epsilon) < 1$. The sum is invertible and

$$n \leq \sum_i \text{rank } R_i \leq \sum_i r_i = n.$$

All ranks attain their bounds, and their images are in direct sum. Continuity and constant rank give the stated subbundles. For $n \geq 2$, every real line bundle on $\mathbb{S}^n$ is trivial, so rank-one summands give independent vector fields; Adams's bound is $\rho(n+1) - 1$ [1]. For $n = 1$ the tangent bundle is already trivial. A complete rank-one splitting forces the familiar parallelizable dimensions 1, 3, 7. For even n , a splitting into two positive intermediate ranks would give $e(T\mathbb{S}^n) = e(E)e(F) = 0$, since intermediate integral cohomology vanishes. This contradicts Euler number two. $\square$

The stronger subset-rank restrictions in Proposition A.1 also apply. The canonical maps in this proof are essential: unrelated pointwise low-rank approximants do not automatically define vector bundles. The hypotheses exclude rays with nonzero selected factors, since for a ray $\delta_i = \|a_i\| \|b_i\| \geq \mu^2$. Allowing a factor to vanish or losing uniform control is a permitted failure of the hypotheses.

C.2 An invariant of fixed block reparameterizations

There is a coordinate-independent version of this sufficient obstruction, although it is still an invariant of the chosen selections and contact parameterization. Retain the global slack identity and the saturated budget $\sum_i (m_i - 2)_+ = n$, now permitting rays. For a fixed family $S_i \in GL(m_i)$, transform

$$A_i \mapsto S_i A_i, \quad B_i \mapsto S_i^{-T} B_i, \quad K_i \mapsto S_i K_i.$$

At each center compute $U_i^S, V_i^S, \alpha_i^S, \beta_i^S, \gamma_i^S$ as in Lemma 20.2 for the transformed maps, using geodesics of a fixed radius r_0 . In particular,

$$\alpha_i^S = \frac{\psi_{r_0}(\delta_i, H_{A,i})}{\|S_i^{-T} b_i\|}, \quad \gamma_i^S = \frac{\delta_i}{\|S_i a_i\| \|S_i^{-T} b_i\|}.$$

The contracted derivative bounds $H_{A,i}, H_{B,i}$ are unchanged by this transformation. Let e_i^S be (20.6), or $\alpha_i^S \beta_i^S$ for a ray, and set

$$\mathfrak{D} = \inf_{S \in \prod_i GL(m_i)} \max \left\{ \sup_{x,i} \gamma_i^S(x), \sup_x \sum_i e_i^S(x) \right\}. \quad (\text{C.2})$$

If a base factor vanishes anywhere, the displayed value for that gauge is defined as $+\infty$.

Corollary C.2 (Gauge-invariant degeneration). *If $T^*\mathbb{S}^n$ admits no splitting with ranks $(m_i - 2)_+$, then $\mathfrak{D} \geq 1$.*

Proof. If $\mathfrak{D} < 1$, some fixed gauge has both suprema below one. Its $\gamma_i < 1$ excludes rays and makes the primal and dual factors nonparallel everywhere. The canonical maps of the preceding proof then have rank at most $m_i - 2$ and an invertible sum, supplying the forbidden splitting. If the original coordinates are changed by T_i , all subsequent gauges S_i range through the same net transformations $S_i T_i$ as before. Thus the infimum is invariant under every fixed blockwise invertible reparameterization. No covariance of the individual Euclidean projectors is needed. $\square$

This index does not assert a lower bound on a barrier Hessian or a Newton-system condition number. The infimum ranges over fixed linear block gauges, not parameter-dependent gauges or changes of the contact sphere itself.

C.3 Rank-one phase dropout

The next statement is stronger than the frame consequence in source sphere dimensions three and seven, but requires nonzero boundary maps. Let M be a closed, connected, simply connected smooth Riemannian manifold. Let $A, B : M \rightarrow \partial Q_3 \setminus \{0\}$ be C^1 and write

$$A = a(1, p), \quad B = b(1, -q), \quad a, b > 0, \quad p, q : M \rightarrow \mathbb{S}^1.$$

Set $\sigma = \langle A, B \rangle$, $C_x(u, v) = -\langle dA_x u, dB_x v \rangle$ and

$$K_A = \sup_M \|d \log a\|, \quad K_B = \sup_M \|d \log b\|, \quad K = \min\{K_A, K_B\}.$$

Bilinear forms need not be symmetric; all norms are operator norms.

Theorem C.3 (Quantitative phase dropout). *If $0 \leq \sigma \leq \epsilon$ and $\|d\sigma\| \leq \eta$, then there is $x_* \in M$ with*

$$\|C_{x_*}\| \leq K\eta + K^2\epsilon.$$

If every C_x is within δ of a positive rank-one form whose nonzero eigenvalue is at least $\mu > 0$, then

$$\delta + K\eta + K^2\epsilon \geq \mu. \quad (\text{C.3})$$

Proof. Simple connectivity lifts p to a real C^1 phase. At a maximum of that phase, $dp = 0$, so $dA = \alpha \otimes A$, where $\alpha = d \log a$. At this point

$$\langle A, dB \rangle = d\sigma - \sigma\alpha, \quad C = -\alpha \otimes (d\sigma - \sigma\alpha).$$

Its norm is at most $K_A\eta + K_A^2\epsilon$. A critical point of the dual phase gives $C = -(d\sigma - \sigma\beta) \otimes \beta$, with $\beta = d \log b$, and the bound using K_B . Taking the smaller bound proves the first assertion. At the corresponding point, any rank-one approximant of norm at least μ has distance at least $\mu - \|C\|$ from C , proving (C.3). $\square$

The hypotheses can be checked using norms of the original factors: $\|A\| = \sqrt{2}a$ and $\|dA(u)\|^2 = 2|da(u)|^2 + a^2\|dp(u)\|^2$, whence $K_A \leq L_A/m_A$ if $\|A\| \geq m_A$ and $\|dA\| \leq L_A$. The dual bound is identical. Reciprocal constant scalar gauges leave K_A, K_B, σ , and C unchanged.

Corollary C.4 (Saturated approximate phase channels). *Let $M = \mathbb{S}^n$, $n \geq 2$, and let n pairs satisfy the preceding theorem with constants ϵ_i, η_i, K_i . Suppose*

$$\left\| \sum_i C_i - I \right\| \leq \xi, \quad \|C_i - R_i\| \leq \delta_i,$$

where R_i is positive of rank one at every point; the choices of R_i need not be continuous. Put $\Delta = \sum_i \delta_i$ and $E_i = K_i\eta_i + K_i^2\epsilon_i$. Then, for every i ,

$$\xi + \Delta + \delta_i + E_i \geq 1.$$

In particular, common bounds give $\xi + (n+1)\delta + E \geq 1$.

Proof. There is nothing to prove if $\xi + \Delta \geq 1$. Otherwise $\|\sum_i R_i - I\| \leq \xi + \Delta < 1$. Write $R_i = \lambda_i \omega_i \otimes \omega_i$ with $\|\omega_i\| = 1$. At any point, choose a unit vector killed by the at most $n-1$ covectors ω_j , $j \neq i$. Evaluation on that vector gives $\lambda_i \geq 1 - \xi - \Delta$. Apply Theorem C.3 to the i th channel with this spectral floor. $\square$

We record explicitly how first-derivative moduli supply η . For r_0 below the injectivity radius, suppose covectors $d\sigma$ at endpoints of geodesic segments of length $t \leq r_0$ differ by at most $\omega(t)$ after parallel transport. Then

$$\|d\sigma\| \leq \inf_{0 < s \leq r_0} \left\{ \frac{\epsilon}{s} + \frac{1}{s} \int_0^s \omega(t) dt \right\}. \quad (\text{C.4})$$

Indeed, along a unit geodesic in the gradient direction, the integral of the derivative is at least $s\|d\sigma\| - \int_0^s \omega(t)dt$, whereas the change in σ is at most ϵ . If $\omega(t) \leq Ht$, this is $\psi_{r_0}(\epsilon, H)$. For $H > 0$ with $\sqrt{2\epsilon/H} \leq r_0$, it gives $\eta \leq \sqrt{2H\epsilon}$. If $H = 0$, $d\sigma$ is locally parallel; its vanishing at a maximum and connectedness give $d\sigma = 0$ everywhere. A common modulus tending to zero at zero makes the right side of (C.4) tend to zero with ϵ .

In a nonnegative kernel $F(x, y) = \sum_i \langle A_i(x), B_i(y) \rangle$ with $F(x, x) \leq \epsilon$, every individual defect satisfies $0 \leq \sigma_i \leq \epsilon$. Thus the preceding inequalities apply separately to each channel when it has its own derivative modulus. A bound on the derivative of the sum does not replace these individual bounds: derivative cancellation is possible.

C.4 Higher-rank dropout from forbidden sphere submersions

Let $1 \leq r < n$, let $K \subset E$ be proper of dimension $r + 2$, and choose $\ell \in \text{int } K^*$. Suppose the normalized base $D = \{z \in K : \ell(z) = 1\}$ has a C^1 boundary J . Let $A : \mathbb{S}^n \rightarrow \partial K \setminus \{0\}$ and $B : \mathbb{S}^n \rightarrow \partial K^* \setminus \{0\}$ be C^1 . Define

$$\begin{aligned} a &= \ell(A), \quad p = A/a : \mathbb{S}^n \rightarrow J, \quad K_A = \sup \|d \log a\|, \\ g_B(v) &= \langle A, dB(v) \rangle, \quad \eta_B = \sup \|g_B\|, \quad C = -dA^* dB. \end{aligned}$$

Write $s_r(C)$ for the r th largest singular value.

Theorem C.5 (Robust submersion obstruction). *If $(n, r) \notin \{(3, 2), (7, 4), (15, 8)\}$, there is a point with*

$$s_r(C) \leq K_A \eta_B.$$

Consequently, if every C is within δ of a rank- r form whose r th singular value is at least $\mu > 0$, then

$$\delta + K_A \eta_B \geq \mu. \tag{C.5}$$

In a Hopf dimension pair, the same conclusion holds if the particular map p has a critical point. If a dual normalized base also has C^1 boundary, the term $K_A \eta_B$ in (C.5) may be replaced by $\min\{K_A \eta_B, K_B \eta_A\}$, with dual definitions.

Proof. Radial projection from a relative interior point of the convex base identifies J C^1 diffeomorphically with $\mathbb{S}^r$. The inverse is C^1 because the supporting normal at each boundary point has strictly positive inner product with its radial vector. Outside the three Hopf pairs, Proposition A.2 rules out an everywhere submersive map p . Hence $\text{rank } dp \leq r - 1$ somewhere. Writing $\alpha = d \log a$, differentiation of $A = ap$ gives

$$C(u, v) = -a \langle dp(u), dB(v) \rangle - \alpha(u) g_B(v).$$

At that point the first term has rank at most $r - 1$ and the second has norm at most $K_A \eta_B$. The singular-value min-max inequality gives $s_r(C) \leq K_A \eta_B$. Singular values change by at most the operator perturbation norm, proving (C.5). Applying the same argument to the normalized dual map supplies the second bound, possibly at a different point; both inequalities hold, so their minimum is valid. $\square$

Here $g_B(x)$ is the one-sided derivative $d_y \langle A(x), B(y) \rangle|_{y=x}$, with x held fixed. It can be bounded from small contact defect and a modulus in that variable. Specifically, if $F(x, y) = \langle A(x), B(y) \rangle \geq 0$, $F(x, x) \leq \epsilon$, and the second-variable derivative has uniform geodesic modulus ω_B , then

$$\eta_B \leq \inf_{0 < s \leq r_0} \left\{ \frac{\epsilon}{s} + \frac{1}{s} \int_0^s \omega_B(t) dt \right\}. \tag{C.6}$$

Move the second argument from x in the direction opposite to $g_B(x)$; the endpoint is nonnegative, so integration gives this inequality. The first-variable bound for η_A is identical. Lipschitz constants H_A, H_B give the corresponding ψ bounds, and, when their optimizing radii fit,

$$\delta + \min\{K_A \sqrt{2H_B \epsilon}, K_B \sqrt{2H_A \epsilon}\} \geq \mu.$$

The finite-radius formula (C.6) remains valid when either Lipschitz constant is zero.

A total diagonal derivative bound is insufficient for this higher-rank argument. With $\sigma = \langle A, B \rangle$,

$$g_B = d\sigma - \sigma d \log a - a dp^* B.$$

At a rank-one phase critical point $dp = 0$, recovering Theorem C.3. For $r > 1$ a critical point only forces rank $dp < r$ and the last term need not vanish. If it is bounded by τ and $\|d\sigma\| \leq \eta$, the valid replacement is

$$\delta + K_A \eta + K_A^2 \epsilon + K_A \tau \geq \mu.$$

Finally, consider a saturated profile $\sum_i r_i = n$ with at least two positive ranks, where every block satisfies these boundary and base-regularity hypotheses and its channel is uniformly δ_i close to rank r_i with singular-value floor μ_i . At least one pair (n, r_i) is forbidden: outside source dimensions 3, 7, 15 all proper pairs are forbidden; in those three dimensions the only permitted proper targets are 2, 4, 8, respectively, and copies of that target dimension cannot sum to n . Thus, under common bounds $\delta_i \leq \delta$, $K_{A,i} \leq K$, $\eta_{B,i} \leq \eta_\partial$, and $\mu_i \geq \mu$, every such profile satisfies

$$\delta + K \eta_\partial \geq \mu.$$

A single full-capacity block is outside this obstruction.

These conclusions combine elementary perturbation estimates with the classical sphere topology already cited in Appendix A. The added statements concern quantitative degeneration of specified global cone-contact selections. They do not follow from the finite slack-matrix approximation framework of [44], and do not obstruct arbitrary nonsmooth, unlabelled, or vanishing-factor approximations.

D Contact sensitivity and Newton conditioning

The approximation obstructions concern derivatives of selected contact maps across labels. A Newton condition number concerns a linear system at one interior point. The following two separations identify why an additional sensitivity theorem is needed to connect them.

D.1 A sparse change of cone coordinates

Suppose $F(Gz) = F(z) + c_G$ for a linear cone automorphism G . In an equality formulation $Ez = b$, put $z' = Gz$, $E' = EG^{-1}$ and $H = \nabla^2 F(z)$. Differentiation gives

$$H' = G^{-T} H G^{-1}, \quad E'(H')^{-1}(E')^T = E H^{-1} E^T. \quad (\text{D.1})$$

The raw augmented system instead changes by congruence:

$$\begin{pmatrix} H' & (E')^T \\ E' & 0 \end{pmatrix} = \begin{pmatrix} G^{-T} & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} H & E^T \\ E & 0 \end{pmatrix} \begin{pmatrix} G^{-1} & 0 \\ 0 & I \end{pmatrix}.$$

Thus the Schur matrix is literally unchanged, whereas the Euclidean spectrum of the augmented matrix need not be.

For Q_m , $m \geq 3$, let $a = e_0 + e_1$, $b = e_0 - e_1$, and let G_λ be the identity on transverse coordinates, with first block

$$\frac{1}{2} \begin{pmatrix} \lambda + \lambda^{-1} & \lambda - \lambda^{-1} \\ \lambda - \lambda^{-1} & \lambda + \lambda^{-1} \end{pmatrix}, \quad \lambda > 0.$$

It preserves the Lorentz quadratic form and its future sheet. Moreover, $G_\lambda a = \lambda a$ and $G_\lambda^{-T} b = \lambda b$. Consequently both primal and dual Euclidean norms can approach zero while all their pairings remain unchanged. This is a genuine contact example:

$$A(u) = (\sqrt{1 + \|u\|^2}, 1, u), \quad B(v) = (\sqrt{1 + \|v\|^2}, -1, -v)$$

satisfy $\langle A(u), B(v) \rangle \geq 0$, with equality at $u = v$. At zero their transverse derivatives are isometries and are unchanged by the boost; the base norms scale by λ . The standard Lorentz barrier is invariant, so (D.1) applies. At e_0 , $H = 2I$, but at $G_\lambda e_0$ it is $2G_\lambda^{-2}$, with condition number λ^{-4} for $0 < \lambda < 1$. This calculation prevents a lower bound involving unnormalized Euclidean factor norms from being read as a lower bound on the intrinsic Schur condition number. The boost only mixes two columns of E but can change their magnitudes, data-access normalization, and the cost of a solver using the raw augmented system.

D.2 Bounded pointwise geometry and unbounded contact derivatives

Transporting the normalization covector avoids the preceding coordinate issue, but leaves a second obstruction. For maps A, B and a strictly positive covector ℓ , the transformations

$$(A, B, \ell) \mapsto (GA, G^{-T}B, G^{-T}\ell)$$

preserve $\langle A, B \rangle$, $-dA^*dB$, A^*dB , and $d \log \ell(A)$. Their cross-label derivatives can nevertheless grow while all pointwise interior geometry stays bounded.

Use the Lorentz Jordan product $(t, z) \circ (s, w) = (ts + z^T w, tw + sz)$ and fix $c_\pm = (1, \pm p_0)/2$ for $p_0 \in S^1$. On S^n , $n \geq 1$, put $a_k(x) = \exp(\sin(kx_1))$ and

$$A_k = a_k c_+, \quad B_k = a_k^{-1} c_-, \quad X_k = a_k(c_+ + \tau c_-), \quad S_k = a_k^{-1}(c_- + \tau c_+), \quad (\text{D.2})$$

where $0 < \tau < 1$ is fixed. The boundary maps are complementary, $e^{-1} \leq a_k \leq e$, and $\|d \log \ell(A_k)\|_\infty = k$ for any positive ℓ . Yet $X_k \circ S_k = \tau e$ exactly. Since $\nabla^2 F(az) = a^{-2} \nabla^2 F(z)$, the primal and dual Hessian norms and inverse norms are uniformly bounded in k . In the common Jordan frame the Nesterov–Todd scaling point has eigenvalues $a_k/\sqrt{\tau}$ and $a_k\sqrt{\tau}$; its spectral ratio is $1/\tau$. Here centrality uses the displayed Jordan-product convention; a constant trace-inner-product normalization has no effect on the conclusion.

For a different example fix $0 < \theta < \pi/2$ and put $p_k(x) = (\cos(kx_1), \sin(kx_1))$, $q_k = R_\theta p_k$, and

$$A_k = (1, p_k), \quad B_k = (1, -q_k), \quad X_k = (1 + \tau, p_k), \quad S_k = (1 + \tau, -q_k). \quad (\text{D.3})$$

The diagonal defect is the constant $1 - \cos \theta$, but

$$d\langle A_k, B_k \rangle = 0, \quad A_k^* dB_k = \sin \theta d(kx_1), \quad \|A_k^* dB_k\|_\infty = k \sin \theta.$$

All interior pairs are common spatial rotations of one fixed pair, so Hessian spectra and scaling-point spectra are independent of k . Their relative Jordan central residual is

$$\frac{2(1 + \tau) \sin(\theta/2)}{2\tau + \tau^2 + 1 - \cos \theta}.$$

For any prescribed positive central-neighborhood radius, sufficiently small fixed $\theta > 0$ puts the entire family inside that neighborhood. Alternatively, $\theta = k^{-1/2}$ gives defect $\Theta(k^{-1})$ and central residual $O(k^{-1/2})$, while the one-sided derivative grows as $\Theta(\sqrt{k})$. The mixed channel is $\cos \theta d(kx_1) \otimes d(kx_1)$; it vanishes at the critical points of x_1 , consistently with the phase obstruction.

These explicit computations prove that pointwise centrality, Hessian conditioning, scaling-point conditioning, and bounded factor scales do not control either radial or one-sided contact sensitivity. The pointwise interpretation of self-scaled geometry is classical [71]; the examples distinguish it from the different, cross-objective quantities used here. One may even impose the identity equality $z = X_k$ at each label. The assembled saddle matrix $\begin{pmatrix} \mu H & I \\ I & 0 \end{pmatrix}$ and its inverse are uniformly bounded whenever μ stays in a fixed positive compact interval, since its inverse is $\begin{pmatrix} 0 & I \\ I & -\mu H \end{pmatrix}$. These artificial instances have changing right-hand sides. They are not a single approximate lift of a fixed body.

D.3 What an additional sensitivity hypothesis would give

In the robust submersion inequality (C.5), suppose the finite-radius estimate reduces to

$$\delta + K_A \sqrt{2H_B \epsilon} \geq \mu, \quad \sqrt{2\epsilon/H_B} \leq r_0,$$

where $\delta < \mu$, $\epsilon > 0$ and $H_B > 0$. Then $H_B \geq (\mu - \delta)^2 / (2K_A^2 \epsilon)$. The finite-radius expression $\inf_{0 < s \leq r_0} (\epsilon/s + H_B s/2)$ remains the valid form when $H_B = 0$ or the optimizing radius does not fit; one cannot divide by H_B in those cases. If a separately established, representation-specific sensitivity theorem provides $K_A \leq c_K \chi^a$ and $H_B \leq c_H \chi^b$, with positive constants and exponents, substitution yields

$$\chi \geq \left(\frac{\mu - \delta}{c_K \sqrt{2c_H \epsilon}} \right)^{1/(a+b/2)}.$$

Likewise, if $C > 0$, $K_0 > 0$, and $q > 0$, then $H_B \leq C 2^{q\beta}$ and $K_A \leq K_0$ imply

$$\beta \geq \frac{1}{q} \log_2 \frac{(\mu - \delta)^2}{2C K_0^2 \epsilon}.$$

These are conditional algebraic consequences. Neither an ordinary Newton condition number nor a coefficient bit count is proved to satisfy these extra hypotheses. In particular, differentiating a KKT solution map requires control of both its inverse Jacobian and the variation of the input data with the contact label.

References

- [1] J. F. Adams. Vector fields on spheres. *Annals of Mathematics*, 75(3):603–632, 1962. doi: 10.2307/1970213. URL <https://www.math.ens.psl.eu/~benoist/refs/Adams.pdf>.
- [2] Andris Ambainis, Andrew M. Childs, François Le Gall, and Seiichiro Tani. The quantum query complexity of certification. *Quantum Information and Computation*, 10(3–4):181–189, 2010. URL <https://www.rintonpress.com/xxqic10/qic-10-34/0181-0189.pdf>.
- [3] Martin S. Andersen, Joachim Dahl, and Lieven Vandenbergh. Logarithmic barriers for sparse matrix cones. *Optimization Methods and Software*, 28(3):396–423, 2013. doi: 10.1080/10556788.2012.684353.
- [4] P. M. Anselone and J. Korevaar. Translation invariant subspaces of finite dimension. *Proceedings of the American Mathematical Society*, 15(5):747–752, 1964. doi: 10.1090/S0002-9939-1964-0169048-7.
- [5] Simon Apers and Sander Gribling. Quantum speedups for linear programming via interior point methods. *SIAM Journal on Computing*, 55(1):93–134, 2026. doi: 10.1137/25M1736098.
- [6] Guillaume Aubrun, Francesca La Piana, and Alexander Müller-Hermes. Factorization through Lorentz cones, 2026. URL <https://arxiv.org/abs/2606.27825>.
- [7] Brandon Augustino, Giacomo Nannicini, Tamás Terlaky, and Luis F. Zuluaga. Quantum interior point methods for semidefinite optimization. *Quantum*, 7:1110, 2023. doi: 10.22331/q-2023-09-11-1110.
- [8] Gennadiy Averkov. Optimal size of linear matrix inequalities in semidefinite approaches to polynomial optimization. *SIAM Journal on Applied Algebra and Geometry*, 3(1):128–151, 2019. doi: 10.1137/18M1201342.
- [9] John C. Baez. The octonions. *Bulletin of the American Mathematical Society*, 39(2):145–205, 2002. doi: 10.1090/S0273-0979-01-00934-X.
- [10] Wayne W. Barrett, Charles R. Johnson, and Michael Lundquist. Determinantal formulae for matrix completions associated with chordal graphs. *Linear Algebra and its Applications*, 121: 265–289, 1989. doi: 10.1016/0024-3795(89)90706-4.
- [11] Robert Beals, Harry Buhrman, Richard Cleve, Michele Mosca, and Ronald de Wolf. Quantum lower bounds by polynomials. *Journal of the ACM*, 48(4):778–797, 2001. doi: 10.1145/502090.502097.
- [12] Edward G. Begle. The Vietoris mapping theorem for bicomact spaces. *Annals of Mathematics*, 51(3):534–543, 1950. doi: 10.2307/1969366. URL <https://webhomes.maths.ed.ac.uk/~v1ranick/papers/begle1.pdf>.
- [13] Aharon Ben-Tal and Arkadi Nemirovski. On polyhedral approximations of the second-order cone. *Mathematics of Operations Research*, 26(2):193–205, 2001. doi: 10.1287/moor.26.2.193.10561. URL https://www2.isye.gatech.edu/~nemirovs/Mor-LorentzAppr_2001.pdf.
- [14] Charles H. Bennett, Ethan Bernstein, Gilles Brassard, and Umesh Vazirani. Strengths and weaknesses of quantum computing. *SIAM Journal on Computing*, 26(5):1510–1523, 1997. doi: 10.1137/S0097539796300933.

- [15] Víctor Blanco and Miguel Martínez-Antón. On minimal extended representations of generalized power cones. *SIAM Journal on Optimization*, 34(3):3088–3111, 2024. doi: 10.1137/23M1617205. URL <https://arxiv.org/abs/2311.10470>.
- [16] Jacek Bochnak, Michel Coste, and Marie-Françoise Roy. *Real Algebraic Geometry*. Springer, 1998. doi: 10.1007/978-3-662-03718-8.
- [17] Stephen Boyd and Lieven Vandenbergh. *Convex Optimization*. Cambridge University Press, 2004. doi: 10.1017/CBO9780511804441.
- [18] Gilles Brassard, Peter Høyer, Michele Mosca, and Alain Tapp. Quantum amplitude amplification and estimation. In *Quantum Computation and Information*, volume 305 of *Contemporary Mathematics*, pages 53–74. American Mathematical Society, 2002. doi: 10.1090/conm/305/05215.
- [19] William Browder. Fiberings of spheres and H -spaces which are rational homology spheres. *Bulletin of the American Mathematical Society*, 68(3):202–203, 1962. doi: 10.1090/S0002-9904-1962-10747-2.
- [20] Domingos Moreira Cardoso and Luís Almeida Vieira. On the optimal parameter of a self-concordant barrier over a symmetric cone. *European Journal of Operational Research*, 169(3): 1148–1157, 2006. doi: 10.1016/j.ejor.2004.11.027. URL <https://optimization-online.org/wp-content/uploads/2003/11/774.pdf>.
- [21] Central-path companion. The cost of following the central path. Unpublished companion manuscript, n.d. Anonymous local manuscript; distribution-sensitive movement, primal–dual geometry, and formulation sections.
- [22] Peter Robert Chares. *Cones and Interior-Point Algorithms for Structured Convex Optimization Involving Powers and Exponentials*. PhD thesis, Université catholique de Louvain, 2009. URL <https://perso.uclouvain.be/francois.glineur/files/theses/Chares-PhD-thesis-2007.pdf>.
- [23] Yuwen Chen and Paul Goulart. An efficient implementation of interior-point methods for a class of nonsymmetric cones. *Journal of Optimization Theory and Applications*, 204:33, 2025. doi: 10.1007/s10957-024-02573-5.
- [24] Chris Coey, Lea Kapelevich, and Juan Pablo Vielma. Solving natural conic formulations with Hypatia.jl. *INFORMS Journal on Computing*, 34(5):2686–2699, 2022. doi: 10.1287/ijoc.2022.1202.
- [25] Christopher Daniel Lang Coey. *Interior Point and Outer Approximation Methods for Conic Optimization*. PhD thesis, Massachusetts Institute of Technology, 2022. URL https://chriscoey.github.io/assets/pdf/phd_thesis.pdf.
- [26] Alexander M. Dalzell, B. David Clader, Grant Salton, Mario Berta, Cedric Yen-Yu Lin, David A. Bader, Nikitas Stamatopoulos, Martin J. A. Schuetz, Fernando G. S. L. Brandão, Helmut G. Katzgraber, and William J. Zeng. End-to-end resource analysis for quantum interior-point methods and portfolio optimization. *PRX Quantum*, 4(4):040325, 2023. doi: 10.1103/PRXQuantum.4.040325.

- [27] Alexander Domahidi, Eric Chu, and Stephen Boyd. ECOS: An SOCP solver for embedded systems. In *2013 European Control Conference (ECC)*, pages 3071–3076, 2013. doi: 10.23919/ECC.2013.6669541.
- [28] Christoph Dürr and Peter Høyer. A quantum algorithm for finding the minimum, 1996. URL <https://arxiv.org/abs/quant-ph/9607014>.
- [29] Charles Ehresmann. Les connexions infinitésimales dans un espace fibré différentiable. In *Séminaire Bourbaki : années 1948/49–1950/51*, number 1 in Séminaire Bourbaki, pages 153–168. Société mathématique de France, 1952. URL https://www.numdam.org/item/SB_1948-1951_1__153_0/. Exposé 24.
- [30] Martin Engert. Finite dimensional translation invariant subspaces. *Pacific Journal of Mathematics*, 32(2):333–343, 1970. URL <https://msp.org/pjm/1970/32-2/pjm-v32-n2-p05-s.pdf>.
- [31] Jacques Faraut and Adam Korányi. *Analysis on Symmetric Cones*. Clarendon Press, Oxford, 1994. doi: 10.1093/oso/9780198534778.001.0001.
- [32] Hamza Fawzi. On representing the positive semidefinite cone using the second-order cone. *Mathematical Programming*, 175:109–118, 2019. doi: 10.1007/s10107-018-1233-0.
- [33] Hamza Fawzi and Pablo A. Parrilo. Exponential lower bounds on fixed-size psd rank and semidefinite extension complexity, 2013. URL <https://arxiv.org/abs/1311.2571>.
- [34] Hamza Fawzi and Mohab Safey El Din. A lower bound on the positive semidefinite rank of convex bodies. *SIAM Journal on Applied Algebra and Geometry*, 2(1):126–139, 2018. doi: 10.1137/17M1142570.
- [35] Hamza Fawzi and James Saunderson. Optimal self-concordant barriers for quantum relative entropies. *SIAM Journal on Optimization*, 33(4):2858–2884, 2023. doi: 10.1137/22M1500216.
- [36] Hamza Fawzi, João Gouveia, Pablo A. Parrilo, Richard Z. Robinson, and Rekha R. Thomas. Positive semidefinite rank. *Mathematical Programming*, 153:133–177, 2015. doi: 10.1007/s10107-015-0922-1.
- [37] Hamza Fawzi, João Gouveia, Pablo A. Parrilo, James Saunderson, and Rekha R. Thomas. Lifting for simplicity: Concise descriptions of convex sets. *SIAM Review*, 64(4):866–918, 2022. doi: 10.1137/20M1324417.
- [38] Yves Félix, Stephen Halperin, and Jean-Claude Thomas. *Rational Homotopy Theory*, volume 205 of *Graduate Texts in Mathematics*. Springer, 2001. doi: 10.1007/978-1-4613-0105-9.
- [39] Martin Fürer, Carlos Hoppen, and Vilmar Trevisan. Fast Gaussian elimination for low treewidth matrices. In *33rd Annual European Symposium on Algorithms (ESA 2025)*, volume 351 of *Leibniz International Proceedings in Informatics*, pages 116:1–116:15, 2025. doi: 10.4230/LIPIcs.ESA.2025.116.
- [40] Michael R. Garey and David S. Johnson. *Computers and Intractability: A Guide to the Theory of NP-Completeness*. W. H. Freeman, San Francisco, 1979.
- [41] François Glineur and Tamás Terlaky. Conic formulation for l_p -norm optimization. *Journal of Optimization Theory and Applications*, 122(2):285–307, 2004. doi: 10.1023/B:JOTA.0000042522.65261.51.

- [42] Jacek Gondzio. Presolve analysis of linear programs prior to applying an interior point method. *INFORMS Journal on Computing*, 9(1):73–91, 1997. doi: 10.1287/ijoc.9.1.73.
- [43] João Gouveia, Pablo A. Parrilo, and Rekha R. Thomas. Lifts of convex sets and cone factorizations. *Mathematics of Operations Research*, 38(2):248–264, 2013. doi: 10.1287/moor.1120.0575.
- [44] João Gouveia, Pablo A. Parrilo, and Rekha R. Thomas. Approximate cone factorizations and lifts of polytopes. *Mathematical Programming*, 151(2):613–637, 2015. doi: 10.1007/s10107-014-0848-z.
- [45] M. Seetharama Gowda and Roman Sznajder. Schur complements, Schur determinantal and Haynsworth inertia formulas in Euclidean Jordan algebras. *Linear Algebra and its Applications*, 432:1553–1559, 2010. doi: 10.1016/j.laa.2009.11.015.
- [46] Robert Grone, Charles R. Johnson, Eduardo M. Sá, and Henry Wolkowicz. Positive definite completions of partial Hermitian matrices. *Linear Algebra and its Applications*, 58:109–124, 1984. doi: 10.1016/0024-3795(84)90207-6.
- [47] Osman Güler and Levent Tunçel. Characterization of the barrier parameter of homogeneous convex cones. *Mathematical Programming*, 81:55–76, 1998. doi: 10.1007/BF01584844.
- [48] Allen Hatcher. *Algebraic Topology*. Cambridge University Press, 2002. URL <https://pi.math.cornell.edu/~hatcher/AT/AT.pdf>.
- [49] Raphael A. Hauser and Osman Güler. Self-scaled barrier functions on symmetric cones and their classification. *Foundations of Computational Mathematics*, 2(2):121–143, 2002. doi: 10.1007/s102080010022.
- [50] Kerry He, James Saunderson, and Hamza Fawzi. Interior point methods for structured quantum relative entropy optimization problems. *Mathematical Programming Computation*, 18(2):481–534, 2026. doi: 10.1007/s12532-025-00296-w.
- [51] Rowena Held, Iva Stavrov, and Brian VanKoten. (semi-)riemannian geometry of (para-)octonionic projective planes. *Differential Geometry and its Applications*, 27(4):464–481, 2009. doi: 10.1016/j.difgeo.2009.01.007. URL <https://arxiv.org/abs/math/0702631>.
- [52] Didier Henrion, Martin Kružík, and Stephan Weis. Extreme points and faces in the moment problem, 2026. URL <https://arxiv.org/abs/2606.21391v1>. Version 1.
- [53] Roland Hildebrand. A lower bound on the barrier parameter of barriers for convex cones. *Mathematical Programming*, 142:311–329, 2013. doi: 10.1007/s10107-012-0576-1.
- [54] Roland Hildebrand. Canonical barriers on convex cones. *Mathematics of Operations Research*, 39(3):841–850, 2014. doi: 10.1287/moor.2013.0640.
- [55] Roland Hildebrand. Projectively self-concordant barriers. *Mathematics of Operations Research*, 47(3):2444–2463, 2022. doi: 10.1287/moor.2021.1215.
- [56] Peter Høyer, Troy Lee, and Robert Špalek. Negative weights make adversaries stronger. In *Proceedings of the Thirty-Ninth Annual ACM Symposium on Theory of Computing*, pages 526–535, 2007. doi: 10.1145/1250790.1250867.

- [57] Pavlo A. Krokhmal and Policarpio Soberanis. Risk optimization with p -order conic constraints: A linear programming approach. *European Journal of Operational Research*, 201(3):653–671, 2010. doi: 10.1016/j.ejor.2009.03.053. URL https://stacks.cdc.gov/view/cdc/226421/cdc_226421_DS1.pdf.
- [58] Mario Kummer. Two results on the size of spectrahedral descriptions. *SIAM Journal on Optimization*, 26(1):589–601, 2016. doi: 10.1137/15M1030789.
- [59] Troy Lee, Rajat Mittal, Ben W. Reichardt, Robert Špalek, and Mario Szegedy. Quantum query complexity of state conversion. In *2011 IEEE 52nd Annual Symposium on Foundations of Computer Science*, pages 344–353, 2011. doi: 10.1109/FOCS.2011.75. URL <https://arxiv.org/abs/1011.3020>.
- [60] Yin Tat Lee and Man-Chung Yue. Universal barrier is n -self-concordant. *Mathematics of Operations Research*, 46(3):1129–1148, 2021. doi: 10.1287/moor.2020.1113. URL <https://arxiv.org/abs/1809.03011>.
- [61] John W. Milnor and James D. Stasheff. *Characteristic Classes*, volume 76 of *Annals of Mathematics Studies*. Princeton University Press, 1974.
- [62] Mohammadhossein Mohammadisiahroudi, Brandon Augustino, Pouya Sampourmahani, and Tamás Terlaky. Quantum computing inspired iterative refinement for semidefinite optimization. *Mathematical Programming*, 217:3–42, 2026. doi: 10.1007/s10107-024-02183-z.
- [63] Ashley Montanaro. Quantum search of partially ordered sets. *Quantum Information and Computation*, 9(7–8):628–647, 2009. URL <https://people.maths.bris.ac.uk/~csxam/papers/0702196.pdf>.
- [64] MOSEK ApS. *MOSEK Modeling Cookbook*, 3.4.0 edition, n.d.. URL <https://docs.mosek.com/modeling-cookbook/expo.html>. Sections 5.2.5–5.2.6; accessed 20 September 2026.
- [65] MOSEK ApS. *MOSEK Modeling Cookbook*, 3.4.0 edition, n.d.. URL <https://docs.mosek.com/modeling-cookbook/powo.html>. Section 4.2.2, p -norm cones; accessed 20 September 2026.
- [66] Ashwin Nayak. Optimal lower bounds for quantum automata and random access codes. In *Proceedings of the 40th Annual IEEE Symposium on Foundations of Computer Science*, pages 369–376, 1999. doi: 10.1109/SFFCS.1999.814608.
- [67] Ashwin Nayak and Felix Wu. The quantum query complexity of approximating the median and related statistics. In *Proceedings of the Thirty-First Annual ACM Symposium on Theory of Computing*, pages 384–393, 1999. doi: 10.1145/301250.301349.
- [68] A. B. Németh. The facial structure of the finite dimensional latticial cone. *Mathematica Pannonica*, 15(2):175–198, 2004. URL <https://mathematica-pannonica.ttk.ptt.hu/articles/mp15-2/mp15-2-175-198.pdf>.
- [69] Yurii Nesterov and Arkadi Nemirovski. Primal central paths and Riemannian distances for convex sets. *Foundations of Computational Mathematics*, 8(5):533–560, 2008. doi: 10.1007/s10208-007-9019-4.
- [70] Yurii Nesterov and Arkadii Nemirovskii. *Interior-Point Polynomial Algorithms in Convex Programming*. SIAM, Philadelphia, 1994. doi: 10.1137/1.9781611970791.

- [71] Yurii E. Nesterov and Michael J. Todd. Self-scaled barriers and interior-point methods for convex programming. *Mathematics of Operations Research*, 22(1):1–42, 1997. doi: 10.1287/moor.22.1.1.
- [72] Yurii E. Nesterov and Michael J. Todd. On the Riemannian geometry defined by self-concordant barriers and interior-point methods. *Foundations of Computational Mathematics*, 2(4):333–361, 2002. doi: 10.1007/s102080010032.
- [73] Takaaki Nomura. Grassmann manifold of a JH-algebra. *Annals of Global Analysis and Geometry*, 12:237–260, 1994. doi: 10.1007/BF02108300.
- [74] Christopher C. Paige and Michael A. Saunders. LSQR: An algorithm for sparse linear equations and sparse least squares. *ACM Transactions on Mathematical Software*, 8(1):43–71, 1982. doi: 10.1145/355984.355989.
- [75] Vitali Pirau and Roland Hildebrand. Convexity measurement of the barrier parameter functional of self-concordant barriers. In *Optimization and Applications*, volume 16426 of *Lecture Notes in Computer Science*, pages 65–75. Springer, 2026. doi: 10.1007/978-3-032-15791-1_5.
- [76] Richard Rado. A theorem on independence relations. *The Quarterly Journal of Mathematics*, os-13(1):83–89, 1942. doi: 10.1093/qmath/os-13.1.83.
- [77] Scott Roy and Lin Xiao. On self-concordant barriers for generalized power cones. *Optimization Letters*, 16(2):681–694, 2022. doi: 10.1007/s11590-021-01748-7.
- [78] Jean Saint Raymond. Local inversion for differentiable functions and the Darboux property. *Mathematika*, 49(1–2):141–158, 2002. doi: 10.1112/S0025579300016132.
- [79] Parameswaran Sankaran and Swagata Sarkar. Degrees of maps between Grassmann manifolds. *Osaka Journal of Mathematics*, 46(4):1143–1161, 2009. URL <https://www.imsc.res.in/~sankaran/Papers/grassojm.pdf>.
- [80] James Saunderson. Limitations on the expressive power of convex cones without long chains of faces. *SIAM Journal on Optimization*, 30(1):1033–1047, 2020. doi: 10.1137/19M1245670.
- [81] Claus Scheiderer. Smooth hyperbolicity cones are second-order cone representable, 2025. URL <https://arxiv.org/abs/2509.17121v2>. Version 2, 5 October 2025.
- [82] Yong Sheng Soh and Antonios Varvitsiotis. Multiplicative updates for symmetric-cone factorizations. *Mathematical Programming*, 207:427–456, 2024. doi: 10.1007/s10107-023-02015-6.
- [83] Norman Steenrod. *The Topology of Fibre Bundles*, volume 14 of *Princeton Mathematical Series*. Princeton University Press, 1951. doi: 10.1515/9781400883875.
- [84] Wim van Dam. Quantum oracle interrogation: Getting all information for almost half the price. In *Proceedings of the 39th Annual IEEE Symposium on Foundations of Computer Science*, pages 362–367, 1998. doi: 10.1109/SFCS.1998.743486.
- [85] Lou van den Dries and Chris Miller. Geometric categories and o-minimal structures. *Duke Mathematical Journal*, 84(2):497–540, 1996. doi: 10.1215/S0012-7094-96-08416-1.
- [86] Robert J. Vanderbei. Symmetric quasidefinite matrices. *SIAM Journal on Optimization*, 5(1):100–113, 1995. doi: 10.1137/0805005.

- [87] Jie Wang. Weighted geometric mean, minimum mediated set, and optimal simple second-order cone representation. *SIAM Journal on Optimization*, 34(2):1490–1514, 2024. doi: 10.1137/22M1531257. URL <https://wangjie212.github.io/jiewang/research/wgm.pdf>.